Contents

Chapter 1

Introduction

1.1 Sound laws as explicit derivations

This book is about a familiar historical-linguistic practice made explicit enough that it can be checked mechanically. Historical linguists have long described the history of languages through ordered sound laws, inherited morphological forms, lexical comparison, and exceptions that turn out to require other explanations (Osthoff and Brugmann 1878--1881). A form is accepted as inherited when its phonological shape follows from the reconstruction and the relevant history of the branch. A form becomes suspicious when the expected development fails. The method is old; the difficulty is that its moving parts are numerous. Once a project contains hundreds of lexical items and dozens of ordered changes, even a careful scholar can lose track of what follows from what.

CAPR, computer-assisted phonological reconstruction, is one way to make this situation tractable. It supplements the comparative method by forcing a proposed historical phonology to take the form of explicit rules that can be applied to concrete reconstructed inputs. A rule either produces the expected output in a particular word or misses it. A rule can be moved earlier or later in an ordered sequence, and the lexical consequences of that movement can be inspected. A proposed input form can be compared with the Old English target it is meant to explain. When the formal system fails, the failure often becomes the most useful part of the inquiry, because it tells us where the regular history ends and where analogy, lexical replacement, borrowing, dialect mixture, uncertain attestation, or a mistaken reconstruction must be considered.

The present book applies that idea to the development from Proto-Germanic and early West Germanic forms into Old English. The empirical object is modest enough to be inspected and large enough to be useful. The implemented Germanic pipeline currently derives the overwhelming majority of the Old English comparison forms in its dataset. Its remaining failures are documented exceptions, so the present write-up explains a stabilized system and leaves further phonological patching aside. It asks which sound changes are being modeled, why they stand in the order they do, which lexical items support those choices, and which words require lexical or morphological explanation beyond the regular cascade.

The book has three parts. This introduction explains the method, the empirical scope, and the conventions used in the following sections. The first main part presents the sound-change sequence itself. It gives the historical discussion, the formal rule, and the chronological evidence for the modeled changes. The second main part turns from the rules to the words. It follows individual lexical items, especially those whose derivation is philologically or morphologically informative, and asks what must be true of the reconstruction, paradigm cell, attestation, or analogical history if the Old English form is to be understood.

The two main parts answer different questions. The sound-change part asks how a coherent chronological system is built. The lexical part asks how that system accounts for individual words. A reader should be able to move in both directions. When a rule chapter says that a particular ordering is supported by *learn*, *field*, *soul*, or *shoulder*, the lexical entries show what those words require. When a lexical entry says that a form is regular, analogical, a reconstructed Old English target, or a known unmodeled exception, the sound-change chapters show what the regular expectation would have been.

1.2 What CAPR formalizes

The central claim behind CAPR is deliberately simple. A regular sound change can be treated as a relation between strings. A reconstructed form is given as an input string. An ordered sequence of sound changes maps that input to an output string. If the output matches the relevant Old English form, the derivation is successful at that level. If it fails, the failure has to be explained.

The string relation is literal. The formal background for treating phonological rules as finite-state relations is the regular-model approach to phonological rule systems developed by Kaplan and Kay (Kaplan and Kay 1994). In the implementation, a sound change is written as a finite-state transducer. A rule has a target, an output, and a conditioning environment. A series of rules is composed into a larger transducer. The resulting cascade represents a historical phonology. The details of the syntax matter for implementation, but the linguistic idea is the ordinary one. The formal rule is a way of making explicit what a historical grammar already says in prose.

A simple toy rule looks like this.

```
define ToyFinalZLoss [{*z} -> 0 || _ .#.];
```

This toy rule is only a compact illustration of the form of an explicit sound law. It says that starred **z* is deleted before the end of a word. It has a target, an output, and an environment. Real rules, written in the Foma finite-state toolkit, are often more complex (Hulden 2009). They may refer to classes of vowels or consonants, stress, syllable weight, word boundaries, or previously defined transducers. The important point is that the prose statement and the formal statement remain accountable to one another.

A CAPR history has two kinds of structure. First, there is a phonological inventory and phonotactic definition for possible ancestral forms. The system must know what counts as a possible input. Without that restriction, backward reconstruction produces many formally possible but linguistically absurd ancestors. Second, there is an ordered list of changes. The order matters because many sound changes feed or bleed one another. If an early change creates the environment for a later change, the order is feeding. If an early change removes the environment for a later change, the order is bleeding. Both relations are central to historical explanation.

The original CAPR work was designed for Burmish and made this point through a web interface that combined cognate assignment, forward projection, backward projection, and visual marking of forms that had become regular or irregular after a change to the transducers. The same methodological lesson carries over to Germanic. A formal system is useful because it makes the consequences of a proposal immediate. A new rule can make a formerly irregular item regular. It can also break a word that used to work. Both outcomes are evidence. The researcher still decides what the evidence means.

For Old English, the input side of the problem differs from the situation in a young or poorly documented language family. The comparative and philological literature is deep. The grammar of Old English has been described repeatedly, and the earlier Germanic stages have been reconstructed with great sophistication by many scholars (Campbell 1959; Hogg 1992; Ringe and Taylor 2014; Fulk 2018). That makes the formal implementation more demanding. The model must respect well-known handbooks and etymological dictionaries, and it must also explain its own deviations from citation forms when a particular paradigm cell is needed to reach the attested Old English comparison form (Orel 2003; Kroonen 2013).

1.3 Scope of this book

This book gives a formal account of a particular historical path from Proto-Germanic and early West Germanic forms into Old English. The historical grammar comes from philology, comparative grammar, and the long tradition of work on Germanic and Old English (Campbell 1959; Hogg 1992; Ringe and Taylor 2014; Fulk 2018). CAPR contributes a way of testing whether a large set of such claims can be made to cohere.

The lexical entries are selective accounts of the words that matter for the formal and historical argument. They do not attempt to give a full history of every English word, every Germanic cognate set, or every semantic development. Some entries are short because the derivation is ordinary. Some are longer because the comparison depends on a paradigm cell, an analogical form, a dialectal or attested variant, or an inherited form that differs from the dictionary citation form.

The model includes morphology where it is necessary to choose the relevant input. It includes orthography where the model must map phonological outputs to Old English surface forms. It remains a targeted derivational model with a narrower scope than a full parser of Old English, a full generator of paradigms, or a theory of all Old English variation.

The most important limitation is also one of the method's strengths. A deterministic sound-change model should keep irregularity visible. If a form is analogical, borrowed, dialectally mixed, or outside the regular path, it should remain visible as such. A model that explains too much by inventing small ad hoc sound laws has ceased to be useful. The aim is to show exactly which words are regular under a stated history, and what kind of explanation the others require.

1.4 Why formalization helps historical linguists

The traditional comparative method already contains an implicit formal discipline (Osthoff and Brugmann 1878--1881; D. Sims-Williams 2018). When we say that Proto-Germanic **p* gives Old English *f*, or that West Germanic **z* becomes *r* in the relevant environment, we are describing a relation between earlier and later forms. When we say that a change must precede another change, we are making a claim about the ordering of operations. When we reject an etymology because it would produce the wrong vowel or consonant, we are using a mental derivation.

The problem is scale. A small derivation can be checked by hand. A single famous etymology can be discussed in a footnote. A network of hundreds of lexical items and many ordered sound changes is different. Sims-Williams's discussion of mechanising historical phonology makes this point directly, showing how explicit forward application of sound changes can uncover new cognates and sharpen relative chronology (U. Sims-Williams 2018; D. Sims-Williams 2018). Errors can persist because each local step is familiar while the whole path is not inspected at once. A reconstruction may be plausible in isolation but fail after three later rules apply. A rule may work in the example for which it was introduced but damage other words. A relative chronology may seem harmless until a witness form forces a different order.

Formalization helps by turning these implicit derivations into inspectable objects. Judgment remains central, but it moves to explicit decisions about the input form, the source evidence, the interpretation of an Old English spelling, the status of a mismatch, and the historical reality of a narrow formal rule. The computer supplies consistency. It applies the rules as written, in the order given, to all relevant forms. It applies conditions uniformly, keeps every word in view, and preserves inconvenient consequences.

This is why CAPR is especially useful in the middle stage of historical work. At the beginning, one may not know enough to formalize much. At the end, one may imagine that the grammar is settled. In the middle, the investigator has many plausible sound changes, many reconstructed forms, and many lexical comparisons, but their interaction is unclear. A formal cascade creates pressure. It asks every part of the hypothesis to meet every other part.

1.5 The Germanic and Old English case

The present Germanic model begins from Proto-Germanic and West Germanic reconstructions and derives Old English comparison forms through an ordered sequence of changes (Ringe and Taylor 2014; Fulk 2018). The sequence includes large familiar developments such as West Germanic rhotacism, Anglo-Frisian brightening, breaking, i-umlaut, high-vowel apocope, and r-metathesis. It also includes smaller steps that matter because they fix particular derivations. Some changes are broad historical processes. Others are narrow formal notes whose value lies in a small group of witness forms.

The empirical dataset contains hundreds of Old English comparison forms. The current project state treats the research phase as complete. The major phonological system has moved out of debugging, and the remaining mismatches are documented as exceptions. This status matters for the prose. The book should read as an explanation of a stabilized model, with the residual problems presented honestly.

The Old English case also brings morphology to the foreground. A dictionary citation form sometimes differs from the form that should be fed to the sound-change cascade. The attested Old English comparison form may correspond to a dative plural, a genitive singular, a present-tense verbal form, or another paradigm cell. If the wrong cell is chosen, the formal derivation may fail for reasons that have little to do with phonology. The lexical entries therefore distinguish the citation reconstruction from the actual transducer input. This distinction is crucial for forms such as *sċuldrum* 'shoulders, dative plural', where a plural case form gives the regular Old English target more directly than a singular citation form.

Old English spelling adds another layer. Some outputs are phonological forms. Some are orthographic normalizations. A form may be attested with a spelling that reflects dialect, manuscript practice, or later reshaping. The model must keep these questions separate. When the lexical section identifies an Old English target, it distinguishes attested forms, normalized forms, reconstructed Old English forms, and selected comparison forms. The formal derivation is strongest when the status of the target is clear.

The result is a model that is both stricter and more modest than a prose grammar. It is stricter because every rule must run. It is more modest because it has to say when an output is only a model output, when a lexical entry requires an analogical account, and when the evidence supports only an approximate placement in the sequence.

1.6 Rules, forms, and accountability

The basic unit in the sound-change section is a rule. Each rule is presented with a historical discussion, a formal definition, and a prose explanation of what the definition does. The historical discussion explains why the change belongs in the grammar. The formal definition states the modeled operation. The chronology paragraph explains what happens if the rule is moved too early or too late, when such tests are informative.

The basic unit in the lexical section is a word or comparison form. Each entry begins with the relevant reconstruction, transducer input, Old English target, and derivation class. It then explains the comparative evidence, the Old English evidence, and the development from the selected input to the target. Complex entries may include paradigm comparison. Simple regular entries may not need one.

The relation between these two units is the main intellectual structure of the book. The rules are justified by the words. The words are explained by the rules. A sound law without lexical consequences is an empty formal gesture. A lexical derivation without an explicit rule sequence is too easy to adjust after the fact. CAPR brings the two kinds of claim into the same space.

This accountability is clearest when a rule is moved. Suppose a change currently stands before breaking. If it is delayed until after breaking, several words may produce the wrong outcome. Those words then become witnesses for a terminus ante quem. The change must have applied before breaking in the modeled sequence. If the same rule can be moved earlier with no effect on any checked form, the model has only a one-sided chronological constraint. The rule may still be historically real, and its placement may still be motivated by handbooks, branch history, or structural coherence. The checked forms by themselves fix only part of its position.

This distinction recurs throughout the sound-change section. Some rules have tight two-sided evidence. Some have a close relation on one side and a broad interval on the other. Some are placed chiefly because the sources treat a development as belonging to a historical cluster, even though the checked forms leave the nearest neighbor unresolved. The prose tries to preserve these differences. A formal model can make chronology look sharper than the evidence warrants. The book therefore separates what the forms prove, what the sources motivate, and what the implementation chooses in order to keep a coherent cascade.

1.7 Relative chronology

Relative chronology is one of the main reasons to formalize sound change. A sound law is rarely just a statement that one segment becomes another. It is also part of a history in which earlier outputs become later inputs. The order of changes determines whether a form survives, feeds a later process, escapes a later process, or becomes opaque.

The sound-change section uses lexical consequences to explain order. The argument comes from the derivational effect of moving the rule; the number assigned to it is only a reference handle. If a delayed rhotacism produces the wrong form for *learn*, then *learn* is evidence about the relative order of rhotacism and the later change that it must precede. If moving an unstressed-vowel rule earlier makes no difference to any checked item, then the earlier side of the rule is not fixed by the current lexical tests.

This method is powerful because it makes negative space visible. Traditional prose often says that one change came before another because the result requires it. CAPR can show how broad that requirement is. A rule may need to precede some later development, but that later development may be far away in the sequence. Such a result is real evidence, although weaker than local adjacency. Conversely, a rule may need to follow an earlier change because the earlier change creates exactly the environment that the later rule consumes. That is a stronger local relation.

The book therefore avoids symbolic order notation in the main prose. It uses ordinary temporal language such as before, after, earlier, and later, then gives the lexical forms that break when the order is changed. This keeps the chronology tied to historical evidence. The formal model supplies the test; the prose explains the linguistic reason.

Some changes remain approximate. This is unavoidable. Handbooks may agree that a development belongs to an early West Germanic stage without fixing its position relative to every other early change. A small rule may be needed for a particular group of forms but lack enough witnesses to anchor both sides of its chronology. A technical stage may be useful in the implementation while lacking the status of an ordinary historical sound law. The draft says so where relevant. Formalization should clarify uncertain history, not convert it into false precision.

1.8 The lexical entries

The lexical section answers a different kind of question. A sound-change chapter may say that a rule is supported by a set of witness forms, but a reader also needs to know what those forms are doing. Is the Proto-Germanic input a citation form or a particular inflectional cell? Is the Old English target a headword, a normalized spelling, an attested variant, or a reconstructed form? Do the handbooks agree on the reconstruction? Does the derivation depend on regular sound change, analogy, or a known unmodeled development?

The lexical entries make those distinctions explicit. Each entry begins with a small amount of metadata because the reader needs to know the selected input and target. The entry then turns to historical prose. The expected structure is simple. First, establish the transducer input and output. Then compare the reconstruction and the relevant source evidence. Then establish the

Old English evidence. Then explain the development to Old English. If the argument depends on morphology, add a paradigm comparison.

The model entry for *shoulder* shows why this structure is needed. The ordinary Old English headword is *sculdor*, but the relevant comparison form for the successful derivation is the dative plural *sċuldrum*. The sources differ in the reconstruction and stem-class implications of the Germanic word. A dative or instrumental plural input of the type *skúldramiz* aligns with the inherited plural ending and produces the Old English target. A singular-oriented input answers a different question. The lexical entry is where the model's choice of input becomes philologically accountable.

The derivation classes help keep the entries honest. A *regular* entry is one in which the selected input and the deterministic rule path produce the Old English target. An `early_analogy` entry uses an input that reflects reshaping before the modeled Old English path. A `late_analogy` entry often depends on a paradigm cell whose form was later generalized or otherwise analogically important. An `attested_variant` entry reaches a real Old English form that may differ from the default dictionary headword. A `reconstructed_oe` entry uses a reconstructed target when that is the useful comparison. A `known_unmodelled` entry names an understood historical process that falls outside the deterministic FST. An `unexplained_unmodelled` entry preserves a real problem without inventing a solution.

These labels are working categories for accountability, deliberately narrower than a typology of all historical change. They tell the reader what kind of explanation the entry is offering. They also prevent the regular system from being quietly expanded to cover phenomena that should remain outside the sound-change cascade.

1.9 The role of failure

A failed derivation is often more informative than a successful one. If the system derives *dæg* from the selected input by the expected sequence, the entry may be straightforward. If it derives *sċoldor* when the relevant Old English form is *sċuldrum*, the mismatch forces a question. Was the wrong input chosen? Is the relevant Old English form a different paradigm cell? Is an analogical development involved? Is the target itself wrong? Is the sound-change rule missing a condition?

CAPR is useful because it makes these questions concrete. A failure has a shape. The output differs in a particular vowel, consonant, ending, or orthographic normalization. That difference points toward a type of problem. A wrong root vowel may indicate an ordering issue or a stem-class issue. A wrong ending may indicate the wrong paradigm cell. A plausible but non-attested output may point to analogy. A form that works only by suppressing a general rule may be a loan or a dialectal intrusion.

The original CAPR Burmish work made this point through the automatic detection of loanwords. If a rigorous model of regular sound change fails for a word while succeeding for the inherited vocabulary around it, the failure can reveal a non-inherited item. The Old English case is different in detail, but the logic is the same. A good regular model isolates irregularity and keeps it visible.

This is why the remaining mismatches in the Germanic model are treated as documented exceptions, which closes the route of adding more phonology for its own sake. A model with a few well-understood failures can be better than a model that achieves total coverage through artificial rules. The governing question is whether each success and each failure has the right kind of explanation.

1.10 Formalization and philology

The formal model and the philological sources have different strengths. The model is good at consistency. The sources are good at historical interpretation, attestation, dialect, morphology, and comparison. The book depends on both.

The rule chapters cite the major handbooks and specialist sources where they are needed. Hogg, Campbell, Ringe and Taylor, Fulk, Luick, Brunner, Kroonen, Orel, and others provide the historical grammar and etymological framework within which the formalization makes sense (Campbell 1959; Hogg 1992; Ringe and Taylor 2014; Fulk 2018; Luick 1914--1940; Sievers 1965; Orel 2003; Kroonen 2013). The FST operationalizes claims from these sources and asks what follows.

The lexical entries likewise begin from dictionaries, grammars, etymological works, and the Old English record. Attestation requires source support beyond appearance as a model target. A reconstruction requires historical support beyond a successful transducer output. When sources differ, the entry should identify the point of disagreement, such as stem class, gender, root vowel, suffix, branch stage, paradigm cell, or Old English attestation. Vague language about "source disagreement" is too weak.

This division of labor also matters for uncertainty. A source may support the historical reality of a change without fixing its exact position in the modeled sequence. A transducer may require a formal step whose historical scope is smaller than its code name suggests. A lexical entry may reach the target only by choosing a morphologically specific input. These facts should be visible. The book's prose therefore tries to write neither as code documentation nor as unformalized narrative. It is historical-linguistic prose under formal constraint.

1.11 The formal sequence

The sound-change section begins with early West Germanic consonant and vowel changes and ends with Old English r-metathesis. The numbering follows the CAPR sound-change inventory. The numbers are useful because they keep the prose, the FST, and the chronology tests aligned. The argument itself lies in the historical discussion, the formal rule, and the lexical consequences.

Several internal numbers are technical or weight-marking stages, and the inventory also contains a numbering gap. This is normal in a formal system. Some implementation steps help the model run without corresponding to a historical sound law that deserves prose treatment. The introduction flags this so that the reader expects discontinuity in the internal labels.

The assembled sound-change section now covers the active manifest-backed sequence from the early West Germanic material through r-metathesis. Its chapters differ in scale. West Germanic rhotacism, Anglo-Frisian brightening, breaking, i-umlaut, apocope, and r-metathesis are large historical developments familiar from the literature. Other chapters treat narrower matters, including vowel adjustments, final-vowel behavior, cluster simplification, and special contexts needed to keep witness words regular. The book preserves this unevenness because the history itself is uneven.

A formal sound-change chapter normally has three obligations. It must identify the historical development, show the formal rule, and say how the modeled order is supported. If one of these obligations is weak, the prose should say so. A rule may be historically secure but chronologically loose. A rule may be chronologically useful but lexically narrow. A rule may be a formal convenience whose historical interpretation is cautious. Such cases are useful precisely because formalization clarifies the evidence.

1.12 The word-by-word sequence

The lexical section gives a complementary view of the same system. It begins from a word and consults the sound laws through the derivation. For each word, the reader sees the selected Proto-Germanic or early input, the Old English comparison form, the derivation class, and the path through the modeled sound changes. Where a word needs further discussion, a lexeme report explains the sources and the reasoning.

This organization is especially important because many words are formally regular but historically interesting. A word may derive successfully only because the selected input is a particular inflected form. A word may be regular in the phonology but require a note about the Old English target. A word may have a straightforward output but a complex comparative background. Conversely, a word may be phonologically uninteresting and need no extended report.

The selective-report policy reflects this. Ordinary regular rows with no note can remain short. Rows with a note, a non-regular derivation class, or a manually supplied report need fuller discussion. The point is to spend attention where the historical argument needs it. A complete book should spare the reader identical descriptions of regular derivations while also explaining difficult words in prose.

The lexical section can therefore be read in two ways. It can be read as a derivation report, showing how the system maps inputs to outputs. It can also be read as a set of philological case studies. The second use is the more important for publication. A formal output is only as good as the input and target it connects. The lexical entries show why those inputs and targets were chosen.

1.13 How to read the formal notation

The book uses several kinds of notation. Reconstructed forms are marked with an asterisk and usually appear in emphasis. Old English forms appear in emphasis in ordinary prose. FST rule names appear in the rule headings and in cross-references. Formal code appears in fenced `foma` blocks. The code blocks are the place where implementation details are visible. The surrounding prose explains the linguistic content.

The formal code should be read as a compact version of a historical claim. A line such as a vowel change before a particular class of following segments means that the model applies the change wherever the stated environment is met. If the rule is too broad, it will damage words. If it is too narrow, it will fail to derive words. If its order is wrong, it may feed or bleed later rules incorrectly.

The derivation tables in the lexical section divide the path into earlier Germanic developments and Old English developments. This division is sometimes approximate, but it helps the reader see the architecture of the model. Some words pass through many changes. Others pass through almost none. The absence of a change in a given word is as important as its presence, because it shows that the rule's environment was absent.

The book avoids treating formal labels as historical explanations by themselves. A label such as `OEIUmlaut` is a reference handle. The explanation lies in the surrounding prose, the code, and the witness forms. This is why the rule chapters give ordinary titles as well as FST names. The reader should be able to understand the linguistic development before inspecting the code.

1.14 What counts as success

A successful derivation is one in which the selected input, passed through the ordered sound-change cascade, yields the target form. That statement needs unpacking. The input must be historically defensible. The target must be identified correctly. The rule path must be regular under the model. A success that depends on a bad input or an uncertain target has only formal value.

For this reason, the book treats success as layered. A purely formal success means that the transducer output matches the target string. A philological success means that the input and target are the right objects to compare. A historical success means that the derivation also makes sense in the broader evidence of Germanic and Old English. Many entries satisfy all three. Some satisfy only the formal condition and therefore need more explanation.

The reverse is also true. A formal failure can be historically informative. If the attested Old English form is analogical, a regular sound-change model should fail to derive it from the inherited input. If the target is a dialectal or later variant, the failure may reflect the limited scope of the modeled path. If a word is borrowed, the inherited cascade is the wrong explanation. Such cases are successful diagnoses when they are labeled correctly.

The derivation classes are therefore part of the argument. They keep different kinds of success and failure apart. A `known_unmodelled` item has a different status from an unexplained mismatch. A `late_analogy` item has a different status from a regular sound-change witness. A `reconstructed_oe` target has a different status from direct attestation. These distinctions make the book more cautious, but also more useful.

1.15 Why a book is needed

CAPR is a database and a set of tools, but the present work also needs book prose. A database can tell us that an output matches or fails. It can store traces, classes, notes, and citations. Prose explains why a particular comparison matters, why a rule is historically plausible, and why an apparently small mismatch changes the analysis of a word.

The prose also lets the reader see the scale of the argument. In a table, a rule with one decisive witness and a rule with dozens of familiar examples may look similar. In a book, the difference can be stated. A narrow note can remain narrow. A broad historical change can be given fuller context. A rule whose placement is source-based can be marked as such. The written form prevents the formal system from projecting false symmetry.

The book format also makes the work contestable. A reader can disagree with a source interpretation, a selected input, a rule environment, or a derivation class. Because the claims are explicit, such disagreement has a place to attach. The ideal outcome is a set of decisions public enough to be corrected.

1.16 Scope and limits of the present draft

The present draft reflects a stabilized phase of the Germanic-to-Old-English work. The main sound-change system is treated as fixed for publication purposes unless a serious error is found. The active work is writing the sound-change chronology, explaining lexical derivations, and making the relation between formal output and philological evidence clear.

The system remains open to future improvement. A future version may add more lexeme reports, improve bibliography, refine the treatment of morphology, or extend the model to dialectal variation. A future version may also separate some compact rule groupings or rewrite some chapters in fuller historical prose. The present book has a narrower goal. It offers a coherent account of the current model and the lexical evidence that matters most for understanding it.

The lexical write-up is also at a different stage from the sound-change write-up. The sound-change sequence has received sustained prose review. The lexical section contains a mixture of generated derivation views and manifest-backed production reports. This introduction is written to support the intended combined draft, which will contain an opening methodological section, a formal chronology section, and a word-by-word section. It should be revised after the combined PDF has been read as a whole.

1.17 A small example of rule evidence

It may help to spell out the logic of a chronology test without making the introduction depend on any one rule. Imagine a rule that changes an inherited consonant before a later vowel development. If the consonant change applies first, the later vowel rule sees the right environment and the word reaches its Old English target. If the consonant change is postponed, the later vowel rule misses that environment, and the output has the wrong vowel. The word is then evidence that the consonant change must be earlier than the vowel rule.

The same reasoning also works in the other direction. If a rule creates a segment that a later rule would incorrectly alter, the rule must be placed after that later rule or else the derivation breaks. If a rule can be moved through a large stretch of the sequence without altering any checked form, the current lexical evidence leaves its position inside that stretch open. The rule may still belong there historically, but the book should report the limit of the order test.

This way of arguing is close to ordinary historical grammar. What CAPR adds is exhaustive application within the chosen dataset. A scholar can always test a rule against one or two examples by hand. The formal system tests it against the whole active set of comparison forms. The resulting evidence may confirm the expected history, expose a bad environment, or reveal that a traditional explanation is correct but less tightly ordered than the prose had suggested.

The recent revision of the sound-change chapters was largely an exercise in this kind of discipline. Earlier wording sometimes treated a broad interval as if it were a close local boundary. The revised prose says more carefully when the checked forms establish only a later relation, only an earlier relation, or a wide range within which source-based placement is still needed. This strengthens the argument by separating evidence from presentation.

1.18 Morphology as part of the input

A sound-change model can only derive the form it is given. This fact is obvious in computation and easy to overlook in etymological prose. A dictionary headword is often a useful citation form, but a real Old English comparison may descend from a particular inflected form. Nouns have case and number. Verbs have tense, mood, person, and number. Stem classes and suffixes matter. If the input form has the wrong ending, the phonology may produce the wrong word for the right historical reason.

The Germanic model therefore distinguishes the citation reconstruction from the selected transducer input. The citation reconstruction identifies the lexical family. The input form identifies the form whose regular development is being tested. Sometimes these are the same. Sometimes they diverge. When they diverge, the lexical entry has to explain why the selected input is legitimate.

This distinction is especially important for analogical cases. A late analogical form may be phonologically regular from a particular paradigm cell while the citation form gives a different result. The entry must then show both sides of the argument. It must say what the ordinary citation comparison would produce, why that output misses the Old English target, and why the chosen cell or reshaped form is a plausible historical source for the attested form. A table of outputs can make the difference clear, while the historical interpretation belongs in prose.

The same point applies to early analogy. If a form was already reshaped before the modeled Old English sound changes, the model should begin from the reshaped input. This is legitimate when the reshaping is independently motivated and stated plainly. A silent adjustment of the input merely to make the output work would be illegitimate. The lexical entries expose the chosen input and ask whether it is historically defensible.

1.19 Orthography and surface form

Old English orthography is another reason why derivational success has to be interpreted carefully. The model may produce a phonological output that then requires orthographic normalization. Some outputs contain symbols used internally by the model to represent phonetic or phonological values. Some Old English targets are normalized editorial forms. Some spellings are dialectal, manuscript-specific, or late. A visible spelling mismatch can be compatible with phonological success, and a visible spelling match may still need historical explanation.

The orthography-and-surface part of the derivation report records the final mapping from model output to Old English comparison form. In simple cases this is almost trivial. In others, it is where the reader sees that the regular phonology has produced a form that still needs editorial interpretation. The lexical entries then decide how much of that interpretation matters for the historical argument.

This matters because computational systems are exact about strings. A human reader may see immediately that two spellings are equivalent for the purpose at hand. The program needs that equivalence to be encoded. Exactness is useful when it catches real problems, but unhelpful when it mistakes notation for history. The book therefore treats orthographic normalization as a distinct layer. It is part of the pipeline and distinct from sound change.

1.20 Loanwords, analogy, and the boundary of the regular

The Neogrammarian commitment to regular sound change is often misunderstood as a claim that every word must obey the same history (Osthoff and Brugmann 1878--1881). In practice, the point is methodological. If sound change is regular, then departures from the regular path are meaningful. They may indicate borrowing, analogy, expressive reshaping, dialect mixture, textual variation, or error. The regular model is the instrument that lets us see those departures.

In the original Burmish CAPR setting, this logic made loanwords stand out. A word that resisted derivation by the inherited sound changes, while its neighbors succeeded, became a candidate for borrowing. In the Old English setting, analogy is more often the central issue. A word may be inherited, but the attested form may reflect leveling from another paradigm cell. A form may preserve a vowel because of a morphological environment that the simple citation form lacks. A verb may require a finite form as input because the infinitive citation form answers a different question.

The book therefore treats the boundary of the regular as a positive result. A known unmodeled item is resolved when the relevant process is named and supported. An unexplained unmodeled item remains a problem, but it is a contained problem. A lexeme retargeting is a substantive historical claim that the previous cognate or target assignment was inferior. These categories keep the non-regular material visible without letting it dissolve the regular grammar.

1.21 Local decisions and global coherence

A historical grammar is often written locally. One chapter discusses breaking. Another discusses i-umlaut. A dictionary entry discusses a word. Each discussion may be correct in isolation. CAPR asks whether all of them can be true at once. This is the global question.

Global coherence allows rules with different status. A model can contain major rules, minor rules, technical stages, and lexical exceptions. The key requirement is that the interaction among them has been tested. A change that works only because an earlier rule has already removed a segment has a different status from a change that can float across twenty positions unnoticed. A word that supports a local order relation carries more chronological weight than a word that merely remains regular throughout a broad interval.

The introduction of explicit rules also encourages a useful kind of humility. It is easy to write prose that sounds settled. It is harder to write code that derives the forms. When the code fails, the prose must change. When the code succeeds too easily because a rule is overbroad, the prose must also change. The formal model is a check on the coherence of philological claims and remains subordinate to philology.

1.22 Reproducibility and disagreement

One of the most important virtues of a formalized reconstruction is that it can be disagreed with precisely. A reader may think a rule is too broad. A different reader may accept the rule but reject its order. Another may object to the selected input for a particular lexical item. Another may think the Old English target has been normalized too aggressively. These are productive disagreements because each one points to an explicit part of the system.

Purely narrative accounts can leave more work implicit. If a grammar says that a development "regularly" gives a certain form, the reader may have to reconstruct the intermediate steps mentally. If an etymological note says that a word is analogical, the relevant paradigm comparison may remain implicit. CAPR aims to make those hidden steps public.

The model remains scholarly and inspectable. Every formalization reflects decisions about the inventory of symbols, the segmentation of forms, the selected reconstructions, the ordering of changes, the treatment of morphology, and the interpretation of Old English evidence. The point is that these decisions are recorded in a way that lets others inspect them. Reproducibility here means the reproducibility of consequences. Given these inputs and these rules in this order, the stated outputs follow.

1.23 From tool to argument

The project began as a tool, but the book must become an argument. Its argument is that traditional rule-based reconstruction gains clarity when its derivations are made explicit. The Old English material is a historically rich case in which the strengths and limits of formalization can be seen.

The sound-change section argues that a large part of the Proto-Germanic to Old English path can be represented as an ordered sequence of explicit rules, with the relative chronology supported by identifiable lexical consequences. The lexical section argues that individual words can be made accountable to that sequence without erasing the philological complexity of reconstruction, attestation, and analogy. The introduction argues that these two forms of accountability belong together.

The book should be judged by both standards. If the formal system derives the words while the prose hides uncertainty, it fails as historical linguistics. If the prose is philologically subtle while the formal system fails to reproduce the derivations, it fails as a formal account. The goal is to hold both demands at once.

1.24 How the parts fit together

The introduction gives the reader the method. The sound-change section gives the rule sequence. The lexical section gives the word evidence. None of the three is complete without the other two.

The method matters because formal derivations can look forbidding without an explanation of their purpose. The rule sequence matters because lexical entries can become arbitrary without an explicit sound-change history. The lexical entries matter because a rule sequence can become detached from evidence unless it is held against actual words.

A reader interested primarily in historical phonology should begin with the sound-change section, then follow the cross-references into the lexical entries. A reader interested in an individual Old English word should begin with the lexical entry and then consult the rule chapters for the changes mentioned in the derivation. A reader interested in computational method should read the whole book as an example of how a traditional historical grammar can be made operational without surrendering philological judgment.

The aspiration is practical. Historical linguistics already has a rich tradition of rigorous argument. CAPR gives that tradition a test bench. It lets us ask whether the rules we say we believe actually derive the words we claim they derive, in the order we claim they occurred. That question is simple. Answering it carefully is the work of the book.

Part I

Sound changes, formalization, and relative chronology

Chapter 2

The ordered sound-change sequence

2.1 Scope and orientation

This section follows a continuous sequence from early West Germanic consonant and vowel changes to the Old English developments that culminate in r-metathesis.

Some chapters treat large familiar changes such as rhotacism, brightening, breaking, umlaut, and apocope. Others are smaller ordered notes whose importance lies in the witness words that fix the sequence.

Not every numbered step has the same historical weight. Each section either represents a meaningful change of its own or helps explain why neighboring changes stand where they do.

Numbering note

The rule numbers follow the CAPR sound-change inventory so that each chapter remains traceable to the implementation and the chronology tests.

A few internal numbers do not become sound-change chapters. SC038, SC062, and SC084 are technical or weight-marking stages, and SC077 is a numbering gap.

2.2 West Germanic rhotacism

Historical discussion

Hogg states that Germanic **z* yielded **r* in intervocalic position in Old English, while final **z* was generally lost (Hogg 1992, 37). Ringe and Taylor argue that this merger of **z* with **r* was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (Ringe and Taylor 2014, 52, 98, 102). Crist likewise places rhotacism after earlier West Germanic **z*-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (Crist 2001, 104–6; 2002, 1, 4).

That historical label matters here. CAPR keeps the implementation name SC003 PGmcRhotacism, but the historical change treated in this chapter is a later West Germanic rhotacism, not a Proto-Germanic one. It must also remain distinct from SC020 PGmcFinalZDeletion, which removes final **z* before the surviving medial consonant is turned into **r*.

SC003. West Germanic rhotacism (`PGmcRhotacism`)

The implementation keeps the rhotacism step explicit.

```
define PGmcRhotacism [
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

In prose, the rule turns surviving medial **z* into **r* in the West Germanic line. CAPR keeps the label SC003 PGmcRhotacism for the modeled rewrite, but the historical interpretation is later than the name suggests.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC044 OEBreaking, PGmc **líznōjaną* yields *lirnian* rather than expected OE *liornian* 'learn', PGmc **líznōθi* yields *lirnaþ* rather than expected *liornaþ*, PGmc **líznô* yields *lirna* rather than expected *liorna*, and PGmc **mízdai* yields *merde* rather than expected OE *meorde* 'meed'. This shows that SC003 PGmcRhotacism must come before SC044 OEBreaking in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources treat West Germanic rhotacism as a later development after the earlier **z*-loss material described above.

2.3 Proto-West-Germanic ai-monophthongization

Historical discussion

Ringe and Taylor treat the reduction of unstressed **ai* as one of the major early vowel shifts shared across the Northwest Germanic area (Ringe and Taylor 2014, 40–41).

That historical support is strongest for the unstressed and especially word-final side of the change. CAPR makes a wider inherited **ai* treatment explicit in one rule, but the broader nonfinal **ai* > **ā* side is more sharply packaged in the implementation than in the current handbook discussion.

SC004. Proto-West-Germanic ai-monophthongization (`PWGmcAiMonophthongization`)

The implementation keeps the monophthongization step explicit.

```
define PWGmcAiMonophthongization [
    [{*ai} -> {*ē} || _ .#.]
    .o.
    [{*ai} -> {*ā}]
    .o.
    [{*ái} -> {*ā}]
];
```

In prose, the rule monophthongizes inherited **ai*. The clearest source support is for the word-final unstressed outcome, where **ai* merges with long mid **ē*; CAPR then keeps the broader inherited **ai* treatment visible in the same modeled step.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC036 OEInterStressRaising, PGmc **sáiwalō* yields *sāwel* rather than expected OE *sāwol* 'soul'. This shows that SC004 PWGmcAiMonophthongization must come before SC036 OEInterStressRaising in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the broader inherited **ai* treatment here because the clearest source support places unstressed **ai* reduction among the early Northwest Germanic vowel shifts.

2.4 Unstressed *a*-raising before final *m*

Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (Campbell 1959, 156, §373). Fulk likewise treats the development of early unstressed **o* to *u* before *m* as one of the important similarities shared by North and West Germanic (Fulk 2018, 16, §5.2).

That makes this a small but real unstressed-vowel development in inflectional material. It belongs here as a short morphophonological note, and the strongest evidence concerns noninitial unstressed material before final **m*. The internal CAPR label is narrower and more technical than the title used here.

SC005. Unstressed **a*-raising before final **m* (`NWGmcAToUBeforeM`)

The implementation keeps the pre-**m* raising step explicit.

```
define NWGmcAToUBeforeM [
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
     ↪  ({*z})? .#.
];
```

In prose, the rule raises unstressed noninitial **a* before final **m* in ending material. It preserves a narrow morphophonological step that remains visible in the `shoulder` family, but the historical case is broader than that single compact-trace witness because the strongest support comes from inflectional endings.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC017 NWGmcULowering, PGmc **skúldramiz* yields *sċoldrum* rather than expected OE *sċuldrum*. This shows that SC005 NWGmcAToUBeforeM must come before SC017 NWGmcULowering in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources place pre-**m* unstressed *u* inside the same early ending history. The note remains a small inflectional development, not a broad lexical sound law.

2.5 Early i-apocope

Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geogują* 'youth' (Sievers 1965, sects 145–146; Ringe and Taylor 2014, 141). Campbell's *dugup* and *geogup* examples belong to the same pattern (Campbell 1959, sect. 332).

This is therefore a specific kind of final-vowel loss. The crucial point is that the ending vowel disappears in a weak suffixal environment early enough to block later umlaut. That anti-umlaut timing is the historical center of the rule.

SC006. Early i-apocope (`PWGmcEarlyIApocope`)

The implementation keeps the early apocope step explicit.

```
define PWGmcEarlyIApocope [
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ .#.,
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ {*z} .#.
];
```

In prose, the rule deletes final **i* in remote unstressed syllables. That timing matters because later umlaut no longer sees the lost ending vowel, which is why forms like *geogują* 'youth' preserve the expected vocalism.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC034 OEAwLongDiphthong, PGmc **skáwōθi* yields *sċēaweþ* rather than expected OE *sċēawaþ*. This shows that SC006 PWGmcEarlyIApocope must come before SC034 OEAwLongDiphthong in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources treat this suffixal loss as an early step in the ending history before the later diphthongal developments.

2.6 Final *ō-lowering before *r

Historical discussion

Ringe and Taylor treat the West Germanic lowering of final bimoric **ō* before word-final **r* as a specific inherited development and illustrate it above all with the families behind *fēower* ‘four’ and *wæter* ‘water’ (Ringe and Taylor 2014, 58–59).

That makes the rule historically real, but narrow. This is not a broad long-vowel chapter. The relevant environment is final or pre-final **ō* before word-final **r*, and the clearest evidence remains concentrated in the `four` and `water` material.

SC007. Lowering of final bimoric *ō before *r (`PWGmcFinalOrLowering`)

The implementation keeps the final-**ō* lowering step explicit.

```
define PWGmcFinalOrLowering [
    {*ō} -> {*a} || _ {*r} .#.
];
```

In prose, the rule lowers final bimoric **ō* before word-final **r*. This is the adjustment that lies behind the West Germanic vocalism of *fēower* ‘four’ and *wæter* ‘water’.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC043 AngloFrisianBrightening, PGmc **wátōr* yields *water* rather than expected OE *wæter* ‘water’. This shows that SC007 PWGmcFinalOrLowering must come before SC043 AngloFrisianBrightening in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the `four` and `water` material belongs to the same early West Germanic final-vowel history described above.

2.7 Coronal-w assimilation

Historical discussion

Ringe and Taylor treat the assimilation of `*dw` and `*zw` to `*ww` as a shared Proto-West-Germanic innovation and support it through the `four` family and plural-pronominal forms such as `you` and `your` (Ringe and Taylor 2014, 56–57).

That historical support is real, but the witness set is small. CAPR models both coronal inputs explicitly before **w*, while the historical prose should keep the reader's attention on the narrow cluster of forms that actually supports the change.

SC008. Assimilation of coronal consonants before **w* (`PWGmcCoronalWAssimilation`)

The implementation keeps the coronal-w assimilation step explicit.

```
define PWGmcCoronalWAssimilation [
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
];
```

In prose, the rule assimilates coronal consonants before **w* so that the sequence behaves as **ww*. The lexical evidence is concentrated in the pathway to *fēower* 'four', while the pronominal material shows that the change is not confined to one isolated noun.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC031 OEWWSimplification, PGmc **fédwōr* yields *fēowwer* rather than expected OE *fēower* 'four'. This shows that SC008 PWGmcCoronalWAssimilation must come before SC031 OEWWSimplification in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the `four` and pronominal material places this assimilation in the same early West Germanic cluster before the later diphthongal developments.

2.8 *ij*-contraction in *friend*

Historical discussion

Ringe and Taylor describe a change of `*ijo` to `*iu` in the `friend` family, with the pathway PGmc **frijond-* > PWGmc **friund* > OE *frēond* 'friend' (Ringe and Taylor 2014, 62). The same source immediately warns that the `*ijo` sequence is unique enough that wider generalization is inadvisable (Ringe and Taylor 2014, 62).

That narrowness is part of the history. This is a short lexical sound-change note on a rare sequence in the `friend` family, and it belongs in a continuous account of the early sequence even though it is not a broadly productive rule.

SC009. *ij*-contraction in *friend* (`PWGmcIjContraction`)

The implementation keeps the contraction step explicit.

```
define PWGmcIjContraction [
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
];
```

In prose, the rule contracts the rare **ijō* sequence in the family behind OE *frēond* 'friend'. The section belongs here because a continuous account of the early sequence should explain that development openly, even though the source base remains effectively one family.

Its chronology is useful but one-sided. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If it is delayed until after SC032 OEDiphthongLeveling, PGmc **fríjōndz* yields *friund* rather than expected OE *frēond*. This shows that SC009 PWGmcIjContraction must come before SC032 OEDiphthongLeveling in the modeled sequence.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources treat the `friend` development as a narrow early change within the same sequence. It remains a one-family note, not a productive sound law. From here the sequence passes into the tighter local seam between SC010 PWGmcJGemination and SC011 PWGmcSyllabicJ.

2.9 West Germanic j-gemination

Historical discussion

Fulk treats West Germanic consonant gemination before `*j` after a short vowel as a regular development and illustrates it with forms such as OE *settan* 'set' and *lecgan* 'lay' (Fulk 2018, 127, §6.15).

That historical support is good, but the environment must stay explicit. This is not a general chapter on doubled consonants. The relevant setting is a short vowel before ***j*.

SC010. West Germanic j-gemination (`PWGmcJGemination`)

The implementation keeps the j-gemination step explicit.

```
define PWGmcJGemination [
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
];
```

In prose, the rule doubles consonants before ***j* after a short vowel. It preserves one of the steps behind OE *nett* 'net' and related West Germanic outcomes in this narrow environment.

Its chronology is useful but asymmetric. If the rule is moved earlier within the tested range, no checked form yields a form different from the expected one. If SC010 PWGmcJGemination is delayed past SC011 PWGmcSyllabicJ, PGmc **nátją* yields *nete* rather than expected OE *nett* 'net'. This shows that SC010 PWGmcJGemination must come before SC011 PWGmcSyllabicJ. The checked forms therefore fix the close relation between SC010 PWGmcJGemination and SC011 PWGmcSyllabicJ, but do not identify a corresponding earlier constraint. CAPR keeps the rule here because the sources treat West Germanic j-gemination as the consonantal step that must already be in place before the following syllabic-*j* development.

2.10 Syllabic j after final-vowel loss

Historical discussion

Ringe and Taylor state directly that after final unstressed `*a` and `*ą` were lost, postconsonantal `*j` became syllabic `*i`, with outcomes behind OE *here* 'army' and *rice* 'kingdom' (Ringe and Taylor 2014, 46).

That source support is good, but the compact trace layer contributes very little live evidence of its own. This therefore stays a modest singleton note. It does not expand into a broad chapter on high-vowel vocalization.

SC011. Syllabic **j* after final-vowel loss (`PWGmcSyllabicJ`)

The implementation keeps the syllabic-j step explicit.

```
define PWGmcSyllabicJ [
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
];
```

In prose, the rule turns postconsonantal **j* into syllabic **i* after final unstressed **a* or **ą* has been lost. It keeps explicit a small but historically real step behind forms such as *here* and *rice*.

Its chronology is mixed. If SC011 PWGmcSyllabicJ is moved before SC010 PWGmcJGemination, PGmc **nátją* yields *nete* rather than expected OE *nett* 'net'. This shows that SC011 PWGmcSyllabicJ must come after SC010 PWGmcJGemination. If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one. The checked forms therefore fix the earlier relation between SC010 PWGmcJGemination and SC011 PWGmcSyllabicJ, but do not identify a corresponding later constraint. CAPR keeps the rule here because the sources treat syllabic **j* as the follower to final-vowel loss once the earlier consonant adjustments are already in place.

The source support is real, but the live trace remains thin. That is why the chapter stays narrow and does not turn into a broader discussion of high-vowel behavior.

2.11 *lþ*-voicing

Historical discussion

Ringe and Taylor treat word-internal **lþ* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan*, *beald*, *wuldor*, and *gylden* (Ringe and Taylor 2014, 170–71). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold*, and *feld* (Campbell 1959, 169, §414).

That makes the change substantial enough for a short chapter, but the scope should stay cautious. The internal CAPR implementation places the rule at this early stage, while the source discussion points most clearly to a northern West Germanic development. It does not support an unqualified pan-PWGmc law.

SC012. *lþ*-voicing (`PWGmcLThVoicing`)

The implementation keeps the *lþ* > *ld* step explicit.

```
define PWGmcLThVoicing [
    {*θ} -> {*d} || {*l} _
];
```

In prose, the rule voices **lþ* to **ld*. This is the development behind families such as `field`, `fold`, `gold`, and `wold`, while the historical discussion keeps the scope cautious about how widely the rule should be projected.

Its chronology is deliberately modest. If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC012 PWGmcLThVoicing before or after any specific neighboring stage.

That does not make the change itself doubtful. The comparative evidence for northern West Germanic *lþ* > *ld* is strong, so CAPR keeps the rule here as an early consonant note. The placement should be read as approximate and source-based, not as a local ordering forced by the checked forms. After this scope-limited note, SC013 PWGmcDentalHardening returns to a broader systemic consonant adjustment.

2.12 Dental hardening

Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative `*ð` became stop `*d` in all positions (Ringe and Taylor 2014, 43).

That makes the change historically clear and systemic. The chapter treats an explicit consonantal adjustment in the early West Germanic sequence, not one narrow lexical family.

SC013. Dental hardening (`PWGmcDentalHardening`)

The implementation keeps the dental-hardening step explicit.

```
define PWGmcDentalHardening [
    {*ð} -> {*d}
];
```

In prose, the rule turns voiced dental fricative **ð* into stop **d*. It preserves a systemic step in the consonant history, not a single isolated lexical anecdote.

Its chronology is deliberately modest. If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC013 PWGmcDentalHardening before or after any specific neighboring stage.

That does not make the change itself doubtful. The comparative history of dental hardening in early West Germanic is clear, so CAPR keeps the rule here as a broad systemic consonant development. The placement should be read as approximate and source-based, while the tested forms leave the exact local neighborhood open. From here the sequence turns to SC014 NWGmcUnstressedAiMonophthongization and SC015 NWGmcILowering, where the first unstable unstressed vowels come into view.

2.13 Early unstressed vowel changes

Historical discussion of the earliest unstressed vowel changes

These two rules stand at the start of the current sequence. One removes the remaining diphthongal quality of unstressed **ai*; the other carries early unstressed front-vowel leveling farther in forms such as *weorold* 'world'. They do not carry equal chronological weight: SC014 NWGmcUnstressedAiMonophthongization is historically clear but not closely fixed by the tested forms, whereas SC015 NWGmcILowering has the stronger diagnostic constraint.

Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the broad Northwest Germanic reduction of unstressed **ai* to a long mid vowel that merges with unstressed **e* (Ringe and Taylor 2014, 37–41). That is enough to make SC014 NWGmcUnstressedAiMonophthongization historically recognizable even though the order test does not by itself determine a closer relative position.

SC014. Monophthongization of unstressed **ai* (`NWGmcUnstressedAiMonophthongization`)

The implementation keeps the monophthongization step explicit.

```
define NWGmcUnstressedAiMonophthongization [
    {*ăi} -> {*ē}
];
```

In prose, the rule removes the unstressed diphthongal quality of **ai* and merges the result with unstressed **e*. It preserves a historically plausible opening step in the early Northwest Germanic vowel history.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC014 NWGmcUnstressedAiMonophthongization before or after any specific neighboring change. CAPR places it at the beginning of the unstressed-vowel prelude because the comparative sources treat unstressed **ai* monophthongization as part of the earliest Northwest Germanic simplification of unstressed vowels. The placement should be read as approximate, not as a local ordering forced by the tested forms.

Historical discussion of early unstressed front-vowel leveling

Campbell treats the merger of unstressed front vowels directly and also records the variation of *weorold* and *weoruld* (Campbell 1959, 141–42, 154–55). That gives SC015 NWGmcILowering a clearer historical center than the change beside it.

SC015. Leveling of early unstressed front vowels (`NWGmcILowering`)

The implementation keeps the lowering step explicit.

```
define NWGmcILowering [
    {*i} -> {*e}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel,
    {*í} -> {*é}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel
];
```

In prose, the rule lowers or levels early front-vowel quality in unstressed syllables. In the current sequence, that adjustment is especially visible in the pathway to *weorold* 'world'.

Its chronology is real but one-sided. If the rule is delayed until after SC036 OEInterStressRaising, PGmc **wír-àldu* yields *wuruld* rather than expected OE *weorold* ‘world’. This shows that SC015 NWGmcILowering must come before SC036 OEInterStressRaising. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one. The checked forms therefore do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the cited `weorold` material places it inside the same early unstressed-vowel sequence.

Together these two early notes hand the sequence forward to SC016 OEWsPalatalGlide and SC017 NWGmcULowering, where the local chronology becomes tighter and the derivations more crowded.

2.14 West Saxon palatal glide and u-lowering

Historical discussion of West Saxon palatal glide and u-lowering

These two rules belong together because the same *ġeoc* 'yoke' derivation passes through both of them. Campbell treats the West Saxon rising-diphthong spellings before back vowels, and the standard lowering of *u* before a following non-high vowel is described separately in the same handbook tradition (Campbell 1959, 17, §44; Campbell 1959, 42–43, §115; Fulk 2018, 56, §4.3).

That relation is enough to justify one paired chapter, but the two changes still need their own historical discussions and rule sections. The first creates the West Saxon *ġeoc* type; the second carries the same material into the broader vowel history that follows.

Historical discussion of West Saxon palatal glide

West Saxon spellings such as *ġeoc* 'yoke', *ġeong* 'young', and *ġeoguþ* 'youth' reflect a real early development before back vowels. Campbell's short account is still the clearest handbook statement of the phenomenon (Campbell 1959, 17, §44).

This makes SC016 OEWsPalatalGlide historically clear even though its current order evidence is one-sided.

SC016. West Saxon palatal glide before back vowels (`OEWsPalatalGlide`)

The implementation states the West Saxon glide insertion directly.

```
define OEWsPalatalGlide [
    {*j} {*u} -> {*j} {*e} {*u} || .#. _,
    {*j} {*ú} -> {*j} {*é} {*u} || .#. _
] .o. [
    {*ʤ} {*u} -> {*ʤ} {*e} {*u} || .#. _,
    {*ʤ} {*ú} -> {*ʤ} {*é} {*u} || .#. _
] .o. [
    {*ʧ} {*u} -> {*ʧ} {*e} {*u} || .#. _,
    {*ʧ} {*ú} -> {*ʧ} {*é} {*u} || .#. _
] .o. [
    {*ʃ} {*u} -> {*ʃ} {*e} {*u} || .#. _,
    {*ʃ} {*ú} -> {*ʃ} {*é} {*u} || .#. _
];
```

In prose, the rule inserts a front glide after an initial palatal before back-vocalic *u*. This is the step that helps produce West Saxon forms such as *ġeoc* 'yoke'.

Its chronology is real but one-sided. If the rule is delayed until after SC017 NWGmcULowering, PGmc **júką* yields *ġoc* rather than expected OE *ġeoc* 'yoke'. This shows that SC016 OEWsPalatalGlide must come before SC017 NWGmcULowering. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one. The checked forms therefore do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the ġeoc / ġeong / ġeoguþ material belongs to the same early West Saxon palatal-glide development described above.

Historical discussion of u-lowering

After the glide-conditioned West Saxon spellings are in place, the broader Northwest Germanic lowering of *u* to *o* before a following non-high vowel provides the clearest standard sound change in this small region. Campbell and Fulk both describe that change directly (Campbell 1959, 42–43, §115; Fulk 2018, 56, §4.3).

This gives SC017 NWGmcULowering a broader source base than the more narrowly West Saxon rule beside it.

SC017. Lowering of **u* before following non-high vowels (`NWGmcULowering`)

The implementation keeps the lowering step explicit.

```
define NWGmcULowering [
    {*u} -> {*o}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
    {*ú} -> {*ó}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel
];
```

In prose, the rule lowers *u* to *o* before a following non-high vowel. This is the change behind forms such as *ġeoc* ‘yoke’, *nosu* ‘nose’, *sċofl* ‘shovel’, and *sorg* ‘sorrow’.

Its chronology is explicit on both sides. If the rule is moved before SC016 OEWsPalatalGlide, PGmc **júką* yields *ġoc* rather than expected OE *ġeoc* ‘yoke’. If it is delayed until after SC019 NWGmcFinalLongORaising, PGmc **núsō* yields *nusu* rather than expected *nosu* ‘nose’, PGmc **skúflō* yields *sċufl* rather than expected *sċofl* ‘shovel’, and PGmc **súrgō* yields *surg* rather than expected *sorg* ‘sorrow’. This shows that SC016 OEWsPalatalGlide must come before SC017 NWGmcULowering, and that SC017 NWGmcULowering must come before SC019 NWGmcFinalLongORaising.

2.15 Stressed monosyllable *ō-raising

Historical discussion

Campbell treats the development of final accented *ō* to *ū* in stressed monosyllables directly, with the familiar outcomes behind *cū* ‘cow’, *hū* ‘how’, *tū* ‘two’, and *bū* ‘both’ (Campbell 1959, 47, §122).

That is enough for a short note. The change is historically legible, but the tested forms do not by themselves determine a closer position for it.

SC018. Raising of final stressed monosyllabic *ō (`NWGmcStressedMonosyllableORaising`)

The implementation keeps the monosyllabic raising step explicit.

```
define NWGmcStressedMonosyllableORaising [
    {*ō} -> {*ū} || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
];
```

In prose, the rule raises final stressed monosyllabic **ō* to **ū*. It preserves a historically recognizable step behind forms such as *cū*, *hū*, and *tū*.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC018 NWGmcStressedMonosyllableORaising before or after any specific neighboring change. The handbooks document the raising of stressed monosyllabic **ō* as part of the early history of long vowels, and CAPR accordingly keeps it in this early vowel section. The placement should be read as approximate, not as a local ordering forced by the tested forms.

2.16 Final long-*o* raising and final *z*-deletion

Historical discussion of final long-*o* raising and final *z*-deletion

These two rules belong together because the same final-syllable structure passes through both. Ringe and Taylor describe the change of unstressed final non-nasalized long **ō* to short **u*, while Hogg and Crist treat word-final *z loss as a separate later step in West Germanic (Ringe and Taylor 2014, 30; Hogg 1992, 37; Crist 2002, 1).

That shared final-syllable history becomes especially visible in *ræste* 'rest': SC019 NWGmcFinalLongORaising must still see final **ō*, and SC020 PGmcFinalZDeletion must remove final **z* only afterward.

Historical discussion of final long-*o* raising

The first change in the pair is the Northwest Germanic raising of unstressed final long **ō* to **u*. Ringe and Taylor state that development directly in comparative terms (Ringe and Taylor 2014, 30).

This is the stage that carries forms such as *nosu*, *sċofl*, and *sorg* into the later Old English sequence.

SC019. Raising of final unstressed long **ō* (`NWGmcFinalLongORaising`)

The implementation states the final-vowel raising directly.

```
define NWGmcFinalLongORaising [
    {*ō} -> {*u}
        || EnglishStarVocalic
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
];
```

In prose, the rule raises final unstressed long **ō* to **u*. This is the step behind forms such as *nosu* 'nose', *sċofl* 'shovel', and *sorg* 'sorrow'.

Its chronology is explicit on both sides. If the rule is moved before SC017 NWGmcULowering, PGmc **núsō* yields *nusu* rather than expected OE *nosu* 'nose', PGmc **skúflō* yields *sċufl* rather than expected *sċofl* 'shovel', and PGmc **súrgō* yields *surg* rather than expected *sorg* 'sorrow'. If it is delayed until after SC020 PGmcFinalZDeletion, PGmc **rástōz* yields *rast* rather than expected *ræste* 'rest'. This shows that SC017 NWGmcULowering must come before SC019 NWGmcFinalLongORaising, and that SC019 NWGmcFinalLongORaising must come before SC020 PGmcFinalZDeletion.

Historical discussion of final *z*-deletion

The second change removes word-final **z*. Standard handbook tradition and Crist's West Germanic discussion both make that development clear, even though CAPR packages it more tightly than the morphology-heavy historical descriptions usually do (Hogg 1992, 37; Crist 2002, 1).

This is the step that closes the small final-syllable sequence and also opens the way to later final-vowel consequences farther to the right.

SC020. Deletion of word-final **z* (`PGmcFinalZDeletion`)

The implementation keeps the deletion step short.

```
define PGmcFinalZDeletion [{*z} -> 0 || _ .#.];
```

In prose, the rule deletes word-final *z. In the current sequence, this is the step that turns the protected final structure of *ræste* into its attested Old English shape after the raising rule has already applied.

Its chronology is explicit on both sides. If the rule is moved before SC019 NWGmcFinalLongORaising, PGmc **rástōz* yields *rast* rather than expected OE *ræste* 'rest'. If it is delayed until after SC040 OEMedUnstressedULowering, PGmc **bébruz* yields *befro* rather than expected *befer* 'beaver', PGmc **kwéðuz* yields *cwedo* rather than expected *cwedu* 'cud', and PGmc **félθuz* yields *feldo* rather than expected *feld* 'field', alongside eight other newly failing rows. This shows that SC019 NWGmcFinalLongORaising must come before SC020 PGmcFinalZDeletion, and that SC020 PGmcFinalZDeletion must come before SC040 OEMedUnstressedULowering.

The checked forms therefore keep the rule within a wider final-syllable interval. The earlier seam with SC019 NWGmcFinalLongORaising is the closer local result, while the later constraint at SC040 OEMedUnstressedULowering mainly shows that word-final *z-loss cannot be postponed indefinitely into the later weak-syllable sequence. CAPR keeps the rule here because the handbooks treat final *z-loss as part of the same West Germanic ending history that follows final **ō*-raising.

2.17 Unstressed **o*-raising

Historical discussion

The older history of *heofon* ‘heaven’ preserves a small but real unstressed-vowel adjustment before the later reshaping of medial vowels in Old English. Campbell derives the visible *-o-* from an earlier unstressed environment, and Ringe and Taylor keep the same family legible in the wider West Germanic record (Campbell 1959, 155–56, §373; Ringe and Taylor 2014, 287).

That is enough for a short note. The change is historically recognizable, but its current order evidence is one-sided and reaches outward to a later chapter.

SC021. Raising of unstressed **o* before later **u* (`NWGmcUnstressedORaising`)

The implementation keeps the unstressed raising step explicit.

```
define NWGmcUnstressedORaising [
    {*o} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _
     ↪  EnglishStarConsonant* {*ų}
];
```

In prose, the rule raises unstressed **o* to **u* before a later **ų*. This is the adjustment that helps keep the *heofon* derivation on its attested path.

Its chronology is real but one-sided. If the rule is delayed until after SC040 OEMedUnstressed-ULowering, PGmc **xémonų* yields *heofun* rather than expected OE *heofon* ‘heaven’. This shows that SC021 NWGmcUnstressedORaising must come before SC040 OEMedUnstressedULowering. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the *heofon* family belongs to the same early unstressed-vowel history described above.

2.18 *mn*-dissimilation

Historical discussion

The history of *mn* sequences is historically legible, but the handbooks describe it more as a small descriptive pattern than as a major isolated sound change. Campbell discusses both loss of unstressed material and later assimilation in forms of this type, including the special status of *month*-type evidence (Campbell 1959, 189, 195, §§470, 484).

That is enough for a short note. The change deserves explicit prose, but the current order evidence does not make it a strong chronological marker.

SC022. Dissimilation of *mn* sequences (`NWGmcMnDissimilation`)

The implementation keeps the dissimilation rule explicit.

```
define NWGmcMnDissimilation [
    {*m} -> {*β}
        || EnglishStarVocalic _
           EnglishStarVocalic EnglishStarConsonant* EnglishStarNasal
];
```

In prose, the rule turns an earlier *m* into *β* when another nasal follows later in the word. It preserves a small but historically recognizable step in the prehistory of forms such as *heofon* and *month*.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC022 NWGmcMnDissimilation before or after any specific neighboring change. The handbooks document *mn*-dissimilation as a real but limited tendency, but they do not give it a close relative chronology. CAPR therefore keeps the note here as a small early consonant adjustment. The placement should be read as approximate, not tightly fixed.

2.19 N-stem *n*-loss

Historical discussion

The broader history is the reduction and leveling of older n-stem endings in West Germanic. Ringe and Taylor describe the resulting syncretism in the n-stems, which is the wider morphological setting for the narrower step isolated here (Ringe and Taylor 2014, 72).

Within the current sequence, the clearest witness is the path to *dōn* 'do'. That makes the change historically legible, but still modest in scope.

SC023. Loss of n-stem **n* in final position (`NWGmcNStemNLoss`)

The implementation states the n-loss directly.

```
define NWGmcNStemNLoss [
    {*ō} {*n} -> {*ǭ} || _ .#.
];
```

In prose, the rule removes the final *n* of the relevant n-stem ending and leaves the nasalized long vowel that later developments can reshape. In the current sequence, this is the step that keeps the derivation of *dōn* on track.

Its chronology is real but one-sided. If the rule is delayed until after SC047 OEHeavySyllableNasalApocope, PGmc **dōną* no longer yields expected OE *dōn* 'do', and the row records no output at all (+*?*). This shows that SC023 NWGmcNStemNLoss must come before SC047 OEHeavySyllableNasalApocope. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources place this n-stem reduction in the same early final-ending history that eventually feeds the later apocope material. The bad outcome on the supported side must still be read as a failed derivation, not as a competing Old English surface form.

2.20 Long *ē*-lowering

Historical discussion

The later West Saxon forms *sċēap* 'sheep' and *ġēar* 'year' imply an earlier lowering of long *ē* before the palatal diphthongal outcomes described more fully later in the sequence. Campbell and Ringe and Taylor discuss those later West Saxon outputs directly (Campbell 1959, 69–70, §185; Ringe and Taylor 2014, 215–16, §6.5.1).

That is enough for a short note. The change remains historically legible, but its positive chronology points outward to a later chapter.

SC024. Lowering of long *ē* before non-nasal consonants (`NWGmcLongELowering`)

The implementation keeps the lowering step explicit.

```
define NWGmcLongELowering [
    {*ē} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal],
    {*ḗ} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal]
];
```

In prose, the rule lowers long *ē* to *ǣ* before non-nasal consonants. This is the earlier adjustment behind the later West Saxon outputs seen in *sċēap* and *ġēar*.

Its chronology is real but one-sided. If the rule is delayed until after SC056 OEWsPalatalDiphthongization, PGmc **skḗpą* yields *sċīep* rather than expected OE *sċēap* 'sheep', and PGmc **jḗrą* yields *ġīer* rather than expected *ġēar* 'year'. This shows that SC024 NWGmcLongELowering must come before SC056 OEWsPalatalDiphthongization. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources treat the lowering as the earlier stage behind the later West Saxon outputs *sċēap* and *ġēar*.

2.21 Long *ē* nasal-rounding

Historical discussion

Before nasals, older long *ē* can round toward the *ō*-vocalism seen later in *mōnaþ* ‘month’ and *mōna* / *mōn*-type material. Campbell treats this split directly in his discussion of Germanic long *ē* before nasal consonants (Campbell 1959, 53, §129).

That is enough for a short note. The change is historically legible, but the tested forms do not make it a close chronological anchor.

SC025. Rounding of long *ē* before nasals (`NWGmcLongENasalRounding`)

The implementation states the rounding step directly.

```
define NWGmcLongENasalRounding [
    {*ē} -> {*ō} || _ EnglishStarNasal,
    {*ḗ} -> {*ō} || _ EnglishStarNasal
];
```

In prose, the rule rounds long *ē* to *ō* before nasals. It preserves a historically intelligible step behind month-type and moon-type outcomes without claiming more chronology than the current testing supports.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC025 NWGmcLongENasalRounding before or after any specific neighboring change. The handbooks document month-type and moon-type outcomes from older long *ē* before nasals, but they do not give the change a close local chronology of its own. CAPR keeps the note here beside the surrounding *ē*-developments for that reason. The placement should be read as approximate and source-based.

2.22 Nasal spirant changes

Historical discussion of nasal loss before spirants and compensatory lengthening

These two rules belong together because they are CAPR's formal articulation of one older development. Campbell describes the process as nasal loss before voiceless spirants with compensatory lengthening and nasalization of the preceding vowel, and Ringe and Taylor treat the same outcomes within inherited northern West Germanic development, not as an isolated late Old English innovation (Campbell 1959, 47, §121; Ringe and Taylor 2014, 140–41).

That shared history also explains the local interaction. SC026 NWGmcNasalSpirantLengthening adjusts the vowel while the nasal plus spirant sequence is still present, and SC027 NWGmcNasalSpirantLoss then removes the nasal. The pair is therefore more than a mere adjacency in the cascade: the first rule prepares the environment that the second rule closes.

SC026. Lengthening before nasal plus spirant (`NWGmcNasalSpirantLengthening`)

The implementation keeps the vowel change explicit across the relevant nasal-spirant environments.

```
define NWGmcNasalSpirantLengthening [
    {*a} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*e} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*i} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*o} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*u} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*æ} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*á} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*é} -> {*ḗ} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*í} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ó} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative
];
```

In prose, the rule lengthens and reshapes the vowel before nasal plus voiceless spirant sequences. This is the stage that helps produce OE *fȳst* 'fist', *gōs* 'goose', and *ġeoguþ* 'youth'.

Its ordinary historical chronology is one-sided. If the rule is delayed until after SC027 NWGmcNasalSpirantLoss, PGmc **fúnxstiz* yields *fyst* rather than expected OE *fȳst* 'fist', PGmc **gánsz* yields *ġeas* rather than expected *gōs* 'goose', and PGmc **júgunθ* yields *ġeogoþ* rather than expected *ġeoguþ* 'youth'. This shows that SC026 NWGmcNasalSpirantLengthening must come before SC027 NWGmcNasalSpirantLoss. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources treat vowel lengthening and nasal loss as two parts of the same inherited nasal-spirant development.

SC027. Loss of the nasal before spirants (`NWGmcNasalSpirantLoss`)

The implementation then removes the nasal from the same environment.

```
define NWGmcNasalSpirantLoss [
    EnglishStarNasal -> 0 || _ EnglishStarVoicelessFricative
];
```

In prose, the rule deletes the nasal before a voiceless spirant after the vowel has already been adjusted. This is the stage that completes the same inherited development behind *fȳst*, *gōs*, and *ġeoguþ*.

Its ordinary historical chronology is one-sided in the opposite direction. If the rule is moved before SC026 NWGmcNasalSpirantLengthening, PGmc **fúnxstiz* yields *fyst* rather than expected OE *fȳst* 'fist', PGmc **gánsz* yields *ġeas* rather than expected *gōs* 'goose', and PGmc **júgunθ* yields *ġeogoþ* rather than expected *ġeoguþ* 'youth'. This shows that SC026 NWGmcNasalSpirantLengthening must come before SC027 NWGmcNasalSpirantLoss. If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only the earlier relation: SC027 NWGmcNasalSpirantLoss must follow SC026 NWGmcNasalSpirantLengthening. They do not identify a corresponding later constraint, and CAPR keeps the rule here because the same inherited development requires the nasal to disappear only after the preceding vowel has already been adjusted.

2.23 Preconsonantal **x*-loss

Historical discussion

Campbell explicitly treats loss of *x* and gives forms such as *fléam* 'flight' and *hēla* 'heel' as examples of the same broad development (Campbell 1959, 186, §461). That is enough to make this a historically legible change.

The present order evidence is much lighter than the historical description. This chapter therefore stays brief: the change belongs in the sequence, but current testing does not make it a strong chronological marker.

SC028. Loss of preconsonantal **x* (`NWGmcPreconsonantalXLoss`)

The implementation keeps the deletion rule explicit.

```
define NWGmcPreconsonantalXLoss [
    {*x} -> 0 || _ {*s} EnglishStarConsonant
];
```

In prose, the rule deletes **x* before **s* plus another consonant. It preserves a historically recognizable part of the older consonant history without assigning it more order-testing force than the current evidence supports.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC028 NWGmcPreconsonantalXLoss before or after any specific neighboring change. The handbooks make preconsonantal *x*-loss historically recognizable, but they do not place it precisely within this local stretch. CAPR therefore keeps it here as a short prefatory note before the better-constrained glide and fronting rules that follow. The placement should be read as approximate, not tightly fixed.

2.24 Awj glide formation and au-fronting

Historical discussion of awj glide formation and au-fronting

These two rules belong together because the same *hay* and *strew* material passes through both of them. SC029 OEAwjGlideFormation reshapes the older *awj* sequence, and SC030 OEAuFronting then fronts the resulting *au*. Campbell's discussion of these outcomes and Ringe and Taylor's derivations of *hīeġ* and *strīeġan* describe the same sequence in ordinary historical terms (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

That relation is close enough to justify one paired chapter, but the two rules still need separate historical discussions and separate chronology paragraphs. The first rule prepares the sequence; the second carries it forward into SC032 OEDiphthongLeveling.

Historical discussion of awj glide formation

Older *awj* sequences are the source of forms such as *hīeġ* 'hay' and *strīeġan* 'strew'. Campbell treats the relevant developments directly, and Ringe and Taylor likewise trace the same material through intermediate *auj*-type stages (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

This makes SC029 OEAwjGlideFormation historically clear even though its current order evidence is one-sided.

SC029. Glide formation in **awj* (`OEAwjGlideFormation`)

The implementation keeps the glide-formation step explicit.

```
define OEAwjGlideFormation [
    {*á} {*w} {*w} {*j} -> {*áu} {*j},
    {*a} {*w} {*w} {*j} -> {*au} {*j},
    {*á} {*w}      {*j} -> {*áu} {*j},
    {*a} {*w}      {*j} -> {*au} {*j}
];
```

In prose, the rule turns older *awj* material into the glide sequence that the following fronting rule can read. This is the step behind forms such as *hīeġ* and *strīeġan*.

Its ordinary historical chronology is one-sided. If the rule is delayed until after SC030 OEAuFronting, PGmc **xáwwją* yields *hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* yields *strauian* rather than expected *strīeġan* 'strew'. This shows that SC029 OEAwjGlideFormation must come before SC030 OEAuFronting. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the sources treat `awj` reshaping as the preparatory step before the fronted diphthongal outcomes develop.

Historical discussion of au-fronting

Once the glide sequence is in place, *au*-fronting produces the fronted diphthongal outcomes that carry this material into the broader West Saxon vowel history. Campbell's account of *au* > *ēa* keeps that larger setting in view (Campbell 1959, 53–54, §135).

That is why SC030 OEAuFronting matters beyond the immediate pair: it reciprocates SC029 OEAwjGlideFormation and then passes a wider set of derivations forward into SC032 OEDiphthongLeveling.

SC030. Fronting of **au* (`OEAuFronting`)

The implementation states the fronting step directly.

```
define OEAuFronting [
    {*au} -> {*aeu},
    {*áu} -> {*áeu}
];
```

In prose, the rule fronts *au* so that later Old English diphthongal outcomes can develop in the expected way. It is the step that connects the *hay* / *strew* material to the wider diphthongal region that follows.

Its chronology is explicit on both sides. If the rule is moved before SC029 OEAwjGlideFormation, PGmc **xáwwją* yields *hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* yields *strauian* rather than expected *strīeġan* 'strew'. If it is delayed until after SC032 OEDiphthongLeveling, PGmc **galáubijaną*, **bráudą*, and **dráugmaz*, together with sixteen other derivations, fail to produce output at all (+*?*) instead of yielding expected OE *ġelīefan* 'believe', *brēad* 'bread', and *drēam* 'dream'. This shows that SC029 OEAwjGlideFormation must come before SC030 OEAuFronting, and that SC030 OEAuFronting must come before SC032 OEDiphthongLeveling.

The later failure set is broad and is best read as failed derivations. It does not present a competing set of Old English surface forms.

2.25 West Saxon diphthong sequence

Historical discussion of the West Saxon diphthong sequence

The four rules gathered here belong to one West Saxon diphthongal zone, but they do not all arise from a single historical event. Campbell discusses inherited *aw/ew* outcomes, palatal-triggered diphthongization, and later Anglian smoothing in connected but separate parts of the vowel history, and Hogg likewise treats the palatal-diphthong side as real yet uneven (Campbell 1959, 46, 53–54, 65–70, 95–96, §§120, 135–136, 170–176, 185, 223–227; Hogg 1992, 106–7, 111–12).

The closest interaction inside the sequence is between SC031 OEWWSimplification and SC034 OEAwLongDiphthong, which together shape *dēaw* 'dew' and *hēawan* 'hew'. SC032 OEDiphthong-Leveling and SC033 OEEwLongDiphthong still belong here, but they point to different parts of the same history: SC032 OEDiphthongLeveling regularizes a wider diphthongal field, while SC033 OEEwLongDiphthong carries the long *ēow* side into the later environment of SC044 OEBreaking.

Historical discussion of WW simplification

West Germanic *ww* sequences lie behind forms such as *dēaw* 'dew' and *hēawan* 'hew', and Campbell treats them as part of the early West Germanic diphthong history (Campbell 1959, 46, §120).

SC031 OEWWSimplification is the first explicit step in that sequence. It is small in form, but the later long-diphthong outcomes depend on it.

SC031. Simplification of **ww* sequences (`OEWWSimplification`)

The implementation states the simplification directly.

```
define OEWWSimplification [
    {*w} {*w} -> {*w}
];
```

In prose, the rule reduces a doubled *w* to a single *w*. That simplification is what allows the later *ēaw* rule to work with the shape seen in *dēaw* and *hēawan*.

Its ordinary historical chronology is one-sided. If the rule is delayed until after SC034 OEAwLong-Diphthong, PGmc **dáwwō* yields *dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* yields *hawan* rather than expected *hēawan* 'hew'. This shows that SC031 OEWWSimplification must come before SC034 OEAwLongDiphthong. If the rule is moved earlier within the ordinary tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier historical constraint, and CAPR keeps the rule here because the `dēaw` / `hēawan` material belongs to the same West Saxon diphthong zone that the following rule develops further.

Historical discussion of diphthong leveling

By the time the sequence reaches forms such as *hēafod* 'head', diphthongal outcomes are already being redistributed across a wider set of words. Campbell's discussion of smoothing and related later monophthongization is the clearest handbook anchor for that layer of the history, even though the rule kept here is more tightly drawn than any single textbook label (Campbell 1959, 95–96, §§223–227).

This makes SC032 OEDiphthongLeveling a real member of the sequence, even if its evidence is less self-contained than the *dēaw* / *hēawan* pair.

SC032. Leveling of diphthongal outputs (`OEDiphthongLeveling`)

The implementation keeps the leveling rule explicit.

```
define OEDiphthongLeveling [
    {*aeu} -> {*ēa},
    {*áeu} -> {*ēa},
    {*eu} -> {*ēo},
    {*éu} -> {*ēo},
    {*iu} -> {*ēo},
    {*íu} -> {*ēo},
    {*e} {*u} -> {*eo},
    {*é} {*u} -> {*éo},
    {*i} {*u} -> {*eo}
];
```

In prose, the rule regularizes several diphthongal outcomes into the West Saxon patterns that appear later in the sequence. It is the step that helps keep forms such as *hēafod* 'head' in their expected shape.

Its chronology is explicit on both sides. If the rule is moved before SC030 OEAuFronting, PGmc **galáubijaną*, **báug*, and **bráudą* fail to produce output at all (+?) instead of yielding expected OE *ġelīefan* 'believe', *bēag* 'bow', and *brēad* 'bread', alongside fifteen other failed derivations. If it is delayed until after SC040 OEMedUnstressedULowering, PGmc **xáubudą* yields *hēafud* rather than expected *hēafod* 'head'. This shows that SC030 OEAuFronting must come before SC032 OEDiphthongLeveling, and that SC032 OEDiphthongLeveling must come before SC040 OEMedUnstressedULowering. The earlier side is real, but it is expressed as failed derivations rather than as alternate surface forms.

Historical discussion of long *ēow*

The long *ēow* forms of *ċēowan* 'chew', *fēower* 'four', and *cnēow* 'knee' belong to the same West Saxon vowel region, though their clearest ordering relation points forward. Campbell's treatment of early *eu* in Old English and Ringe and Taylor's examples from *chew*, *four*, and *knee* show that this is a real part of the diphthong history (Campbell 1959, 53–54, §136; Ringe and Taylor 2014, 188, 202).

That makes SC033 OEEwLongDiphthong an essential member of the sequence, even though its strongest boundary lies ahead at SC044 OEBreaking.

SC033. Long *ēow* before following vowels and weak endings (`OEEwLongDiphthong`)

The implementation states the long-diphthong development directly.

```
define OEEwLongDiphthong [
    {*e} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*i} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*é} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*í} {*w} -> {*ēo} {*w} || _ OEEwLongContext
];
```

In prose, the rule turns *ew* and *iw* sequences into long *ēow*. This is the step behind forms such as *ċēowan*, *fēower*, and *cnēow*.

Its ordinary historical chronology is one-sided. If the rule is delayed until after SC044 OEBreaking, PGmc **kéwwaną* yields *ċeowan* rather than expected OE *ċēowan* 'chew', PGmc **fédwōr* yields *feower* rather than expected *fēower* 'four', and PGmc **knéwą* yields *cneow* rather than expected *cnēow* 'knee'. This shows that SC033 OEEwLongDiphthong must come before SC044 OEBreaking. If the rule is moved earlier within the ordinary tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier historical constraint, and CAPR keeps the rule here because the sources treat long *ēow* as part of the same diphthong region even though its strongest ordering relation points forward to breaking.

Historical discussion of long *ēaw*

After SC031 OEWWSimplification has reduced *ww* to single *w*, the remaining *aw* sequence can develop into the long *ēaw* seen in *dēaw* and *hēawan*. Campbell treats these outputs in the early diphthong history of West Germanic and Old English (Campbell 1959, 46, 53–54, §§120, 135–136).

SC034 OEAwLongDiphthong therefore closes the nearest local pair in the chapter and also points onward to SC043 AngloFrisianBrightening.

SC034. Long *ēaw* before following vowels (`OEAwLongDiphthong`)

The implementation keeps the long-*ēaw* step explicit.

```
define OEAwLongDiphthong [
    {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
    {*á} {*w} -> {*ḗa} {*w} || _ [EnglishStarVocalic | {*ô}]
];
```

In prose, the rule turns *aw* before a following vowel into long *ēaw*. This is the stage that yields forms such as *dēaw* and *hēawan*.

Its chronology is explicit on both sides. If the rule is moved before SC031 OEWWSimplification, PGmc **dáwwō* yields *dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* yields *hawan* rather than expected *hēawan* 'hew'. If it is delayed until after SC043 AngloFrisianBrightening, PGmc **skáwōjaną* yields *sċawian* rather than expected OE *sċēawian* 'show', PGmc **skáwōθi* yields *sċawaþ* rather than expected *sċēawaþ*, and PGmc **stráwą* yields *strœw* rather than expected *strēaw* 'straw'. This shows that SC031 OEWWSimplification must come before SC034 OEAwLongDiphthong, and that SC034 OEAwLongDiphthong must come before SC043 AngloFrisianBrightening.

The checked forms therefore place the rule within a wider West Saxon diphthong interval. The earlier relation to SC031 OEWWSimplification is the closer local seam, while the later edge at SC043 AngloFrisianBrightening chiefly shows that the *dēaw* / *hēawan* / *sċēawian* material must be in place before the brightening region begins. CAPR keeps the rule here because the handbooks treat those forms as one part of the West Saxon diphthong zone before the later brightening development.

2.26 Prefix and compound adjustments

Historical discussion of prefixal **a*-reduction

Weakly stressed prefixes can lose their older low vowel early in Old English, and that is the historical setting for SC035 OEPrefixAReduction. Campbell treats the small but real class of pretonic losses directly, while Ringe and Taylor's derivation of **galaubijana* gives the clearest comparative witness for the same development (Campbell 1959, 147, §354; Ringe and Taylor 2014, 245; 2014, 267).

The result is a modest rule with a narrow historical range. It matters because it gives prefixed forms the weak vowel shape that later vocalic rules inherit.

SC035. Reduction of prefixal **a* (`OEPrefixAReduction`)

The implementation states the prefixal reduction directly.

```
define OEPrefixAReduction [
    {*a} -> {*ĕ}
        || .#. {*g} _
           [EnglishStarConsonant | EnglishPalatalConsonant]
           EnglishStarVocalic
];
```

In prose, the rule reduces prefixal **ga-* to unstressed **ge-*. This is the step that gives forms such as *ġelīefan* 'believe' their expected prefix vowel.

Its chronology is one-sided but concrete. If the rule is delayed until after SC043 AngloFrisianBrightening, PGmc **galáubijanǭ* yields *ġealīefan* rather than expected OE *ġelīefan* 'believe'. This shows that SC035 OEPrefixAReduction must come before SC043 AngloFrisianBrightening. The earlier direction is not yet fixed by the checked forms, so the card does not yet show what this rule must follow.

Historical discussion of inter-stress raising

The strongest member of this chapter is SC036 OEInterStressRaising. Campbell's discussion of *weorold* / *weoruld* and Ringe and Taylor's derivation of **weraldu* > **weruldu* > OE *weorold* place the rule squarely in the history of low-stress medial vowels (Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 322, §6.3.3).

This is more than a small spelling adjustment. The rule changes the vowel that stands between stronger stress peaks, which is why its witnesses remain so useful for chronology.

SC036. Raising of medial **a* between stress peaks (`OEInterStressRaising`)

The implementation keeps both parts of the raising rule together.

```
define OEInterStressRaising [
    {*a} -> {*u}
        || PGmcStarVowel EnglishStarConsonant* _
           [EnglishStarConsonant - {*j}]+ [{*u}|{*ū}],
    {*à} -> {*u}
];
```

In prose, the rule raises medial unstressed **a* to **u* in the low-stress position between stronger syllables. This is the stage behind forms such as *sāwol* 'soul' and *weorold* 'world'.

Its chronology is explicit on both sides. If the rule is moved before SC019 NWGmcFinalLongO-Raising, PGmc **sáiwalō* yields *sāwel* rather than expected OE *sāwol* 'soul'. If it is delayed until after SC040 OEMedUnstressedULowering, PGmc **sáiwalō* yields *sāwul* rather than expected *sāwol*, and PGmc **wír-àldu* yields *weoruld* rather than expected *weorold* 'world'. This shows that SC019 NWGmcFinalLongORaising must come before SC036 OEInterStressRaising, and that SC036 OEInterStressRaising must come before SC040 OEMedUnstressedULowering.

The checked forms therefore place the rule within a broader low-stress interval. The later boundary at SC040 OEMedUnstressedULowering is the more local result inside this stretch, while the earlier relation to SC019 NWGmcFinalLongORaising mainly shows that *world*- and *soul*-type vocalism belongs after the earlier final-vowel developments. CAPR keeps the rule here because the handbooks treat these medial unstressed vowels as one historical grouping.

Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and obscured compounds (Campbell 1959, 148–49, §§356–357; Campbell 1959, 153, §367; Campbell 1959, 159, §§386–387).

That is the historical setting for SC037 OECompoundLinkingSyncope. The rule is worth stating explicitly because compounds such as *reġnboga* 'rainbow' depend on it, even though its chronology is narrower and less ordinary-historical than the rule beside it.

SC037. Syncope of compound linking vowels (`OECompoundLinkingSyncope`)

The implementation deletes the weak linking vowel in the relevant compound environment.

```
define OECompoundLinkingSyncope [
    [{*a}|{*i}|{*u}] -> 0
        || PGmcStarAcuteVowel OEAnyConsonant+ _
           OEAnyConsonant+ PGmcStarGraveVowel
];
```

In prose, the rule removes a weak linking vowel inside compounds before a following grave-stressed member. This is the step that yields forms such as *reġnboga* 'rainbow'.

The order test does not yet identify an ordinary historical stage that this rule must follow. If it is delayed until after SC038 OEStripSecondaryStress, PGmc **régna-bùgô* yields *reġnefoga* rather than expected OE *reġnboga* 'rainbow'. That result shows only that compound-linking syncope must precede the later technical stress-stripping stage built into the implementation. Because SC038 OEStripSecondaryStress is not an ordinary sound change, this is not a historical local order in its own right. CAPR keeps the rule here because the handbooks treat reduced compound junctures and unstable linking vowels as part of the same weakened-compound behavior discussed around SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

2.27 Medial unstressed vowel changes

Historical discussion of medial unstressed vowel changes

These two rules belong together because the same low-stress vocalic region supplies their witnesses, and the order evidence ties them together through *wuduwe* 'widow'. Campbell discusses both the *w*-conditioned *u* forms and the later *weorold* / *weoruld* alternation, while Ringe and Taylor give the same connection comparatively in **widuwon-*, **weraldu*, and **jugunþi* (Campbell 1959, 92, §218; Campbell 1959, 140, §332; Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 267; Ringe and Taylor 2014, 322, §6.3.3).

The pair is therefore historically tighter than a merely adjacent grouping. SC039 OEWICombinativeUUmlaut is the narrower rule, but it feeds the exact vowel sequence that SC040 OEMedUnstressedULowering must then reshape.

SC039. Combinative **u*-umlaut in *wi*-forms (`OEWICombinativeUUmlaut`)

The implementation keeps the *w*-conditioned adjustment very small.

```
define OEWICombinativeUUmlaut [
    {*í} -> {*ú}
        || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
];
```

In prose, the rule changes the first vowel of *wi*-forms under the following back-vowel conditions. This is the step that helps produce OE *wuduwe* 'widow'.

Its chronology is clear on the later side. If the rule is delayed until after SC040 OEMedUnstressedULowering, PGmc **wíduwōn* yields *wudowe* rather than expected OE *wuduwe* 'widow'. This shows that SC039 OEWICombinativeUUmlaut must come before SC040 OEMedUnstressedULowering. If the rule is moved earlier within the tested sequence, no checked form yields a form different from the expected one.

The checked forms therefore fix only that later relation. They do not identify a corresponding earlier constraint, and CAPR keeps the rule here because the `widow` material belongs to the same low-stress vocalic sequence as the following medial lowering.

SC040. Lowering of medial unstressed **u* (`OEMedUnstressedULowering`)

The implementation states the lowering rule directly.

```
define OEMedUnstressedULowering [
    {*u} -> {*o}
        || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]]
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _
           [[EnglishStarConsonant | EnglishPalatalConsonant] - {*m}]
];
```

In prose, the rule lowers medial unstressed **u* to **o* in the relevant consonantal environment. This is the stage behind forms such as *weorold* 'world'.

Its chronology is explicit on both sides. If the rule is moved before SC039 OEWICombinativeUUmlaut, PGmc **wíduwōn* yields *wudowe* rather than expected OE *wuduwe* 'widow'. If it is delayed until after SC072 OEUnstressedLongVowelShortening, PGmc **júgunθ* yields *ġeogoþ* rather than expected *ġeoguþ* 'youth'. This shows that SC039 OEWICombinativeUUmlaut must come before SC040 OEMedUnstressedULowering, and that SC040 OEMedUnstressedULowering must come before SC072 OEUnstressedLongVowelShortening.

The later relation to SC072 OEUnstressedLongVowelShortening is real, but it is much broader than the local *widow* pair. The closest chronological result inside this chapter is still the reciprocal relation between SC039 OEWICombinativeUUmlaut and SC040 OEMedUnstressedULowering.

2.28 Final bare-*a* loss

Historical discussion

The handbooks treat loss of final short low vowels as part of a broader erosion of final syllables, but that broader background still supports a short explicit rule here (Campbell 1959, 143, §341; Ringe and Taylor 2014, 60–61).

This change belongs after the medial unstressed vowel changes because it affects final syllables and leaves the low-stress interior of the word behind. It also belongs before restoration because later fronted forms depend on the environment it leaves behind.

SC041. Loss of final bare **a* (`PWGmcFinalBareALoss`)

The implementation keeps the loss of the final vowel explicit.

```
define PWGmcFinalBareALoss [
    {*a} -> 0 || _ .#.
];
```

In prose, the rule deletes a surviving final bare **a*. This is the step that prevents a large class of words from carrying a spurious final vowel into Old English.

Its chronology is broad on the left and sharper on the right. If the rule is moved before SC020 PGmcFinalZDeletion, PGmc **bárdaz* yields *bearda* rather than expected OE *beard* 'beard', and PGmc **kámbaz* yields *camba* rather than expected *camb* 'comb'. If it is delayed until after SC046 OEARestoration, PGmc **kráftaz* yields *craft* rather than expected OE *cræft* 'craft', and PGmc **dágaz* yields *dag* rather than expected *dæġ* 'day'. This shows that SC020 PGmcFinalZDeletion must come before SC041 PWGmcFinalBareALoss, and that SC041 PWGmcFinalBareALoss must come before SC046 OEARestoration.

The earlier boundary reaches across a wide stretch of the cascade and is best read as a broad limit, not a local pair. The later boundary is the nearer result: restoration needs final bare-*a* loss to have happened already.

2.29 Surviving bimoric **ō* unrounding

Historical discussion

This is a narrow prefatory rule. The handbooks do not isolate one large independent sound change under exactly this label. Still, the surviving bimoric **ō* pathway behind forms such as *ræste* 'rest' needs to be stated explicitly if the sequence is to begin cleanly before SC043 AngloFrisianBrightening. Campbell, Hogg, and Ringe and Taylor all make the surrounding fronting and restoration region historically intelligible even when this particular feeder step remains model-shaped (Campbell 1959, 52, 60, §§131, 157–158; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90).

That is enough for a short reader-facing note. The rule belongs here because it closes a small architectural seam on the left side of the brightening chapter, not because it should rival the broader historical weight of the chapters that follow.

SC042. Unrounding of the surviving bimoric **ō* (`PWGmcSurvivingBimoricOUnrounding`)

The implementation keeps the step very small and explicit.

```
define PWGmcSurvivingBimoricOUnrounding [
    {*ō} -> {*ā} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ .#.
];
```

In prose, the rule unrounds a surviving bimoric **ō* to **ā* in the environment that later feeds the fronted and restored outcome in forms such as *ræste* 'rest'.

Its chronology is real on both sides, but not equally local. If the rule is moved before SC020 PGmcFinalZDeletion, PGmc **rástōz* yields *rasta* rather than expected OE *ræste* 'rest'. If the rule is delayed until after SC043 AngloFrisianBrightening, the same PGmc form again yields *rasta* instead of *ræste*. This shows that SC020 PGmcFinalZDeletion must come before SC042 PWGmcSurvivingBimoricOUnrounding, and that SC042 PWGmcSurvivingBimoricOUnrounding must come before SC043 AngloFrisianBrightening.

The later relation to SC043 AngloFrisianBrightening is the closer local handoff. The earlier constraint at SC020 PGmcFinalZDeletion mainly shows that this feeder step belongs somewhere after the earlier final-*z sequence. CAPR keeps the rule here as a short context note immediately before brightening, and its entire chronology is still carried by the single *rest* derivation.

2.30 Anglo-Frisian brightening

Historical discussion

This chapter carries more historical weight than the narrow note before it. The change usually called Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments, and later Old English developments repeatedly presuppose that fronted stage even when they partly conceal it. Campbell gives the classical statement of the fronting itself, Hogg supplies the standard modern label pair, and Ringe and Taylor make the local chronology with breaking and restoration unusually clear (Campbell 1959, 52, §131; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90; Fulk 2018, 73–74, §§4.12–4.13).

That is why the chapter is more than a general handbook excursus. The finite-state evidence shows that the rule fronts a vowel and also creates the input that SC044 OEBreaking must read and that SC046 OEARestoration later partly reverses before back vowels.

SC043. Fronting of low **a* outside nasal environments (`AngloFrisianBrightening`)

The implementation keeps the brightening as one composed rule.

```
define AngloFrisianBrightening [
    AngloFrisianBrighteningUnstressed .o.
    AngloFrisianBrighteningStressed .o.
    AngloFrisianBrighteningLongFinal
];
```

In prose, the rule fronts low **a* to **æ*-type outcomes outside nasal environments. The composed definition reflects the fact that the transducer handles stressed, unstressed, and long-final branches separately even though the historical rule is normally discussed more compactly.

Its chronology is explicit on both sides. If the rule is moved before SC042 PWGmcSurvivingBimoricOUnrounding, PGmc **rástōz* yields *rasta* rather than expected OE *ræste* ‘rest’. If it is delayed until after SC044 OEBreaking, PGmc **sláxaną* yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. This shows that SC042 PWGmcSurvivingBimoricOUnrounding must come before SC043 AngloFrisianBrightening, and that SC043 AngloFrisianBrightening must come before SC044 OEBreaking.

That position is historically apt. The rule is early enough to feed later breaking, but not so early that the surviving-bimoric **ō* pathway on its left can be ignored. It is one of the main vocalic pivots of this part of the sequence.

2.31 Breaking and velar-fricative palatalization

Historical discussion of breaking and velar-fricative palatalization

These two rules belong together because the first establishes the local vocalic environment that the second must read. Breaking creates the *eo*-type outputs before *h*, *rC*, and *lC*, and the following velar-fricative palatalization then operates in that already reshaped environment. Campbell, Ringe and Taylor, and Fulk all make breaking a standard part of the post-brightening sequence, while the local fricative palatalization is historically narrower but still clear enough to stand beside it (Campbell 1959, 54, 166, §§139, 405–406; Ringe and Taylor 2014, 168–69, 213–14, §§6.2.1–6.2.3, 6.4.1–6.4.2; Fulk 2018, 73–74, §4.13).

That interaction is close enough to justify a shared historical discussion. Even so, the hierarchy remains uneven. Breaking is the clearer handbook center, while velar-fricative palatalization is the tighter local follower whose chronology becomes especially visible through the *feoh* and *feohtan* type derivations.

SC044. Breaking before *h*, *rC*, and *lC* (`OEBreaking`)

The implementation keeps the breaking stage as one composed rule.

```
define OEBreaking OEBreakingA
    .o. OEBreakingE
    .o. OEBreakingI;
```

In prose, the rule breaks front vowels into diphthongal outcomes before the relevant consonantal environments. This is the step that yields forms such as *feoh* 'fee' and *feohtan* 'fight'.

Its chronology is concrete on both sides. If the rule is moved before SC043 AngloFrisianBrightening, PGmc **sláxanę̨* yields *sleaan* | *slēaan* rather than expected OE *slēan* 'slay'. If it is delayed until after SC045 OEVelarFricativePalatalization, PGmc **féxu* yields *fehu* rather than expected OE *feoh* 'fee', and PGmc **féxtaną* yields *fehtan* rather than expected *feohtan* 'fight'. This shows that SC043 AngloFrisianBrightening must come before SC044 OEBreaking, and that SC044 OEBreaking must come before SC045 OEVelarFricativePalatalization.

That two-sided local seam is why SC044 OEBreaking works so well as the main center of the pair.

SC045. Palatalization of velar fricatives beside front vowels (`OEVelarFricativePalatal`

The following rule handles the local fricative palatalization.

```
define OEVelarFricativePalatalization [
    {*x} -> {*ç} || _ EnglishStarFrontVowel,
    {*ɣ} -> {*j} || _ EnglishStarFrontVowel,
    {*x} -> {*ç} || EnglishStarFrontVowel _,
    {*ɣ} -> {*j} || EnglishStarFrontVowel _,
    {*x} -> {*ç} || _ {*j},
    {*ɣ} -> {*j} || _ {*j}
]
    .o. EnglishStarAlphabet*;
```

In prose, the rule palatalizes **x* and **ɣ* beside front vowels or before **j*. In this chapter it is the local follower to breaking, not a general article on all Old English palatalization.

Its chronology is explicit on both sides. If the rule is moved before SC044 OEBreaking, PGmc **féxu* yields *fehu* rather than expected OE *feoh*, and PGmc **féxtaną* yields *fehtan* rather than expected *feohtan*. If it is delayed until after SC060 OEWsPalatalUmlaut, PGmc **séxs* yields *sihs* rather than expected OE *six*. This shows that SC044 OEBreaking must come before SC045 OEVelarFricative-Palatalization, and that SC045 OEVelarFricativePalatalization must come before SC060 OEWsPalatalUmlaut.

The later relation to SC060 OEWsPalatalUmlaut remains a cross-reference, not a reason to enlarge the chapter. The core local pair is still SC044 OEBreaking and SC045 OEVelarFricativePalatalization.

2.32 A-restoration and nasal changes

Historical discussion of A-restoration

The first member of this chapter is the clearest historical hinge in the post-brightening region. Campbell's restoration of *a* before following back vowels and Ringe and Taylor's discussion of later retraction describe the same phenomenon that the transducer keeps explicit here (Campbell 1959, 60–61, §§157–159; Ringe and Taylor 2014, 189–90, §6.3.1; Fulk 2018, 74, §4.13). The rule matters because Anglo-Frisian fronting is often visible only through the later environments that restore some of its outcomes to back *a*.

That makes SC046 OEARestoration the source-backed hinge of the chapter. The nasal rules that follow belong in the same neighborhood, but they do not carry quite the same historical weight in the handbooks.

SC046. Restoration of **a* before following back vowels (`OEARestoration`)

The implementation keeps the restoration step explicit.

```
define OEARestoration (
    {*æ} -> {*a} || _
        OEARestorationIntervening OEARestorationTriggerVowel
        - OEARestorationIntervening OEARestorationWeakTailVowel
);
```

In prose, the rule changes earlier fronted **æ* back to **a* before the relevant following back-vowel environments. This is the step that turns fronted forms such as *bæcan* back into the attested OE *bacan* 'bake'.

Its chronology is explicit on both sides. If the rule is moved before SC043 AngloFrisianBrightening, PGmc **bákaną* yields *bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáraną* yields *færan* rather than expected *faran* 'fare'. If it is delayed until after SC048 OESecondaryNasalization, PGmc **bákaną* again yields *bæcan* instead of *bacan*, and PGmc **wádaną* yields *wædan* instead of *wadan* 'wade'. This shows that SC043 AngloFrisianBrightening must come before SC046 OEARestoration, and that SC046 OEARestoration must come before SC048 OESecondaryNasalization.

The rule is therefore not a decorative aftereffect of brightening. It is a real restoration hinge with a positive local window on both sides.

Historical discussion of heavy-syllable nasal loss and secondary nasalization

The remaining two rules are more tightly paired inside the model than they are in ordinary handbook naming. Their connection is derivational and broad. SC047 OEHeavySyllableNasalApocope removes the final nasalized vowel in heavy syllables, while SC048 OESecondaryNasalization marks the preceding *a* before final *n*. The result is a large reciprocal failure set if the two are inverted. Campbell's discussion of later nasal loss and the later back-mutation environment gives the broader background, while Ringe and Taylor help with the later cross-reference toward SC059 OEBackMutation (Campbell 1959, 86, 166, §§205–206, 403; Ringe and Taylor 2014, 319, §6.9.4).

That shared discussion is justified because the two rules interact directly inside the derivation. Even so, the hierarchy remains visible: the pair is a strong computational core, but less like a classical textbook chapter than SC046 OEARestoration.

SC047. Heavy-syllable nasal apocope of final **ą* (`OEHeavySyllableNasalApocope`)

The implementation keeps the apocope step short.

```
define OEHeavySyllableNasalApocope [
    {*ą} -> 0 || OEAnyConsonant _ .#.
];
```

In prose, the rule deletes final nasalized **ą* after a heavy syllable. This is the step that prevents a large class of forms from retaining spurious weak final vowels.

Its chronology is real on both sides, though not equally local. If the rule is moved before SC034 OEAwLongDiphthong, PGmc **stráwą* yields *strœw* rather than expected OE *strēaw* 'straw'. If it is delayed until after SC048 OESecondaryNasalization, PGmc **bákaną* yields *bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* yields *binden* rather than expected *bindan* 'bind', alongside a very broad *-en* failure set. This shows that SC034 OEAwLongDiphthong must come before SC047 OEHeavySyllableNasalApocope, and that SC047 OEHeavySyllableNasalApocope must come before SC048 OESecondaryNasalization.

The earlier side is narrow, but the later side is one of the broadest reciprocal failure sets in this part of the model.

SC048. Secondary nasalization before final **n* (`OESecondaryNasalization`)

The following rule states the nasalization step directly.

```
define OESecondaryNasalization [
    {*a} -> {*ą} || _ {*n} .#.
];
```

In prose, the rule nasalizes **a* before final *n*. This is the step that helps keep the live *-an* outcomes distinct from the spurious *-en* forms that appear if the late nasal rules are misordered.

Its chronology is explicit on both sides. If the rule is moved before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* yields *bacen* rather than expected OE *bacan*, and PGmc **bíndaną* yields *binden* rather than expected *bindan*, representing the same broad reciprocal failure set. If it is delayed until after SC059 OEBackMutation, PGmc **stélaną* yields *steolan* rather than expected OE *stelan* 'steal', and PGmc **wébaną* yields *weofan* rather than expected *wefan* 'weave'. This shows that SC047 OEHeavySyllableNasalApocope must come before SC048 OESecondaryNasalization, and that SC048 OESecondaryNasalization must come before SC059 OEBackMutation.

That combination explains the chapter's internal hierarchy. SC046 OEARestoration is the clearest historical hinge, while SC047 OEHeavySyllableNasalApocope and SC048 OESecondaryNasalization form the stronger reciprocal nasal core.

2.33 B allophony and Sievers-law syncope

Historical discussion of B allophony

The first change is the positional alternation of Germanic **b*. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992, 101–2). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick's spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

This is a narrow consonantal distribution with limited independent scope, but it matters because later derivations already assume that the alternation is in place.

SC049. Distribution of **b* after vowels and liquids (`PGmcBAllophony`)

The first rule formalizes the stop-fricative alternation of Germanic **b*.

```
define PGmcBAllophony [
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
];
```

In prose, the rule says that **b* becomes a fricative after vowels and liquids, while geminate **bb* keeps the stop value.

Historically, this is the sort of narrow distributional statement that the handbooks place within the consonant system and discuss only briefly on its own. Even so, it matters because later derivations assume that the alternation is already in place. The clearest tested consequence appears in *reġnboga* 'rainbow'. If the rule is moved before the earlier linking-vowel adjustment, the derivation yields *reġnfoga* 'rainbow' rather than expected OE *reġnboga* 'rainbow'. This shows that SC037 OECompoundLinkingSyncope must come before SC049 PGmcBAllophony. No equally sharp later lexical breakpoint emerges within the tested sequence, so the rule has no explicit later boundary within the present sequence.

Historical discussion of Sievers-law syncope

Sievers' Law belongs to a different historical problem. It is a prosodic and morphological adjustment in heavy stems, not a distributional allophone of a stop consonant. Adamczyk treats the Old English reflexes of the law as real historical material in weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15).

That makes the change historically narrower but chronologically important. It is the last small feeder before the palatalization sequence begins in earnest, and its place in the cascade is clearer than that of the preceding allophony rule.

SC050. Sievers-law syncope (`SieversLawSyncope`)

The second rule removes the Sievers-law **i* before **j* after a consonant.

```
define SieversLawSyncope [
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
];
```

In plain language, the rule contracts the heavier **-CijV-** sequence to **-CjV-**. That is why it belongs to the historical aftermath of Sievers' Law and stands apart from the earlier stop-fricative distribution.

Its place in the sequence is clearer than that of the allophony rule. If the change is delayed until after SC052 OEVelarPalatalization, the cluster behind *streċċan* 'stretch' is affected too late. With PGmc **strákkijaną* in the wrong order, the derivation yields *strecċan* 'stretch'. The expected Old English form is *streċċan* 'stretch'. That is a real chronological consequence. No equally precise earlier lexical breakpoint fixes how far back the syncope must stand, so the historical picture remains asymmetric. The rule is secure as an immediate feeder into the palatalization zone, even though its earlier limit is less sharply bounded. The evidence therefore places SC050 SieversLaw-Syncope before SC052 OEVelarPalatalization.

2.34 Palatalization of **sk* to **sc*

Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992, 106–7, 111–12). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick is especially useful for the larger frame. He treats the cluster change as part of a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an excuse to merge the whole palatalization and umlaut region into one undivided chapter.

SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

The implementation states the **sk* rule explicitly.

```
define OESkPalatalization [
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
];
```

In prose, the rule turns **sk* into a palatal outcome in the environments that lead to Old English **sc*.

Its historical place is between the earlier restoration and the later palatal vowel developments. If it is moved too early, the forms behind *flasce* 'flask' and *wascan* 'wash' are fronted too soon, yielding *flæsce* 'flask' and *wæscan* 'wash' rather than expected OE *flasce* and *wascan*. This gives the earlier result. This shows that SC046 OEARestoration must come before SC051 OESkPalatalization. If it is moved too late, the cluster no longer feeds the later West-Saxon diphthongal outcomes that appear in *sċeaft* 'shaft', *sċēar* 'shear', *sċēaþ* 'sheath', *sċēap* 'sheep', and *sċield* 'shield'. That is why the rule sits naturally beside SC052 OEVelarPalatalization and before SC056 OEWsPalatalDiphthongization.

No single later wrong form is isolated for the whole group of **sċea-** / **sċie-** witnesses, but the current notes do show that the cluster must already be palatalized before the later West-Saxon diphthongal rule applies. This places SC051 OESkPalatalization before SC056 OEWsPalatalDiphthongization.

The narrower chapter shape matters. The cluster rule is real and historically visible, but it is still only one part of the broader palatalizing sequence. The change should therefore be read as a distinct cluster development inside that sequence, not as a complete account of Old English palatalization.

2.35 Velar palatalization before front vowels

Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading "Frühe Verschiebungen in palataler Richtung," he treats English `k` and `g` before bright vowels together with the larger field of palatal effects (Luick 1914--1940, 157, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological sequence. The examples associated with that sequence, such as *ceaster* 'town', *geaf* 'gave', *giefan* 'give', and *giest* 'guest', already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (Luick 1914--1940, 157–67, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone `sk` cluster. His remark that "[sk] is more prone to palatalization and assibilation than [k]" is brief, but it makes clear that different members of the larger palatal field need not behave identically (Campbell 1959, 278, §440). Elsewhere in the same part of the grammar he uses forms such as *cild* 'child', *dæg* 'day', *giefan* 'give', and *giest* 'guest', which show how palatalized velars, palatal influence, and later umlautal outcomes meet in the same region of the lexicon without collapsing into one process (Campbell 1959, 69–72, 89, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant `j` (Hogg 1992, 103–4). This formulation is important because it moves the discussion from a broad list of palatal outcomes to a more precise phonological environment involving adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that "after initial velars and **sk* had been palatalized" West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1). Their own examples of the plain-velar rule, such as *weccan* 'wake', *licgan* 'lie', *lecgan* 'lay', *secg* 'retainer', *ecg* 'edge', *wicg* 'horse', and *brycg* 'bridge', illustrate the same point in lexical detail: front vowels and `j` create the palatal environment in which plain `k` and `g` cease to behave as plain velars (Ringe and Taylor 2014, 213–14, §6.4.1).

Taken together, these accounts show a gradual tightening of focus. Luick treats palatalization as a broad early movement. Campbell distinguishes more sharply between plain velars and the `sk` complex. Hogg specifies the adjacency and syllable conditions more directly. Ringe and Taylor then place the plain velar change in an explicit sequence that leads forward to later West-Saxon diphthongization. The literature therefore supports two claims at once: the change belongs to a larger palatalizing environment, and it must be kept distinct from neighboring processes if the sequence of developments is to be described accurately.

SC052. Palatalization of **k* before front vowels and **j* (`OEVelarPalatalizationKFront`)

The first part of the implementation isolates the `k`-side environments of the change.

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*ī́},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*ī́} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*ī́} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
] ;
```

In prose, the rule turns plain `k` into a palatal outcome before front vowels and `j`, including the geminated environment before `j`.

Historically, this section corresponds to the core of the older discussion of palatalized velars. It captures the environments behind forms such as *weccan* 'wake', *licgan* 'lie', and *lecgan* 'lay', where front vowels or `j` trigger the palatal outcome in the first place (Ringe and Taylor 2014, 213–14, §6.4.1). It is also the part of the process that prepares forms later assumed by SC052 OEVelarPalatalization and, farther on, by SC055 OEIUmlautFronting.

Within the present implementation, this helper rule is not ordered separately from the broader velar-palatalization rule below. Its chronology is therefore that of the larger rule it feeds. If the palatalization complex is moved before Sievers-law syncope, PGmc **strákkijaną* yields *strećċan* 'stretch' rather than expected OE *strećċan* 'stretch'. If it is delayed beyond the umlautal core, PGmc **kūi* and **lúnganjō* yield *ċȳ* 'cows' and *lunġen* 'lungs' rather than expected OE *cȳ* and *lungen*. The shared boundary pattern is therefore clear. SC050 SieversLawSyncope must come before SC052 OEVelarPalatalizationKFront, and the palatalization complex must in turn come before SC055 OEIUmlautFronting.

SC052. Velar palatalization before front vowels (`OEVelarPalatalization`)

The broader rule adds the `g` environments and composes them with the `k` palatalization rule above.

```
define OEVelarPalatalization [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

In prose, the rule palatalizes plain `k` and `g` in front-vocalic and `j`-adjacent environments. Writing it as a separate rule clarifies the relative order of plain-velar palatalization, `sk`-palatalization, and umlautal developments.

The rule belongs after the earlier syncope that prepares forms like *strećċan* 'stretch' and before the later umlautal rules that would otherwise over-palatalize forms such as *cȳ* 'cows' and *lungen* 'lungs'. See SC055 OEIUmlautFronting and SC055 OEIUmlaut below.

If the rule is moved too early, before the syncope that prepares the consonant cluster, it breaks the derivation that should yield *strećċan* 'stretch'. With PGmc **strákkijaną* in the wrong order, the model produces *streccan* 'stretch'; the expected Old English form is *strećċan* 'stretch'.

If it is moved too late, after i-umlaut, it over-palatalizes forms such as *cȳ* 'cows' and *lungen* 'lungs'. PGmc **kūi* then yields *ċȳ* 'cows'; the expected form is *cȳ* 'cows'. PGmc **lúnganjō* yields *lunġen* 'lungs'; the expected form is *lungen* 'lungs'.

These lexical failures show that SC050 SieversLawSyncope must come before SC052 OEVelarPalatalization and that SC052 OEVelarPalatalization must come before SC055 OEIUmlaut.

Once the rule is in place, plain velars before front vowels and `j` no longer remain plain. They become the palatal outcomes presupposed by later developments, including the umlautal rules discussed in SC055 OEIUmlautFronting. That matters for dictionary-like forms such as *cild* 'child' or *dæg* 'day' and for the broader relation between consonantal palatalization and later vowel-fronting processes (Luick 1914--1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1).

The evidence places the rule within a wider palatalizing environment, but it does not require every neighboring palatal process to be merged with it. `sk` belongs to a related but distinct development, and the later umlautal material poses a different historical problem. The relation to the earlier syncope rule is likewise specific and limited: the *strećċan* ‘stretch’ evidence shows a real dependency without turning the feeder process into a coequal sound law of the same scope.

2.36 Post-velar **w*-loss and loss of **w* before final **i*

Historical discussion of early **w*-loss before umlaut

The two rules gathered here are unequal in weight. The first is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor make the historical core clear when they derive PGmc **singwan* to Old English *singan* 'sing' (Ringe and Taylor 2014, 214, §6.4.2). That gives the change a real comparative anchor, but it does not turn it into a large chapter of its own. It is the kind of small local sound change that needs a place in the sequence without claiming the status of a major handbook law.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* 'sea' from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The chapter therefore belongs in the stretch between plain palatalization and the umlautal core, but it should keep the asymmetry visible: the first rule is a narrow loss in the **ngw* sequence, and the second is a stronger glide-loss development with a specific lexical witness.

SC053. Loss of **w* after velars (`OEPostVelarWLoss`)

The first rule handles the **ngw* simplification.

```
define OEPostVelarWLoss [
    {*w} -> 0 || {*n} {*g} _
];
```

In prose, the rule removes **w* after the velar cluster in forms of the **singwan* type.

Historically, this is a very small rule. It keeps developments such as *singan* 'sing' visible in the sequence, but it does not create a large family of lexical breakpoints. If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC053 OEPostVelarWLoss before or after any specific neighboring change. CAPR keeps it here because the comparative evidence for `*singwan > singan` makes a narrow post-velar **w*-loss historically plausible in this pre-umlaut stretch. Even so, the placement should be read as approximate: the rule is a small prefatory note before the better-attested glide-loss and umlautal developments to the right.

SC054. Loss of **w* before final **i* (`OEWLossBeforeI`)

The second rule is the more historically legible member of the pair.

```
define OEWLossBeforeI [
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
];
```

In prose, the rule removes non-initial **w* before final unstressed **i*.

The best witness is *sǣ* 'sea'. Campbell's discussion of the loss of **w* before **i*, Ringe and Taylor's derivation from earlier **saiwi-* / **sawi-*, and Luick's parallel account all point to the same historical consequence (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). The glide has to disappear early enough for the preceding vowel to continue into the later fronted and lengthened outcome. If the glide survives too long, the derivation retains **w* and misses *sǣ* 'sea'. If the rule is moved before SC020 PGmcFinalZDeletion, the same witness yields *sǣw* 'sea' rather than expected OE *sǣ*. This shows that SC020 PGmcFinalZDeletion must come before SC054 OEWLossBeforeI. If the rule is delayed until after SC063 OEHighVowelApocope, the same witness again yields *sǣw* rather than expected *sǣ*. This places SC054 OEWLossBeforeI before SC063 OEHighVowelApocope.

The checked forms therefore place the rule within a wide pre-umlaut interval: after SC020 PGmcFinalZDeletion and before SC063 OEHighVowelApocope, without fixing close neighbors on both sides. CAPR keeps it here because the handbooks treat the loss of **w* before unstressed **i* as part of the pre-umlaut history behind *sǣ* ‘sea’. The modeled placement should be read as a source-based choice within that interval, with the chapter serving as a lead-in to the umlautal material and not as a locally pinned pair on both sides.

2.37 The Old English i-umlaut and West Saxon palatal diphthongization

Historical discussion of i-umlaut and West Saxon palatal diphthongization

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that "the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE" (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* 'guest', *giefan* 'give', *hierde* 'shepherd', and *ieldra* 'older' (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: "we come now to a change which is almost as uncontroversial as it is important" (Hogg 1992, 112). His examples, such as *bryd* 'bride', *trymman* 'strengthen', *bedd* 'bed', *ciest* 'chest', and *wiersa* 'worse', likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992, 112–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it "diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)" (Fulk 2018, 74, §4.13). Ringe and Taylor's examples such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

The sequence of discussion is fairly clear. Luick, Campbell, and Hogg all give i-umlaut primary importance. Ringe and Taylor and Fulk then help separate that major change from the narrower West-Saxon diphthongization that stands beside it. The literature therefore establishes a large, system-wide umlautal change and a narrower adjoining process affecting words after initial palatals. That distinction matters because the two processes act in different environments and produce different lexical consequences.

SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

The first component of the implementation handles the broad fronting of vowels under the influence of following `i` or `j`.

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In prose, the rule fronts and raises the relevant simple vowels when a following `i` or `j` provides the trigger.

Historically, this is the most central part of the umlautal development described by Luick, Campbell, Hogg, Ringe and Taylor, and Fulk. Within the present implementation it stands after SC052 OEVelarPalatalization and before the narrower West-Saxon palatal-diphthongization rule discussed below.

The handbooks describe the same conditioning environment in different ways but with the same phonological consequence: a following high front vocoid triggers the fronting of earlier back vowels. That is why forms such as *fylgan* 'follow', *gylden* 'golden', *wyrm* 'worm', and *giest* 'guest' can all be treated inside the same formal rule even though they belong to different lexical classes (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The same ordering logic that governs the umlaut complex governs this component. If the umlautal rule set is moved before SC052 OEVelarPalatalization, PGmc **kūi* yields *ċȳ* 'cows' rather than expected OE *cȳ*, and **lúnganjō* yields *lunġen* 'lungs' rather than expected OE *lungen*. At the other edge, the later West-Saxon diphthongization must follow the umlautal rule set: if that later rule is moved too early, PGmc **géftiz* yields *ġieft* 'gift' rather than expected OE *ġift*, and **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. This shows that SC052 OEVelarPalatalization must come before SC055 OEIUmlautFronting, and that SC055 OEIUmlautFronting must come before SC056 OEWsPalatalDiphthongization.

As a component rule, it shares the chronology of SC055 OEIUmlaut.

SC055. Raising under i-umlaut (`OEIUmlautRaising`)

The second component handles the raising of umlauted æ to e.

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In plain language, this rule takes the fronted low vowel created by the earlier fronting rule and raises it further where the same umlaut trigger still holds.

Historically, this belongs inside the same broad i-umlaut development. It is part of the same chronological development and shares the evidence base of SC055 OEIUmlaut.

Like the fronting component, this raising rule falls between SC052 OEVelarPalatalization and SC056 OEWsPalatalDiphthongization. If the umlaut complex is moved before SC052 OEVelarPalatalization, **kūi* yields *ċȳ* instead of expected *cȳ* and **lúnganjō* yields *lunġen* instead of expected *lungen*. If the later West-Saxon diphthongization is moved too early, **géftiz* yields *ġieft* rather than expected *ġift*, and **skáiθiz* yields *sċǣþ* rather than expected *sċēaþ*.

These outcomes show that SC052 OEVelarPalatalization must come before SC055 OEIUmlautRaising, and that SC055 OEIUmlautRaising must come before SC056 OEWsPalatalDiphthongization.

This narrower subrule matters because the sources do not describe umlaut as simple fronting alone. Campbell explicitly notes that the low front vowel changes again before `m` and `n` in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992, 112).

SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

The third component handles the diphthongal outcomes that also undergo i-umlaut.

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
```

```
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

In prose, the rule states that diphthongal inputs are subject to umlaut as well: the vowel change is not confined to simple vowels.

This matters historically because the handbooks describe i-umlaut as a system-wide assimilatory development. The rule therefore stands inside the same chronological bracket as SC055 OEIUmlautFronting and SC055 OEIUmlautRaising, even though its outputs are shaped differently.

The relevant examples are the recurring West-Saxon `ie` forms cited in the handbooks, including *giest* 'guest', *giefan* 'give', and *hierde* 'shepherd' in Campbell and *ciest* 'chest' in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992, 112–14). The present formalization keeps those diphthongal outcomes visible as a distinct part of the general umlautal development and does not leave them implicit under the broad description of fronting.

Chronologically, this component also shares the same evidence as the umlaut complex as a whole. If the umlaut complex is moved before SC052 OEVelarPalatalization, it over-palatalizes **kūi* and **lúnganjō*; too-early West-Saxon diphthongization yields *ġieft* and *sċǣþ* instead of expected *ġift* and *sċēaþ*. The rule therefore belongs between SC052 OEVelarPalatalization and SC056 OEWsPalatalDiphthongization. This places SC052 OEVelarPalatalization before SC055 OEIUmlautDiphthong, and it places SC055 OEIUmlautDiphthong before SC056 OEWsPalatalDiphthongization.

SC055. The composite i-umlaut rule (`OEIUmlaut`)

The implementation also defines a composite rule that composes the three preceding parts.

```
define OEIUmlaut OEIUmlautFronting
    .o. OEIUmlautRaising
    .o. OEIUmlautDiphthong;
```

In prose, this says that the implementation treats the umlaut as a sequence of fronting, raising, and diphthongal adjustments composed in order.

Chronologically, the composite rule must follow SC052 OEVelarPalatalization. If it is moved too early, forms such as *cȳ* 'cows' and *lungen* 'lungs' become over-palatalized. PGmc **kūi* yields *ċȳ* 'cows'; the expected form is *cȳ* 'cows'. PGmc **lúnganjō* yields *lunġen* 'lungs'; the expected form is *lungen* 'lungs'.

The same local network gives the later boundary. If West-Saxon palatal diphthongization is moved too early, PGmc **géftiz* yields *ġieft* 'gift' rather than expected OE *ġift*, and **skáiθiz* yields *sċǣþ* 'sheath' rather than expected *sċēaþ*. The composite umlaut rule therefore must apply after SC052 OEVelarPalatalization and before SC056 OEWsPalatalDiphthongization.

Those failures show that the broad umlautal rule needs an earlier terminus post quem in the palatalization sequence, even though it remains the main vowel change within the present chapter.

The composite rule is important because the literature presents the umlaut as a single historical development even while the implementation decomposes it into formal parts. The composite definition is the point at which the separate fronting, raising, and diphthongal effects are treated as one chronological event in the Old English sequence.

SC056. West Saxon palatal diphthongization (`OEWsPalatalDiphthongization`)

The narrower West-Saxon rule is treated separately from the broader umlautal complex.

```
define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ǣ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
```

```
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.]
];
```

In prose, this rule diphthongizes certain vowels after already palatal consonants in West Saxon. It therefore has a narrower dialectal and chronological scope than the broader umlaut rule.

The historical evidence for that narrower scope is concrete. Ringe and Taylor illustrate the rule with forms such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create', where an already palatal consonant triggers the diphthongal outcome (Ringe and Taylor 2014, 215–16, §6.5.1). Hogg's *giefan* 'give' and *sceap* 'sheep' material belongs to the same phonological zone (Hogg 1992, 108–9), while Fulk distinguishes this palatal-consonant-triggered diphthongization from the broad front-mutation process (Fulk 2018, 74, §4.13).

Its place is later than SC055 OEIUmlaut. If this rule is moved too early, the later ordering is constrained by forms such as *ġift* 'gift' and *sċēaþ* 'sheath'. PGmc **géftiz* then yields *ġieft* 'gift'; the expected form is *ġift* 'gift'. PGmc **skáiθiz* yields *sċǣþ* 'sheath'; the expected form is *sċēaþ* 'sheath'.

This shows that SC055 OEIUmlaut must come before SC056 OEWsPalatalDiphthongization. No comparably sharp later boundary is available.

No tested lexical item provides a comparably precise later terminus ante quem. The available evidence therefore establishes the rule's relation to the earlier umlautal process much more clearly than it fixes a later point by which it must already have applied.

The two rules should accordingly be kept distinct. The broad umlautal rule accounts for a system-wide assimilatory change; the West-Saxon rule accounts for a narrower palatal-consonant-conditioned diphthongization whose chronological and dialectal scope is more restricted.

2.38 J-cluster coalescence

Historical discussion

This chapter belongs to the later part of the palatalization and fronting region. Campbell, Ringe and Taylor, and Fulk all discuss the same neighborhood of palatalized and fronted outcomes that underlies forms such as *bīeġan* 'bend' and *sēċan* 'seek' (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13). None of them turns this later cluster adjustment into a major independent headline. The historical interest lies in the fact that it remains a real part of the sequence even though the larger palatalization and umlaut chapters carry more of the explanatory weight.

That narrower scale matters. Earlier chapters have already established the plain velar and **sk* palatalizations, and the umlaut chapter has already handled the major vowel consequences. The present rule is a later coalescence inside that same neighborhood. It deserves explicit prose because the lexical outcomes are clear, not because it eclipses the larger processes around it.

SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

The implementation keeps the later cluster coalescence very small and explicit.

```
define OEJClusterCoalescence (
    [{*g} {*j} -> {*ʤ}]
    .o. [{*k} {*j} -> {*ʧ}]
);
```

In prose, the rule coalesces **gj* and **kj* into the palatal outcomes that later surface in forms such as *bīeġan* 'bend' and *sēċan* 'seek'.

Its earlier dependency is clearer than its later limit. If the rule is moved before SC052 OEVelarPalatalization, the developments behind *bīeġan* 'bend' and *sēċan* 'seek' are lost. Related forms such as *fylġan* 'follow', *heċġ* 'hedge', and *sengan* 'singe' fail in the same broader palatalization zone. PGmc `*báugijaną` yields *bēaġan* 'bend' rather than expected OE *bīeġan*, and PGmc `*sōkijaną` yields *sōċan* 'seek' rather than expected *sēċan*. This shows that SC052 OEVelarPalatalization must come before SC057 OEJClusterCoalescence. No comparably sharp later lexical breakpoint emerges within the remaining sequence, so the chronology remains short and one-sided.

That modest shape is historically appropriate. The rule is a real later member of the palatalization region, but it does not need to absorb the umlautal chapter behind it or the nasal-dissimilation chapter that follows it. The later coalescence remains visible in the sequence once the larger neighboring chapters are already in place.

2.39 Nasal dissimilation

Historical discussion

Luick preserves individual outcomes such as *enetre* 'yearling' (with the spelling *enitre* in his text) without isolating a separate law around them (Luick 1914--1940, 166). Campbell likewise reaches forms such as *heofon* 'heaven' in a discussion of suffixal variation and does not set them off in any special section on nasal dissimilation (Campbell 1959, 155). Hogg mentions *heofon* 'heaven' in the course of his account of back mutation, again without isolating a separate law (Hogg 1992, 112).

Fulk supplies the clearest general formulation: "In the cluster mn, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular" (Fulk 2018, 121, §6.11). Ringe and Taylor stay close to the lexical evidence and note that *enetre* 'yearling' reflects "loss of the second **n* by dissimilation" (Ringe and Taylor 2014, 282).

The discussion therefore develops from scattered lexical observations to a more explicit but still cautious generalization. Luick preserves the kind of form the rule is meant to capture. Campbell and Hogg show that related outcomes enter the handbooks, but only incidentally, as part of larger accounts of other changes. Fulk makes the recurrent `mn` tendency explicit, while Ringe and Taylor provide an exact lexical case in *enetre* 'yearling'. What emerges is a limited but recurring dissimilatory pattern whose scope is far smaller than that of the major Old English vowel laws.

SC058. Nasal dissimilation in short-vowel environments (`OENasalDissimilation`)

The implementation formalizes the change as a narrow rule applying in short vowel environments before a following `n`.

```
define OENasalDissimilation [
    {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n}
     ↪  [EnglishStarShortVowel | .#.]
];
```

In plain language, the rule turns medial `m` into `f` in a restricted short-vowel environment before a following syllable containing `n`.

Historically, the rule captures the limited type of dissimilation reflected in forms such as *heofon* 'heaven', *fæstenn* 'fasting', and *enetre* 'yearling'. It is much narrower than the major vowel changes and is best understood as a recurring but partly lexicalized pattern.

The relation between the sources and the formalization is correspondingly close but not exact. Fulk formulates the tendency at the level of `mn` clusters and illustrates it with *heofon* 'heaven' and *fæstenn* 'fasting' (Fulk 2018, 121, §6.11). Ringe and Taylor show the same kind of development in *enetre* 'yearling' (Ringe and Taylor 2014, 282). Campbell's "heofon is for older hefzen" and Hogg's sequence **hefon* > *heofon* preserve outcomes of the same kind as those modeled here (Campbell 1959, 155; Hogg 1992, 112). The formal rule is therefore narrower than the total set of handbook remarks: it models one plausible recurrent environment and does not claim to exhaust every dissimilatory development involving nasals.

Chronologically, the order test does not by itself determine a sharper position within the Old English sequence. If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC058 OENasalDissimilation before or after any specific neighboring change.

Even so, the rule has real interpretative consequences. It provides a place in the implementation for outcomes of the *heofon* 'heaven', *fæstenn* 'fasting', and *enetre* 'yearling' type discussed in the literature (Fulk 2018, 121, §6.11; Ringe and Taylor 2014, 282; Campbell 1959, 155; Luick 1914--1940, 166; Hogg 1992, 112). Without an explicit rule, those outcomes would be left to diffuse analogy or to unexplained exception lists.

The evidence points to a narrow dissimilatory tendency, especially in `mn`-type clusters and a small group of lexical outcomes. There is no support for a regular change operating across a broad phonological field. The rule is secure enough to model, but the available tests leave its position within the Old English sequence underdetermined. CAPR keeps it in this middle Old English stretch because the relevant lexical outcomes are discussed alongside surrounding weak-vowel and suffixal developments, not because the handbooks fix a closer relative chronology.

2.40 Back mutation

Historical discussion

Back mutation is the substantive center of this part of the sequence. Campbell treats it as a later Old English diphthongizing development before following back vowels, and his examples already show why forms such as *heofon* 'heaven' are historically legible outcomes in their own right (Campbell 1959, 86, §207). Hogg treats the same development as a later change with clear parallels to breaking (Hogg 1992, 112). Ringe and Taylor sharpen the picture by distinguishing West Saxon forms such as *giefan* 'give' and *wefan* 'weave' from non-West-Saxon forms such as *geofad* and *weofan* (Ringe and Taylor 2014, 319, §6.9.4). Fulk likewise treats back mutation as a distinct historical phenomenon with its own profile beside the earlier umlautal changes (Fulk 2018, 69, §4.8).

That makes back mutation different from the short notes that follow it. Back mutation belongs to the same local stretch of the sequence, but it carries more historical weight and clearer lexical consequences. Even so, its later relation lies beyond this immediate stretch of the sequence, and the later weak-tail region is best kept as a forward reference only.

SC059. Back mutation before labials and liquids (`OEBackMutation`)

The implementation keeps the change as one explicit rule.

```
define OEBackMutation [
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

In prose, the rule backs and diphthongizes earlier front vowels before a following labial or liquid plus a back-vocalic trigger.

Its chronology is real on both sides, but not equally local. The earlier side is already fixed by the preceding vowel and weak-tail material. If the rule is moved too early, forms such as **gébaną* produce *ġeofan* 'give'; the expected form is *ġiefan* 'give'. **stélaną* likewise produces *steolan* 'steal'; the expected form is *stelan* 'steal'. The later side is different. If the rule is pushed too far to the right, **wébaną* yields *weofan* 'weave'; the expected form is *wefan* 'weave'. That later edge is real, but it points beyond the present stretch of the sequence into the later weak-tail reductions, so here it should remain only a forward reference.

These lexical failures show that SC048 OESecondaryNasalization must come before SC059 OEBackMutation and that SC059 OEBackMutation must come before SC078 OEWeakTailReduction.

The checked forms therefore place the rule within a wider later-vowel interval. The nearer earlier constraint is SC048 OESecondaryNasalization; the later relation to SC078 OEWeakTailReduction mainly shows that back mutation must precede the wider weak-tail reductions. CAPR keeps the rule here because the handbooks treat back mutation as a distinct later vowel development between the earlier mutation/restoration material and the closing weak-tail reductions.

2.41 West Saxon palatal umlaut

Historical discussion

The evidence is narrow enough that the discussion can stay brief. Campbell and Ringe and Taylor both support the development behind forms such as *miht* ‘might’ and *niht* ‘night’, while Fulk’s broader chronology makes clear that this material belongs beside the umlaut and palatal-vowel region as a subordinate note beside it (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

That is why the note belongs here after back mutation even though its clearest historical tie still reaches back to the earlier umlautal chapter. The phenomenon is real, yet its place in the sequence is one-sided. The evidence is clear enough to state and narrow enough to remain brief.

SC060. West Saxon palatal umlaut before **h*-clusters (`OEWsPalatalUmlaut`)

The implementation treats the West Saxon change as one explicit rule.

```
define OEWsPalatalUmlaut [
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

In prose, the rule reduces short diphthongs to **i* before the relevant **h* clusters.

The crucial point is its earlier dependency. The rule must follow SC055 OEIUmlaut, because if it is moved too early the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage *mieht* and *nieht* rather than expected OE *miht* and *niht*. No comparably sharp later lexical breakpoint emerges within the remainder of the section. The note therefore belongs here as a short afterpiece to the umlaut chapter, not as the start of a new larger unit.

This shows that SC055 OEIUmlaut must come before SC060 OEWsPalatalUmlaut. No comparably sharp later boundary is available.

2.42 Weak-tail nasal loss

Historical discussion

The development belongs to the narrower end of the later weak-tail sequence. It is historically legible through the pathway that leads to *dōn* ‘do’, and the broader late weak-tail setting is supported by the usual handbook discussions of apocope and related reduction (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120–21; Fulk 2018, 91, §5.6). But the decisive lexical tie lies much farther back in the sequence, in the older development of **dōną*. That keeps the note real, while also keeping it small.

Within this later run of changes it follows back mutation and West Saxon palatal umlaut, but the evidence remains slighter than theirs.

SC061. Reduction of final nasal weak-tail endings (`OEWeakTailNasalLoss`)

The implementation keeps the change as one short rule.

```
define OEWeakTailNasalLoss [
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
];
```

In prose, the rule reduces final weak-tail endings of the type **-ną* and **-mą* to plain final **-n* and **-m*.

The clearest lexical witness is the pathway to *dōn* ‘do’. If the rule is moved too early, before the older reduction that already shapes the **dōną* sequence, the derivation records no output instead of expected OE *dōn* ‘do’. No equally sharp later breakpoint appears within the tested sequence. That is why the note remains one-sided and why its earlier relation should be understood as a distant cross-reference only and should not reshape the broader sequence.

This shows that SC023 NWGmcNStemNLoss must come before SC061 OEWeakTailNasalLoss. No comparably sharp later boundary is available.

The development is best treated as a small late weak-tail adjustment. It remains visible in the sequence because it affects the pathway to *dōn* ‘do’, but the evidence does not support treating it as the center of a wider historical development.

2.43 High-vowel apocope

Historical discussion

By this point in the sequence the main palatal and umlautal changes are already in place, but weak-tail reduction is not finished. Final high vowels still survive in many forms until a late apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk all describe this as a real Old English development, even when they differ over how much of the surrounding syncope material should be grouped with it (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120; Ringe and Taylor 2014, 284–303, §§6.8.1, 6.8.4; Fulk 2018, 91, §5.6).

The rule matters because it makes many familiar Old English forms look abruptly shorter than their earlier stages. It is also a good place to show how finite-state chronology works. The derivation can say exactly which forms fail if apocope is moved too early or too late, so the late weak-tail sequence becomes visible through concrete lexical breakpoints and explicit ordering statements.

SC063. High-vowel apocope after heavy syllables and in trisyllables (`OEHighVowelApocope`)

The implementation keeps the whole apocope system in one explicit rule.

```
define OEHighVowelApocope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
```

```
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u̯} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u̯} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel _ .#.,
    {*u} -> 0 || {*x} _ .#.,
    {*u̯} -> 0 || {*x} _ .#.,
    {*i} -> 0 || {*x} _ .#.
];
```

In prose, the rule deletes final **i*, **u*, and **ų* when the preceding structure is heavy enough, or when a trisyllabic form behaves as equivalent to a heavy environment. The longer code box makes visible how many separate environments the transducer has to distinguish in order to realize what the handbooks describe more compactly.

Its chronology is explicit on both sides. If the rule is moved before SC055 OEIUmlaut, PGmc **kūi* yields *cū* rather than expected OE *cȳ* 'cow', and PGmc **brūdiz* yields *brūd* rather than expected OE *brȳd* 'bride'. If the rule is delayed until after SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* yields *fyrht* rather than expected OE *fyrhte* 'fright'. This means that SC055 OEI-Umlaut must come before SC063 OEHighVowelApocope, and that SC063 OEHighVowelApocope must come before SC072 OEUnstressedLongVowelShortening.

That placement is historically apt. The rule must come late enough for umlautal effects to have already been created, but it is not the last weak-tail event in the language. Apocope removes a major set of final high vowels, yet later weak-tail reductions still remain.

2.44 Post-apocope **n*-loss and medial syncope

Historical discussion of post-apocope **n*-loss and medial syncope

After high-vowel apocope the weak tail is still not entirely settled. Hogg, Ringe and Taylor, and Fulk all describe a late region in which further medial reduction and cluster pressure remain active, even though the evidence is much less even than it was for the main apocope rule (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6). The inherited **furht-* family adds one especially narrow witness of its own, because it shows that a single surviving nasal can still decide whether the weak-tail output is right or wrong (Kroonen 2013, 201).

This chapter is therefore intentionally modest. One rule has real positive chronology on both sides, but only through a single witness family. The other belongs naturally to the same late region without yet producing a comparably sharp first-break result. Keeping both visible makes the weak-tail aftermath more honest than either silence or overstatement would.

SC064. Loss of stem-final **n* after long **ī* (`NWGmcInStemNLoss`)

The first rule is extremely narrow in form.

```
define NWGmcInStemNLoss [{*n} -> 0 || {*ī} _ .#.];
```

In prose, it removes a final **n* after long **ī*. That looks tiny on the page, but the effect is real in the inherited family behind *fyrhte* 'fright'.

The chronology is two-sided even though the witness base is not broad. If the rule is moved before SC041 PWGmcFinalBareALoss, PGmc **fúrxtīnaz* yields *fyrhten* rather than expected OE *fyrhte* 'fright'. If the rule is delayed until after SC072 OEUnstressedLongVowelShortening, the same PGmc form again yields *fyrhten* rather than expected *fyrhte*. This shows that SC041 PWGmcFinalBareALoss must come before SC064 NWGmcInStemNLoss, and it places SC064 NWGmcInStemNLoss before SC072 OEUnstressedLongVowelShortening.

That symmetry does not make the rule large. Both boundaries are carried by the same witness family, so the evidence is real but narrow. The checked forms therefore place SC064 NWGmcInStemNLoss within a wider post-apocope interval between earlier final-loss material and later unstressed-vowel shortening. CAPR keeps the rule here because the `fright` family belongs to that broader weak-tail aftermath.

SC065. Medial syncope before dentals after heavy syllables (`OEMedialSyncope`)

The second rule formalizes one narrower slice of late medial syncope.

```
define OEMedialSyncope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
     ↪  [{*θ}|{*ð}|{*d}|{*t}]
];
```

In prose, it deletes medial **i* before a following dental after a heavy syllable. The broader historical background is secure enough, since the handbooks do treat late medial syncope as part of the same weak-tail region (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

The finite-state chronology is much weaker, however. If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC065 OEMedialSyncope before or after any specific neighboring change. CAPR places it here because the handbooks treat late medial syncope as part of the post-apocope weak-tail sequence that continues into the later syncope-and-cluster-simplification material. The placement should be read as approximate, not as a local ordering forced by the tested forms.

That limitation is worth stating plainly. Late medial syncope belongs in the history of the weak tail, but this particular rule does not yet have a diagnostic constraint of its own on either side.

2.45 Late syncope and degemination

Historical discussion of late syncope and degemination

Once later medial syncope begins to bite, the language inherits new consonant clusters that do not always remain stable. Hogg and Ringe and Taylor both describe this connection between vowel loss and later consonant simplification, while Brunner's discussion of *netle* 'nettle' beside later *netele* keeps the syncope evidence tied to a concrete lexical type (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–96, §§6.7.3–6.8.2; Sievers 1965, 144–45, §§158–159). Fulk is especially useful for the larger timing, because he places this syncope after i-umlaut (Fulk 2018, 91, §5.6).

The resulting chapter has an uneven center of gravity. Syncope itself is well motivated, one downstream degemination rule has a clear lexical breakpoint, and the dental assimilation step between them is plausible without yet being independently well anchored. That imbalance is part of the point. The sequence shows how the transducer can make a narrow chain of consequences explicit without pretending that every member has the same evidential weight.

SC066. L-adjacent syncope in medial syllables (`OELAdjacentSyncope`)

The syncope rule is stated directly.

```
define OELAdjacentSyncope [
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
];
```

In prose, it deletes medial **i* before **l*, creating forms such as *netle* 'nettle' and *spinl* 'spindle'.

Its chronology is explicit on both sides. If the rule is moved before SC055 OEIUmlaut, PGmc **nátilōn* yields *nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* yields *spenl* rather than expected *spinl* 'spindle'. If the rule is delayed until after SC068 OEPreconsonantalDegemination, PGmc **spénnilō* yields *spinnl* rather than expected *spinl*. This shows that SC055 OEIUmlaut must come before SC066 OELAdjacentSyncope, and that SC066 OELAdjacentSyncope must come before SC068 OEPreconsonantalDegemination.

The checked forms therefore place the rule in a wider late-syncope interval. The later relation to SC068 OEPreconsonantalDegemination is the nearer local result; the earlier boundary at SC055 OEIUmlaut mainly shows that this syncope belongs after the umlautal phase described in the handbooks. CAPR keeps it here as the opening step in the syncope-and-cluster-simplification sequence.

SC067. Dental assimilation in newly formed clusters (`OEDentalAssimilation`)

The dental repair step is formally very short.

```
define OEDentalAssimilation [
    {*θ} -> 0 || {*t} _
];
```

In prose, it removes **θ* after **t* when syncope has created an over-heavy dental cluster. That kind of cluster simplification is historically plausible as part of the same late sequence that follows syncope (Hogg 1992, 120–21; Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC067 OEDentalAssimilation before or after any specific neighboring change.

That makes the rule best read as a narrow intermediate step inside the late syncope sequence. It is useful in the derivation, but the present evidence does not justify treating it as a stronger chronology anchor than it is. The handbooks support the broader pattern of syncope followed by cluster simplification, while CAPR states this dental simplification as a separate step. The placement is therefore historically plausible but approximate, not a tightly fixed local ordering.

SC068. Preconsonantal degemination before sonorants (`OEPreconsonantalDegeminatio`

The final degemination rule is written as one composed definition.

```
define OEPreconsonantalDegemination OEPreconsonantalDegemTT .o.
 ↪  OEPreconsonantalDegemNN;
```

In prose, it simplifies doubled **tt* or **nn* before a following sonorant. The historical logic is straightforward enough. Once syncope has created a cluster such as the one behind *spinl* ‘spindle’, the doubled consonant does not remain (Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

Its positive evidence is one-sided but exact. If the rule is moved before SC066 OELAdjacentSyncope, PGmc **spénnilō* yields *spinnl* rather than expected OE *spinl* ‘spindle’. This shows that SC066 OELAdjacentSyncope must come before SC068 OEPreconsonantalDegemination. If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one.

That one-sided profile is still meaningful. The checked forms fix the earlier relation but do not identify a corresponding later constraint. CAPR keeps the rule here because the sources treat this simplification as a follower to the syncope-created cluster sequence.

2.46 Early o-shortening

Historical discussion

By the time the sequence reaches this point, the language has already undergone the larger palatal and umlautal reorganizations to the left. What now comes into view is a later weak-tail region in which unstressed vowels are shortened, fronted, merged, and in some forms lost altogether. Campbell's discussion of early shortening of unaccented long vowels helps place this material in the larger history, while Hogg, Ringe and Taylor, and Fulk all describe the same late region through the intertwined history of apocope, syncope, shortening, and later reductions (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

Early o-shortening belongs at the opening of that region, but it is not its strongest hinge. The evidence is broader and more distant than it is for the rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening. The rule therefore works best as an opening note that makes the chronology legible without pretending that the whole late weak tail begins and ends here.

SC069. Early shortening of unstressed **ō* before nasals (`OEEarlyOShortening`)

The implementation isolates the early shortening step as one rule.

```
define OEEarlyOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ EnglishStarNasal
];
```

In prose, the rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Its chronology is real, but it is one-sided. If the rule is moved before SC023 NWGmcNStemNLoss, PGmc **nḗdrōn* yields *nǣdran* rather than expected OE *nǣdre* 'adder', PGmc **érθōn* yields *eorþan* rather than expected *eorþe* 'earth', and PGmc **fláskōn* yields *flascan* rather than expected *flasce* 'flask'. The same earlier shift also disrupts forms such as *heorte* 'heart' and *līne* 'line'. This broad set of failures shows that SC023 NWGmcNStemNLoss must come before SC069 OEEarlyOShortening.

If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one. The checked forms therefore do not identify a corresponding later constraint. CAPR keeps the rule here because the sources treat early **ō*-shortening as an opening step in the later weak-tail sequence, not as the central chronology seam of the region.

2.47 Early unstressed fronting and later o-shortening

Historical discussion of early unstressed fronting and later o-shortening

The next pair forms a clearer local hinge. Campbell's account of shortening of unaccented long vowels is still relevant here, but the real value of the pair lies in the way the finite-state derivation separates an earlier fronting stage from a later shortening stage. Hogg, Ringe and Taylor, and Fulk all place these developments inside the same late weak-tail region in which shortening, syncope, and final-vowel adjustment continue to interact (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

The hierarchy inside the pair is not flat. SC070 OEUnstressedFrontingEarly is the stronger hinge because it has an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms the same seam from the right, but its later side remains open within the tested range. That imbalance is historically useful: it shows how the late weak tail is held together by small but concrete lexical breakpoints, not by one single undifferentiated rule.

SC070. Early fronting of unstressed **a* (`OEUnstressedFrontingEarly`)

The implementation gives the early fronting stage its own named step.

```
define OEUnstressedFrontingEarly OEUnstressedAFronting;
```

In prose, the rule fronts unstressed **a* to **æ* at the point where the earlier shortening has already created a frontable vowel, but the later shortening of unstressed **ō* has not yet happened. This is the step that makes endings such as OE *-en* possible in forms like *lungen* 'lungs'.

Its chronology is explicit on both sides. If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* yields *lunġen* rather than expected OE *lungen* 'lungs'. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* yields *boreþ* rather than expected OE *boraþ* 'bears', and PGmc **mḗnōθz* yields *mōneþ* rather than expected *mōnaþ* 'month'. This shows that SC052 OEVelarPalatalization must come before SC070 OEUnstressedFrontingEarly, and that SC070 OEUnstressedFrontingEarly must come before SC071 OELateOShortening.

That two-sided pattern is why SC070 OEUnstressedFrontingEarly serves as the hinge of the pair. The later relation to SC071 OELateOShortening is the closer local result, while the earlier boundary at SC052 OEVelarPalatalization mainly shows that this fronting belongs after the older palatal developments. CAPR keeps it here as an early stage inside the later unstressed-vowel sequence.

SC071. Later shortening of unstressed **ō* (`OELateOShortening`)

The following rule handles the later shortening stage.

```
define OELateOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]*
];
```

In prose, the rule shortens the remaining unstressed long **ō* after the earlier fronting stage has already done its work. This is the stage that leaves the later "stable a" endings behind forms such as OE *boraþ* 'bears' and *liornaþ* 'learns'.

Its earlier boundary is the reciprocal side of the SC070 OEUnstressedFrontingEarly relation. If the rule is moved before SC070 OEUnstressedFrontingEarly, PGmc **búrōθi* yields *boreþ* rather than expected OE *boraþ*, and PGmc **líznōθi* yields *liorneþ* rather than expected *liornaþ*. No equally sharp later breakpoint appears within the tested range, so the available evidence shows only that SC070 OEUnstressedFrontingEarly must come before SC071 OELateOShortening.

This one-sided profile is appropriate to the chapter. SC071 OELateOShortening is a real follower in the same pair, but it does not need to carry more chronology than the evidence supports.

2.48 Unstressed long-vowel shortening and ae-merger

Historical discussion of unstressed long-vowel shortening and ae-merger

This pair is the strongest internal seam in the late weak tail. Campbell's discussion of shortening of unaccented long vowels gives the classical background, while Ringe and Taylor place shortening of unstressed long vowels among the last prehistoric Old English changes and then carry the story forward into the immediately following developments (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7). What the finite-state derivation adds is a very sharp distinction between the shortening itself and the later merger of unstressed **æ* with **e*.

That is why this chapter can be more substantial than the opening note or the earlier pair. SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a real reciprocal relation in the cards, and the chapter can show both sides of it directly. The pair also keeps its outward relations in view: SC064 NWGmcInStemNLoss remains the earlier prerequisite for shortening, while SC085 OEHLoss remains the later outward handoff from the merger.

SC072. Shortening of unstressed long vowels (`OEUnstressedLongVowelShortening`)

The implementation keeps the shortening stage as one composed rule.

```
define OEUnstressedLongVowelShortening OEUnstressedLongVowelShortening1
    .o. OEUnstressedLongVowelShortening2
    .o. OEUnstressedLongVowelShortening3
    .o. OEUnstressedLongVowelShortening5
    .o. OEUnstressedLongVowelShortening6
    .o. OEUnstressedLongVowelShortening7
    .o. OEUnstressedLongVowelShortening8;
```

In prose, the rule shortens the remaining unstressed long vowels before the weak final outcomes settle into their later forms. The broad effect is visible in many weak endings, but the chronology can still be pinned down by a few particularly clear witnesses.

Its chronology is explicit on both sides. If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* yields *fyrhten* rather than expected OE *fyrhte* 'fright'. If the rule is delayed until after SC073 OEUnstressedAEMerger, PGmc **nḗdrōn* yields *nǣdræ* rather than expected OE *nǣdre* 'adder', and PGmc **fádēr* yields *fædœr* rather than expected *fœder* 'father'. This shows that SC064 NWGmcInStemNLoss must come before SC072 OEUnstressedLongVowelShortening, and that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAE-Merger.

That two-sided relation makes SC072 OEUnstressedLongVowelShortening the historical center of the pair. It still depends on earlier weak-tail preparation to the left, but within the local chapter it is the shortening stage that creates the strongest seam.

SC073. Merger of unstressed **æ* with **e* (`OEUnstressedAEMerger`)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger OEWeakTailReduction3;
```

In prose, the rule merges unstressed **æ* with **e* after shortening has already produced the vulnerable weak final vowels. This is the stage that turns a broad set of final outcomes toward the ordinary OE *-e* spellings.

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressed-LongVowelShortening, PGmc **nḗdrōn* yields *nǣdræ* rather than expected OE *nǣdre*, and PGmc **fádēr* yields *fædœr* rather than expected *fœder*. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* yields *tāœ* rather than expected OE *tā* ‘toe’. This means that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The checked forms therefore give the pair a close internal seam on the left and a broader outward limit on the right. SC072 OEUnstressedLongVowelShortening is the adjacent partner that fixes the local order; the later relation to SC085 OEHLoss mainly shows that the merger must precede the closing h-loss and contraction region. That is why this pair works as the strongest local core in the late weak tail without absorbing the later closing cluster into the chapter.

2.49 Medial unstressed-i lowering

Historical discussion of medial unstressed-i lowering and **ng* retention

The next pair belongs to the same late weak-tail region as the shortening and merger chapter to the left, but it is smaller and more locally conditioned. Hogg and Ringe and Taylor both treat the late weakening and merger of unstressed vowels as part of a continuing history, and that background helps explain why the present chapter reads best as a narrow follow-on, not a new center of gravity (Hogg 1992, 120–21; Ringe and Taylor 2014, 327–32, §§6.9.5–6.9.6). The specific value of the pair is derivational. SC074 OEMedUnstressedILowering1 generalizes a medial unstressed-*i* lowering, while SC075 OEMedUnstressedILowering immediately narrows that result by preserving *i* before **ng* in words of the *sċilling* 'shilling' type.

That close interaction is why the two rules still belong in one small chapter. The history is not simply adjacency in the cascade. The second rule directly repairs the overbroad outcome that the first would otherwise leave behind in the **ng* environment. Even so, the pair remains narrower and more witness-limited than SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

SC074. First medial unstressed-*i* lowering (`OEMedUnstressedILowering1`)

The implementation gives the first lowering step its own rule.

```
define OEMedUnstressedILowering1 [
    {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _
];
```

In prose, the rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. This is the broader step that would spread the *e*-outcome through the late weak tail if it were left uncorrected.

Its chronology is explicit on both sides. If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* yields *fyrhti* rather than expected OE *fyrhte* 'fright'. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skíllingaz* yields *sċilleng* rather than expected *sċilling* 'shilling'. This shows that SC072 OEUnstressedLongVowelShortening must come before SC074 OEMedUnstressedILowering1, and that SC074 OEMedUnstressedILowering1 must come before SC075 OEMedUnstressedILowering.

The evidence is narrow on each side, but it is still real. The rule belongs between the stronger shortening/merger chapter and the more specific **ng* preservation that follows it.

SC075. Preservation of medial unstressed **i* before **ng* (`OEMedUnstressedILowering`)

The following rule gives the local **ng* restriction its own explicit step.

```
define OEMedUnstressedILowering [
    {*e} -> {*i} || _ {*n} {*g}
];
```

In prose, the rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *sċilling* 'shilling'.

Its earlier boundary is the reciprocal side of the SC074 OEMedUnstressedILowering1 relation. If the rule is moved before SC074 OEMedUnstressedILowering1, PGmc **skíllingaz* yields *sċilleng* rather than expected OE *sċilling*. No equally sharp later breakpoint appears within the tested range, so the available evidence shows only that SC074 OEMedUnstressedILowering1 must come before SC075 OEMedUnstressedILowering.

That one-sided profile is enough for a follower rule of this kind. It is historically useful because it keeps the **ng* forms from being swallowed by the broader lowering, but it does not need to carry more chronology than the evidence supplies.

2.50 Prefix i-reduction

Historical discussion

Late weak-tail reduction does not affect only inflectional endings and medial vowels. Unstressed prefixes also weaken, and that smaller development deserves a visible place in the sequence even though its chronology is much less sharply fixed. Fulk is the clearest source here, since his discussion of vowels in prefixes makes forms like OE **be-* and **ne-* historically legible outcomes in their own right (Fulk 2018, 97, §5.7). Hogg and Ringe and Taylor supply the broader late environment in which such weakening belongs, even though they do not isolate this rule as a major center of the late-tail history (Hogg 1992, 120–21; Ringe and Taylor 2014, 298–332, §§6.8.3–6.9.6).

That is enough for a short note, but not for a major chronology anchor. Prefix reduction belongs in the late weak tail, yet the tested forms do not by themselves determine a closer position for this specific rule.

SC076. Reduction of prefixal **i* in unstressed position (`OEPrefixIReduction`)

The implementation keeps the prefixal reduction as one rule.

```
define OEPrefixIReduction [
    {*i} -> {*ĕ} || .#. [{*b} | {*n}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] EnglishStarVocalic
];
```

In prose, the rule reduces unstressed prefixal **i* to a weaker vowel in the *bi-* and *ni-* type prefixes before a consonant plus a following vowel. This is the development that helps make later prefix spellings such as OE **be-* and **ne-* historically intelligible.

If the rule is moved earlier or later within the tested sequence, no checked form yields a form different from the expected one. The tested forms therefore do not place SC076 OEPrefixIReduction before or after any specific neighboring change.

That modest result is still useful. The handbooks give real support for late prefix-vowel weakening, and CAPR places the rule in this late weak-tail stretch on those historical grounds. The placement should be read as approximate and source-based, not as a local ordering forced by the tested forms.

2.51 Weak-tail reduction

Historical discussion

The last rule in the present late weak-tail cluster is stronger than the small prefix note that precedes it. Campbell, Hogg, Ringe and Taylor, and Fulk all support a late region in which apocope, shortening, contraction, and further weak-tail reductions continue to reshape final syllables (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–91, §5.6). What makes the present rule stand out is that the finite-state chronology gives it a real boundary on both sides.

That does not make it a license to absorb later material. The later relation to SC086 OEContraction is meaningful, but it remains a cross-reference to the next cluster, not a reason to pull that cluster into the present chapter.

SC078. Reduction of remaining weak-tail vowels (`OEWeakTailReduction`)

The implementation keeps the last weak-tail reduction as one explicit step.

```
define OEWeakTailReduction OEWeakTailReduction1;
```

In prose, the rule carries the remaining weak-tail reductions that prevent a broad class of spurious *-en* or extra-vowel outcomes from surviving too late in the derivation.

Its chronology is real on both sides, though the two sides are not equally local. If the rule is moved before SC070 OEUnstressedFrontingEarly, PGmc **bákaną* yields *bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* yields *binden* rather than expected *bindan* 'bind', alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc **fléuxaną* yields *flēoan* rather than expected OE *flēon* 'flee', and PGmc **sláxaną* yields *sleaan* rather than expected *slēan* 'slay'. This shows that SC070 OEUnstressedFrontingEarly must come before SC078 OEWeakTailReduction, and that SC078 OEWeakTailReduction must come before SC086 OEContraction.

The asymmetry of those two boundaries is important. The earlier side covers a very wide interval and should be read as a broad diagnostic range, not as a close neighboring constraint. The later side is narrower and more directly interpretable. Together they make SC078 OEWeakTailReduction substantial enough for its own chapter, but still not a reason to merge the next cluster into the present section.

2.52 Final-j loss and final geminate simplification

Historical discussion of final-j loss and final geminate simplification

The first closing pair belongs to the late verbal and weak-tail region that follows SC078 OEWeakTailReduction, but it is not yet the strongest center of the closing cluster. Its coherence comes from a genuine derivational interaction. Once SC079 OEJLossAfterHeavy removes **j* after the relevant heavy environments, forms such as *lungen* 'lungs' can end up with an unwanted final geminate that SC080 OEFinalGeminateSimplification immediately removes. That interaction is close enough to justify one shared historical discussion.

The hierarchy inside the pair is still uneven. The heavier historical load lies on SC079 OEJLossAfterHeavy, whose broad earlier relation reaches back to SC055 OEIUmlaut, while SC080 OEFinalGeminateSimplification is the narrower follower that resolves the final *nn* outcome in one sharply diagnostic derivation. The chapter therefore remains compact and explicit.

SC079. Loss of **j* after heavy syllables (`OEJLossAfterHeavy`)

The implementation gives the **j*-loss step its own rule.

```
define OEJLossAfterHeavy [
    {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
     ↪  [EnglishStarConsonantNoR | EnglishPalatalConsonant] _,
    {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] [EnglishStarConsonantNoR |
     ↪  EnglishPalatalConsonant] _
];
```

In prose, the rule removes **j* after the relevant heavy-syllable configurations. This is the step that lets a broad set of late verbal forms move beyond earlier umlaut-sensitive vocalism.

Its chronology is explicit on both sides. If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijaną* yields *ġelēafan* rather than expected OE *ġelīefan* 'believe', PGmc **báugijaną* yields *bēaġan* rather than expected *bīeġan* 'bow', and PGmc **fúlgijaną* yields *fulġan* rather than expected *fylġan* 'follow'. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* yields *lungenn* rather than expected OE *lungen* 'lungs'. This shows that SC055 OEIUmlaut must come before SC079 OEJLossAfterHeavy, and that SC079 OEJLossAfterHeavy must come before SC080 OEFinalGeminateSimplification.

The left side is broad, but the right side is sharply local. Together they explain why SC079 OEJLossAfterHeavy is the stronger member of the pair.

SC080. Simplification of final geminates (`OEFinalGeminateSimplification`)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
    {*n} -> 0 || {*n} _ .#.
];
```

In prose, the rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Its earlier boundary is the reciprocal side of the SC079 OEJLossAfterHeavy relation. If the rule is moved before SC079 OEJLossAfterHeavy, PGmc **lúnganjō* yields *lungenn* rather than expected OE *lungen*. No later real break appears within the tested range before SC087 OERMetathesis, so the available evidence shows only that SC079 OEJLossAfterHeavy must come before SC080 OEFinalGeminateSimplification.

That is enough for a follower rule of this kind. It is historically useful because it prevents the unwanted final geminate from surviving, but it does not need to carry more chronology than the evidence supplies.

2.53 J-strengthening, vocalization, and ei-contraction

Historical discussion of j-strengthening, vocalization, and ei-contraction

The middle closing sequence is technically tighter than the opening pair, but it is also more internally uneven. Its real center is SC082 OEIntervocalicJVocalization. SC081 OEJStrengtheningAfterFrontDiphthong prepares the consonantal stage that the later vocalization must not erase too early, and SC083 OEUnstressedEIContraction then removes the extra *ei*-like sequence that would otherwise survive too long in the resulting weak verbal endings.

That hierarchy is historically meaningful. The three rules form one local chain because the output of each immediately conditions the next, but the chain is not flat. SC082 OEIntervocalicJVocalization is the strongest member because it has the clearest local evidence on both sides, while SC081 OEJStrengtheningAfterFrontDiphthong is the broad earlier flank and SC083 OEUnstressedEIContraction is the one-sided follower on the right.

SC081. Strengthening of **j* after front diphthongs (`OEJStrengtheningAfterFrontDiphth`

The implementation keeps the strengthening step as one explicit rule.

```
define OEJStrengtheningAfterFrontDiphthong [
    {*j} -> {*ʒ} || [{*ēa}|{*ḗa}|{*íe}|{*īe}|{*éa}] _ EnglishStarVocalic
];
```

In prose, the rule keeps **j* as a strengthened consonantal outcome after the relevant front diphthongs and so prevents too-early vocalization.

Its chronology is explicit on both sides. If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjaną* yields *strēaġan* rather than expected OE *strīeġan* ‘strew’. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields *strīeian* rather than *strīeġan*. This shows that SC055 OEIUmlaut must come before SC081 OEJStrengtheningAfterFrontDiphthong, and that SC081 OEJStrengtheningAfterFrontDiphthong must come before SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval, not a close neighboring pair. The later relation to SC082 OEIntervocalicJVocalization is the real local seam in the *strīeġan* derivation, which is why CAPR keeps SC081 OEJStrengtheningAfterFrontDiphthong here as the flank on the left side of the chain.

SC082. Intervocalic vocalization of **j* (`OEIntervocalicJVocalization`)

The implementation then turns the consonantal **j* into a vocalic outcome between vowels.

```
define OEIntervocalicJVocalization [
    {*j} -> {*i} || EnglishStarVocalic _ EnglishStarVocalic
];
```

In prose, the rule vocalizes intervocalic **j* to **i*. This is the step that creates the extra *ei*-like sequence later removed by SC083 OEUnstressedEIContraction in many weak verb forms.

Its chronology is concrete on both sides. If the rule is moved before SC081 OEJStrengtheningAfterFrontDiphthong, PGmc **stráwjaną* yields *strīeian* rather than expected OE *strīeġan* ‘strew’. If it is delayed until after SC083 OEUnstressedEIContraction, PGmc **búrōjaną* yields *boreian* rather than expected OE *borian* ‘bore’, PGmc **xándlōjaną* yields *handleian* rather than expected *handlian* ‘handle’, and PGmc **mákōjaną* yields *maceian* rather than expected *macian* ‘make’. This shows that SC081 OEJStrengtheningAfterFrontDiphthong must come before SC082 OEIntervocalicJVocalization, and that SC082 OEIntervocalicJVocalization must come before SC083 OEUnstressedEIContraction.

That two-sided local seam is why SC082 OEIntervocalicJVocalization is the center of the three-rule chain.

SC083. Contraction of unstressed *ei* (`OEUnstressedEIContraction`)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
    {*e} -> 0 || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ {*i}
];
```

In prose, the rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* ‘bore’ and *liccian* ‘lick’.

Its earlier boundary is the reciprocal side of the SC082 OEIntervocalicJVocalization relation. If the rule is moved before SC082 OEIntervocalicJVocalization, PGmc **búrōjaną* yields *boreian* rather than expected OE *borian*, PGmc **líznōjaną* yields *liorneian* rather than expected *liornian*, and PGmc **líkkōjaną* yields *licceian* rather than expected *liccian*. No later real break appears within the tested range before SC087 OERMetathesis, so the available evidence shows only that SC082 OEIntervocalicJVocalization must come before SC083 OEUnstressedEIContraction.

That one-sided profile is appropriate to the right follower in this chain. The rule is historically real, but it does not need to carry a stronger later boundary than the evidence provides.

2.54 H-loss and contraction

Historical discussion of h-loss and contraction

This adjacent pair is the clearest compact core in the closing cluster. The interaction is direct. Once SC085 OEHLoss removes intervocalic **h*, the derivation is left with hiatus that SC086 OEContraction immediately resolves. That derivational dependence is exactly the kind of close interaction that justifies one shared historical discussion.

The pair is also stronger and more book-legible than the more technical three-rule chain to its left. Ringe and Taylor give the clearest modern account of the late sequence of *h*-loss and contraction (Ringe and Taylor 2014, 305–14, §§6.9.1–6.9.3). Fulk's discussion of contracted verbs places the same outcomes into a broader Germanic context (Fulk 2018, 270, §12.21), and Luick's treatment of West Germanic contractions gives older grammatical support for the same family of outcomes (Luick 1914--1940, 165).

SC085. Loss of intervocalic **h* (`OEHLoss`)

The implementation keeps the consonant loss as one explicit rule.

```
define OEHLoss [
    {*x} -> 0 || EnglishStarVocalic _ EnglishStarVocalic
];
```

In prose, the rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

Its chronology is explicit on both sides. If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* yields *tāæ* rather than expected OE *tā* 'toe'. If it is delayed until after SC086 OEContraction, PGmc **fléuxaną* yields *flēoan* rather than expected OE *flēon* 'flee', PGmc **sláxaną* yields *sleaan* rather than expected *slēan* 'slay', PGmc **téxun* yields *teoon* rather than expected *tēon* 'draw', and PGmc **táixōn* yields *tāe* rather than expected *tā*. This shows that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss, and that SC085 OEHLoss must come before SC086 OEContraction.

The earlier side is narrow, but the later side is a tight four-row reciprocal seam that clearly feeds the following contraction rule.

SC086. Contraction of the resulting hiatus (`OEContraction`)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
    {*a} {*a} -> {*ā},
    {*e} {*e} -> {*ē},
    {*i} {*i} -> {*ī},
    {*o} {*o} -> {*ō},
    {*u} {*u} -> {*ū},
    {*ea} {*a} -> {*ēa},
    {*ēa} {*a} -> {*ēa},
    {*eo} {*a} -> {*ēo},
    {*ēo} {*a} -> {*ēo},
    {*eo} {*o} -> {*ēo},
    {*ēo} {*o} -> {*ēo},
    {*éo} {*o} -> {*ḗo},
    {*ḗo} {*o} -> {*ḗo},
    {*ā} {*a} -> {*ā},
    {*ā} {*e} -> {*ā},
    {*ē} {*a} -> {*ē},
    {*ē} {*e} -> {*ē},
    {*ḗ} {*a} -> {*ḗ},
    {*ḗ} {*e} -> {*ḗ},
```

```
    {*ī} {*a} -> {*ī},
    {*ī} {*e} -> {*ī},
    {*ī́} {*a} -> {*ī́},
    {*ī́} {*e} -> {*ī́},
    {*ō} {*a} -> {*ō},
    {*ō} {*e} -> {*ō},
    {*ū} {*a} -> {*ū},
    {*ū} {*e} -> {*ū}
];
```

In prose, the rule contracts the vowel sequences created after *h*-loss. This is the step that turns over-long transitional forms into outcomes such as *flēon* 'flee', *slēan* 'slay', and *tēon* 'draw'.

Its earlier boundary is the reciprocal side of the SC085 OEHLoss relation. If the rule is moved before SC085 OEHLoss, PGmc **fléuxaną* yields *flēoan* rather than expected OE *flēon*, PGmc **sláxaną* yields *sleaan* rather than expected *slēan*, PGmc **téxun* yields *teoon* rather than expected *tēon*, and PGmc **táixōn* yields *tāe* rather than expected *tā*. No later real break appears within the tested range before SC087 OERMetathesis, so the available evidence shows only that SC085 OEHLoss must come before SC086 OEContraction.

That one-sided profile is still substantial because the earlier reciprocal seam is so clear. The already visible SC078 OEWeakTailReduction relation also points here, but it remains a cross-reference, not a reason to absorb SC078 OEWeakTailReduction into the same chapter.

2.55 R-metathesis

Historical discussion

R-metathesis closes the present sequence, but it does not behave like the second half of a tidy local pair. The historical process is real enough to deserve explicit prose, yet its chronology reaches much farther back on the left than it does on the right. Sievers-Brunner gives a clear page-safe grammatical statement of the phenomenon through forms such as *berstan* ‘burst’, *forst* ‘frost’, and *cærse* ‘cress’ (Sievers 1965, 159, §179). Luick likewise treats metathesis as a later rearrangement whose interaction with breaking remains variable and not tightly local (Luick 1914--1940, 201).

That is why the chapter stays short. The note belongs after the contraction chapter in the assembled order, but the evidence does not justify inventing a positive claim that SC086 OEContraction must come before SC087 OERMetathesis simply because the two are adjacent.

SC087. Metathesis of **r* with a following short vowel (`OERMetathesis`)

The implementation states the metathesis directly.

```
define OERMetathesis [
    {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

In prose, the rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* ‘burst’ where an earlier order would still show a broken vowel sequence.

Its chronology is one-sided. If the rule is moved before SC044 OEBreaking, PGmc **bréstaną* yields *beorstan* rather than expected OE *berstan* ‘burst’. That shows that SC044 OEBreaking must come before SC087 OERMetathesis. If the rule is moved later within the tested sequence, no checked form yields a form different from the expected one.

That profile is exactly why the chapter remains modest. The checked forms fix the earlier relation but do not identify a corresponding later constraint. CAPR keeps the rule here because the sources treat r-metathesis as a late rearrangement that follows the earlier breaking and contraction history without being fixed immediately beside either one.

Part II

Lexical derivations

Chapter 3

Word-by-word derivations

3.1 Introduction

The lexical catalogue is organized by derivation class rather than as a single undifferentiated list. This makes the interpretive burden of each entry explicit. Regular derivations establish the baseline relation between earlier Germanic forms and Old English reflexes under the sound history followed here. The variant, analogy, reconstructed-comparator, and exception classes then show where that baseline does not by itself explain the attested form.

This lexical catalogue is a word-centered volume. It traces individual earlier Germanic forms to Old English reflexes and groups them by the kind of comparison each entry requires. A separate sound-change study would remain rule-centered, treating chronology, rule interaction, and broader exception patterns.

3.2 Data and sources

This volume assembles the lexical corpus from the aligned Germanic dataset and the compact derivation traces that accompany each entry. Comparative dictionaries, Old English dictionaries, and historical grammars are cited in the prose where they bear on particular lexical arguments.

The result is a lexical catalogue rather than a separate report on citation method or trace machinery.

3.3 Transducer and derivation method

Each lexical entry keeps the pilot structure: a generated derivation summary, a boxed derivation trace split into Earlier Germanic changes and Old English changes, and the current entry prose. The summary distinguishes citation reconstruction, selected input, transducer outcome, and selected target where those differ, and the boxed trace remains a compact PDF-oriented rendering of the current compact trace data.

3.4 Derivation classes

The lexical catalogue is ordered by seven derivation classes in the current manifest. Counts in this alpha are:

- Regular derivations: **70**
- Attested variants: **4**
- Early analogy: **35**
- Late analogy: **28**
- Reconstructed Old English comparators: **3**

- Known but unmodelled remodellings: **2**
- Unexplained or deliberately unmodelled exceptions: **5**

3.5 Regular derivations

Regular derivations are entries where the earlier Germanic form and the Old English reflex stand in a straightforward relation under the sound history followed here. These entries form the baseline against which the analogy and exception classes are interpreted.

adder — OE nǣdre

Derivation: **nḗdrōn* > *nǣdre* (regular).

Derivation trace

Proto input: **nḗdrōn*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**nǣdræ*
[no change]		OE Unstressed AE Merger	**nǣdre*
Northwest Germanic			
NWGmc N Stem N Loss	**nḗdrǭ*		
NWGmc Long E Lowering	**nǣdrǭ*		

Old English form: *nǣdre*

Reconstruction and comparative evidence

Kroonen distinguishes the masculine snake word **nadra-* from a feminine ablauting formation **nēdrōn-*, and gives Old English *nǣdre*, *næddre* under the latter (Kroonen 2013, 426). Orel likewise points from the masculine entry to a feminine **nēdrōn* ~ **nadrōn* type (Orel 2003, 325).

The derivational input therefore is not a reshaped convenience form. It is the comparative reconstruction that specifically underlies the Old English noun.

Old English evidence

The Old English word is securely represented by *nǣdre*, with *næddre* as a secondary variant. Clark Hall cross-references *næddre* to *nædre*, and Fulk treats *næddre* as the later geminated form beside the older base (Clark Hall 1960, 225; Fulk 2018, 149).

Development to Old English

From **nḗdrōn*, the stressed long mid vowel develops to Old English *nǣdre*, and the weak feminine ending remains as final *-e*, giving *nǣdre*. The doubled consonant of *næddre* is secondary and does not alter the inherited base form.

bake — OE bacan

Derivation: **bákaną* > *bacan* (regular).

Derivation trace

Proto input: **bákaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**bækaną*
[no change]	OE A Restoration	**bakaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**bakan*
[no change]	OE Secondary Nasalization	**bakąn*
	OE Weak Tail Reduction	**bakan*

Old English form: *bacan*

Reconstruction and comparative evidence

Orel reconstructs the verb as **bakanan* and cites Old English *bacan* beside Old High German *backan, bahhan* (Orel 2003). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from **bakan* to Old English *bacan* (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall both record *bacan* as the ordinary Old English verb 'to bake' (Bosworth and Toller 1898, 72; Clark Hall 1960). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

Development to Old English

From **bákaną*, Anglo-Frisian brightening first gives **bækaną*. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* (Campbell 1959, 61; Ringe and Taylor 2014). The development is therefore straightforward: **bákaną* > *bacan*.

beech — OE bōc

Derivation: **bōkō* > *bōc* (regular).

Derivation trace

Proto input: **bōkō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**bōk*
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	**bōku*		

Old English form: *bōc*

Reconstruction and comparative evidence

Kroonen gives the beech noun as **bōk(j)ō-* and cites Old English *boc*, *bēce* among its reflexes (Kroonen 2013). The form followed here, **bōkō*, is the nominative-singular shape of that family, which is the relevant comparison form here.

Old English evidence

Kroonen's Old English evidence already separates the paradigm material: *boc* as the nominative form and *bēce* as an oblique form (Kroonen 2013). The relevant comparator is therefore *bōc*; *bēċe* remains related paradigm evidence rather than the form chosen for this comparison.

Development to Old English

With nominative input **bōkō*, the development is compact. Northwest Germanic final long *ō* raises to *u*, and later high-vowel apocope leaves *bōc*. The regular comparison is therefore **bōkō* > *bōc*.

begin — OE beġinnan

Derivation: **bigínnaną* > *beġinnan* (regular).

Derivation trace

Proto input: **bigínnaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**bigínnan*
[no change]	OE Secondary Nasalization	**bigínnąn*
Northwest Germanic	OE Velar Palatalization	**biʤínnąn*
[no change]	OE Prefix I Reduction	**bĕʤínnąn*
	OE Weak Tail Reduction	**bĕʤínnan*

Old English form: *beġinnan*

Reconstruction and comparative evidence

The verb is modeled here as inherited **bigínnaną*. Ringe and Taylor state that intervocalic **g* is palatalized between front vowels in Old English (Ringe and Taylor 2014), and Campbell lists *ginnan* among familiar examples of palatal *g* in this verb family (Campbell 1959, 174).

Old English evidence

Bosworth-Toller and Clark Hall lemmatize the verb as *be-ginnan* / *beginnan* (Bosworth and Toller 1898, 84; Clark Hall 1960). Those plain-*g* dictionary spellings support the same verb that appears here in normalized form as *beġinnan*.

Development note

The prefix deserves separate notice. Ringe and Taylor explicitly cite *bi-* > *be-* as an Old English unstressed-prefix development (Ringe and Taylor 2014).

Development to Old English

From **bigínnaną*, heavy-syllable nasal apocope yields **bigínnan*. Intervocalic **g* between front vowels then palatalizes to *ġ*, and the unstressed prefix reduces *bi-* to *be-*, giving *beġinnan*.

bier — OE bǣr

Derivation: **bḗrō* > *bǣr* (regular).

Derivation trace

Proto input: **bḗrō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**bǣr*
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	**bḗru*		
NWGmc Long E Lowering	**bǣ́ru*		

Old English form: *bǣr*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **bērō-* f. 'bier' and cites Old English *bar*, *bær* among the reflexes (Kroonen 2013, 717). The derivational input **bḗrō* is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller lemmatize the noun as *bær*, and Kroonen also records *bar* beside it (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717). The target *bǣr* is therefore a normalized long-vowel spelling of the same noun.

Source note

Lexicographic spellings vary between *bær* and *bar*. The normalized target *bǣr* simply marks the same long vowel explicitly (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717).

Development to Old English

From **bḗrō*, Northwest Germanic final long *ō* raises to *u*, long *ē* lowers to *ǣ*, and high-vowel apocope yields *bǣr*. The resulting noun matches the normalized Old English target.

birth — OE byrd

Derivation: **búrdiz* > *byrd* (regular).

Derivation trace

Proto input: **búrdiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**byrdi*
[no change]		OE High Vowel Apocope	**byrd*
Northwest Germanic			
PGmc Final Z Deletion	**búrdi*		

Old English form: *byrd*

Reconstruction and comparative evidence

Kroonen cites the noun under stem-level **burdi-* and gives Old English *(ge-)byrd* among the reflexes (Kroonen 2013). The form followed here, **búrdiz*, is the nominative-style form that stands behind that stem label.

Old English evidence

Clark Hall and Bosworth-Toller both attest simplex *byrd* as an Old English noun meaning ‘birth’ (Clark Hall 1960; Bosworth and Toller 1898, 125). The prefixed form *gebyrd* is also well established in the tradition: Kroonen lists *(ge-)byrd*, Bosworth-Toller has a separate *ge-byrd* entry, and Campbell cites *gebyrd* and *gebyrdu* in his grammatical discussion (Kroonen 2013; Bosworth and Toller 1898, 125; Campbell 1959).

Form note

The relevant comparator here is the simplex noun *byrd*. The prefixed forms remain related attested material within the same lexical family, and Hogg’s discussion of deverbal feminines provides the broader derivational setting (Hogg 1992).

Development to Old English

From **búrdiz*, loss of final *z* gives **búrdi*. I-umlaut fronts *u* to *y*, and high-vowel apocope then yields *byrd*. The result is the ordinary simplex Old English noun.

bone — OE bān

Derivation: **báiną* > *bān* (regular).

Derivation trace

Proto input: **báiną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**bān*
PWGmc Ai Monophthongization	**bāną*		
Northwest Germanic			
[no change]			

Old English form: *bān*

Reconstruction and comparative evidence

Kroonen cites the noun as **baina-*, and Orel gives the same lexeme under **bainan* (Kroonen 2013; Orel 2003). Both are comparative headword conventions for the same neuter noun whose Old English reflex is *bān*.

Old English evidence

Clark Hall and Bosworth-Toller record *bān* as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898). Bright’s glossary also distinguishes citation-form *bān* from oblique *bāne*, which keeps the nominative-accusative singular separate from the rest of the paradigm (Bright 1971).

Source note

The comparative headwords **baina-* and **bainan* provide lexeme background. The relevant comparison form here is the nominative-accusative singular **báiną*.

Development to Old English

West Germanic monophthongization turns stressed **ai* into *ā*, giving **bāną*; heavy-syllable nasal apocope then yields *bān*. The resulting form matches the attested Old English citation noun.

both — OE bū

Derivation: **bō* > *bū* (regular).

Derivation trace

Proto input: **bō*

Earlier Germanic changes		Old English changes
West Germanic		[no change]
[no change]		
Northwest Germanic		
NWGmc Stressed Monosyllable O Raising	**bū*	

Old English form: *bū*

Reconstruction and comparative evidence

Kroonen treats the Germanic numeral under **ba-* and gives the inherited paradigm **bai*, **bans*, **bōz*/**bōns*, **bō*, with Old English *bēġen*, *bā*, and neuter *bū* (Kroonen 2013, 47). For the present entry, the relevant inherited form is the unextended neuter dual **bō*.

The older explanation of *bēġen* derives it from **bō-jen-*, and Orel still gives OE *bezen* (< **bō-jenō)* beside *ON báðir*, *OFris bēthe*, *OS be-thia*, and *OHG bēde* (Orel 2003, 65). Fulk reports that explanation cautiously and notes Seebold's preference for a **bō-þ-* analysis instead (Fulk 2018, sect. 10.1). That debate matters for *bēġen* and for the extended forms behind Modern English *both*, German *beide*, and Dutch *beide*; it does not displace the inherited neuter **bō* > *bū* treated here.

Old English evidence

The Old English dual paradigm is well established. Brunner gives masculine *bēġen*, feminine *bā*, and neuter *bū* beside *bā*, with compounds such as *bā twā*, *bū tū*, and *bām twām* (Sievers 1965, sect. 324 Anm. 2). Campbell and Fulk present the same basic pattern: masculine *bēġen*, feminine *bā*, neuter *bā*, *bū*, genitive *bēġra*, *bēġ(e)a*, and dative *bǣm* (Campbell 1959, sect. 683; Fulk 2018, sect. 10.1).

bū is therefore an attested neuter dual form, not a reconstruction. It is the cleanest target for this entry because *bēġen* belongs to the historically more contested **bō-jen-* / analogical zone, while *bā* remains a partner form within the dual paradigm rather than the most straightforward monosyllabic comparison.

Development to Old English

**bō* is a stressed monosyllabic form. Campbell cites *cū*, *hū*, *tū*, and *bū* as examples of final accented *ō* > *ū* in the West Germanic stage leading to Old English (Campbell 1959, sect. 122). Brunner states the same development more directly: Auslautendes *ō* erscheint als û in *bū* … cu … *hū*, *tū* (Sievers 1965, sect. 69).

The development is therefore straightforward: **bō* > *bū*.

Form comparison

The comparison below sets the relevant forms side by side. It separates the inherited OE target from the other forms that belong to the same broader lexical history.

Form	Source / stage	Status	Relevance to this entry
**bō > bū*	PGmc neuter dual > OE neuter dual	selected regular comparison	main line of the entry
bēġen	OE masculine dual	attested, but historically contested and at least partly analogical in Kroonen	real OE evidence, not the Old English form here
bā	OE feminine dual; also neuter variant	attested partner form	part of the OE paradigm, but not the chosen monosyllabic comparator
báðir, German *beide*, Dutch *beide*, Modern English *both*	Norse, continental West Germanic, Modern English extended forms	related but different formation	useful background, not the direct continuation of OE *bū*

bow — OE bīeġan

Derivation: **báugijaną* > *bīeġan* (regular).

Derivation trace

Proto input: **báugijaną*

Earlier Germanic changes
West Germanic
[no change]
Northwest Germanic
[no change]

Old English changes	
OE Au Fronting	**báeugijaną*
OE Diphthong Leveling	**bēagijaną*
OE Heavy Syllable Nasal Apocope	**bēagijan*
OE Secondary Nasalization	**bēagiją n*
Sievers Law Syncope	**bēagją n*
OE Velar Palatalization	**bēaʤją n*
OE I Umlaut	**bīeʤją n*
OE Weak Tail Reduction	**bīeʤjan*
OE J Loss After Heavy	**bīeʤan*

Old English form: *bīeġan*

Reconstruction and comparative evidence

Kroonen reconstructs the weak verb as **baugjan-* ‘to (make) bend’ and cites Old English *biegan* among its reflexes (Kroonen 2013). Ringe and Taylor give the northwest Germanic preform **baugijana*, with later **béagjan* behind West Saxon Old English *biegan* (Ringe and Taylor 2014). The entry therefore concerns the weak causative member of the bend-family, alongside the related strong verb and noun.

Old English evidence

Clark Hall lemmatizes *biegan*, and Bosworth-Toller records *bigan* with examples such as Ic *bēge mīne cneówa* and Se ord *bīgde* upp *tō þām* hiltum (Clark Hall 1960; Bosworth and Toller 1898, 102). The form *bīeġan* used here is a normalized spelling of that attested Old English weak verb.

Development to Old English

From **báugijaną*, the stem reaches pre-Old-English **bēagjan*, after which palatalization of **gj* and i-umlaut yield West Saxon *biegan*; Campbell lists *biegan* among the regular *ie* outcomes of

**éa* under i-umlaut (Ringe and Taylor 2014; Campbell 1959, 80). The development is therefore straightforward: **báugijaną* > *bīeġan*.

breeches — OE brēċ

Derivation: **brōkiz* > *brēċ* (regular).

Derivation trace

Proto input: **brōkiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Palatalization	**brōtʃi*
[no change]		OE I Umlaut	**brētʃi*
Northwest Germanic		OE High Vowel Apocope	**brētʃ*
PGmc Final Z Deletion	**brōki*		

Old English form: *brēċ*

Reconstruction and comparative evidence

Kroonen cites the noun under **brōk-*, with Old English *brōc* and plural 'breeches' among its reflexes (Kroonen 2013). Ringe and Taylor give the plural development directly as PNWGmc **brokiz* > **breeci* > OE *bréc* (Ringe and Taylor 2014). The deeper verbal base belongs to the noun's etymological background, while the derivational input here is the plural noun form **brōkiz*.

Old English evidence

Bright notes *brōc* with plural *brēc*, and Clark Hall gives *brēc* fp. breeches while also listing *broc* as a feminine noun probably represented chiefly in the plural (Bright 1971; Clark Hall 1960, 64). The spelling *brēċ* used here makes the long vowel and palatal consonant explicit; the Old English evidence itself is the attested plural *brēc*.

Development to Old English

After loss of final *-z*, the stem ends in *-ki*, so the velar palatalizes and *ō* undergoes i-umlaut to *ē*; final high-vowel apocope then yields *brēċ* (Ringe and Taylor 2014). The development is therefore regular: **brōkiz* > *brēċ*.

calf — OE ċealf

Derivation: **kálbaz* > *ċealf* (regular).

Derivation trace

Proto input: **kálbaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**kálb*
[no change]		Anglo Frisian Brightening	**kœlb*
		OE Breaking	**kealb*
Northwest Germanic		PGmc B Allophony	**kealβ*
PGmc Final Z Deletion	**kálba*	OE Velar Palatalization	**tʃealβ*

Old English form: *ċealf*

Reconstruction and comparative evidence

Kroonen treats the noun under **kalbiz-* and notes an older s-stem **kalbaz*, pl. **kalbizō*, while Orel cites **kalbaz* as the citation form and Ringe and Taylor derive West Saxon *Cealf* from **kalbaz*, **kalbiz-* (Kroonen 2013; Orel 2003, 248; Ringe and Taylor 2014, 220). The derivational input here is the singular **kálbaz*, since the entry concerns the citation-form noun.

Old English evidence

Clark Hall gives *cealf* I. (æ, e) nm. (nap. *cealfru*), and Bosworth-Toller likewise records *Caelf* / *Cealf* beside plural forms such as *calfur* and *cealfru* (Clark Hall 1960; Bosworth and Toller 1898, 131). Campbell and Fulk show the same singular-plus--*r*- plural pattern (Campbell 1959; Fulk 2018, 193). The spelling *ċealf* used here makes the palatalized initial explicit; the ordinary attested dictionary headword is *cealf*.

Development to Old English

After loss of final *-z* and bare *-a*, Anglo-Frisian brightening gives **kælb*, and breaking before *l* plus consonant yields **kealb*. Ringe and Taylor's account of the lexeme and their rule for initial *k* in front-vocalic environments support the West Saxon palatalized onset represented here as *ċ-*, so **kálbaz* develops regularly to *ċealf* (Ringe and Taylor 2014, 220).

corn — OE corn

Derivation: **kúrną* > *corn* (regular).

Derivation trace

Proto input: **kúrną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**kórn*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**kórną*		

Old English form: *corn*

Reconstruction and comparative evidence

Kroonen cites the noun as **kurna-*, and Orel gives the citation form **kurnan*, both with Old English *corn* among the reflexes (Kroonen 2013; Orel 2003, 264). The singular form **kúrną* is the nominative-accusative singular appropriate to the citation noun.

Old English evidence

Clark Hall gives *corn n. 'corn,' grain*, Bright's glossary lists *corn, n.* with genitive singular *cornes*, and Bosworth-Toller treats *corn* as an ordinary noun headword (Clark Hall 1960; Bright 1971, 347; Bosworth and Toller 1898, 144). The target is therefore an attested citation form, while forms such as *cornes* simply provide paradigm background.

Development to Old English

With northwest Germanic lowering, **kúrną* becomes **kórną*, and later loss of final nasal after a heavy syllable yields **kórn*, whence *corn*. The oblique form **kurnăn* belongs to comparative background rather than to the derivational input of this entry.

deed — OE dǣd

Derivation: **dḗdiz* > *dǣd* (regular).

Derivation trace

Proto input: **dḗdiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**dǣd*
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	**dḗdi*		
NWGmc Long E Lowering	**dǣdi*		

Old English form: *dǣd*

Reconstruction and comparative evidence

Orel reconstructs the noun as **dēdiz*, and Ringe and Taylor derive the same inherited i-stem from Proto-Germanic **dédiz* through northwest Germanic **dadiz* (Orel 2003; Ringe and Taylor 2014). The stress-marked form **dḗdiz* represents that same inherited noun in a notation that keeps the stressed long vowel explicit.

Old English evidence

Campbell states that Primitive Germanic *ē* appears as West Saxon *ǣ* but in other Old English dialects mostly as *ē*, and Brunner gives the contrast explicitly as West Saxon *dǣd* beside non-West-Saxon *dēd* (Campbell 1959; Sievers 1965). Clark Hall likewise lists *dæd* and cross-refers Anglian *dēd* to it (Clark Hall 1960). West Saxon *dǣd* is therefore the relevant Old English form here, with Anglian *dēd* as a dialectal doublet.

Development to Old English

From inherited **dēdiz*, loss of final *-z* and the West Saxon lowering of stressed long *ē* yield *dǣd*; Anglian *dēd* preserves the non-West-Saxon outcome (Campbell 1959; Sievers 1965). The development treated here is therefore the regular West Saxon line.

door — OE dor

Derivation: **dúrą* > *dor* (regular).

Derivation trace

Proto input: **dúrą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**dór*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**dórą*		

Old English form: *dor*

Reconstruction and comparative evidence

Kroonen reconstructs a neuter **dura-* 'gate, (single) door' and cites Old English *dor* among its reflexes. In the same entry he separates Old English *duru*, Old Frisian *dore*, and Old High German *tura* as reflexes of **durō-* instead (Kroonen 2013).

Old English evidence

Clark Hall records *dor* as a neuter noun and separately records feminine *duru* with its own inflection (Clark Hall 1960). Ringe and Taylor likewise treat *duru* as an early Old English u-stem, originally a root noun shifted into that class (Ringe and Taylor 2014). The Old English form here is therefore the attested neuter *dor*, while *duru* remains a parallel Old English reflex from another stem history.

Development to Old English

From **dúrą*, Northwest Germanic u-lowering gives **dórą*, and heavy-syllable nasal apocope then yields *dor*. The regular development treated in this entry is therefore **dúrą* > *dor*; the feminine *duru* belongs to the separate line identified by Kroonen and Ringe-Taylor (Kroonen 2013; Ringe and Taylor 2014).

fare — OE faran

Derivation: **fáraną* > *faran* (regular).

Derivation trace

Proto input: **fáraną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**færaną*
[no change]	OE A Restoration	**faraną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**faran*
[no change]	OE Secondary Nasalization	**farąn*
	OE Weak Tail Reduction	**faran*

Old English form: *faran*

Reconstruction and comparative evidence

Kroonen gives the inherited strong verb as **faran-*, and Orel gives the same lexeme as **faranan*, both with Old English *faran* among the reflexes (Kroonen 2013; Orel 2003, 132). Campbell also uses *faran* as a standard example of Old English A-restoration (Campbell 1959, 61).

Old English evidence

Clark Hall lemmatizes the strong verb as *faran* and separately records weak *færan* 'to frighten'; *fære*, *færst*, and *færð* belong to present-tense forms of *faran* rather than to the infinitive itself (Clark Hall 1960). Bosworth-Toller preserves the same distinction (Bosworth and Toller 1898, 108). The Old English form here is therefore the attested citation infinitive *faran*.

Development to Old English

From **fáraną*, Anglo-Frisian brightening first gives **færaną*, but A-restoration before single *r* returns **faraną*; later apocope and weak-tail reduction yield *faran* (Campbell 1959, 61). Fulk's contrast with participial faren- < **farœn-* < **faran-* shows why fronting elsewhere in the paradigm does not alter the infinitive headword (Fulk 2018).

fell — OE fell

Derivation: _*féllą_ > _fell_ (regular).

Derivation trace

Proto input: _*féllą_

Earlier Germanic changes		**Old English changes**	
West Germanic		OE Heavy Syllable Nasal Apocope	_*féll_
[no change]			
Northwest Germanic			
[no change]			

Old English form: _fell_

Reconstruction and comparative evidence

Kroonen reconstructs the noun as _*fella-_ ‘membrane, skin, hide’ and cites Old English _fell_ beside Dutch _vel_ and German _Fell_ (Kroonen 2013). The form followed here, _*féllą_, is the derivable singular form of that same inherited noun.

Old English evidence

Clark Hall records _fell_ as the noun ‘fell, skin, hide’, and Bright’s glossary likewise gives _fell_ with inflected forms such as accusative singular _fel_ and dative plural _fellum_ (Clark Hall 1960; Bright 1971). The target is therefore the attested noun _fell_, not the verb _fellan_ or the preterite _feoll_.

Development to Old English

With _*féllą_, no special earlier reshaping is needed: heavy-syllable nasal apocope yields _*féll_, surfacing as _fell_. The regular development treated here is therefore _*féllą_ > _fell_.

fern — OE fearn

Derivation: _*fárnaz_ > _fearn_ (regular).

Derivation trace

Proto input: _*fárnaz_

Earlier Germanic changes		**Old English changes**	
West Germanic		PWGmc Final Bare A Loss	_*fárn_
[no change]		Anglo Frisian Brightening	_*fǽrn_
Northwest Germanic		OE Breaking	_*fearn_
PGmc Final Z Deletion	_*fárna_		

Old English form: _fearn_

Reconstruction and comparative evidence

Kroonen cites the noun as masculine _*farna-_ and gives Old English _fearn, fern_, while Orel gives the same lexeme as neuter _*farnan_ with Old English _fearn_ (Kroonen 2013; Orel 2003, 133). Those are comparative headword conventions rather than competing Old English outcomes; the modeled input here is the nominative-style _*fárnaz_.

Old English evidence

Clark Hall gives _fearn_ as an Old English noun, and Bosworth-Toller records _fearn_ with inflected forms such as _fearnes_, _fearna_, and _fearne_ (Clark Hall 1960; Bosworth and Toller 1898, 219). Kroonen's additional _fern_ remains useful comparative background, but the best-supported citation target in the local lexical sources is _fearn_ (Kroonen 2013).

Development to Old English

From _*fárnaz_, loss of final _-z_ and final _-a_ gives _*fárn_; Anglo-Frisian brightening then yields _*fœrn_, and breaking before _r_ plus consonant gives _fearn_ (Campbell 1959; Ringe and Taylor 2014). The development treated here is therefore the regular _rC_-breaking line.

field — OE feld

Derivation: _*félθuz_ > _feld_ (regular).

Derivation trace

Proto input: _*félθuz_

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	_*féld_
PWGmc L Th Voicing	_*félduz_		
Northwest Germanic			
PGmc Final Z Deletion	_*féldu_		

Old English form: _feld_

Reconstruction and comparative evidence

Ringe and Taylor treat Old English _feld_ as one of the cases where earlier _*felþu-_ ~ _*feldu-_ may reflect either inherited _*þ_ ~ _*d_ alternation or the regular West Germanic development _*lþ_ > _ld_ (Ringe and Taylor 2014, 170). The broader proto label _*félθuz_ can remain as comparative background, while that narrower historical ambiguity does not affect the regular classification.

Old English evidence

Clark Hall records _feld_ with oblique forms such as _felda_ and _felde_, and Campbell notes early place-name spellings in _-felth_ beside the later standard form (Clark Hall 1960, 114; Campbell 1959, 169). The Old English form here is therefore the attested citation noun _feld_, with the older _-felth_ spellings as historical support rather than as rival targets.

Development to Old English

In the modeled pathway, medial _*lþ_ becomes _ld_, final _-z_ is lost, and high-vowel apocope then yields _feld_. Whether the voiced dental ultimately reflects inherited alternation or the regular _*lþ_ > _ld_ development, both accounts converge on the same Old English form (Ringe and Taylor 2014, 170).

fly — OE flēogan

Derivation: _*fléuganą_ > _flēogan_ (regular).

Derivation trace

Proto input: **fléuganą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**flēoganą*
[no change]	OE Heavy Syllable Nasal Apocope	**flēogan*
Northwest Germanic	OE Secondary Nasalization	**flēogą̄n*
[no change]	OE Weak Tail Reduction	**flēogan*

Old English form: *flēogan*

Reconstruction and comparative evidence

Ringe and Taylor derive the verb as **fleugana* > OE *fléogan* and elsewhere contrast West Saxon *fléogan* with Anglian *flégan*, alongside related forms *fléoge* / *flége* (Ringe and Taylor 2014). The form followed here, **fléuganą*, represents that inherited strong verb in the notation used here.

Old English evidence

Clark Hall and Bosworth-Toller record *flēogan* as the ordinary Old English strong verb, and Bright gives the familiar paradigm *flēag*, flugon, flogen with present *fleogeð* (Clark Hall 1960; Bosworth and Toller 1898; Bright 1971). The target in this entry is therefore the attested verbal infinitive itself.

Form note

Ringe and Taylor also list related *fléoge* / *flége* and Anglian *flégan*, which belong to the same family but do not replace the infinitive *flēogan* treated here (Ringe and Taylor 2014).

Development to Old English

From **fléuganą*, Old English diphthong leveling gives **flēoganą*; heavy-syllable nasal apocope and weak-tail reduction then yield *flēogan* (Ringe and Taylor 2014). The development is therefore regular: **fléuganą* > *flēogan*.

forlorn — OE lēosan

Derivation: **léusanąą* > *lēosan* (regular).

Derivation trace

Proto input: **léusaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**lēosaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēosan*
Northwest Germanic	OE Secondary Nasalization	**lēosą̄n*
[no change]	OE Weak Tail Reduction	**lēosan*

Old English form: *lēosan*

Reconstruction and comparative evidence

Kroonen reconstructs the verb under **leusan-* and cites prefixed daughters such as Gothic *fra-liusan* and Old English *for-lēosan*; Orel likewise gives Old English *for-leósan* (Kroonen 2013; Orel

2003). The inherited verbal base is therefore clear, though the daughter set often appears with the prefix.

Old English evidence

The direct Old English evidence behind English *forlorn* lies in the prefixed verb *forlēosan* and especially in the participle *forloren*, recorded by Ringe and Taylor and in the dictionaries (Ringe and Taylor 2014; Clark Hall 1960; Bosworth and Toller 1898). The simplex infinitive *lēosan* represents the verbal base itself.

Form note

As a base-form comparison, the simplex infinitive is *lēosan*, while the English adjective continues the prefixed Old English family *forlēosan* / *forloren* (Ringe and Taylor 2014).

Development to Old English

From **léusaną*, Old English diphthong leveling gives **lēosaną*, and later nasal apocope and weak-tail reduction yield *lēosan* (Ringe and Taylor 2014). The prefixed forms follow the same verbal base with added *for-*.

gang — OE gang

Derivation: **gángaz* > *gang* (regular).

Derivation trace

Proto input: **gángaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**gáng*
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	**gánga*		

Old English form: *gang*

Reconstruction and comparative evidence

Orel reconstructs the noun as **gangaz* and cites Old English *gang* beside Old Norse *gangr*, Old Frisian *gang* / *gong*, Old Saxon *gang*, and Old High German *gang* (Orel 2003). The form followed here, **gángaz*, is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller both record *gang* as the noun 'going, journey, way', and Bright's glossary gives *gong (gang), m., path, course* (Clark Hall 1960; Bosworth and Toller 1898, 159; Bright 1971, 392). The target is therefore the attested noun headword itself.

Form note

This entry concerns the noun *gang*, not the separate verb *gangan* (Clark Hall 1960; Bosworth and Toller 1898, 159).

Development to Old English

From **gángaz*, loss of final *-z* gives **gánga*, and later loss of final bare *-a* yields *gang*. The development is therefore regular: **gángaz* > *gang*.

give — OE ġiefan

Derivation: **gébaną* > *ġiefan* (regular).

Derivation trace

Proto input: **gébaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**géban*
[no change]	OE Secondary Nasalization	**gébąn*
	PGmc B Allophony	**géβąn*
Northwest Germanic	OE Velar Palatalization	**ʤéβąn*
[no change]	OE Ws Palatal Diphthongization	**ʤíeβąn*
	OE Weak Tail Reduction	**ʤíeβan*

Old English form: *ġiefan*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **geban-* and cites Old English *giefan* among its reflexes (Kroonen 2013). Ringe and Taylor contrast West Saxon *giefan* with Mercian *for-geofan* and Northumbrian *geafa*, showing that the inherited verb takes different later dialectal shapes (Ringe and Taylor 2014).

Old English evidence

Campbell gives *gefan* (W-S *giefan*) among examples of initial palatalization, and Clark Hall records the verb under plain *giefan* with forms such as *geaf* and *giefen* (Campbell 1959; Clark Hall 1960). The spelling *ġiefan* used here makes the palatal initial explicit.

Dialect note

West Saxon *ie* here reflects palatal diphthongization after initial palatalization; non-West-Saxon forms such as *geafa* or *for-geofan* continue the same verb without the West Saxon vocalism (Ringe and Taylor 2014).

Development to Old English

From **gébaną*, initial *g* palatalizes before *e*; West Saxon palatal diphthongization then yields *ie*, and later tail reduction gives *giefan* (Campbell 1959; Ringe and Taylor 2014). The result is therefore the regular West Saxon infinitive.

gold — OE gold

Derivation: **gúlθą* > *gold* (regular).

Derivation trace

Proto input: **gúlθą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**góld*
PWGmc L Th Voicing	**gúldą*		
Northwest Germanic			
NWGmc U Lowering	**góldą*		

Old English form: *gold*

Reconstruction and comparative evidence

Ringe and Taylor cite the noun as **gulþa-* / **gulda-*, and Kroonen gives the same pair (Ringe and Taylor 2014, 42; Kroonen 2013). The form followed here, **gúlθą*, preserves the older consonantal form while leaving open whether the medial stop reflects inherited alternation or regular West Germanic development.

Old English evidence

Bosworth-Toller and Clark Hall both record *gold* as the ordinary Old English neuter noun (Bosworth and Toller 1898, 121; Clark Hall 1960, 152). The target is therefore the attested citation form itself.

Development note

Ringe and Taylor note that the medial stop can be understood either as alternation **gulþa-* / **gulda-* or as the ordinary West Germanic change **lþ* > *ld*; both routes lead to the same Old English consonantism (Ringe and Taylor 2014, 42).

Development to Old English

From **gúlθą*, the regular consonant development gives **gúldą*; Northwest Germanic / Old English lowering then yields **góldą*, and apocope gives *gold* (Campbell 1959; Ringe and Taylor 2014, 42).

grave — OE grafan

Derivation: **grábaną* > *grafan* (regular).

Derivation trace

Proto input: **grábaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**græbaną*
[no change]	OE A Restoration	**grabaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**graban*
[no change]	OE Secondary Nasalization	**grabąn*
	PGmc B Allophony	**graβąn*
	OE Weak Tail Reduction	**graβan*

Old English form: *grafan*

Reconstruction and comparative evidence

Campbell gives *grafan* among the standard examples of Old English a-restoration before a single consonant and a following back vowel, and Ringe and Taylor describe the same development for Class VI infinitives (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence

Clark Hall records *grafan* as the verb ‘to dig, grave’ and separately records noun *græf* ‘grave, trench’ (Clark Hall 1960). The target here is the attested infinitive headword of the verb.

Development to Old English

From _*grábaną_, Anglo-Frisian brightening first gives a fronted stem vowel. A-restoration then returns *a* before single *b* plus the back-vocalic infinitive ending, and later apocope and weak-tail reduction yield *grafan* (Campbell 1959, 61; Ringe and Taylor 2014).

Form note

Noun *græf* and verbal forms such as *græfð* or past participial *græfen* belong to other lexical or paradigm positions and do not replace the infinitive *grafan* as the target here (Clark Hall 1960).

guest — OE ġiest

Derivation: _*gástiz_ > *ġiest* (regular).

Derivation trace

Proto input: _*gástiz_

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	_*gæsti_
[no change]		OE Velar Palatalization	_*ʤæsti_
Northwest Germanic		OE I Umlaut	_*ʤesti_
PGmc Final Z Deletion	_*gásti_	OE Ws Palatal Diphthongization	_*ʤiesti_
		OE High Vowel Apocope	_*ʤiest_

Old English form: *ġiest*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the noun as an ordinary i-stem whose West Saxon development shows palatal diphthongization, while non-West-Saxon evidence preserves forms of the *gest* type (Campbell 1959; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall record the word under forms such as *gist*, *gest*, *giest*, and *gyst* (Bosworth and Toller 1898; Clark Hall 1960). The Old English form here, *ġiest*, is the normalized West Saxon form within that attested family.

Development to Old English

From _*gástiz_, Anglo-Frisian brightening gives a *gæst-* stage, and i-mutation affects the front vowel before the lost high-vocalic ending. In West Saxon the initial palatal environment then produces *ie*, so the regular outcome is *ġiest* (Campbell 1959; Ringe and Taylor 2014).

Dialect note

West Saxon *ġiest* is the Old English form here. Anglian *gest* and related spellings remain real Old English comparators rather than corrections to that choice (Ringe and Taylor 2014; Bosworth and Toller 1898).

hair — OE hǣr

Derivation: **xḗrą* > *hǣr* (regular).

Derivation trace

Proto input: **xḗrą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	**çǣrą*
[no change]		OE Heavy Syllable Nasal Apocope	**çǣr*
Northwest Germanic			
NWGmc Long E Lowering	**xǣrą*		

Old English form: *hǣr*

Reconstruction and comparative evidence

Kroonen cites the ordinary Proto-Germanic hair word as **hēra-* (Kroonen 2013). The form followed here, **xḗrą*, represents that same long-ē stem in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hær* / *hǣr* as the ordinary Old English noun 'hair' (Clark Hall 1960, 158; Bosworth and Toller 1898, 510). The target is therefore the attested headword itself.

Development to Old English

From **xḗrą*, Northwest Germanic lowering gives a long front vowel, and later loss of the final nasal leaves the Old English form *hǣr*. The development treated here is straightforward and does not require any special paradigm choice.

Form note

Older references to **xazwăz* belong to a different lexeme, and the separate *haddr* / *heordan* / *hād-* material does not displace the ordinary simplex *hǣr* treated here (Kroonen 2013; Clark Hall 1960, 158).

harvest — OE hierfest

Derivation: **xárbistuz* > *hierfest* (regular).

Derivation trace

Proto input: **xárbistuz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xærbistu*
[no change]		OE Breaking	**xearbistu*
Northwest Germanic		OE Velar Fricative Palatalization	**çearbistu*
PGmc Final Z Deletion	**xárbistu*	PGmc B Allophony	**çearβistu*
		OE I Umlaut	**çierβistu*
		OE High Vowel Apocope	**çierβist*
		OE Med Unstressed I Lowering1	**çierβest*

Old English form: *hierfest*

Reconstruction and comparative evidence

Bammesberger and Ringe-Taylor treat **harbist-* as the inherited base and explain that the regular native West Saxon development would be of the *hierfest* / *hyrfest* type (Bammesberger 1997; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall record *hærfest*, with *herfest* as a variant in the lexical tradition (Bosworth and Toller 1898; Clark Hall 1960). Those attested forms remain the main dictionary evidence for the noun.

Development to Old English

From **xárbistuz*, Anglo-Frisian brightening, breaking, and i-mutation produce a *hierbist-* stage, and later lowering of unstressed medial *i* to *e* gives *hierfest*. That is the regular West Saxon development treated here (Ringe and Taylor 2014; Campbell 1959).

Source note

The Old English form here, *hierfest*, represents the regular native West Saxon outcome discussed by Bammesberger and Ringe-Taylor. The attested Old English lexical tradition, however, is chiefly *hærfest* / *herfest*, commonly treated as non-West-Saxon or Anglian material in West Saxon transmission (Bammesberger 1997; Ringe and Taylor 2014).

hedge — OE heġġ

Derivation: **xágjaz* > *heġġ* (regular).

Derivation trace

Proto input: **xágjaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**xággj*
PWGmc J Gemination	**xággjaz*	Anglo Frisian Brightening	**xæggj*
Northwest Germanic		OE Velar Fricative Palatalization	**çæggj*
PGmc Final Z Deletion	**xággja*	OE Velar Palatalization	**çæʤʤj*
		OE I Umlaut	**çeʤʤj*
		OE J Loss After Heavy	**çeʤʤ*

Old English form: *heġġ*

Reconstruction and comparative evidence

The derivational input models a palatal **-gj-* noun whose Old English development includes gemination, palatalization, and i-mutation. The current derivation therefore reaches a palatal-geminate outcome of the *heġġ* type (Campbell 1959).

Old English evidence

Bosworth-Toller and Clark Hall record the noun under standard spellings *hecg* / *heċġ* (Bosworth and Toller 1898; Clark Hall 1960). The lexical item itself is therefore well attested even though the form compared here is normalized.

Development to Old English

From **xágjaz*, West Germanic j-gemination first yields a geminate stop, and later Old English palatalization and loss of final *j* produce *heġġ*. The development is treated as regular rather than exceptional.

Form note

Standard dictionary spelling is *heċġ* or *hecg*. Normalized *heġġ* is the Old English form here, while the ordinary lexicographic forms remain the main Old English citation evidence (Bosworth and Toller 1898; Clark Hall 1960).

helm — OE helm

Derivation: **xélmaz* > *helm* (regular).

Derivation trace

Proto input: **xélmaz*

Earlier Germanic changes		**Old English changes**	
West Germanic		PWGmc Final Bare A Loss	**xélm*
[no change]		OE Velar Fricative Palatalization	**çélm*
Northwest Germanic			
PGmc Final Z Deletion	**xélma*		

Old English form: *helm*

Reconstruction and comparative evidence

Kroonen cites the helmet noun as **helma-* and separately distinguishes a different lexeme **helman-* ‘rudder’ (Kroonen 2013). The form followed here, **xélmaz*, is the nominative-style form used for the helmet noun itself.

Old English evidence

Clark Hall and Bosworth-Toller record *helm* as the ordinary Old English noun for ‘helmet’, while *helma* belongs to a separate rudder lexeme (Clark Hall 1960; Bosworth and Toller 1898, 542).

Development to Old English

From **xélmaz*, loss of final *z* and later loss of the short final vowel yield *helm*. The development is therefore a straightforward citation-form match.

Form note

Comparative **helma-* is headword notation for the helmet cognate set. It should not be confused with Old English *helma*, which is a different noun meaning ‘helm, rudder’ (Kroonen 2013; Clark Hall 1960).

help — OE helpan

Derivation: **xélpaną* > *helpan* (regular).

Derivation trace

Proto input: **xélpaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	**çélpaną*
[no change]		OE Heavy Syllable Nasal Apocope	**çélpan*
Northwest Germanic		OE Secondary Nasalization	**çélpąn*
[no change]		OE Weak Tail Reduction	**çélpan*

Old English form: *helpan*

Reconstruction and comparative evidence

The entry treats the strong verb itself rather than the separate noun *help*. Bright's principal parts *helpan, healp, hulpon, holpen* show the ordinary Old English strong-verb family continued by this input (Bright 1971).

Old English evidence

Clark Hall and Bosworth-Toller record *helpan* as the verbal headword 'to help' (Clark Hall 1960; Bosworth and Toller 1898, 542). The target is therefore the attested infinitive citation form.

Development to Old English

From **xélpaną*, no special repair is needed beyond the ordinary reduction of the infinitive ending. The derivation therefore reaches *helpan* directly.

Form note

Noun *help* belongs to a separate lexical line and should not replace verbal *helpan* as the target here (Clark Hall 1960; Bosworth and Toller 1898, 542).

hind — OE hind

Derivation: **xéndjō* > *hind* (regular).

Derivation trace

Proto input: **xéndjō*

Earlier Germanic changes		Old English changes	
West Germanic		OE Velar Fricative Palatalization	**çéndju*
[no change]		OE I Umlaut	**çindju*
Northwest Germanic		OE High Vowel Apocope	**çindj*
NWGmc Final Long O Raising	**xéndju*	OE J Loss After Heavy	**çind*

Old English form: *hind*

Reconstruction and comparative evidence

Kroonen cites the animal name as **hindō-* f. 'hind' (Kroonen 2013). The form followed here, **xéndjō*, represents that same noun in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hind* as the noun 'hind, female deer' (Clark Hall 1960; Bosworth and Toller 1898, 554). The target is therefore the attested lexical item itself.

Development to Old English

From **xéndjō*, i-mutation produces the front-vocalic Old English stem, and later apocope plus loss of final *j* yield *hind*. The outcome is therefore a regular citation-form derivation.

Form note

hindan 'from behind, behind' is a different Old English lexeme and does not belong to the noun history of *hind* (Clark Hall 1960; Bosworth and Toller 1898, 554).

hold — OE healdan

Derivation: **xáldaną* > *healdan* (regular).

Derivation trace

Proto input: **xáldaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**xældaną*
[no change]	OE Breaking	**xealdaną*
Northwest Germanic	OE Velar Fricative Palatalization	**çealdaną*
[no change]	OE Heavy Syllable Nasal Apocope	**çealdan*
	OE Secondary Nasalization	**çealdą̄n*
	OE Weak Tail Reduction	**çealdan*

Old English form: *healdan*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the verb as a regular **a* + lC breaking case, with West Saxon *healdan* opposed to Anglian and Mercian *haldan* (Campbell 1959; Ringe and Taylor 2014).

Old English evidence

Bright gives the ordinary strong-verb citation form and principal parts *healdan, heold, heoldon, healden* (Bright 1971). The target is therefore the attested infinitive headword itself.

Development to Old English

From **xáldaną*, Anglo-Frisian brightening first yields a fronted vowel, and West Saxon breaking then produces *ea* before *ld*. Later reduction of the infinitive ending gives *healdan* (Campbell 1959; Ringe and Taylor 2014).

Dialect note

West Saxon *healdan* is the Old English form here. Anglian and Mercian *haldan* are genuine non-West-Saxon doublets rather than corrections to that choice (Campbell 1959; Ringe and Taylor 2014).

horn — OE horn

Derivation: **xúrną* > *horn* (regular).

Derivation trace

Proto input: **xúrną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**xórn*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**xórną*		

Old English form: *horn*

Reconstruction and comparative evidence

Kroonen and Orel cite lemma-style Proto-Germanic headwords of the **hurna-* / **xurnan* type for this noun (Kroonen 2013; Orel 2003, 234). The form followed here, **xúrną*, is the nominative-style form used in the derivation here.

Old English evidence

Clark Hall, Bosworth-Toller, and Bright all record *horn* as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898, 108; Bright 1971). The target is therefore the attested citation form.

Development to Old English

From **xúrną*, Northwest Germanic u-lowering gives **xórną*, and later loss of the final nasal leaves *horn*. The development treated here is fully regular.

Form note

The note's oblique **xurnăn* belongs to comparative stem background only. It does not replace the derivational input **xúrną* as the derivational form used here (Kroonen 2013; Orel 2003, 234).

lead — OE lǣdan

Derivation: **láidijaną* > *lǣdan* (regular).

Derivation trace

Proto input: **láidijaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**lādijan*
PWGmc Ai Monophthongization	**lādijaną*	OE Secondary Nasalization	**lādiją̄n*
		Sievers Law Syncope	**lādją̄n*
		OE I Umlaut	**lǣdją̄n*
Northwest Germanic		OE Weak Tail Reduction	**lǣdjan*
[no change]		OE J Loss After Heavy	**lǣdan*

Old English form: *lǣdan*

Reconstruction and comparative evidence

Ringe and Taylor derive Old English *lǣdan* from Proto-Germanic **laidijaną*, and Kroonen likewise cites a weak verb of the **laidjan-* type for 'lead' (Ringe and Taylor 2014; Kroonen 2013, 363).

Old English evidence

Clark Hall and Bosworth-Toller both record *lædan* / *lǣdan* as the ordinary Old English verb 'to lead, guide, conduct' (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **láidijaną*, monophthongization of **ai* first gives a **lād-* stage. Later syncope, i-mutation, weak-tail reduction, and loss of *j* after a heavy stem yield *lǣdan*, so the development represented here is fully regular (Ringe and Taylor 2014).

learn — OE liornian

Derivation: **líznōjaną* > *liornian* (regular).

Derivation trace

Proto input: **líznōjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**líornōjaną*
[no change]	OE Heavy Syllable Nasal Apocope	**líornōjan*
	OE Secondary Nasalization	**líornōjąn*
Northwest Germanic	OE I Umlaut	**líornējąn*
[no change]	OE Unstressed Long Vowel Shortening	**líornejąn*
	OE Weak Tail Reduction	**líornejan*
	OE Intervocalic J Vocalization	**líorneian*
	OE Unstressed EI Contraction	**líornian*

Old English form: *liornian*

Reconstruction and comparative evidence

Ringe and Taylor place the verb in a class-II weak family of the **liznō-* type (Ringe and Taylor 2014), and Kroonen keeps the same comparative base for the Germanic lexeme (Kroonen 2013). The derivational input therefore requires no change of stem class or paradigm cell.

The non-obvious issue is dialectal. Campbell records Northumbrian *liornian* beside *leornian*, and he states that the *eo* of *leornian* is secondary, while Northumbrian preserves forms with *io*, where original *eo* and *io* remain distinct (Campbell 1959, sect. 123 n. 2).

Old English evidence

The form modeled here is *liornian*, the Northumbrian member of the Old English family. Dictionary practice more often privileges *leornian* as the ordinary headword (Clark Hall 1960; Bright 1971), but Campbell's dialect evidence shows that *liornian* is a genuine Old English form (Campbell 1959, sect. 123 n. 2).

This entry therefore remains compact. The point is to state clearly that the Old English form here belongs to the Northumbrian side of the OE evidence rather than to the leveled *leornian* headword tradition.

Development to Old English

From **líznōjaną*, the expected Old English developments include rhotacism of *z*, followed by breaking before *r* plus consonant, and the ordinary reduction of the weak verbal ending. The result is *liornian*, preserving the *io* spelling that Campbell associates with Northumbrian where original *eo* and *io* remain distinct (Campbell 1959, sect. 123 n. 2).

Leornian reflects the later *eo* development that Campbell treats as secondary [Campbell (1959), §123 n. 2; §296]. The selected Northumbrian target is therefore the regular comparison form for the *i*-grade member of the family.

Form comparison

The comparison below sets the relevant forms side by side. It distinguishes the regular Northumbrian form modeled here from the better-known West Saxon headword.

Form	Status	Relevance to this entry
**líznōjanq̨ > liornian*	computed regular output; attested Northumbrian comparison form	selected comparison
leornian	attested later *eo* form and dictionary headword	useful control, but not the target of this entry

lid — OE hlid

Derivation: **xlídą* > *hlid* (regular).

Derivation trace

Proto input: **xlídą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**xlíd*
[no change]		
Northwest Germanic		
[no change]		

Old English form: *hlid*

Reconstruction and comparative evidence

Orel cites a neuter lexeme of the **xliđ-* type with Old English *hlid*, and Lloyd includes OE *hlid* beside ON *hliþó* and OHG *(h)lit* among forms that retain *i* (Orel 2003; Lloyd 1966).

Old English evidence

Clark Hall and Bosworth-Toller record *hlid* as the noun ‘lid, cover, door, gate’ (Clark Hall 1960; Bosworth and Toller 1898, 563).

Development to Old English

The derivational input already represents the later Germanic *hliđ-* stage used for the derivation here. From **xlídą*, heavy-syllable apocope yields *hlid*, and the form belongs to the retained-*i* set noted by Lloyd rather than to the lowered *e* type (Lloyd 1966).

Form note

An earlier etymological stage **liþuz* belongs to comparative background only. The form represented here is the later **xlídą* > *hlid* line that matches the attested Old English noun (Orel 2003; Lloyd 1966).

light — OE līehtan

Derivation: **léuxtijaną* > *līehtan* (regular).

Derivation trace

Proto input: **léuxtijaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Diphthong Leveling	**lēoxtijaną*
[no change]		OE Heavy Syllable Nasal Apocope	**lēoxtijan*
Northwest Germanic		OE Secondary Nasalization	**lēoxtijąn*
[no change]		Sievers Law Syncope	**lēoxtjąn*
		OE I Umlaut	**līextjąn*
		OE Weak Tail Reduction	**līextjan*
		OE J Loss After Heavy	**līextan*

Old English form: *līehtan*

Reconstruction and comparative evidence

Fulk gives Proto-Germanic **liuxtijanan* with Old English *līehtan* ‘illuminate’, and Ringe and Taylor likewise derive West Saxon *liehtan* from the same weak-verb formation (Fulk 2018; Ringe and Taylor 2014).

Old English evidence

Clark Hall and Bosworth-Toller preserve the verb family under spellings such as *liehtan*, *lihtan*, and *līhtan*, distinct from the related noun *lēoht* and adjective *leoht*/*liht* (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **léuxtijaną*, the regular verbal line preserves **xt*, passes through a West Saxon *liehtan* stage, and is represented here by normalized *līehtan*. The word treated in this entry is therefore the verb ‘to light, illuminate’, not the related noun from **leuxtą* (Fulk 2018; Ringe and Taylor 2014).

Dialect note

Ringe and Taylor and Campbell distinguish West Saxon *liehtan* from Anglian *lihtan*, while later West Saxon also shows *lyhtan* (Ringe and Taylor 2014; Campbell 1959).

linden — OE lind

Derivation: **líndō* > *lind* (regular).

Derivation trace

Proto input: **líndō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**línd*
[no change]			
Northwest Germanic			
NWGmc Final Long O Raising	**líndu*		

Old English form: *lind*

Reconstruction and comparative evidence

Kroonen cites **lindō-* 'lime tree' and gives Old English *lind* as the relevant reflex (Kroonen 2013).

Old English evidence

Clark Hall and Bosworth-Toller both record *lind* as the noun 'lime-tree, linden' (Clark Hall 1960; Bosworth and Toller 1898, 630).

Development to Old English

From **líndō*, Northwest Germanic final **ō* raising first gives a **líndu* stage, and later high-vowel apocope yields *lind*. The development is therefore straightforward and regular.

Form note

The Old English noun represented here is *lind*. Clark Hall also has a separate adjectival *linden* 'made of linden-wood', but that is not the noun counterpart for this entry (Clark Hall 1960).

milk — OE meoloc

Derivation: **mélukz* > *meoloc* (regular).

Derivation trace

Proto input: **mélukz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Med Unstressed U Lowering	**mélok*
[no change]		OE Back Mutation	**méolok*
Northwest Germanic			
PGmc Final Z Deletion	**méluk*		

Old English form: *meoloc*

Reconstruction and comparative evidence

Kroonen and Orel reconstruct the noun as **meluk-* / **melukz*, and the nominative-style input used here is **mélukz* (Kroonen 2013; Orel 2003, 306).

Old English evidence

Old English preserves a mixed dossier for this noun. Ringe and Taylor describe West Saxon *meolc* < *meoluc* < **meluk*, Campbell likewise discusses *meoluc* and *meoloc*, and Anglian shows *milc* (Ringe and Taylor 2014; Campbell 1959).

Development to Old English

The unsyncopated line from **mélukz* loses final **z*, lowers unstressed *u* to *o*, and with back mutation yields *meoloc*. That fuller unsyncopated outcome is the form represented here (Ringe and Taylor 2014).

Form comparison

Syncopated *meolc* and Anglian *milc* belong to the competing leveled tradition associated with oblique forms, whereas *meoloc* / *meoluc* preserves the fuller nominal shape (Campbell 1959; Ringe and Taylor 2014).

mother — OE mōder

Derivation: **mōdēr* > *mōder* (regular).

Derivation trace

Proto input: **mōdēr*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**mōdær*
[no change]		OE Unstressed AE Merger	**mōder*
Northwest Germanic			
NWGmc Long E Lowering	**mōdǣr*		

Old English form: *mōder*

Reconstruction and comparative evidence

Kroonen and Orel cite the Proto-Germanic r-stem kinship noun as **mōder-* / **mōdēr* (Kroonen 2013; Orel 2003).

Old English evidence

The transmitted Old English headword tradition is *mōdor* / *modor*, with oblique *mēder* in the paradigm. Clark Hall, Campbell, and Ringe and Taylor all preserve that contrast (Clark Hall 1960; Campbell 1959; Ringe and Taylor 2014).

Development to Old English

From **mōdēr*, the regular suffixal development yields *mōder*. That regular nominative reflex is the form represented here, while the more familiar citation form *mōdor* reflects later levelling within the r-stem paradigm (Campbell 1959; Ringe and Taylor 2014).

Form comparison

The note therefore concerns inherited vocalism rather than a different lexeme: *mōder* is the regularized nominative represented here, but dictionaries usually print *mōdor* / *modor*, and the oblique evidence survives in *mēder* (Clark Hall 1960; Ringe and Taylor 2014).

net — OE nett

Derivation: **nátją* > *nett* (regular).

Derivation trace

Proto input: **nátją*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**nættją*
PWGmc J Gemination	**náttją*	OE Heavy Syllable Nasal Apocope	**nættj*
Northwest Germanic		OE I Umlaut	**nettj*
[no change]		OE J Loss After Heavy	**nett*

Old English form: *nett*

Reconstruction and comparative evidence

Orel gives **natjan* with Old English *nett*, and Fulk's account of West Germanic gemination before *j* explains the geminate outcome after a short vowel (Orel 2003; Fulk 2018).

Old English evidence

Clark Hall and Bosworth-Toller record *nett* as the noun, and Campbell notes that final geminates are often graphically simplified in Old English spelling (Clark Hall 1960; Bosworth and Toller 1898, 29; Campbell 1959).

Development to Old English

From **nátją*, West Germanic j-gemination first gives **náttją*. Later brightening, loss of the weak ending, and loss of final *j* after a heavy stem yield *nett*, so the development represented here is regular (Fulk 2018).

Form note

Spellings in *net* can therefore be graphic simplifications, but the lexical target supported by the dictionary evidence is *nett* (Campbell 1959; Orel 2003).

nightmare — OE mare

Derivation: **márōn* > *mare* (regular).

Derivation trace

Proto input: **márōn*

Earlier Germanic changes		**Old English changes**	
West Germanic		Anglo Frisian Brightening	**mærǭ*
[no change]		OE A Restoration	**marǭ*
Northwest Germanic		OE Unstressed Long Vowel Shortening	**maræ*
NWGmc N Stem N Loss	**márǭ*	OE Unstressed AE Merger	**mare*

Old English form: *mare*

Reconstruction and comparative evidence

Ringe and Taylor treat the lexeme as Proto-Germanic / Proto-Northwest-Germanic **marōn-*, with Old English *mare, maran*, and variant *mere*; Orel preserves the same comparative lemma though with a different Old English headword tradition (Ringe and Taylor 2014; Orel 2003).

Old English evidence

Clark Hall records *mare* 'nightmare, monster' and also preserves related variant forms *mera* / *mere* (Clark Hall 1960, 213).

Development to Old English

The selected simplex input **márōn* regularly gives *mare* after brightening, A-restoration before the n-stem ending, and later reduction of the final vowel. The word represented here is the attested simplex noun, not an attested compound (Ringe and Taylor 2014).

Form note

The concept corresponds to an unattested compound **nihtmare*, but the Old English lexical evidence is for simplex *mare*, with oblique *maran* and variant *mere* / *mera* (Ringe and Taylor 2014; Clark Hall 1960, 213).

coat — OE rocc

Derivation: **rúkkaz* > *rocc* (regular).

Derivation trace

Proto input: **rúkkaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**rókk*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**rókkaz*		
PGmc Final Z Deletion	**rókka*		

Old English form: *rocc*

Reconstruction and comparative evidence

Orel cites a masculine **rukkaz* for the garment word, while Kroonen gives **hrukka-*. Both treat this as the garment lexeme and not as the separate stone word (Orel 2003; Kroonen 2013, 290).

Old English evidence

Clark Hall and Bosworth-Toller record *rocc* as an over-garment or tunic and preserve compounds such as *bisceoprocc* and *breóstrocc* (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

With **rúkkaz* as the derivational input, Northwest Germanic u-lowering and later loss of final *-a* yield *rocc* as a regular outcome.

Source note

This entry concerns the garment noun only. The stone word seen in *stānrocc* belongs to a different lexical history (Clark Hall 1960; Bosworth and Toller 1898).

sheep — OE sċēap

Derivation: **skḗpą* > *sċēap* (regular).

Derivation trace

Proto input: **skḗpą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**skǣp*
[no change]		OE Sk Palatalization	**ʃǣp*
Northwest Germanic		OE Ws Palatal Diphthongization	**ʃēap*
NWGmc Long E Lowering	**skǣpą*		

Old English form: *sċēap*

Reconstruction and comparative evidence

Ringe and Taylor cite a later West Germanic **skap* > WS *scéap*, while Orel preserves a Proto-Germanic noun of the **skēp-* type for the same lexeme (Ringe and Taylor 2014; Orel 2003).

Old English evidence

Clark Hall records *scēap* with spelling variation, and Campbell likewise lists West Saxon *scéap* among the palatal-diphthongized forms (Clark Hall 1960; Campbell 1959).

Development to Old English

From **skḗpą*, Northwest Germanic lowering gives **skǣpą*; after apocope and palatalization the West Saxon branch diphthongizes to *sċēap*. The development represented here is therefore fully regular.

Dialect note

Ringe and Taylor contrast West Saxon *scéap* with Mercian and Kentish *scép*, and Campbell also notes Northumbrian *scip*. The form represented here is the West Saxon headword (Ringe and Taylor 2014; Campbell 1959).

shilling — OE sċilling

Derivation: **skíllingaz* > *sċilling* (regular).

Derivation trace

Proto input: **skíllingaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**skílling*
[no change]		OE Sk Palatalization	**ʃílling*
Northwest Germanic		OE Med Unstressed I Lowering1	**ʃílleng*
PGmc Final Z Deletion	**skíllinga*	OE Med Unstressed I Lowering	**ʃílling*

Old English form: *sċilling*

Reconstruction and comparative evidence

Kroonen treats the cognate set under **skellinga-* ~ **skillinga-* and connects it with **skeld-linga-*, while Orel likewise gives the coin word with OE *scilling* among the reflexes (Kroonen 2013; Orel 2003). The form followed here, **skíllingaz*, is the nominative-style form used here to represent that inherited **-ing-* derivative.

Old English evidence

Clark Hall records *scilling*, and Campbell cites *scilling* among nouns whose derivational *-ing* keeps *i* in unstressed syllables (Clark Hall 1960; Campbell 1959). The target represented here is the ordinary OE citation form, normalized as *sċilling*.

Development to Old English

From **skíllingaz*, loss of final *-az* yields **skílling*. Old English palatalization of initial *sk* before front vocalism then gives *sċilling*. The note matters because derivational *-ing-* keeps *i*, so the regular outcome is *sċilling*, not **sċilleng* (Campbell 1959; Hogg 1992).

Form note

Kroonen's **skellinga-* ~ **skillinga-* and his internal analysis **skeld-linga-* belong to the etymological background of the cognate set. The form followed here, **skíllingaz*, is the specific form used for the derivation represented here (Kroonen 2013).

show — OE sċēawian

Derivation: **skáwōjaną* > *sċēawian* (regular).

Derivation trace

Proto input: **skáwōjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Aw Long Diphthong	**skḗawōjaną*
[no change]	OE Heavy Syllable Nasal Apocope	**skḗawōjan*
Northwest Germanic	OE Secondary Nasalization	**skḗawōjąn*
[no change]	OE Sk Palatalization	**ʃḗawōjąn*
	OE I Umlaut	**ʃḗawējąn*
	OE Unstressed Long Vowel Shortening	**ʃḗawejąn*
	OE Weak Tail Reduction	**ʃḗawejan*
	OE Intervocalic J Vocalization	**ʃḗaweian*
	OE Unstressed EI Contraction	**ʃḗawian*

Old English form: *sċēawian*

Reconstruction and comparative evidence

Orel and Kroonen cite a Class II verb of the type **skawōjan-*, with OE *scēawian* among the reflexes (Orel 2003; Kroonen 2013, 482). Brunner likewise records the Old English family as *scēawian*, *scāwian*, which places this entry in the ordinary show-verb set rather than in a special finite-cell workaround (Sievers 1965).

Old English evidence

Bright lists *scēawian* (W. II.) and also the related form *scēawa* (Bright 1971). The source tradition therefore uses *scēawian*, while the target represented here is the normalized project spelling *sċēawian*.

Development to Old English

From **skáwōjaną*, Old English *aw* before a following vowel yields *ēaw*, and the Class II suffix keeps **ō* between **w* and **j*. The development therefore runs regularly to *sċēawian*, without the direct **aw+j* problem seen in other verb types (Campbell 1959; Orel 2003).

Form note

The difference between *scēawian* and *sċēawian* is orthographic normalization of initial <*sc*>, not a difference of lexeme or paradigm cell (Campbell 1959; Hogg 1992).

sleep — OE slǣpan

Derivation: **slḗpaną* > *slǣpan* (regular).

Derivation trace

Proto input: **slḗpaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**slǣpan*
[no change]		OE Secondary Nasalization	**slǣpąn*
Northwest Germanic		OE Weak Tail Reduction	**slǣpan*
NWGmc Long E Lowering	**slǣpaną*		

Old English form: *slǣpan*

Reconstruction and comparative evidence

Kroonen preserves the comparative verb as **slēpan-*, and Fulk cites the same family under root **slēb-* (Kroonen 2013; Fulk 2018, 120). The form followed here, **slḗpaną*, is the infinitive-style form used here for that inherited sleep-verb.

Old English evidence

Clark Hall gives *slæpan* with preterite *slēp*, *slēap*, and Bright likewise lists *slæpan (slāpan)*, *slēp slēpon slēpen* (Clark Hall 1960; Bright 1971, 435). The target represented here is therefore the normalized infinitive *slǣpan*, not the preterite forms and not the separate noun *slǣp*.

Development to Old English

From **slḗpaną*, Northwest Germanic lowering gives **slǣpaną*. The later OE tail developments then yield *slǣpan* regularly. Brunner and Bülbring show that the OE tradition also has variant spellings such as West Saxon *slāpan*/*slæpan* and Anglian or Kentish *slēpan*, but those do not displace the infinitive chosen here (Sievers 1965; Bülbring 1902).

Form note

The note concerns lemma type rather than a special derivational problem: this row represents the verb *slǣpan*, whereas *slǣp* belongs to noun or lookup background and *slēp*/*slēap* are preterite forms (Clark Hall 1960; Bright 1971, 435).

smear — OE smierwan

Derivation: **smérwijaną* > *smierwan* (regular).

Derivation trace

Proto input: **smérwijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**sméorwijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**sméorwijan*
Northwest Germanic	OE Secondary Nasalization	**sméorwijąn*
[no change]	Sievers Law Syncope	**sméorwjąn*
	OE I Umlaut	**smíerwjąn*
	OE Weak Tail Reduction	**smíerwjan*
	OE J Loss After Heavy	**smíerwan*

Old English form: *smierwan*

Reconstruction and comparative evidence

Kroonen gives the comparative headword as **smerwjan-* (Kroonen 2013). Ringe and Taylor instead cite a later-stage **smirwijana*, from which they derive West Saxon *smierwan*, Mercian *smirwan*, and Northumbrian *smiriga* (Ringe and Taylor 2014). The form followed here, **smérwijaną*, therefore represents the Kroonen-aligned PGmc layer, while the later-stage dialect split belongs to a different chronological level.

Old English evidence

The target represented here is the West Saxon citation form *smierwan*. Campbell's Anglian discussion explains the contrasting *smirwan*, and Clark Hall, Brunner, and Bright show that the same lexical family later also includes forms such as *smirian*, *smyrian*, and preterite *smyrode* (Campbell 1959; Clark Hall 1960; Sievers 1965; Bright 1971).

Development to Old English

From **smérwijaną*, breaking before *r + consonant* yields *eo*, and later i-umlaut produces *ie*. The result is West Saxon *smierwan*. Anglian *smirwan* reflects the well-known failure of breaking in this environment, not a different lexeme (Campbell 1959; Ringe and Taylor 2014).

Dialect note

The entry therefore represents the West Saxon member of a broader OE family: *smierwan* in West Saxon, *smirwan* in Anglian or Mercian, and related later class-II forms such as *smirian* or *smyrian* in the same lexical field (Ringe and Taylor 2014; Clark Hall 1960).

span — OE spannan

Derivation: **spánnaną* > *spannan* (regular).

Derivation trace

Proto input: **spánnaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**spánnan*
[no change]	OE Secondary Nasalization	**spánnąn*
Northwest Germanic	OE Weak Tail Reduction	**spánnan*
[no change]		

Old English form: *spannan*

Reconstruction and comparative evidence

Kroonen cites the inherited verb as **spannan-*, with OE *spannan* among the reflexes (Kroonen 2013). The form followed here, **spánnaną*, is the infinitive-style form used here for that same verbal lexeme.

Old English evidence

Clark Hall keeps noun *spann* and verb *spannan* separate, and Brunner likewise records *sponnan, spannan stv.* (Clark Hall 1960; Sievers 1965). This entry treats the strong-verb infinitive, not the separate noun.

Development to Old English

From **spánnaną*, the final nasal ending is lost and the regular OE weak-tail steps surface *spannan*. No paradigm-cell substitution is needed: the current derivation already lands on the infinitive directly.

Form note

The note matters because English *span* can also reach noun *spann* in local lookup material. The entry represented here is the verb *spannan*, with the noun treated elsewhere (Clark Hall 1960).

spar — OE spearra

Derivation: **spárrô* > *spearra* (regular).

Derivation trace

Proto input: **spárrô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**spærrô*
[no change]	OE Breaking	**spearrô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**spearra*
[no change]		

Old English form: *spearra*

Reconstruction and comparative evidence

Kroonen and Orel place this noun in the beam or rafter set **spar(r)an-*, with cognates such as Old Saxon and Old High German *sparro* (Kroonen 2013; Orel 2003). The form followed here, **spárrô*, is the OE-facing nominal form used here for that same lexeme.

Old English evidence

The noun represented here is *spearra*. The important lexical point is negative: English gloss overlap also reaches the unrelated verb *sperran* ‘to bar’, but that verb does not belong to this row.

Development to Old English

From **spárrô*, Anglo-Frisian brightening gives **spærrô*, and OE breaking before geminate *rr* yields **spearrô*, later *spearra*. The development is therefore regular for a breaking-conditioned noun of this type (Luick 1914--1940).

Form note

This entry concerns the noun *spearra* only. It should be kept separate from verb *sperran*, even though the Modern English glosses overlap (Kroonen 2013; Orel 2003).

still — OE stillan

Derivation: **stéllijaną* > *stillan* (regular).

Derivation trace

Proto input: **stéllijaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**stéllijan*
[no change]		OE Secondary Nasalization	**stéllijąn*
Northwest Germanic		Sievers Law Syncope	**stélljąn*
[no change]		OE I Umlaut	**stilljąn*
		OE Weak Tail Reduction	**stilljan*
		OE J Loss After Heavy	**stillan*

Old English form: *stillan*

Reconstruction and comparative evidence

The wider West Germanic family includes adjective *still* and verb *stillen* (Kluge 2011). The form followed here, **stéllijaną*, represents the verbal j-formation used for the OE row.

Old English evidence

Clark Hall gives *stillan* as the verb and separately *stille* as the adjective (Clark Hall 1960). Bosworth-Toller likewise preserves a substantial prefixed verbal family under *ge-stillan* and related forms (Bosworth and Toller 1898, 724). The Old English form here is the verb *stillan*, not the adjective.

Development to Old English

As a heavy-stem Class I weak verb, **stéllijaną* undergoes the expected syncope and i-umlaut, and later loss of *j* after a heavy stem yields *stillan*. The development represented here is regular.

Form note

The note concerns lexical framing rather than sound law: *stillan* is the verb represented here, while *stille* belongs to the related adjectival branch of the family (Clark Hall 1960; Kluge 2011).

summer — OE sumer

Derivation: **súmaraz* > *sumer* (regular).

Derivation trace

Proto input: **súmaraz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**súmar*
[no change]		Anglo Frisian Brightening	**súmær*
Northwest Germanic		OE Unstressed AE Merger	**súmer*
PGmc Final Z Deletion	**súmara*		

Old English form: *sumer*

Reconstruction and comparative evidence

Kroonen gives the lexeme as **sumara-*, and Ringe and Taylor likewise use **sumaraz*, while Orel preserves an alternate **sumeraz* (Kroonen 2013; Ringe and Taylor 2014; Orel 2003, 425). The form followed here, **súmaraz*, follows the **a* vocalism that underlies the regular development represented here.

Old English evidence

Clark Hall gives *sumor m., gs. sumeres, ds. sumera, sumere*, and Bright likewise lists *sumor (sumer)* with genitive *sumeres* (Clark Hall 1960; Bright 1971, 440). The tradition therefore preserves both *sumor* and *sumer*, with the oblique forms strongly supporting second-syllable *e*.

Development to Old English

From **súmaraz*, loss of final *-az* is followed by fronting and merger in the unstressed second syllable, yielding *sumer*. The form compared here is the regularized *e*-form, while the common citation form *sumor* remains part of the attested OE tradition (Ringe and Taylor 2014).

Form note

The entry does not deny *sumor*. It represents *sumer* as the regular outcome chosen here, while *sumor* remains a common headword spelling and *sumeres*/*sumere* show that the *e*-vocalism was also real in Old English (Clark Hall 1960; Bright 1971, 440).

sunder — OE sundrian

Derivation: **súndrōjanǫ* > *sundrian* (regular).

Derivation trace

Proto input: **súndrōjanǫ*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**súndrōjan*
[no change]	OE Secondary Nasalization	**súndrōjąn*
Northwest Germanic	OE I Umlaut	**súndrējąn*
[no change]	OE Unstressed Long Vowel Shortening	**súndrejąn*
	OE Weak Tail Reduction	**súndrejan*
	OE Intervocalic J Vocalization	**súndreian*
	OE Unstressed EI Contraction	**súndrian*

Old English form: *sundrian*

Reconstruction and comparative evidence

Orel distinguishes three related formations: adverbial **sunþraz* > *sundor*, Class I verbal **sunþrjanan* > *syndrian*, and Class II verbal **sunþrōjanan* > *sundrian* (Orel 2003). Kluge-Seebold aligns the cognate set with German *sondern* and OE *gesundrian*, so this entry belongs with the Class II verb, not the adverb (Kluge 2011).

Old English evidence

Clark Hall and Bosworth-Toller keep *sundrian* and *syndrian* separate from adverbial *sundor*, and both preserve the prefixed verbal family *ā-sundrian* (Clark Hall 1960, 296; Bosworth and Toller 1898). The target represented here is therefore the weak verb *sundrian*.

Development to Old English

From **súndrōjanǫ*, the Class II weak-verb suffix yields regular OE *-ian*, producing *sundrian*. Because this is the **-ōjan-* verb and not the Class I **-jan-* formation, the form represented here does not belong to the umlauted *syndrian* branch.

Form note

The earlier confusion was lexical, not phonological: *sundor* is the separate adverb, and *syndrian* is a related but different verb. The verb treated here is the Class II verb *sundrian* (Orel 2003; Clark Hall 1960, 296).

swallow — OE swealwe

Derivation: **swálwōn* > *swealwe* (regular).

Derivation trace

Proto input: **swálwōn*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**swælwǭ*
[no change]		OE Breaking	**swealwǭ*
Northwest Germanic		OE Unstressed Long Vowel Shortening	**swealwæ*
NWGmc N Stem N Loss	**swálwǭ*	OE Unstressed AE Merger	**swealwe*

Old English form: *swealwe*

Reconstruction and comparative evidence

Kroonen gives the bird name as **swalwōn-*, and Ringe and Taylor cite the later West Germanic stage **swalwa*, from which West Saxon *swealwe* and Mercian *swalwe* develop (Kroonen 2013, 535; Ringe and Taylor 2014, 200). The selected etymological comparison belongs to the swallow-bird family, not to the verb *swelgan*.

Old English evidence

Clark Hall records *swealwe (a, o)* as the noun headword (Clark Hall 1960). Campbell and Brunner also preserve later or oblique-family forms such as *swaluwe*, *swalewan*, and *swealuwe*, but those belong to wider variation around the noun rather than to the citation form represented here (Campbell 1959; Sievers 1965).

Development to Old English

From **swálwōn*, brightening yields **swælw-*, and breaking before *lw* gives **swealw-*. The later noun ending develops regularly to *swealwe*. The relevant point is that the bird name has no inherited **g*: that consonant belongs to the separate verb *swelgan* (Ringe and Taylor 2014, 200; Kroonen 2013, 535).

Form note

The final prose keeps the citation form *swealwe* separate from two different kinds of background material: the unrelated verb *swelgan*, and later or oblique spellings such as *swaluwe* or *swalewan* (Clark Hall 1960; Campbell 1959).

swine — OE swīn

Derivation: citation reconstruction **swī́ną*; form followed here **swī́ną* > *swīn* (regular).

Derivation trace

Proto input: **swī́ną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**swī́n*
[no change]		
Northwest Germanic		
[no change]		

Old English form: *swīn*

Reconstruction and comparative evidence

Kroonen cites the noun as **swina-* 'pig' (Kroonen 2013). The selected comparative form here is the bare neuter citation cell **swī́ną*, which matches the singular noun aimed at in Old English rather than an oblique stem form.

Old English evidence

Clark Hall records *swin (y)* as the ordinary noun headword (Clark Hall 1960). The target here is that singular citation form *swīn*, not a plural glossed in Modern English as *swine*.

Development to Old English

From **swī́ną*, loss of the final nasal vowel yields *swīn*. The outcome is therefore the regular monosyllabic noun with preserved long root *ī*.

Source note

The derivational input writes stressed long *ī* as **ī́*, so comparative **swī́ną* and derivational **swī́ną* represent the same lexical form.

think — OE þenċan

Derivation: **θánkijaną* > *þenċan* (regular).

Derivation trace

Proto input: **θánkijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**θánkijan*
[no change]	OE Secondary Nasalization	**θánkijąn*
Northwest Germanic	Sievers Law Syncope	**θánkjąn*
[no change]	OE Velar Palatalization	**θánʧjąn*
	OE I Umlaut	**θenʧjąn*
	OE Weak Tail Reduction	**θenʧjan*
	OE J Loss After Heavy	**θenʧan*

Old English form: *þenċan*

Reconstruction and comparative evidence

Kroonen gives the verb as **þankjan-* 'to think', and Ringe and Taylor cite fully inflected **þanki-janą* beside OE *þenċan* (Kroonen 2013; Ringe and Taylor 2014). The noun **þankaz* belongs only to the wider derivational background.

Old English evidence

Bosworth-Toller preserves the verb under *þencan/geþencan*, and the citation form here is the ordinary infinitive *þenċan* (Bosworth and Toller 1898).

Development to Old English

From **θánkijaną*, palatalization before **j* and i-umlaut produce *þenċan*. The infinitive is therefore a straightforward weak-verb outcome.

Lexical note

Campbell's assibilation discussion uses the same verb *þencan*; the class-III relic *hycgan* is a different lexeme (Campbell 1959; Hogg 1992).

thorn — OE þorn

Derivation: **θúrnaz* > *þorn* (regular).

Derivation trace

Proto input: **θúrnaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**θórn*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**θórnaz*		
PGmc Final Z Deletion	**θórna*		

Old English form: *þorn*

Reconstruction and comparative evidence

Kroonen gives **þurna-* 'thorn, briar', while Orel preserves the masculine pair **þurnuz* ~ **þurnaz* (Kroonen 2013; Orel 2003). The form followed here, **θúrnaz*, belongs to that same comparative family.

Old English evidence

Bright lists *þorn*, m., and Clark Hall likewise treats *þorn* as the ordinary noun headword (Bright 1971; Clark Hall 1960).

Development to Old English

The inherited stem shows regular lowering of *u* to *o* before *r*, and final loss yields *þorn*. The noun is therefore a regular Old English continuation of the Proto-Germanic thorn-family.

Source note

The comparative sources preserve more than one stem formation, but the Old English target itself is simply the citation form *þorn*.

tide — OE tīd

Derivation: citation reconstruction **tī́diz*; form followed here **tī́diz* > *tīd* (regular).

Derivation trace

Proto input: **tī́diz*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**tī́d*
[no change]			
Northwest Germanic			
PGmc Final Z Deletion	**tī́di*		

Old English form: *tīd*

Reconstruction and comparative evidence

Kroonen gives **tīdi-* ‘time’, and Orel’s **tīđiz* points to the same feminine noun (Kroonen 2013; Orel 2003). The related verb *tīdan* is separate.

Old English evidence

Bright records *tīd* with singular *tīde* and plural *tīda*, and Clark Hall treats *tīd* as the ordinary noun ‘time, period, season’ (Bright 1971; Clark Hall 1960, 309).

Development to Old English

From **tī́diz*, final *z* is lost and the high final vowel drops, leaving *tīd*. The development is straightforward for a feminine i-stem.

Lexical note

The note matters only because English *tide* can pull in the separate weak verb *tīdan*; the noun targeted here is *tīd*.

token — OE tācn

Derivation: **táiknq̨* > *tācn* (regular).

Derivation trace

Proto input: **táikną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**tākn*
PWGmc Ai Monophthongization	**tākną*		
Northwest Germanic			
[no change]			

Old English form: *tācn*

Reconstruction and comparative evidence

Kroonen cites **taikna-* and Orel **taiknan* for the noun ‘sign, token’ (Kroonen 2013; Orel 2003, 438). The form followed here, **táikną*, is the simple citation-form noun used for the derivation.

Old English evidence

Campbell and Sievers-Brunner preserve both unbroken *tācn* and broken *tācen*, with oblique *tācnes* remaining unbroken (Campbell 1959; Sievers 1965).

Development to Old English

Monophthongization of *ai* yields *ā*, and loss of the final nasal vowel leaves *tācn*. The unbroken form is therefore a regular Old English outcome.

Form note

tācn is the attested unbroken citation form selected here. Later West Saxon prose often prefers *tācen*, but that does not displace the older unbroken form (Campbell 1959; Sievers 1965).

town — OE tūn

Derivation: **tūną* > *tūn* (regular).

Derivation trace

Proto input: **tūną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**tūn*
[no change]		
Northwest Germanic		
[no change]		

Old English form: *tūn*

Reconstruction and comparative evidence

Kroonen cites **tūna-* ‘fenced area’, while Orel gives **tūnan* ~ **tūnaz* (Kroonen 2013; Orel 2003, 452). The form followed here, **tūną*, is the simple citation-form noun used in the derivation.

Old English evidence

Clark Hall records *tūn* as the ordinary headword ‘enclosure, yard, village, town’ (Clark Hall 1960).

Development to Old English

The inherited long *ū* is preserved, and loss of the final nasal vowel yields *tūn* regularly (Sievers 1965).

Source note

The comparative headwords vary, but the Old English target here is the direct citation form *tūn*, not an oblique **tūnăn*.

wade — OE wadan

Derivation: **wádaną* > *wadan* (regular).

Derivation trace

Proto input: **wádaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wædaną*
[no change]	OE A Restoration	**wadaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**wadan*
[no change]	OE Secondary Nasalization	**wadąn*
	OE Weak Tail Reduction	**wadan*

Old English form: *wadan*

Reconstruction and comparative evidence

Campbell and Ringe and Taylor describe A-restoration before a following back vowel, and Luick explicitly includes *wadan* among the standard open-syllable examples (Campbell 1959; Ringe and Taylor 2014; Luick 1914--1940, 239).

Old English evidence

Clark Hall gives *wadan* as the verb ‘to go, move, stride, advance’, and Bright lists the same infinitive in the strong-verb paradigm (Clark Hall 1960; Bright 1971).

Development to Old English

From **wádaną*, Anglo-Frisian brightening first gives **wædaną*. A-restoration before the back-vocalic infinitive ending then returns *a*, and later reduction yields *wadan* (Campbell 1959; Ringe and Taylor 2014).

Development note

The note matters because this infinitive belongs to the A-restoration class. The citation form is therefore *wadan*, not a fronted *wæden*-type output.

warp — OE weorpan

Derivation: **wérpaną* > *weorpan* (regular).

Derivation trace

Proto input: **wérpaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Breaking	**wéorpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**wéorpan*
Northwest Germanic	OE Secondary Nasalization	**wéorpąn*
[no change]	OE Weak Tail Reduction	**wéorpan*

Old English form: *weorpan*

Reconstruction and comparative evidence

Ringe and Taylor distinguish preterite **warp* from infinitive **werpana*, and the derivational input here is the verbal form **wérpaną* (Ringe and Taylor 2014).

Old English evidence

Clark Hall records *weorpan* as the strong verb headword and separately lists *wearp* as both noun and preterite. Bright gives the paradigm *weorpan, wearp, wurpon, worpen* (Clark Hall 1960; Bright 1971).

Development to Old English

Breaking before *r* + *C* yields *weor-*, and the infinitive develops regularly to *weorpan* (Campbell 1959; Hogg 1992).

Lexical note

The note matters because English *warp* also points to related *wearp* material. Here the target is specifically the infinitive *weorpan*.

wash — OE wascan

Derivation: **wáskaną* > *wascan* (regular).

Derivation trace

Proto input: **wáskaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wæskaną*
[no change]	OE A Restoration	**waskaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**waskan*
[no change]	OE Secondary Nasalization	**waską̄n*
	OE Weak Tail Reduction	**waskan*

Old English form: *wascan*

Reconstruction and comparative evidence

Kroonen cites **waskan-*, Orel **waskanan*, and Ringe and Taylor likewise derive Old English *wascan* from the same verb family (Kroonen 2013; Orel 2003, 489; Ringe and Taylor 2014, 142).

Old English evidence

Clark Hall heads the verb as *wascan*, while Sievers-Brunner also notes the variant *wæscan* (Clark Hall 1960; Sievers 1965).

Development to Old English

From **wáskaną*, brightening gives **wæskaną*. A-restoration before the *sC* cluster restores *a*, and medial *sc* remains unpalatalized before the following back vowel, yielding *wascan* (Campbell 1959; Ringe and Taylor 2014, 142).

Form note

The conservative citation form *wascan* is selected here. Spellings such as *wæscan* or *wasċan* belong to variant or normalized background rather than to the target of this entry.

wax — OE weaxan

Derivation: **wáxsaną* > *weaxan* (regular).

Derivation trace

Proto input: **wáxsaną*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**wæxsaną*
[no change]		OE Breaking	**weaxsaną*
Northwest Germanic		OE Heavy Syllable Nasal Apocope	**weaxsan*
[no change]		OE Secondary Nasalization	**weaxsąn*
		OE Weak Tail Reduction	**weaxsan*
		OE Xs Merge	**weaXSan*

Old English form: *weaxan*

Reconstruction and comparative evidence

Kroonen cites the verb as **wahs(j)an-*, Orel as **waxsanan*, and Ringe and Taylor discuss the prehistory of Old English *weaxan* within the same verbal family (Kroonen 2013; Orel 2003, 478; Ringe and Taylor 2014).

Old English evidence

Clark Hall gives *weaxan* as the verb headword and separately records bare *wax* as a preterite form; Bright likewise treats *weaxan* as the infinitive (Clark Hall 1960; Bright 1971).

Development to Old English

From **wáxsaną*, brightening and breaking yield *weax-*, and the infinitive develops regularly to *weaxan*. The cluster is preserved here, since *xs* > *s* belongs to forms where another consonant follows, such as *wæstm*, not to the infinitive itself (Campbell 1959; Sievers 1965).

Lexical note

The target here is the infinitive *weaxan*. Noun *weax* and preterite *wax*/*wēox* belong to different lexical or paradigm slots.

way — OE weġ

Derivation: **wégaz* > *weġ* (regular).

Derivation trace

Proto input: **wégaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**wég*
[no change]		OE Velar Palatalization	**wéʤ*
Northwest Germanic			
PGmc Final Z Deletion	**wéga*		

Old English form: *weġ*

Reconstruction and comparative evidence

Kroonen cites the noun as **wega-* ‘way, road’, while the selected derivational form here is nominative-singular **wégaz* (Kroonen 2013). Campbell, Hogg, and Ringe and Taylor use the same word as the standard contrast between singular palatal *weġ* and inflected *wegas*/*wegum* (Campbell 1959; Hogg 1992; Ringe and Taylor 2014, 341).

Old English evidence

The Old English singular is the ordinary noun *weg*, here normalized as *weġ* to show the palatal final. The contrasting plural and oblique forms *wegas*, *wegum* keep a velar stop before the following back vowel (Campbell 1959; Ringe and Taylor 2014, 341).

Development to Old English

From **wégaz*, final **z* is lost and the weak tail apocopates, leaving word-final **g* after a front vowel. In that environment Old English palatalization yields *weġ*, whereas *wegas* remains velar because the following *a* blocks the same outcome (Campbell 1959; Hogg 1992; Ringe and Taylor 2014, 341).

Form note

Normalized *weġ* and dictionary *weg* represent the same noun. *wē* is not supported in the checked Old English evidence for 'way'.

weapon — OE wǣpn

Derivation: **wḗpną* > *wǣpn* (regular).

Derivation trace

Proto input: **wḗpną*

Earlier Germanic changes		**Old English changes**	
West Germanic		OE Heavy Syllable Nasal Apocope	**wǣpn*
[no change]			
Northwest Germanic			
NWGmc Long E Lowering	**wǣpną*		

Old English form: *wǣpn*

Reconstruction and comparative evidence

Kroonen reconstructs a double-stem noun **wēbna-* ~ **wēpna-* and cites OE *wæpn* among its reflexes (Kroonen 2013, 617). The form followed here, **wḗpną*, represents the unbroken citation-form noun rather than the later broken simplex.

Old English evidence

Campbell's cluster-noun discussion preserves unbroken *wépn* beside broken *wépen*-type forms (Campbell 1959, 150; 1959, 226–27). Bright contrasts broken nominative *wǣpen*/*wapen* with unbroken oblique *wǣpnes*, while Clark Hall lemmatizes the noun under *wapen* and also preserves unbroken forms in compounds and related spellings (Bright 1971, 29; Clark Hall 1960, 355).

Development to Old English

Northwest Germanic lowering gives *wǣpn*, and loss of the final nasal vowel leaves the unbroken cluster word-finally. The Old English form here is the attested unbroken form *wǣpn*.

Form note

The ordinary late West Saxon simplex headword is *wǣpen*, but Campbell's noun-class discussion also preserves unbroken *wépn* beside broken *wépen*-type forms (Campbell 1959, 150; 1959, 226–27). *wǣpnes* remains the regular unbroken oblique comparator (Bright 1971, 29; Clark Hall 1960, 355).

will — OE willa

Derivation: **wéljô* > *willa* (regular).

Derivation trace

Proto input: **wéljô*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**willjô*
PWGmc J Gemination	**wélljô*	OE Unstressed Long Vowel Shortening	**willja*
Northwest Germanic		OE J Loss After Heavy	**willa*
[no change]			

Old English form: *willa*

Reconstruction and comparative evidence

Kroonen separates noun **weljan-* 2 'will, wish' from verb **weljan-* 1 'to want', while Orel and Kluge represent the noun as **weljōn* (Kroonen 2013; Orel 2003; Kluge 2011). The selected derivational form **wéljô* is the noun-side input used for this row.

Old English evidence

Clark Hall lemmatizes noun *willa m.* separately from verb *willan* (Clark Hall 1960, 368). The Old English form here is the noun citation form, not the related verb.

Development to Old English

From **wéljô*, j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa*. The noun is therefore a regular weak masculine outcome.

Lexical note

The target here is the noun *willa* 'will, wish'. Related verb *willan* belongs to a separate lexeme and should not be substituted for the noun row (Kroonen 2013; Clark Hall 1960, 368).

wind — OE windan

Derivation: **wíndaną* > *windan* (regular).

Derivation trace

Proto input: **wíndaną*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**wíndan*
[no change]		OE Secondary Nasalization	**wíndąn*
Northwest Germanic		OE Weak Tail Reduction	**wíndan*
[no change]			

Old English form: *windan*

Reconstruction and comparative evidence

Kroonen distinguishes noun **winda-* from verb **windan-*, and the present row belongs to the verb (Kroonen 2013). Later handbook discussion keeps the dental original from PIE **wendh-*, not a Verner alternant (Fulk 2018; Ringe and Taylor 2014).

Old English evidence

Clark Hall and Bosworth-Toller record *windan* as the verb headword (Clark Hall 1960; Bosworth and Toller 1898, 101). The Old English form here is the ordinary infinitive of the strong verb.

Development to Old English

The form followed here, **wíndaną*, yields the regular infinitive *windan* by ordinary heavy-syllable apocope and weak-tail reduction. The form is therefore a straightforward strong-verb outcome.

Lexical note

The note matters because English *wind* also names the noun. This row targets the class-III verb, not the noun (Kroonen 2013; Clark Hall 1960).

wold — OE weald

Derivation: **wálθuz* > *weald* (regular).

Derivation trace

Proto input: **wálθuz*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**wældu*
PWGmc L Th Voicing	**wálduz*	OE Breaking	**wealdu*
Northwest Germanic		OE High Vowel Apocope	**weald*
PGmc Final Z Deletion	**wáldu*		

Old English form: *weald*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **walþu-* and gives OE *weald* beside other West Germanic *wald* forms (Kroonen 2013). The form followed here, **wálθuz*, is the nominative singular used for the derivation.

Old English evidence

Clark Hall makes *weald* the main noun headword and cross-refers *wald* and *wold* to it (Clark Hall 1960). The Anglian-looking *wald* therefore remains variant background rather than the main target.

Development to Old English

**lþ* voices to *ld*, Anglo-Frisian brightening yields *wæld-*, and breaking before the cluster gives *weald-*; apocope then yields *weald*. The noun is therefore a regular breaking outcome.

Dialect note

The note matters because *wald* survives as an Anglian-type variant in the same family. The Old English form here is normalized *weald*, not the variant form (Clark Hall 1960; Ringe and Taylor 2014).

yarn — OE ġearn

Derivation: **gárną* > *ġearn* (regular).

Derivation trace

Proto input: **gárną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**gærną*
[no change]	OE Breaking	**gearną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**gearn*
[no change]	OE Velar Palatalization	**ʤearn*

Old English form: *ġearn*

Reconstruction and comparative evidence

Kroonen cites the noun as **garna-*, and Ringe and Taylor give the early chain **garna* > **geern* > **gearn* > OE *gearn* (Kroonen 2013; Ringe and Taylor 2014, 220). The form followed here, **gárną*, is the nominal citation form used here, while oblique **garnăn* belongs only to comparative background.

Old English evidence

Clark Hall records *gearn (e) n.* 'yarn, spun wool', and Bosworth-Toller glosses *gearn* as *filatum* (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **gárną*, brightening and breaking before *rn* yield *gearn*; palatalization of initial *g* before the resulting front-vocalic sequence gives normalized *ġearn*. The derivation is regular.

Form note

Dictionary *gearn* and normalized *ġearn* refer to the same noun. The comparative stem **garna-* and oblique **garnăn* do not replace the derivational input **gárną*.

3.6 Attested variants and comparison forms

These entries treat the attested Old English form as one member of an attested or historically documented variant set. The lexical comparison must therefore account for variation rather than for a single unproblematic citation form.

cud — OE cwedu

Derivation: citation reconstruction _*kwíθuz_; form followed here _*kwéðuz_ > _cwedu_ (attested variant).

Derivation trace

Proto input: _*kwéðuz_

Earlier Germanic changes		Old English changes
West Germanic		[no change]
PWGmc Dental Hardening	_*kwéduz_	
Northwest Germanic		
PGmc Final Z Deletion	_*kwédu_	

Old English form: _cwedu_

Reconstruction and comparative evidence

Kroonen reconstructs the resin word as _*kwedu-2_ and gives Old English variants _cwidu_, _cweodu_, and _c(w)udu_ (Kroonen 2013, 355). Orel likewise lists _cwidu_ under the cognate set (Orel 2003, 266). The derivational input _*kwéðuz_ therefore represents the older e-grade, voiced-dental form behind the chosen variant _cwedu_.

Old English evidence

The Old English word survives in a wider variant set than one dictionary headword suggests. Ringe and Taylor discuss _cwidu_ > _cwudu_ > _cudu_ and also note late West Saxon _cweodu_; Clark Hall gives _cwudu_, _cweodu_, and _cudu_ (Ringe and Taylor 2014, 338; Clark Hall 1960, 84). Attested _cwedu_ is treated here as the conservative variant within that set.

Development to Old English

From _*kwéðuz_, the West Germanic voiced dental hardens in the expected way and the regular Old English development yields _cwedu_. The other Old English spellings belong to the same lexical family, but reflect later leveling, back-umlaut, or further reduction rather than a need to replace the selected input.

Variant comparison

Variant type	Old English form	Comment
conservative target	_cwedu_	selected attested variant represented here
leveled i-grade form	_cwidu_	common lexical variant in the same family
back-umlauted forms	_cweodu_, _cwudu_	later developments within the same OE tradition
reduced form	_cudu_	further reduced member of the same variant set

ten — OE tēon

Derivation: **téxun* > *tēon* (attested variant).

Derivation trace

Proto input: **téxun*

Earlier Germanic changes	Old English changes	
West Germanic	OE Med Unstressed U Lowering	**téxon*
[no change]	OE Breaking	**téoxon*
Northwest Germanic	OE H Loss	**téoon*
[no change]	OE Contraction	**tḗon*

Old English form: *tēon*

Reconstruction and comparative evidence

Fulk states that Old English *tien* shows umlaut from the inflected forms, whereas the uninflected form without umlaut is reflected in *hund-tēon-tig* (Fulk 2018, sect. 10.2). Brunner gives the same contrast more broadly: *tēon* develops from **tëhun*, while West Saxon tien, *tȳn* belong to a different, umlauted branch of the numeral history (Sievers 1965, sects 129.2, 129 Anm. 6, 234).

The comparison form *tēon* therefore represents the bare cardinal's un-umlauted line, not the later umlauted simplex tradition.

Old English evidence

The attested simplex forms are varied. Campbell gives *tien*, north-western West Saxon *tēn*, and late Northumbrian *tēo*, *tēa* (Campbell 1959, sect. 682). Brunner likewise lists West Saxon tien, *tȳn* beside *tēn*, *tēo*, *tēa* in other dialects (Sievers 1965, sect. 325).

Exact simplex *tēon* is weaker as a directly cited headword than those spellings. The un-umlauted stem is, however, explicit in *tēoða* and *-tēontig* (Sievers 1965, sect. 129.2; Fulk 2018, sect. 10.2). The comparison form *tēon* is therefore a normalized spelling of that un-umlauted base.

Development to Old English

From **téxun*, lowering of medial unstressed *u* gives **téxon*, breaking gives **téoxon*, loss of intervocalic *h*/*x* yields **téoon*, and contraction produces **tḗon*, written *tēon*. This is the regular bare-cardinal path.

The umlauted forms *tien* / *tīen* belong to a different branch, created when the numeral was levelled from inflected forms with a front-vocalic trigger (Fulk 2018, sect. 10.2; Sievers 1965, sect. 129 Anm. 6).

Variant comparison

The comparison below sets the relevant forms side by side. It distinguishes the normalized un-umlauted base from the attested simplex variants.

Form or branch	Status	Relevance to this entry
tēon	normalized un-umlauted comparison form; trace-supported	Old English form here
tien / *tīen*	attested West Saxon umlauted simplex forms	genuine OE variants, but not the bare-cardinal line modeled here

Form or branch	Status	Relevance to this entry
tēn / *tēo* / *tēa*	attested un-umlauted simplex variants in other dialects	support the same branch as the comparison form

three — OE þrīe

Derivation: **θréjez* > *þrīe* (attested variant).

Derivation trace

Proto input: **θréjez*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**θrije*
[no change]		OE Intervocalic J Vocalization	**θriie*
Northwest Germanic		OE Contraction	**θrīe*
PGmc Final Z Deletion	**θréje*		

Old English form: *þrīe*

Reconstruction and comparative evidence

Kroonen cites the numeral under a broader stem-style reconstruction rather than under one Old English-ready paradigm cell (Kroonen 2013, 586). The input **θréjez* is therefore best understood as the inherited masculine nominative-accusative singular.

That distinction matters because the Old English numeral does not have one uniform citation form across the paradigm. The masculine singular line must be kept apart from feminine-neuter *þrēo* and from later reduced spellings of the masculine form.

Old English evidence

Campbell gives masculine nominative-accusative *þrīe*, feminine and neuter nominative-accusative *þrēo*, genitive *þrēora*, and dative *þrim*, adding that late West Saxon has *þry*, *þri* for *þrīe* (Campbell 1959, sect. 683). Fulk presents the same masculine *þrīe* beside the wider numeral paradigm (Fulk 2018, sect. 10.1).

The target is therefore an attested Old English paradigm cell. *þrī* belongs to later reduction or headword-style citation, whereas *þrīe* is the conservative masculine nominative-accusative form.

Development to Old English

From **θréjez*, loss of final *-z* leaves a form of the **θréje* type. The following *j* fronts the stem vowel, then vocalizes between vowels, and contraction yields *þrīe*. The compact trace records the same sequence as **θrije* > **θriie* > *þrīe*.

Variant comparison

The comparison below sets the relevant forms side by side. It separates the selected masculine cell from the later reduced form and from the rest of the numeral paradigm.

Form	Status	Relevance to this entry
þrīe	attested masculine nom./acc.; regular output	Old English form here

Form	Status	Relevance to this entry
þrī / *þry*	later reduced masculine variant	genuine OE variant, but not the conservative comparison form
þrēo	attested feminine-neuter nom./acc.	same numeral, different paradigm cell
þrēora, *þrim*	attested genitive and dative forms	confirm the wider paradigm, not the cell compared here

wasp — OE wæfs

Derivation: **wábsaz* > *wæfs* (attested variant).

Derivation trace

Proto input: **wábsaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**wábs*
[no change]		Anglo Frisian Brightening	**wæbs*
Northwest Germanic		PGmc B Allophony	**wæβs*
PGmc Final Z Deletion	**wábsa*		

Old English form: *wæfs*

Reconstruction and comparative evidence

The Proto-Germanic form **wábsaz* reaches Old English without any special change of stem or paradigm cell. The question in this entry is instead which attested Old English member of the variant set should serve as the comparison form.

Fulk presents the Old English forms together as *wæfs* with variants *wæsp* and *wæps* (Fulk 2018, sect. 6.5). Bülbring and Brunner then make the chronology more explicit by deriving later *wæps* and late West Saxon *wasp* from earlier *waefs* / *wæfs* through restricted metatheses (Bülbring 1902, sect. 484 Anm. 3; Sievers 1965, sects 193, 204).

Old English evidence

The earliest directly cited Old English form is *wæfs*, written *waefs* in the Épinal-Corpus material discussed by Bülbring and Brunner (Bülbring 1902, sect. 484 Anm. 3; Sievers 1965, sect. 193). Later Old English also shows *wæps* and *wæsp* / *wasp*, and dictionary practice often favors *wæps* or later spellings as headwords (Clark Hall 1960, 341).

This entry therefore distinguishes chronological priority from headword habit. *wæfs* is not a convenient reconstruction: it is an attested Old English form and also the one that matches the regular development most closely.

Development to Old English

From **wábsaz*, the regular Old English path passes through loss of final *z*, Anglo-Frisian fronting, and the allophonic development of *b* to a fricative before *s*, yielding *wæfs*.

The later forms *wæps* and *wæsp* / *wasp* belong to subsequent, lexically restricted metatheses. They are genuine Old English forms, but they are later within the variant history.

Variant comparison

The comparison below sets the relevant forms side by side. It separates the earliest attested and regular form from the later metathesized doublets.

Form	Status	Relevance to this entry
wæfs	earliest attested OE form; regular output	Old English form here
wæps	later attested metathesized variant	genuine OE doublet, but secondary
wæsp / *wasp*	later West Saxon metathesized variant	genuine OE doublet, but not the form compared here

3.7 Early analogy and pre-Old-English input selection

These entries involve a distinction between the lexeme-level citation reconstruction and the earlier form carried through the Old English derivation. The issue is upstream of Old English: the form compared here is the pre-Old-English shape that yields the attested target under the sound history followed here.

bottom — OE botm

Derivation: citation reconstruction **búdmaz*; form followed here **búttmaz* > *botm* (early analogy).

Derivation trace

Proto input: **búttmaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**bóttm*
[no change]		OE Preconsonantal Degemination	**bótm*
Northwest Germanic			
NWGmc U Lowering	**bóttmaz*		
PGmc Final Z Deletion	**bóttma*		

Old English form: *botm*

Reconstruction and comparative evidence

Kroonen reconstructs the word as a stem complex **budmō*, gen. **buttaz*, summarized as **budman-* ~ **buttman-*, and gives Old English *botm* as the reflex (Kroonen 2013, 120). The comparative label **búdmaz* names the lexeme-level stem complex, while the derivational input **búttmaz* represents the pre-Old-English form with oblique **butt-* generalized into the nominative formation.

Orel likewise preserves both sides of the comparison under **budmaz *butmaz* (Orel 2003, 100). The derivational input is thus a historical stem choice, not an arbitrary respelling.

Old English evidence

The Old English noun itself is secure. Clark Hall gives *botm* (Clark Hall 1960, 63). Bosworth-Toller cross-references *bodan* to *botm*, showing the wider reflex family without weakening the attested lemma (Bosworth and Toller 1898, 112).

Development to Old English

Once the oblique **butt-* stem has been generalized, the derivational input **búttmaz* develops regularly to *botm*. The analogical step is therefore early: it belongs to pre-Old-English stem formation rather than to a later choice among Old English paradigm cells.

brand — OE brandes

Derivation: citation reconstruction **brándaz*; form followed here **brándas* > *brandes* (early analogy).

Derivation trace

Proto input: **brándas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**brándæs*
[no change]	OE Unstressed AE Merger	**brándes*
Northwest Germanic		
[no change]		

Old English form: *brandes*

Reconstruction and comparative evidence

The inherited noun is the masculine a-stem **brándaz*, continued by Old English *brand* and its continental cognates (Orel 2003, 53). The selected input **brándas* is not a different lexeme but the genitive singular of that same a-stem noun.

What matters here is therefore not a stem-class disagreement but the difference between the citation form and a specific inherited inflectional cell. The derivational input preserves the same root and declension as the headword while making the oblique ending explicit.

Old English evidence

Old English dictionaries lemmatize the noun as *brand* (Clark Hall 1960, 49; Bosworth and Toller 1898, 116). Bosworth-Toller also records inflectional forms such as *brandas*, *branda*, and *brandum* under the same entry (Bosworth and Toller 1898, 116).

The specific comparison form in this entry, *brandes*, is the expected genitive singular of that a-stem noun. It is therefore an inferred Old English paradigm form rather than the ordinary dictionary headword.

Development to Old English

From **brándas*, the regular Old English development passes through the usual unstressed-vowel weakening of the inflectional ending, yielding *brandes*. Nothing in the stem itself requires a special repair. The root consonants and the stressed vowel are the same as in the citation lemma *brand*.

The analytical weight of the entry lies in the ending. By choosing the oblique singular rather than the nominative citation form, the entry presents the same lexeme in a different inherited cell.

Form comparison

The comparison below sets the relevant forms side by side. It separates the citation lemma from the selected oblique singular.

Form / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**brándaz*	expected OE lemma *brand*	*brand*	regular headword-level outcome
genitive singular	**brándas*	regular output: *brandes*	*brandes*	exact match for the oblique cell

The noun itself is straightforwardly inherited. The main point of the entry is that *brandes* belongs to the same regular a-stem paradigm as *brand*, even though the citation lemma remains the nominative singular.

breast — OE brēost

Derivation: citation reconstruction **brústz*; form followed here **bréustą* > *brēost* (early analogy).

Derivation trace

Proto input: **bréustą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Diphthong Leveling	**brēostą*
[no change]	OE Heavy Syllable Nasal Apocope	**brēost*
Northwest Germanic		
[no change]		

Old English form: *brēost*

Reconstruction and comparative evidence

The word family shows two related but distinct Proto-Germanic formations. The root noun **brust-* lies behind forms such as Gothic *brusts*, whereas Old English *brēost* belongs to a thematic formation **breusta-*, alongside Old Norse *brjóst* and Old Saxon *briost* (Kroonen 2013, 114; Orel 2003, 95; Ringe and Taylor 2014, 43).

The derivational input **bréustą* therefore differs from the citation label **brústz* because Old English reflects the thematic branch rather than the root noun. The morphological choice comes before the Old English sound changes themselves.

Old English evidence

Clark Hall records the noun as *brēost* / *breóst* (Clark Hall 1960, 65). The form is an established Old English lexeme, not a reconstructed target assembled from comparative evidence alone.

What requires explanation is not the Old English attestation but the relation between that attested noun and the broader Germanic word family. The relevant comparison form is therefore the thematic Old English noun *brēost*.

Development to Old English

From **bréustą*, the regular Old English development gives *brēost*, with the expected *eu* > *ēo* vowel history (Campbell 1959, sect. 115). No special repair is needed once the correct thematic formation is chosen.

The earlier mismatch arose only if the word was forced into the root-noun line. The Old English noun itself continues the thematic branch cleanly and directly.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader root-noun family label from the thematic formation actually continued in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader root-noun family	**brústz*	root-noun type outcomes outside OE	non-OE comparanda	useful family label, but not the direct source of *brēost*
selected thematic formation	**bréustą*	regular output: *brēost*	*brēost*	exact match between formation and attested OE noun

The relevant point is the formation split. *brēost* is the regular Old English outcome of the thematic **breusta-* branch, not of the root noun **brust-*.

craft — OE cræft

Derivation: citation reconstruction **kráftiz*; form followed here **kráftaz* > *cræft* (early analogy).

Derivation trace

Proto input: **kráftaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**kráft*
[no change]		Anglo Frisian Brightening	**kræft*
Northwest Germanic			
PGmc Final Z Deletion	**kráfta*		

Old English form: *cræft*

Reconstruction and comparative evidence

Comparative sources disagree about the older stem class. Kroonen gives a u-stem **kraftu-*, while Orel prints **kraftiz* ~ **kraftuz* (Kroonen 2013, 340; Orel 2003, 259). The comparative label **kráftiz* remains in view as a lexeme-level shorthand, while **kráftaz* is the pre-Old-English form used for the Old English derivation.

Old English evidence

The Old English noun itself is secure. Clark Hall and Bosworth-Toller both give *cræft* as the headword (Clark Hall 1960, 19; Bosworth and Toller 1898, 145).

Development to Old English

The comparison is between possible pre-Old-English inputs. The i-stem comparator **kráftiz* gives *creft*, while the u-stem comparator **kráftuz* gives *craft*. The a-stem-shaped input **kráftaz* yields *cræft* and is therefore the form used for the Old English derivation. This does not require treating the comparative dictionaries as identical; it shows the narrower point that the Old English derivation needs a pre-Old-English form without the i-umlaut trigger of **-iz* and without the back-vowel outcome associated with the u-stem comparator.

Form comparison

Candidate input	OE output	Result
*kráftiz	*creft*	non-match; i-stem comparator
*kráftuz	*craft*	non-match; u-stem comparator
*kráftaz	*crœft*	exact match; selected pre-OE input

dill — OE dile

Derivation: citation reconstruction **déljaz*; form followed here **déliz* > *dile* (early analogy).

Derivation trace

Proto input: **déliz*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**dili*
[no change]		OE Med Unstressed I Lowering1	**dile*
Northwest Germanic			
PGmc Final Z Deletion	**déli*		

Old English form: *dile*

Reconstruction and comparative evidence

Comparative evidence preserves both an i-stem and a ja-stem formation, with Old English *dile* on one side and continental forms such as Old Saxon *dilli* and Old High German *tilli* on the other (Fulk 2018, 170). The derivational input **déliz* therefore represents the i-stem side of the paradigm, whereas the citation label **déljaz* is a broader comparative headword.

That stem-class distinction matters for the Old English consonant shape. A ja-stem with **-lj-* would be expected to produce gemination, but the Old English noun shows a single *l*. Fulk's discussion of ja-stems transferred to the i-stems provides the relevant morphological background for the OE side (Fulk 2018, 170).

Old English evidence

Old English dictionaries record the plant name as *dile*, alongside the variant *dili* (Bosworth and Toller 1898, 164; Clark Hall 1960, 95). The form discussed here is therefore an attested Old English noun with single *l*.

The Old English evidence is the relevant point. Whatever broader comparative headword is chosen for the family, the inherited form reflected in OE is the i-stem type *dile*, not a geminated *dill* outcome.

Development to Old English

From **déliz*, regular loss of final *z* and the later lowering of unstressed *i* yield *dile*. The stem itself remains ungeminated throughout that path.

The important contrast is negative rather than phonological. If the word were forced through a ja-stem **-lj-* pathway, the expected result would show *ll*. The attested Old English noun instead matches the i-stem development.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the stem class actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative ja-stem label	**déljaz*	ja-stem type outcome with gemination	*dill*-type comparison	useful comparative label, but not the OE form
selected i-stem formation	**déliz*	regular output: *dile*	*dile*	exact match between formation and attested OE noun

The single *l* is the decisive diagnostic. It identifies *dile* with the i-stem formation rather than with the continental ja-stem branch.

fast — OE festan

Derivation: citation reconstruction **fastēną*; form followed here **fástijaną* > *festan* (early analogy).

Derivation trace

Proto input: **fástijaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**fæstijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**fæstijan*
Northwest Germanic	OE Secondary Nasalization	**fæstijąn*
[no change]	Sievers Law Syncope	**fæstjąn*
	OE I Umlaut	**festjąn*
	OE Weak Tail Reduction	**festjan*
	OE J Loss After Heavy	**festan*

Old English form: *festan*

Reconstruction and comparative evidence

Kroonen places the verb within the wider **fastu-* adjective family and its derived **fasten-* verbal line, the comparative background behind Old English 'to fast' (Kroonen 2013, 171). Ringe and Taylor, however, distinguish the Old English verb more closely: they treat OE 'to fast' as originally a class-I weak verb that later acquired the stative meaning through lexical confusion (Ringe and Taylor 2014, 110).

The derivational input **fástijaną* therefore represents the inherited class-I formation reflected in Old English, whereas the citation label **fastēną* belongs to the broader comparative presentation of the lexeme.

Old English evidence

Old English dictionaries record forms such as *festan*, alongside related *fæstan* / *fǣstan* spellings and meanings (Bosworth and Toller 1898, 213). The form selected here is *festan*, which fits the regular class-I phonological development.

The *æ*-forms remain relevant, but they do not control the entry. In the present analysis they belong to a later analogical reshaping under the adjective *fæst*, whereas *festan* is the regular inherited class-I comparison form.

Development to Old English

From **fástijaną*, Anglo-Frisian brightening and subsequent i-umlaut produce the fronted vowel seen in *festan*. The later weak-tail reductions and loss of *j* after a heavy syllable complete the regular Old English outcome.

What makes the entry non-regular is not the phonology of *festan* itself, but the choice of formation. Old English continues the class-I verb, even though the comparative headword is often given under the parallel **fastēn-* family.

Class comparison

The comparison below sets the relevant forms side by side. It distinguishes the comparative class-III headword from the class-I formation actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative class-III headword	**fastēną*	class-III type outcome, not *festan*	wider family context	useful family label, but not the direct source of the target
selected class-I weak verb	**fástijaną*	regular output: *festan*	*festan*	exact match between formation and attested OE verb
later analogical reshaping	adjective-driven *fæst* influence	*fæstan* / *fǣstan* type spellings	fæstan-type evidence	genuine later OE reshaping, but secondary to the Old English form here

The relevant point is the class split. *festan* is the regular Old English outcome of the class-I formation, while the better-known *æ*-forms belong to a later analogical layer.

flask — OE flasce

Derivation: citation reconstruction **flaskō*; form followed here **fláskōn* > *flasce* (early analogy).

Derivation trace

Proto input: **fláskōn*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

NWGmc N Stem N Loss	**fláskǭ*

Old English changes

Anglo Frisian Brightening	**flæskǭ*
OE A Restoration	**flaskǭ*
OE Unstressed Long Vowel Shortening	**flaskæ*
OE Unstressed AE Merger	**flaske*

Old English form: *flasce*

Reconstruction and comparative evidence

The wider Germanic family is often cited under a form such as **flaskō*, but the evidence relevant for Old English points instead to a weak feminine formation **fláskōn* / **flaskǭ* (Orel 2003, 104). That distinction is crucial for the suffixal history of the noun.

The derivational input therefore differs from the citation label in stem class. Old English *flasce* belongs with the weak feminine line, and the plural or oblique forms *flascan* support that analysis (Ringe and Taylor 2014, 192).

Old English evidence

Old English dictionaries record the noun as *flasce*, with inflectional support from forms such as *flascan*; a later West Saxon *flaxe* is also noted as a secondary variant (Bosworth and Toller 1898, 235; Clark Hall 1960, 121).

The relevant comparison form is therefore the weak feminine noun *flasce*. The plural and oblique evidence matters because it helps explain why the vowel and ending are preserved as they are in the singular.

Development to Old English

From **fláskōn*, the weak feminine passes through the expected loss of *n* and the later Old English development of the unstressed ending, reaching *flasce*. Campbell cites restored *a* in exactly this environment, including *flasce* after inflected *flascan* (Campbell 1959, sect. 158). Once the weak feminine formation is chosen, the noun follows a regular path to its Old English shape.

The decisive issue is morphological rather than phonological. A simple strong feminine citation form does not capture the OE weak noun as cleanly as the selected **fláskōn* does.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak feminine formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader comparative headword	**flaskō*	broader family label	wider family context	useful lexeme label, but not the cleanest OE-facing derivation
selected weak feminine formation	**fláskōn*	regular output: *flasce*	*flasce*	exact match between formation and attested OE noun

The weak feminine suffix is the relevant point. It aligns the inherited form with attested *flasce* and its supporting paradigm forms.

follow — OE fylġan

Derivation: citation reconstruction **fulgēną*; form followed here **fúlgijaną* > *fylġan* (early analogy).

Derivation trace

Proto input: **fúlgijaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**fúlgijan*
[no change]	OE Secondary Nasalization	**fúlgijąn*
	Sievers Law Syncope	**fúlgjąn*
Northwest Germanic	OE Velar Palatalization	**fúlʤjąn*
[no change]	OE I Umlaut	**fylʤjąn*
	OE Weak Tail Reduction	**fylʤjan*
	OE J Loss After Heavy	**fylʤan*

Old English form: *fylġan*

Reconstruction and comparative evidence

Kroonen keeps the verb under **fulgen-* and gives Old English *fylgan*, *folgian*, adding that Old Norse *fylgja* and Old English *fylg(e)an* continue a formation **fulgjan-* (Kroonen 2013, 159). The comparative headword and the class-I formation are therefore related but not identical.

Ringe and Taylor make the split explicit as PNWGmc **fulgija-* ~ **fulgai-* > OE *fylgan* ~ *folgian* and describe it as a dual formation that probably reflects an older alternation between j-present and e-stative (Ringe and Taylor 2014, 293–94). This is a stem-class choice, not a spelling choice. The derivational input **fúlgijaną* belongs to the class-I **fulgija-* / **fulgjan-* branch; the citation form **fulgēną* belongs to the parallel class-II history behind *folgian*.

Old English evidence

The Old English evidence preserves both formations. Clark Hall lists *fylgan* with variant spellings *fylgian* and *fyligan* (Clark Hall 1960, 125). Bosworth-Toller likewise has a separate *fylgean* entry (Bosworth and Toller 1898, 275).

Bright notes traces of the older conjugation in *fylg(e)an* (Bright 1971, 77) and lists *folgian* (*fylgean*) in the glossary (Bright 1971, 364). The relevant comparison form in this entry is therefore the class-I verb *fylgan* / *fylgean*, here normalized as *fylġan*. The spelling with ġ represents the palatalized velar before a front-vocalic environment.

Development to Old English

**fúlgijaną* is a class-I weak-verb formation. In the class-I branch the **j* blocks NWGmc lowering of *u* to *o*, since Ringe and Taylor formulate that lowering for environments in which no **j* intervened (Ringe and Taylor 2014, 96). The same front-vocalic environment then triggers i-umlaut, so *u* becomes *y* (Ringe and Taylor 2014, sect. 6.6.2).

The subsequent Old English developments are palatalization of the velar, weak-tail reduction, and loss of *j* after a heavy syllable, yielding *fylġan*. This is the regular outcome of the class-I formation. The class-II form *folgian* belongs to the parallel **-ē-* / **-ai-* branch and is not the form modeled here.

Class comparison

A class comparison identifies which inherited formation corresponds to the established Old English form under discussion. The comparison below is manual; no full automatic class probe is presented here.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation class-II formation	**fulgēną*	probe output: *folgon*	*folgian*	mismatch: the regular output is not the remodeled infinitive *folgian*
parallel class-II branch	PNWGmc **fulgai-*	Ringe-Taylor: OE *folgian*	*folgian*	documents the separate class-II branch, but not the target of this entry
selected class-I formation	**fúlgijaną*	regular output: *fylġan*	*fylġan* / *fylgan*	exact match between input, output, and class

The relevant point is the class split. *fylġan* is the regular Old English outcome of the class-I **fulgija-* / **fulgjan-* formation, whereas *folgian* belongs to the parallel class-II branch.

gall — OE ġealla

Derivation: citation reconstruction **gállą*; form followed here **gállô* > *ġealla* (early analogy).

Derivation trace

Proto input: **gállô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**gællô*
[no change]	OE Breaking	**geallô*
Northwest Germanic	OE Velar Palatalization	**ʤeallô*
[no change]	OE Unstressed Long Vowel Shortening	**ʤealla*

Old English form: *ġealla*

Reconstruction and comparative evidence

The wider cognate family can be presented under a form such as **gállą*, but the Old English noun itself belongs with a weak noun **gallōn-*, cited here as **gállô* (Kroonen 2013, 165). The derivational input therefore differs from the broader comparative headword in stem class.

That stem-class distinction matters directly for the Old English shape. The weak masculine pathway preserves the ending needed for *ġealla*, whereas a simple strong-noun headword does not align as closely with the attested OE noun.

Old English evidence

Old English dictionaries record the noun as *gealla*, and Bright also gives the dative *geallan*, confirming a weak-noun paradigm (Bosworth and Toller 1898, 297; Clark Hall 1960, 145; Bright 1971, 372). The form used here, *ġealla*, is a normalized spelling with macrons omitted and palatal ġ made explicit.

Campbell also notes dialectal variation, contrasting West Saxon or Kentish *gealla* with Anglian *galla* (Campbell 1959, sect. 486). The target of this entry is the West Saxon type *ġealla*.

Development to Old English

From **gállô*, the weak noun develops through the expected Old English history of the suffix and the regular breaking environment before *ll*, yielding *ġealla* (Campbell 1959, sect. 486). Once the weak masculine input is chosen, the noun follows a regular path to its attested Old English form.

The decisive issue is therefore morphological. Old English reflects the weak noun, while the broader family label belongs to a different way of presenting the cognate set.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak noun formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader family label	*gállą	broader cognate-set headword	wider family context	useful lexeme label, but not the direct source of *ġealla*
selected weak noun	*gállô	regular output: *ġealla*	ġealla	exact match between formation and attested OE noun
dialectal Anglian continuation	weak noun branch	Anglian *galla* type	galla	genuine OE variant, but not the West Saxon form used here

The weak-noun stem class is the relevant point. It gives a direct route to attested *ġealla*, while the broader comparative label serves only as a family heading.

knight — OE cniht

Derivation: citation reconstruction **kníxtaz*; form followed here **knéxtaz* > *cniht* (early analogy).

Derivation trace

Proto input: **knéxtaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**knéxt*
[no change]		OE Breaking	**knéoxt*
Northwest Germanic		OE Ws Palatal Umlaut	**knixt*
PGmc Final Z Deletion	**knéxta*		

Old English form: *cniht*

Reconstruction and comparative evidence

The comparative sources align on an *e*-grade reconstruction for this noun. Ringe and Taylor cite **kneht*, and Orel gives **knextaz* (Ringe and Taylor 2014, 142; Orel 2003, 256). Kluge-Seebold likewise points to **knehta-* (Kluge 2011, 506). The derivational input **knéxtaz* follows that comparative evidence.

A competing citation reconstruction **kníxtaz* remains possible as a label for the word family, but it is not the reconstruction followed here. The Old English development discussed below is based on **knéxtaz*.

Old English evidence

Old English dictionaries record the noun as *cniht* (Clark Hall 1960, 63; Bosworth and Toller 1898, 71). Campbell cites plural *cneohtas* among the broken forms, showing the same vowel environment from another point in the paradigm (Campbell 1959, sect. 146).

The target is therefore an ordinary attested Old English noun. No reconstructed OE comparator is needed here.

Development to Old English

From **knéxtaz*, the relevant Old English changes include breaking before the velar cluster and then the later reduction that yields *cniht*. Campbell later notes the early West-Saxon alternation *cniht* beside plural *cneohtas* (Campbell 1959, sect. 305). Sievers-Brunner gives the same contrast as *cniht* ... *cneohtas* (Sievers 1965, sect. 122). With that corrected input, the derivation is straightforward.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the handbook-supported *e*-grade input from a competing citation reconstruction.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing citation reconstruction	**kníxtaz*	not the reconstruction followed here	broader citation tradition	useful as a competing label, but not the source-based choice used for the OE derivation
handbook-supported reconstruction	**knéxtaz*	regular output: *cniht*	*cniht*	exact match between comparative reconstruction and attested OE noun
related plural evidence	same stem family	plural *cneohtas* type background	*cneohtas*	supports the vowel environment, but not the Old English form here cell

lade — OE hladan

Derivation: citation reconstruction **laθōjaną*; form followed here **xláðaną* > *hladan* (early analogy).

Derivation trace

Proto input: **xláðaną*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xlædaną*
PWGmc Dental Hardening	**xládaną*	OE A Restoration	**xladaną*
Northwest Germanic		OE Heavy Syllable Nasal Apocope	**xladan*
[no change]		OE Secondary Nasalization	**xladą̄n*
		OE Weak Tail Reduction	**xladan*

Old English form: *hladan*

Reconstruction and comparative evidence

Ringe and Taylor cite the strong verb *hladan* directly (Ringe and Taylor 2014, 248). The citation label **laθōjaną* is used here only as a broader family heading, not as the direct source of the OE derivation.

The derivational input therefore marks an early stem choice. The entry follows the strong Verner-grade form that reaches Old English *hladan* directly.

Old English evidence

Bosworth-Toller records the verb as *hladan* and preserves the expected strong-verb paradigm material around it (Bosworth and Toller 1898, 559). Clark Hall likewise records *hladan* (Clark Hall 1960, 159). The target is an attested infinitive rather than a reconstructed paradigm cell.

For this entry the relevant comparison form is the infinitive *hladan* itself. The question is how that attested strong verb relates to the broader comparative family.

Development to Old English

From **xláðaną*, the verb passes through the expected early voiced stop stage, Anglo-Frisian brightening, and the later A-restoration that returns the root vowel to *a* before the full infinitival ending. Campbell treats verbs such as *hladan* as showing restored *a* in the present system (Campbell 1959, sect. 744). Ringe and Taylor likewise derive *hladan* from the voiced strong-verb line (Ringe and Taylor 2014, 248). The resulting Old English infinitive is *hladan*.

Class comparison

The comparison below sets the relevant forms side by side. It separates the wider weak-verb family label from the strong verb actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative weak-verb label	*laθōjaną	wider family background	broader family context	useful family label, but not the direct source of *hladan*
selected strong Verner-grade input	*xláðaną	regular output: *hladan*	hladan	exact match between formation and attested OE infinitive

lap — OE lappa

Derivation: citation reconstruction **lábbaz*; form followed here **láppô* > *lappa* (early analogy).

Derivation trace

Proto input: **láppô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**læppô*
[no change]	OE A Restoration	**lappô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**lappa*
[no change]		

Old English form: *lappa*

Reconstruction and comparative evidence

The comparative sources point to a weak noun with *pp*: Orel gives **lappōn*, and the Old English dictionary tradition preserves variant *læppa* (Orel 2003, 236; Clark Hall 1960, 180). The derivational input **láppô* follows that evidence.

A competing comparative label **lábbaz* has also circulated for the word family, but the cited handbooks do not make it the direct source of the Old English weak noun. The form relevant to the OE development is the weak masculine input **láppô*.

Old English evidence

Campbell cites *lappa* as a case of restored *a* (Campbell 1959, sect. 158). Sievers-Brunner records *lappa* beside variant *læppa* (Sievers 1965, sect. 10). The dictionary tradition also preserves *læppa* (Clark Hall 1960, 180; Bosworth and Toller 1898, 613).

The target of this entry is the restored singular *lappa*. The variant *læppa* and the oblique or plural *leappan* remain part of the Old English record and help frame the noun's vowel history.

Development to Old English

Campbell explicitly lists *lappa* among the forms with restored *a* (Campbell 1959, sect. 158). Sievers-Brunner records *lappa* beside variant *læppa* at the same Old English stage (Sievers 1965, sect. 10). With the weak masculine input chosen, the selected *lappa* outcome is therefore the regular project comparison.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the weak masculine formation from a competing voiced comparative label.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing voiced comparative label	*lábbaz	not the form followed for the OE weak-noun derivation	broader comparative background	useful as a competing label, but not the source-based choice used here
selected weak masculine noun	*láppô	regular output: *lappa*	lappa	exact match between formation and attested OE noun

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
attested OE variant line	same noun family	*læppa, leappan*	læppa / leappan	useful control forms within the same OE tradition

laugh — OE hliehhan

Derivation: citation reconstruction **lákaną*; form followed here **xláxjaną* > *hliehhan* (early analogy).

Derivation trace

Proto input: **xláxjaną*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xlæxxjaną*
PWGmc J Gemination	**xláxxjaną*	OE Breaking	**xleaxxjaną*
Northwest Germanic		OE Velar Fricative Palatalization	**xleaxçjaną*
[no change]		OE Heavy Syllable Nasal Apocope	**xleaxçjan*
		OE Secondary Nasalization	**xleaxçjąn*
		OE I Umlaut	**xliexçjąn*
		OE Weak Tail Reduction	**xliexçjan*
		OE J Loss After Heavy	**xliexçan*

Old English form: *hliehhan*

Reconstruction and comparative evidence

The wider Germanic family includes a non-j branch represented here by the citation label **lákaną*, while the derivational input **xláxjaną* reflects the j-present line behind Old English *hliehhan*.

This branch choice matters because it brings with it the geminate fricative and the vowel development characteristic of the Old English verb. The comparative family label and the OE-facing input are therefore related but not identical.

Old English evidence

Bosworth-Toller records *hlihhan* as the verb 'to laugh' (Bosworth and Toller 1898, 551). Clark Hall cross-references *hlæhan*, *hlehhan*, and *hlihhan* to *hliehhan* (Clark Hall 1960, 160–61). Bright's glossary likewise gives *hlihhan (hliehhan, hlyhhan)* (Bright 1971, 315). The target of this entry is the West Saxon *hliehhan*.

The variant set matters as background, but the argument of the entry rests on the attested lemma *hliehhan* itself.

Development to Old English

From **xláxjaną*, West Germanic j-gemination yields the doubled consonant (Campbell 1959, sect. 407). Ringe and Taylor derive Old English *hliehhan* from the j-present branch via breaking before the palatalized geminate (Ringe and Taylor 2014, 240). They separately compare the related noun *hleahtor* as the outcome of **hlahtraz* (Ringe and Taylor 2014, 328).

Branch comparison

The comparison below sets the relevant forms side by side. It separates the wider non-j family label from the j-present branch actually reflected in Old English.

Formation / branch	Candidate input	Expected or documented OE outcome	OE comparison form	Result
wider non-j family	*lákaną	comparative background outside the selected OE line	wider family context	useful family label, but not the direct source of *hliehhan*
selected j-present branch	*xláxjaną	regular output: *hliehhan*	hliehhan	exact match between branch and attested OE lemma
attested OE variants	same OE verb line	*hlæhhan*, *hlehhan*	hlæhhan / hlehhan	genuine variant evidence, but secondary to the form compared here

loam — OE lām

Derivation: citation reconstruction **laimōn*; form followed here **láimą* > *lām* (early analogy).

Derivation trace

Proto input: **láimą*

Earlier Germanic changes		Old English changes	
West Germanic		OE Heavy Syllable Nasal Apocope	**lām*
PWGmc Ai Monophthongization	**lāmą*		
Northwest Germanic			
[no change]			

Old English form: *lām*

Reconstruction and comparative evidence

The inherited comparative noun is given as **laimōn* or **laiman-*, and both Orel and Kroonen identify Old English *lām* as a neuter reflex of that family (Orel 2003, 272; Kroonen 2013, 363). The form followed here, **láimą*, differs from the comparative headword because it represents the stem class that matches the Old English noun most directly.

This is therefore a class shift within the history of the English branch rather than a dispute about the OE target itself.

Old English evidence

Old English dictionaries record the noun as *lām*, a neuter word for ‘loam, clay, mud’ (Bosworth and Toller 1898, 604; Clark Hall 1960, 196). The target is an attested citation form rather than a reconstructed comparator.

The relevant question is not whether *lām* is Old English, but which inherited formation best accounts for that attested neuter noun.

Development to Old English

From **láimą*, regular monophthongization of *ai* and the later loss of the final nasal syllable yield *lām*. With that OE-facing input, the phonological development is straightforward.

Class comparison

The comparison below sets the inherited comparative n-stem label beside the OE-facing stem class used to derive the attested noun.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited comparative noun	*laimōn	comparative family background	wider family context	useful headword, but not the direct OE-facing input
OE-facing stem class followed here	*láimą	regular output: *lām*	lām	exact match between the form followed here and the attested OE noun

lung — OE lungen

Derivation: citation reconstruction **lungō*; form followed here **lúnganjō* > *lungen* (early analogy).

Derivation trace

Proto input: **lúnganjō*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**lúngennju*
PWGmc J Gemination	**lúngannjō*	OE High Vowel Apocope	**lúngennj*
Northwest Germanic		OE J Loss After Heavy	**lúngenn*
NWGmc Final Long O Raising	**lúngannju*	OE Final Geminate Simplification	**lúngen*

Old English form: *lungen*

Reconstruction and comparative evidence

Kroonen treats the basic noun as **lungōn-* and also cites an OE-facing derivative **lungunjō-*, continued by Old English *lungen* and close West Germanic cognates (Kroonen 2013, 384). The form followed here, **lúnganjō*, models that derived feminine formation rather than the base noun. The notation differs slightly from Kroonen's **lungunjō-*, but both point to the same derived feminine line.

The difference between the citation label and the derivational input is therefore derivational. Old English *lungen* is not a direct reflex of the bare base noun **lungō*; it belongs to an expanded feminine formation.

Old English evidence

Old English dictionaries record the noun as *lungen*, with inflected forms such as *lungenne* and *lungena* (Bosworth and Toller 1898, 634). Clark Hall also preserves a small family of compounds such as *lungenādl*, *lungensealf*, and *lungenwyrt* (Clark Hall 1960, 191).

The target is an attested Old English lexeme with its own paradigm, not a rescued inflectional cell.

Development to Old English

From the selected derived input, the expected derivational consonant and vowel adjustments lead to *lungen*. Once the expanded feminine formation is chosen, the Old English outcome is regular.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the base noun from the derived feminine formation reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
base noun	*lungō	base-noun outcome without the OE derivative suffix	broader family context	useful headword, but not the direct source of *lungen*
derived OE-facing formation	*lúnganjō	regular output: *lungen*	lungen	exact match between the derived formation and the attested OE noun
Kroonen's cited derivative	**lungunjō-*	comparative support for the same OE-facing formation	lungen and cognate set	supports the derived feminine formation, with notation differing from the normalized input form used here

navel — OE nafola

Derivation: citation reconstruction **nablô*; form followed here **nábulô* > *nafola* (early analogy).

Derivation trace

Proto input: **nábulô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Med Unstressed U Lowering	**nábolô*
[no change]	Anglo Frisian Brightening	**næbolô*
Northwest Germanic	OE A Restoration	**nabolô*
[no change]	PGmc B Allophony	**naβolô*
	OE Unstressed Long Vowel Shortening	**naβola*

Old English form: *nafola*

Reconstruction and comparative evidence

Kroonen instead gives a nasal-suffix navel formation with Old English *nafela* among its reflexes (Kroonen 2013, 420), while Ringe and Taylor give the derivational pathway **nabulō* > **næbula* > *nafola* (Ringe and Taylor 2014, 270). The difference is one of stage and notation rather than of lexeme identity: the derivational input **nábulô* is the pre-syncope form needed for the Old English development.

For the Old English comparison, the crucial point is simply that the pre-OE form still contains a medial vowel.

Old English evidence

Ringe and Taylor note the early West Saxon shift *nafola* > *nafela* (Ringe and Taylor 2014, 336). Campbell likewise records *nafela* beside Corpus *nabula* (Campbell 1959, sect. 159). The target of this entry is the nominative singular *nafola*, the form that matches the selected derivational pathway most directly.

nafela is the better-known later West Saxon spelling, while *nabula* preserves a less reduced medial vowel. These forms belong to the same lexical history, but this entry is centered on *nafola*.

Development to Old English

Ringe and Taylor give the pre-OE line **nabulō* > **næbula* > OE *nafola* (Ringe and Taylor 2014, 270). The trace represents that same development with stress-marked notation and explicit intermediate weakening. Intervocalic *b* then surfaces as *f*, and final weak-tail shortening gives *nafola*.

The medial vowel is still present when A-restoration applies. That is why the selected pre-syncope input differs from the syncopated comparative headword.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the comparative citation form from the pre-syncope input and from the later OE spellings.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
syncopated comparative headword	**nablô*	reduced *næfla*-type outcome rather than *nafola*	not the Old English form here	useful citation form, but too reduced for the pathway modeled here
selected pre-syncope input	**nábulô*	regular output: *nafola*	*nafola*	exact match between derivational input and target
later OE reduction stages	same lexical history	attested *nafela*; Corpus *nabula*	*nafela* / *nabula*	related OE spellings, but not the chosen comparator

neck — OE hnecca

Derivation: citation reconstruction **xnákkaz*; form followed here **xnékkô* > *hnecca* (early analogy).

Derivation trace

Proto input: **xnékkô*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**xnékka*
[no change]			
Northwest Germanic			
[no change]			

Old English form: *hnecca*

Reconstruction and comparative evidence

The noun belongs to an ablauting n-stem family. Kroonen reconstructs a paradigm with nominative **hnekkō*, genitive **hnukkaz*, and accusative plural **hnakkuns*, and he places Old English *hnecca* among the e-grade descendants (Kroonen 2011, 167). Kluge-Seebold likewise identifies *ae. hnecca* as an ablaut partner of the a-grade *Nacken* family (Kluge 2011, 347).

A competing comparative label **xnákkaz* remains useful for the wider family, and Orel also gives an a-grade headword line (Orel 2003, 218). The derivational input **xnékkô*, however, is the form that matches the Old English branch.

Old English evidence

Clark Hall records the weak masculine noun *hnecca* (Clark Hall 1960, 162). Bosworth-Toller likewise records *hnecca* (Bosworth and Toller 1898, 567). The target is therefore an attested citation form, not an oblique cell or a reconstructed lemma.

The phonological question is upstream of the Old English evidence. The attested noun already shows that the branch continued an e-grade form rather than the a-grade seen in much of the continental material.

Development to Old English

From **xnékkô*, the derivation is straightforward. The trace shortens the final long vowel to **xnékka*, and Old English orthography gives *hnecca*.

The derivation depends on the earlier selection of the e-grade weak-noun form continued by Old English.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the wider a-grade family from the selected e-grade Old English branch.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing comparative label	**xnákkaz*	broader a-grade family rather than the selected OE source	continental *Nacken* line	useful family label, but not the input followed for the Old English derivation
weak noun with a-grade	**xnakkô*	expected *hnacca* type outcome	*hnacca*	fixes the class, but not the vowel grade

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
selected e-grade nominative	**xnékkô*	regular output: *hnecca*	*hnecca*	exact match between derivational input and attested OE noun
oblique paradigm background	**hnukkaz*, **hnakkuns*	ON/OHG/German a-grade continuation	*hnakki* / *Nacken*	shows the wider ablaut family, but not the chosen OE branch

needle — OE nǣdl

Derivation: citation reconstruction **nḗθlō*; form followed here **nḗðlō* > *nǣdl* (early analogy).

Derivation trace

Proto input: **nḗðlō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**nǣdl*
PWGmc Dental Hardening	**nḗdlō*		
Northwest Germanic			
NWGmc Final Long O Raising	**nḗdlu*		
NWGmc Long E Lowering	**nǣdlu*		

Old English form: *nǣdl*

Reconstruction and comparative evidence

Ringe and Taylor treat the word as a voiced/voiceless alternant, citing PGmc **nēþlō*, **nēdlō-* 'needle' … > OE *nédl* (Ringe and Taylor 2014, 329). The form followed here, **nḗðlō*, is the voiced Verner-grade form used for the Old English comparison, while the citation form **nḗθlō* remains the broader lexeme label.

The development discussed here follows the Ringe-Taylor alternant framework.

Old English evidence

Clark Hall records the attested citation form *nǣdl* (Clark Hall 1960, 210). Campbell lists *nédl* among the expected unbroken forms after *t* and *d* (Campbell 1959, sect. 367). Hogg also includes *nidi* / *nǣdl* in the same broader cluster history (Hogg 1992, 95).

The target is therefore an attested citation form. No oblique-cell substitution is involved in this entry.

Development to Old English

Ringe and Taylor give the historical line **nēþlō*, **nēdlō-* … > OE *nédl* (Ringe and Taylor 2014, 329). Campbell likewise lists *nédl* among the expected unbroken forms after *t* and *d* (Campbell 1959, sect. 367). The trace expresses the same pathway with the voiced alternant followed here for the Old English comparison.

The essential choice lies in which Proto-Germanic alternant is taken as the starting point. Once the voiced form is chosen, the rest of the pathway is regular.

Alternant comparison

The comparison below sets the relevant forms side by side. It separates the broader citation headword from the voiced alternant used for Old English.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative voiceless headword	**nḗθlō*	broader word-family label rather than the OE-facing alternant	**nēþlō* line	useful citation form, but not the derivational input for the Old English comparison
voiced Verner alternant followed here	**nḗðlō*	regular output: *nǣdl*	*nǣdl*	exact match between the form followed here and the attested OE noun
later hardening stage	*nḗdlō	intermediate pre-OE stage in the same derivation	nǣdl	genuine stage in the pathway, but not the Proto-Germanic form followed here

nose — OE nosu

Derivation: citation reconstruction **nasō*; form followed here **núsō* > *nosu* (early analogy).

Derivation trace

Proto input: **núsō*

Earlier Germanic changes		Old English changes
West Germanic		[no change]
[no change]		
Northwest Germanic		
NWGmc U Lowering	**nósō*	
NWGmc Final Long O Raising	**nósu*	

Old English form: *nosu*

Reconstruction and comparative evidence

Kroonen reconstructs a Germanic ablaut pair **nasō-* ~ **nusō-* and adds that the root **nus-* is likely to have arisen as a secondary zero grade after a remodeling of the older paradigm (Kroonen 2013, 423). Campbell is more specific for Old English, citing *nosu* < **nusō* (Campbell 1959, 44).

The citation reconstruction **nasō* is therefore best treated as the full-grade comparative headword, while the derivational input **núsō* represents the remodeled zero-grade line continued by the Old English form discussed here. Orel's **nasō* … OE *nasu* preserves the competing full-grade notation and shows that the two lines should not be collapsed without comment (Orel 2003, 320).

Old English evidence

Nosu is an attested Old English noun. Ringe and Taylor list it among the few surviving early Old English feminine u-stems (Ringe and Taylor 2014, 385). Clark Hall likewise gives *nosu f.*, with genitive-dative singular *nosa*, and cross-refers *nasu* to *nosu* (Clark Hall 1960, 810).

The selected OE target is therefore an attested *nosu*, not a reconstructed placeholder. At the same time, the lexicographical record keeps *nasu* visible as a parallel notation belonging to the full-grade side of the tradition.

Development to Old English

From **núsō*, the regular path is the one documented by the current trace: **núsō* > **nósō* > **nósu* > *nosu*. The early special step lies in the choice of the zero-grade input, not in any late Old English repair.

With that input chosen, the OE development is straightforward. The full-grade line behind **nasō* instead points toward *nasu*, not to the form treated here.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the full-grade comparative line from the remodeled zero-grade input that yields the Old English form.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
full-grade comparative line	*nasō	expected full-grade continuation *nasu*	nasu	useful comparative background, but not the Old English-facing input
remodeled zero-grade line	*núsō	regular output: *nosu*	nosu	exact match between derivational input and attested OE noun

sap — OE sæp

Derivation: citation reconstruction **sapōn*; form followed here **sápą* > *sæp* (early analogy).

Derivation trace

Proto input: **sápą*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**sæpą*
[no change]	OE Heavy Syllable Nasal Apocope	**sæp*
Northwest Germanic		
[no change]		

Old English form: *sæp*

Reconstruction and comparative evidence

The comparative sources do not give one uniform inherited stem. Kroonen preserves the word family as **saf*/ppan-, with Old English *sæp* m. (Kroonen 2013, 420). Orel preserves the comparative notation **sapōn* ~ **sapan* (Orel 2003, 319).

The form followed here, **sápą*, therefore does not replace those comparative labels. It identifies the OE-facing stem shape that yields the attested noun treated here.

Old English evidence

Clark Hall records *sæp* (e) n. (Clark Hall 1960, 247). The target is therefore an attested neuter Old English noun. Orel's plain *sap* notation belongs to comparative normalization, not to the spelling adopted here for the Old English form (Orel 2003, 319).

Development to Old English

From **sápą*, Anglo-Frisian brightening yields *sæ*, and heavy-syllable nasal apocope then produces *sæp*. That is the regular path documented by the current trace.

The competing comparative lines do not give the same result. The inherited n-stem notation **sapōn* yields *sape*, while an i-stem continuation from the **sapi-* line leads to *sep* / *sepe* rather than to *sæp*. The special step in this entry is therefore the early stem choice, not a late OE paradigm-cell selection.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative stem lines from the Old English-facing input.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative n-stem line	*sapōn	local comparator output: *sape*	sape	useful comparative background, but not the source of attested *sæp*
inferred i-stem comparator from **sapi-*	*sapiz	local comparator output: *sepe*	sepe	confirms that an i-triggering stem does not reach the target
selected a-stem input	*sápą	regular output: *sæp*	sæp	exact match between derivational input and attested OE noun

sea — OE sǣ

Derivation: citation reconstruction **sái*; form followed here **sáiwiz* > *sǣ* (early analogy).

Derivation trace

Proto input: **sáiwiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE W Loss Before I	**sāi*
PWGmc Ai Monophthongization	**sāwiz*	OE I Umlaut	**sǣi*
Northwest Germanic		OE High Vowel Apocope	**sǣ*
PGmc Final Z Deletion	**sāwi*		

Old English form: *sǣ*

Reconstruction and comparative evidence

Kroonen gives the noun in stem notation as **saiwi-*, an i-stem whose English reflex is cited as OE *sæ* (Kroonen 2013, 423). Ringe and Taylor write the fuller form **saiwiz* and derive it through **sawi* > **sei* > OE *sǣ* (Ringe and Taylor 2014, sect. 6.7.1). The comparative headword is therefore shorter than the form required for the English history: **sái* names the lexeme, but **sáiwiz* preserves the medial **w* and the final high vowel that control the later development.

Old English evidence

The Old English noun is the ordinary word for 'sea'. Kroonen cites it as *sæ*; the normalized form here is *sǣ* (Kroonen 2013, 423). Campbell likewise treats *sea* as continuing the same **saiui-* > **sǣi* history, with loss of *u*/*w* before *i* (Campbell 1959, sect. 406).

Development to Old English

Once the fuller i-stem input is chosen, the development is regular. After Proto-West-Germanic monophthongization **sáiwiz* > **sāwiz* and final **-z* loss **sāwiz* > **sāwi*, the non-initial **w* disappears before unstressed **i*, and the following high vowel fronts the root vowel before final apocope. The documented chain is **sáiwiz* > **sāwiz* > **sāwi* > **sāi* > **sǣi* > *sǣ* (Ringe and Taylor 2014, sect. 6.7.1; Campbell 1959, sect. 406).

Stem and stage comparison

The comparison below sets the relevant forms side by side. It separates the abbreviated comparative headword from the fuller i-stem input that yields the Old English form.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
abbreviated comparative headword	*sái	too short to preserve the **w* … **i* environment needed for the documented chronology	sǣ	useful comparative label, but not the Old English-facing input
selected i-stem input	*sáiwiz	documented regular output: *sǣ*	sǣ	exact match between derivational input and Old English target

sieve — OE sife

Derivation: citation reconstruction **síbaz*; form followed here **síbi* > *sife* (early analogy).

Derivation trace

Proto input: **síbi*

Earlier Germanic changes	Old English changes	
West Germanic	PGmc B Allophony	**síβi*
[no change]	OE Med Unstressed I Lowering1	**síβe*
Northwest Germanic		
[no change]		

Old English form: *sife*

Reconstruction and comparative evidence

Kluge-Seebold gives wg. **sibi-* n. ... ae. sife, and Campbell groups *sife* with short neuter i-stems such as *spere* (Kluge 2011, 847; Campbell 1959, sect. 609). The older morphological background is the s-stem **sib-iz*, but the derivational input is the normalized i-stem form **síbi*.

Kroonen's nearby **sebjō-* entry belongs to the separate kinship lexeme that yields Old English *sibb*, not to the sieve word. Orel's **sibaz* ... OE *sife* preserves a broader handbook notation, but that a-stem shape does not fit the Old English form treated here (Orel 2003, 328).

Old English evidence

Clark Hall gives *sibi (GL)* ... = *sife* and also *sife n. 'sieve'* (Clark Hall 1960, 263). Campbell likewise cites Corpus Glossary *sibi* and treats *sife* as a short neuter i-stem (Campbell 1959, sects 444, 609). The normalized Old English target is therefore *sife*, while *sibi* is an attested earlier spelling rather than a separate lexeme.

Development to Old English

From **síbi*, the regular derivation gives **síβi* > **síβe* > *sife*. Medial *b* is realized as a spirant and later written *f*, while the final unstressed *i* lowers to *e*. The older s-stem background **sib-iz* explains the morphology, but the derivational input **síbi* is the immediate pre-Old-English form.

Stem comparison

The comparison below sets the relevant forms side by side. It distinguishes the accepted i-stem line from its rejected competitors.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
ja-stem kinship line	Kroonen **sebjō-* / comparator **sibja*	OE *sibb*	sibb	separate lexeme, not the target treated here
a-stem handbook line	*síbaz	expected *sif*	sif	wrong ending for the attested noun
selected i-stem line from older **sib-iz*	*síbi	documented regular output: *sife*	sife; early spelling *sibi*	exact match between derivational input and Old English evidence

spare — OE sparian

Derivation: citation reconstruction **sparēną*; form followed here **spárōjaną* > *sparian* (early analogy).

Derivation trace

Proto input: **spárōjaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**spærōjaną*
[no change]	OE A Restoration	**sparōjaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**sparōjan*
[no change]	OE Secondary Nasalization	**sparōją̄n*
	OE I Umlaut	**sparēją̄n*
	OE Unstressed Long Vowel Shortening	**spareją̄n*
	OE Weak Tail Reduction	**sparejan*
	OE Intervocalic J Vocalization	**spareian*
	OE Unstressed EI Contraction	**sparian*

Old English form: *sparian*

Reconstruction and comparative evidence

Kroonen keeps the inherited verb under class-III **sparēn-* (Kroonen 2013, 465). Orel similarly preserves **sparēnan* (Orel 2003, 362). Ringe and Taylor, however, reconstruct **sparai-* ~ **sparja-* for the English branch and derive the citation verb from a class-II line (Ringe and Taylor 2014, 162, 191). The derivational input **spárōjaną* therefore represents the refashioned class-II formation behind Old English *sparian*, while the citation reconstruction **sparēną* remains the inherited comparative headword.

Old English evidence

Campbell says that *sparian* does not show the ordinary class-III characteristics, but the Ritual forms, normalized here as *spæria*, *spær*, and *spærede*, together with Vespasian Psalter *spearad*, point to primitive Old English forms both with and without back vowels (Campbell 1959, sect. 764). Brunner likewise records Northumbrian *spæria*, *spærede* beside common Old English *sparian* and Vespasian Psalter *spearad* (Sievers 1965, sect. 364 Anm. 11). The citation form treated here is *sparian*; the Anglian forms are relics of the older formation, not alternative headwords of equal status.

Development to Old English

Once the class-II formation **spárōjaną* is chosen, the remaining development is regular. The regular derivation shows brightening, restoration of *a* before the back vocalism of the suffix, later i-mutation within the weak ending, weak-tail reduction, and contraction to *sparian*. By contrast, Brunner's rule against further apocope of final *-e* explains why Ritual *spær* cannot be the regular continuation of inherited **spárē* (Sievers 1965, sect. 150).

Formation comparison

The comparison below sets the relevant forms side by side. It contrasts the inherited class-III formation with the refashioned class-II one that yields the citation verb.

Formation / comparison	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited class-III infinitive	**spárēną*	paradigm comparison / probe output: *sparen*	*sparian*	wrong class and wrong ending for the citation verb
inherited class-III imperative singular	**spárē*	paradigm comparison / probe output: *spære*	Ritual *spær*	loss of final *-e* is not regular, so the relic form cannot control the entry
inherited class-III finite present	**spárēθi*	paradigm comparison / probe output: *spæreþ*	*spearad*	attested form is mixed, not a direct continuation of the inherited cell
selected class-II formation	**spárōjaną*	documented regular output: *sparian*	*sparian*	exact match between derivational input and Old English citation form

staff — OE stæf

Derivation: citation reconstruction **stábiz*; form followed here **stábaz* > *stæf* (early analogy).

Derivation trace

Proto input: **stábaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**stáb*
[no change]		Anglo Frisian Brightening	**stæb*
Northwest Germanic		PGmc B Allophony	**stæβ*
PGmc Final Z Deletion	**stába*		

Old English form: *stæf*

Reconstruction and comparative evidence

The comparative dictionaries do not give one uniform stem class. Kroonen reconstructs an a-stem **staba-* (Kroonen 2013, 471). Orel writes **stábiz* ~ **stábaz* (Orel 2003, 368). That disagreement matters because a direct i-stem input in *-iz* would predict i-mutation in Old English, whereas the attested noun keeps *æ*.

Old English evidence

The Old English noun itself is the ordinary citation form *stæf*. Luick lists *stæf* among closed monosyllables with *æ* (Luick 1914--1940, 176). Ringe and Taylor pair singular *stæf* with plural *stafas* (Ringe and Taylor 2014, 193). The normalized form here is therefore *stæf*; later English *staff* with *a* belongs to a later stage of the word's history.

Development to Old English

With the selected a-stem input, the development is regular. Final *-z* disappears, bare final -*a* is lost, Anglo-Frisian brightening gives *æ* in the closed monosyllable, and medial *b* surfaces as a fricative written *f*. The documented chain is **stábaz* > **stába* > **stáb* > **stæb* > *stæf*. A direct continuation of **stábiz*, by contrast, would produce i-mutated *stefe* rather than the attested singular.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the rejected i-stem line from the selected a-stem input.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
comparative i-stem line	**stábiz*	expected *stefe* after i-mutation	*stæf*	wrong vowel for the attested singular
mixed comparative notation	Orel **stabiz* ~ **stabaz*; Kluge **stabi-*/a-	source-level stem-class uncertainty	stæf	useful comparative background, but not a single OE-facing input
selected a-stem input	**stábaz*	documented regular output: *stæf*	*stæf*	exact match between derivational input and Old English target

stem — OE stefn

Derivation: citation reconstruction **stámnaz*; form followed here **stébnō* > *stefn* (early analogy).

Derivation trace

Proto input: **stébnō*

Earlier Germanic changes		Old English changes	
West Germanic		PGmc B Allophony	**stéβnu*
[no change]		OE High Vowel Apocope	**stéβn*
Northwest Germanic			
NWGmc Final Long O Raising	**stébnu*		

Old English form: *stefn*

Reconstruction and comparative evidence

The source tradition behind *stefn* is not the same as the comparative label **stámnaz*. Ringe and Taylor cite **stebnō* for the noun continued by Gothic *stibna* and Old English *stebn* > *stefn* > *stemn* (Ringe and Taylor 2014, 330). Orel likewise gives **stebnō* ~ **stemnō*, whereas Kroonen prefers **stimnō-*, and Fulk describes the etymology of *stefn, stemn* as insecure (Orel 2003, 374; Kroonen 2013, 480; Fulk 2018, sect. 6.11 n. 6).

These forms belong to the Old English noun *stefn* 'voice, sound'. The derivational input **stébnō* is therefore best treated as the OE-facing transponent supported by that source tradition. It does not settle the deeper comparative reconstruction implied by the citation label **stámnaz*.

Old English evidence

Clark Hall records *stefn* as the noun 'voice, sound' and cross-refers *stemn* to the same word (Clark Hall 1960, 276). Ringe and Taylor give the OE chronology directly as *stebn* > *stefn* > *stemn* (Ringe and Taylor 2014, 330).

Bülbring and Luick treat *stemn* as a later West Saxon development from older *stefn*, produced by *fn* > *mn* only after the earlier period of nasal influence on *e* (Bülbring 1902, sect. 62 Anm. 3, 445; Luick 1914--1940, sect. 75 Anm. 1). The relevant comparison form is therefore the conservative *stefn*, not the later West Saxon doublet *stemn*.

Development to Old English

From **stébnō*, raising of final long *ō* gives a **stébnu* stage. Regular fricativization of *b* before *n* then yields **stéβnu*, and loss of the final high vowel leaves **stéβn*, written *stefn* in Old English. The later form *stemn* belongs to a separate West Saxon assimilation after this stage (Ringe and Taylor 2014, 330; Bülbring 1902, sect. 445).

Source comparison

The comparison below sets the relevant forms side by side. It keeps apart the broader comparative label, the OE-facing transponent, and the later West Saxon variant history.

Form or label	Status	OE relation	Result
**stámnaz*	comparative citation label for the broader stem/trunk family	does not itself control the *stefn* derivation discussed here	broader lexical label only
**stébnō*	voice-noun transponent	regular output: *stefn*	Old English-facing input
stemn	later attested West Saxon doublet	secondary form from *stefn* by *fn* > *mn*	real OE variant, but not the selected comparator

swan — OE swanes

Derivation: citation reconstruction **swánaz*; form followed here **swánas* > *swanes* (early analogy).

Derivation trace

Proto input: **swánas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**swánæs*
[no change]	OE Unstressed AE Merger	**swánes*
Northwest Germanic		
[no change]		

Old English form: *swanes*

Reconstruction and comparative evidence

The Germanic noun is ordinarily cited as the masculine a-stem **swanaz* (Orel 2003, 367). The form followed here, **swánas*, is not a competing lexeme reconstruction. It is the genitive singular of the same paradigm.

The question here is therefore one of paradigm cell rather than stem history. The citation form remains **swanaz* > *swan*; the comparison form is the genitive singular **swánas* > *swanes*.

Old English evidence

Bright's glossary records the ordinary noun as *swan, m.* and also gives the exact inflected form *swanes*, citing the phrase *swanes feðre* (Bright 1971, 441).

The target is therefore an attested Old English genitive singular, not a reconstruction. It is also not the ordinary citation lemma. The entry must keep those two facts distinct.

Development to Old English

From **swánas*, Anglo-Frisian brightening gives **swánæs*, and subsequent merger of unstressed *æ* with *e* yields *swanes*. The comparison is straightforward once the genitive singular is chosen as the relevant cell.

Paradigm-cell comparison

The comparison below sets the relevant forms side by side. It separates the ordinary citation form from the selected inflected cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*swánaz	OE headword *swan*	swan	ordinary lexeme line
genitive singular	*swánas	regular output: *swanes*	swanes	attested cell

thousand — OE þūsend

Derivation: citation reconstruction **θūs-undī*; form followed here **θūs-èndi* > *þūsend* (early analogy).

Derivation trace

Proto input: **θūsèndi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Strip Secondary Stress	**θūsend*
PWGmc Early I Apocope	**θūsènd*		
Northwest Germanic			
[no change]			

Old English form: *þūsend*

Reconstruction and comparative evidence

Kroonen reconstructs the Germanic numeral as **þūsundī-* and cites Old English *þūsend* among its continuations (Kroonen 2013, 554). The derivational input **θūs-èndi* is not the same claim. It is an OE-oriented transponent with the second-member vowel already resolved to *e* and the final high vowel already shortened for apocope.

The important question is therefore chronological. Why does Old English show *þūsend*, while related languages such as Old Saxon and Old High German keep *u* in the second syllable? (Kroonen 2013, 554).

Old English evidence

Old English *þūsend* is an ordinary citation form, not a selected oblique or paradigm cell. Campbell treats it as a neuter noun with normal case forms (Campbell 1959, sect. 689). The problem lies in the internal history of the word, not in its lexical status.

Development to Old English

If the old final -*ī* had remained long enough to trigger ordinary double umlaut, Campbell's rule would point toward a form of **þȳsend* type rather than attested *þūsend* (Campbell 1959, sect. 203). Preserved root *ū* therefore argues that the umlaut-triggering vowel was lost or neutralized before the ordinary OE umlaut outcome could develop.

That early loss, however, does not by itself explain the medial *e*. Luick compares the word with *ærende* and later groups *thousand* with forms reshaped on that pattern (Luick 1914--1940, sects 198, 492). Viredaz is more cautious, arguing that Old English *e* in this weak position may simply write schwa and so need not prove a unique *ærende*-type analogy (Viredaz 2025, sect. 2.1.4).

The selected transponent **θūs-èndi* captures the OE-side state from which the regular derivation reaches *þūsend*.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the secure chronology from the more interpretive account of the second-syllable vowel.

Stage / interpretation	Candidate form	OE relation	Result
surviving -*ī* with ordinary double umlaut	**þūsundī-* treated as still umlaut-active in OE	would point toward **þȳsend*	excluded by preserved *ū*
early loss of the trigger without further reshaping	**þūsund-* type	explains *ū*, but not why OE alone has medial *e*	incomplete account
selected OE-oriented transponent	**θūs-èndi*	regular output: *þūsend*	selected modeling input

timber — OE timber

Derivation: citation reconstruction **tímrą*; form followed here **tímbrą* > *timber* (early analogy).

Derivation trace

Proto input: **tímbrą*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**tímbr*
[no change]	OE Epenthetic Vowel	**tímber*
Northwest Germanic		
[no change]		

Old English form: *timber*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **timbra-* and cites Old English *timber* among its continuations (Kroonen 2013, 517). Ringe and Taylor instead state the history from PGmc **timra* through West Germanic **timbr* to Old English *timber* (Ringe and Taylor 2014, 327).

The difference is therefore not over the Old English noun itself. It concerns whether medial *b* belongs in the comparative citation form or appears in an early pre-Old-English stage of the cluster.

Old English evidence

Clark Hall lemmatizes the noun as *timber* and also records *timbor* as a variant spelling (Clark Hall 1960, 294). The Old English form is thus an ordinary citation noun, not a selected oblique cell or a reconstructed convenience form.

Development to Old English

With the consonantal frame *timbr-* in place, the rest of the development is straightforward. Loss of final -*ą* leaves **tímbr*, and epenthetic *e* in the final cluster yields *timber*. Ringe and Taylor's **timra* > **timbr* > OE *timber* and the handbook treatment of this epenthetic vowel point to the same Old English result (Ringe and Taylor 2014, 327; Campbell 1959, sects 463–464).

Formation comparison

The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing consonantal input.

Formation or notation	Candidate form	OE relation	Result
Kroonen's comparative citation	**timbra-*	already matches the consonantal frame of OE *timber*	closest comparative support for the derivational input
Ringe-Taylor citation line	**timra* > **timbr*	reaches the same OE noun through early cluster expansion	compatible comparative background
modeled input	**tímbrą*	regular output: *timber*	Old English-facing input

wake — OE wacan

Derivation: citation reconstruction **wakēną*; form followed here **wákaną* > *wacan* (early analogy).

Derivation trace

Proto input: **wákaną*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wækaną*
[no change]	OE A Restoration	**wakaną*
Northwest Germanic	OE Heavy Syllable Nasal Apocope	**wakan*
[no change]	OE Secondary Nasalization	**wakąn*
	OE Weak Tail Reduction	**wakan*

Old English form: *wacan*

Reconstruction and comparative evidence

Kroonen gives the strong verb as **wakan-* with Old English *wacan* (Kroonen 2013, 568). Ringe and Taylor separately derive Old English *wacian* from weak **wakai-* ~ **wakja-* (Ringe and Taylor 2014, sect. 3.3.2).

The difference is therefore lexical and class-based, not graphic. Strong *wacan* 'wake up, arise' and weak *wacian* 'be awake, watch' belong to related but distinct histories.

Old English evidence

Clark Hall keeps *wacan* and *wacian* as separate headwords (Clark Hall 1960, 338). Bosworth-Toller adds an important caution under *wacan*: the simplex infinitive itself does not occur, its place seeming to be taken by *wæcnan* (Bosworth and Toller 1898, 226).

The target *wacan* is therefore best understood as a normalized strong headword for the verb family, not as a directly quoted simplex infinitive. It still remains the correct Old English comparison form for the strong branch.

Development to Old English

With strong **wákaną*, Anglo-Frisian brightening first gives a form of the **wækaną* type. A-restoration then returns *a*, and the ordinary tail reductions yield *wacan*. The weak verb *wacian* belongs to a different prehistory and is not the expected outcome of this input.

Class comparison

The comparison below sets the relevant forms side by side. It separates the strong and weak verb lines.

Formation / class	Candidate input	OE outcome or comparison	Result
weak class-III / class-II branch	**wakēną*, **wakai-* ~ **wakja-*	OE *wacian* and related weak forms	related lexeme, but not the target of this entry
strong class-VI branch	**wákaną*	regular output: *wacan*	Old English-facing input
strong normalized headword	*wacan*	dictionary comparison form beside attested strong-family forms	correct Old English comparator, though not a directly quoted simplex infinitive

water — OE wæter

Derivation: citation reconstruction **wátną*; form followed here **wátōr* > *wæter* (early analogy).

Derivation trace

Proto input: **wátōr*

Earlier Germanic changes		**Old English changes**	
West Germanic		Anglo Frisian Brightening	**wætær*
PWGmc Final Or Lowering	**wátar*	OE Unstressed AE Merger	**wæter*
Northwest Germanic			
[no change]			

Old English form: *wæter*

Reconstruction and comparative evidence

Kroonen reconstructs a heteroclitic noun **watar- ~ *watan-* and states that the Proto-Germanic material points to **watōr*, **watenaz* (Kroonen 2013, 616). Ringe and Taylor likewise start from singular **wator* before the Old English branch (Ringe and Taylor 2014, sect. 3.1.4).

The generalized comparative label is therefore broader than the singular form that actually corresponds to Old English *wæter*. The relevant comparator is the inherited nominative-accusative singular **wátōr*.

Old English evidence

Bright gives the noun as *wæter* with the regular paradigm *wæteres*, *wætere*, *wæter(u)*, *wætera*, *wæterum* (Bright 1971, 29). Ringe and Taylor add the dialectal contrast between West Saxon *weeter* and Mercian *weter* (Ringe and Taylor 2014, sect. 6.5.2).

The target is therefore an attested Old English citation form within a normal paradigm. The complication lies on the comparative side of the lexeme, not in Old English attestation.

Development to Old English

From **wátōr*, pre-final **ō* becomes *a* before final *r*, yielding **watar* (Ringe and Taylor 2014, sect. 3.1.4). Anglo-Frisian brightening then gives **wætær*, and merger of unstressed *æ/e* yields *wæter*.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the generalized lexeme label from the singular input that matches the Old English citation form.

Stage or notation	Candidate form	OE relation	Result
generalized comparative label	**wátną*	broader lexeme shorthand, not the singular that corresponds directly to *wæter*	useful background only
heteroclitic stem notation	**watar- ~ *watan-*	source-faithful comparative reconstruction	explains why a singular comparator is needed
inherited singular input	**wátōr*	regular output: *wæter*	Old English-facing input

whale — OE hwæl

Derivation: citation reconstruction **wálaz*; form followed here **xwálaz* > *hwæl* (early analogy).

Derivation trace

Proto input: **xwálaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**xwál*
[no change]		Anglo Frisian Brightening	**xwæl*
Northwest Germanic			
PGmc Final Z Deletion	**xwála*		

Old English form: *hwæl*

Reconstruction and comparative evidence

The comparative sources are not uniform. Orel gives **xwalaz* and notes some mixed **xwaliz* evidence (Orel 2003, 197). Kroonen instead cites **hwali-* (Kroonen 2013, 262).

Both notations agree on inherited initial *hw-/xw-*, but they differ in stem label. The a-stem-like input followed here is closer to Orel's notation than to Kroonen's citation form.

Old English evidence

Clark Hall lemmatizes the noun as *hwal*, and Bosworth-Toller preserves the plural *hwalas* (Clark Hall 1960, 170; Bosworth and Toller 1898, 326). The comparison form is normalized here as *hwæl* for the singular citation form with Anglo- Frisian fronting.

The plural *hwalas* remains important control evidence. It shows the same lexeme with *a* in an open syllable, beside singular *hwæl* in the closed monosyllable.

Development to Old English

From **xwálaz*, final *-z* disappears and bare final *-a* is lost. Anglo-Frisian fronting then yields *æ* in the closed monosyllable, and Old English orthography writes *hwæl*.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative notations from the normalized Old English singular.

Comparative line	Candidate form	OE relation	Result
Orel's citation	**xwalaz*	same stem notation as the modeled singular line	closest comparative support for the derivational input
Kroonen's citation	**hwali-*	same initial cluster, different stem label	important comparative rival, but not the notation followed here
modeled input	**xwálaz*	regular output: *hwæl*	Old English-facing input
plural control	*hwalas*	attested open-syllable plural beside singular *hwæl*	confirms that the lexeme also preserves an *a*-vocalism branch

whine — OE hwīnan

Derivation: citation reconstruction **wainōjanį*; form followed here **xwī́naną* > *hwīnan* (early analogy).

Derivation trace

Proto input: **xwī́naną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Heavy Syllable Nasal Apocope	**xwī́nan*
[no change]	OE Secondary Nasalization	**xwī́nąn*
Northwest Germanic	OE Weak Tail Reduction	**xwī́nan*
[no change]		

Old English form: *hwīnan*

Reconstruction and comparative evidence

The citation reconstruction preserved in the header belongs to the lament-family verb seen in German *weinen* and Old English *wānian*. Kroonen instead separates Old English *hwīnan* under **hwinan-* (Kroonen 2013, 267). Orel likewise distinguishes strong **xwinanan* from weak **wainōjanan* (Orel 2003, 201). Ringe and Taylor make the same split at the Northwest Germanic level, linking Old Norse *hvina* and Old English *hwinan* to the same strong verb (Ringe and Taylor 2014, 130).

The two families also differ phonologically and morphologically. The lament family has initial *w-*, diphthongal *ai*, and weak-II morphology, whereas the verb behind Old English *hwīnan* has initial *hw-/xw-*, long *ī*, and strong-verb inflection. The derivational input **xwī́naną* therefore represents a competing comparative identification rather than a hidden cell of **wainōjaną*.

Old English evidence

Clark Hall records *hwinan* with the gloss ‘to hiss, whizz, whistle’ (Clark Hall 1960, 171). Seebold keeps the verb among the strong verbs and notes that only a present-tense attestation is directly preserved (Seebold 1970, 280).

The Old English form is normalized here as *hwīnan*. That normalization adds the usual vowel length marking to the dictionary spelling *hwinan*; it does not turn an unattested verb into a reconstructed one.

Development to Old English

Once the strong-verb input **xwī́naną* is selected, the path to Old English is straightforward. The compact trace shows heavy-syllable nasal apocope, secondary nasalization, and weak-tail reduction, after which the form surfaces as *hwīnan*.

No special paradigm maneuver is needed for this verb. The comparison is between two different Germanic verb families: the Old English form belongs with the strong verb **hwīnan-*, not with the weak lament verb.

Verb-family comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative labels that stand behind the inherited Old English forms.

Verb family / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lament-family weak verb	**wainōjaną*	comparative continuation in OE *wānian*	*wānian*	competing citation reconstruction, but not the source of *hwīnan*
selected strong verb	**xwī́naną*	regular output: *hwīnan*	*hwīnan*	exact match between derivational input and OE verb
comparative North Germanic cognate	Northwest Germanic strong verb behind ON *hvina* / OE *hwinan*	ON *hvina* / OE *hwinan*	*hwīnan*	supports the strong-verb identification

withy — OE wīþiġ

Derivation: citation reconstruction **wáiθiz*; form followed here **wī́θagą* > *wīþiġ* (early analogy).

Derivation trace

Proto input: **wī́θagą*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**wī́θægą*
[no change]	OE Heavy Syllable Nasal Apocope	**wī́θæg*
Northwest Germanic	OE Velar Palatalization	**wī́θæʤ*
[no change]	OE Unstressed AE Merger	**wī́θeʤ*
	OE Late Unstressed Ag Suffix	**wī́θiʤ*

Old English form: *wīþiġ*

Reconstruction and comparative evidence

The comparative evidence groups the word with Germanic forms of the **wīþja/ō-* or **wiþ-* type (Orel 2003, 503). That material is useful for the cognate set, but it does not by itself explain the Old English suffix of *wīþiġ*.

For Old English, the relevant point is the suffix history. Campbell's account of OE *-ig*, including forms such as *hunig*, supports an analysis in which the *-iġ* of *wīþiġ* continues a derivational **-ag-* sequence rather than a heavy ja-stem **-ij-* (Campbell 1959, sects 275, 376). The form followed here, **wī́θagą*, is thus a formation choice rather than a mere respelling of the comparative headword.

Old English evidence

Clark Hall records the noun as *wiðig* (Clark Hall 1960, 358). The form used here, *wīþiġ*, is a normalized Old English spelling with macrons and palatal ġ marked explicitly.

The relevant comparison form is therefore not a reconstructed dictionary convenience but an established Old English noun. What requires explanation is why the selected Proto-Germanic input is **wī́θagą* rather than a comparative headword of the **wīþja-* type.

Development to Old English

From **wī́θagą*, Anglo-Frisian brightening gives a fronted vowel in the suffixal syllable, and, on the Campbell analysis adopted here, the later Old English development of **-ag-* yields *-iġ* (Campbell 1959, sects 275, 376). Palatalization supplies the final *ġ*, and the full development reaches *wīþiġ*.

This derivation is regular for the form compared here. The central claim of the entry is therefore morphological: Old English *wīþiġ* belongs with an **-ag-* derivative, whereas the comparative **wīþja-* label belongs to a different way of presenting the cognate family.

Formation comparison

The comparison below sets the relevant formations side by side. It distinguishes the comparative headword from the Old English-facing formation that actually yields the attested noun.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative family label	*wáiθiz	broader cognate-set headword	OE family context	useful lexeme label, but not the direct source of *wīþiġ*
heavy ja-stem analysis	**wīþja-* type	Campbell/Adamczyk-style heavy ja-stem -*e* / zero outcome	*wīþiġ*	does not account cleanly for the OE suffix
**-ag-* derivative followed here	*wī́θagą	regular output: *wīþiġ*	wīþiġ	exact match between formation and target

world — OE weorold

Derivation: citation reconstruction **wíra-àldiz*; form followed here **wír-àldu* > *weorold* (early analogy).

Derivation trace

Proto input: **wíràldu*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

NWGmc I Lowering	**wéràldu*

Old English changes

OE Inter Stress Raising	**wéruldu*
OE Med Unstressed U Lowering	**wéroldu*
OE Back Mutation	**wéoroldu*
OE High Vowel Apocope	**wéorold*

Old English form: *weorold*

Reconstruction and comparative evidence

The word is the old compound 'age of men'. Comparative sources preserve two slightly different views of its first element. Orel and the **wira-* tradition keep the older *i*-vocalism, while Ringe and Taylor discuss the lowered form **weraldiz* and its pre-Old-English chain **weraldu* > **weruld* (Orel 2003, 501; Ringe and Taylor 2014, 341). Kluge-Seebold likewise gives the compound **wira-aldō* and explicitly includes Old English *weorold* (Kluge 2011, 981).

The derivational input **wír-àldu* therefore differs from the citation label in two ways. It keeps the older **wir-* vowel of the comparative headword, but it also presupposes the early shift of the compound into the ō-stems that Ringe and Taylor note for this lexeme (Ringe and Taylor 2014, 341). The early analogical step lies in that stem-class reassignment; the later phonological developments can then run regularly.

Old English evidence

Old English does not preserve a single isolated form. Ringe and Taylor give West Saxon *weorold* ~ *worold*, Mercian *weoruld*, Northumbrian *woruld*, and Kentish *wiarald* (Ringe and Taylor 2014, 341). Sievers-Brunner and Bright present the same wider set, including the syncopated *world* and later rounded *wurold* (Sievers 1965, sect. 113; Bright 1971, 465).

The Old English form used here is the West Saxon form *weorold*. It is an attested Old English form within that broader variant cluster, not the only form the lexeme ever shows.

Development to Old English

From **wír-àldu*, Northwest Germanic *i*-lowering gives **wér-àldu*. Inter-stress raising then changes the medial *a* to *u*, producing **wér-uldu*. In the Old English branch that unstressed *u* lowers to *o*, and back mutation yields **wéor-oldu*; final high-vowel apocope then gives *weorold*.

This sequence matches the comparative background in Ringe and Taylor's **weraldiz* > **weraldu* > **weruld* chain while preserving the **wir-* notation of the comparative label (Ringe and Taylor 2014, 341). The modeled Old English form therefore stands at the meeting point of an early stem-class reshaping and later regular sound change.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing stage chosen for the derivation.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
comparative compound with older first-element vowel	**wíra-àldiz*	citation reconstruction / lexeme label	preserves the older **wir-* tradition of the compound
literature-stage lowered compound after early stem-class shift	**weraldiz* > **weraldu* > **weruld*	Ringe-Taylor background chain to OE *weorold* ~ *worold*	explains the older comparative literature cited for the word
Old English-facing input	**wír-àldu*	regular output: *weorold*	exact match for the West Saxon form used here
broader OE variant cluster	—	*worold*, *weoruld*, *woruld*, *wiarald*, *world*	real attested comparanda that remain outside that West Saxon line

youth — OE ġeoguþ

Derivation: citation reconstruction **júgunθiz*; form followed here **júgunθ* > *ġeoguþ* (early analogy).

Derivation trace

Proto input: **júgunθ*

Earlier Germanic changes		Old English changes	
West Germanic		OE Diphthong Leveling	**jéogūθ*
[no change]		OE Unstressed Long Vowel Shortening	**jéoguθ*
Northwest Germanic			
OE Ws Palatal Glide	**jéugunθ*		
NWGmc Nasal Spirant Lengthening	**jéugūnθ*		
NWGmc Nasal Spirant Loss	**jéugūθ*		

Old English form: *ġeoguþ*

Reconstruction and comparative evidence

The wider etymological tradition reconstructs an earlier form of the word as **ju(w)unþi-* (Kroonen 2013, 316). The comparative label **júgunθiz* already stands at a later Germanic stage with *g*, and the derivational input **júgunθ* is later again: it represents the form after final *-i* has been lost.

That staging matters because Ringe and Taylor explicitly give the sequence **jugunþi* > **juguþ* > OE *geoguþ* ~ *iuguþ* (Ringe and Taylor 2014, 141). The derivational input therefore differs from the broader comparative headword because the Old English development must begin after early loss of final *-i*.

Old English evidence

The Old English noun is attested with varying spellings. Ringe and Taylor cite *geoguþ* ~ *iuguþ* (Ringe and Taylor 2014, 141). The form is normalized here as *ġeoguþ*: the initial palatal is written with *ġ*, and the attested spelling variation is treated as orthographic rather than lexical.

Nothing in the source stack suggests that a different paradigm cell should be chosen. The relevant Old English comparison form is the noun *ġeoguþ* itself.

Development to Old English

The decisive early step is the loss of final *-i* before the Old English umlaut stage. If that high vowel remained, the word would develop an over-umlauted *y*-type vowel instead of the attested form (Ringe and Taylor 2014, 141).

From **júgunθ*, the later development is regular: palatal fronting yields **jéugunθ*; nasal-spirant lengthening and loss give **jéogūθ* (Fulk 2018, 109); unstressed long-vowel shortening then produces **jéoguθ*, which surfaces as *ġeoguþ*. Campbell preserves *u* after accented *u* in forms such as *duguþ* and *munuc* (Campbell 1959, sect. 374). Brunner likewise cites *iuzuð Jugend* and *munuc Mönch* in the same environment (Sievers 1965, sect. 150.3).

Stage comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the later stages relevant to the Old English noun.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
earlier etymological headword	**ju(w)unþi-*	comparative family background	older comparative reconstruction of the lexeme
later g-bearing comparative label	**júgunθiz*	citation reconstruction / lexeme label	preserves the later Germanic stage behind the selected entry
Old English-facing input	**júgunθ*	regular output: *ġeoguþ*	exact match for the Old English form used here
full *-i* stage retained too long	**jugunþi*	expected over-umlauted *y*-type result	negative control showing why early *-i* loss must precede the OE umlaut stage

3.8 Late analogy and paradigm-cell selection

These entries involve a later paradigm-cell or analogical comparison. The citation reconstruction remains relevant to the lexeme, but the attested Old English form is best explained through a particular inflectional or analogical form rather than through the citation form alone.

ban — OE bannes

Derivation: citation reconstruction **bánną*; form followed here **bánnas* > *bannes* (late analogy).

Derivation trace

Proto input: **bánnas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**bánnæs*
[no change]	OE Unstressed AE Merger	**bánnes*
Northwest Germanic		
[no change]		

Old English form: *bannes*

Reconstruction and comparative evidence

Orel cites a bann-noun under **bannan*, while Seebold distinguishes bann-stems of both masculine and neuter type and gives Old English *gebann* as the noun reflex (Orel 2003, 35; Seebold 1970, 89). The citation reconstruction **bánną* names the lexeme, but the comparison here turns on the genitive singular **bánnas*.

That distinction matters because the analysis depends on medial, not final, gemination.

Old English evidence

Old English lexicographic evidence securely supports the noun itself. Bosworth-Toller records the noun under nominative *ge-bann*, with oblique usage such as *gebanne* (Bosworth and Toller 1898, 303). The exact unprefixed genitive *bannes* is less directly cited in the dictionaries, so it is best treated here as the regular genitive form used for comparison rather than as a dictionary headword.

Development to Old English

From **bánnas*, the geminate remains medial before the case ending and the unstressed vowel develops regularly to give *bannes*. The paradigm comparison therefore sets the genitive against nominative *ban*, the ordinary nominative form of the same noun, rather than against a directly cited genitive headword.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows why the genitive singular is the conservative cell used for the entry.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bánną*	regular output: *ban*	ban	regular nominative outcome, but not the Old English form here
genitive singular	**bánnas*	regular output: *bannes*	*bannes*	direct match for the conservative genitive

berry — OE berġes

Derivation: citation reconstruction **bázją*; form followed here **bázjas* > *berġes* (late analogy).

Derivation trace

Proto input: **bázjas*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

[no change]

Old English changes

Anglo Frisian Brightening	**bærjæs*
OE I Umlaut	**berjæs*
OE Unstressed AE Merger	**berjes*

Old English form: *berġes*

Reconstruction and comparative evidence

Kroonen reconstructs the berry noun as **basja-* ~ **bazja-* (Kroonen 2013, 54). The derivational input **bázjas* is therefore not a rival lexeme headword, but a specific genitive singular cell drawn from that paradigm.

The relevant point is that **rj* did not geminate in Proto-West Germanic. Ringe and Taylor's *here*, *herges* comparison shows the same *rj* environment in an Old English paradigm without any hidden gemination repair (Ringe and Taylor 2014, 181).

Old English evidence

Campbell cites feminine *berige* 'berry' and notes that *-j-* is retained after *r* in this type (Campbell 1959, 250). The reviewed evidence therefore supports the citation form more directly than the exact genitive *berġes*, which is best read here as the regular genitive comparison form rather than as a dictionary headword.

Development to Old English

Citation **bázją* gives *bere*, not the Old English form here. The genitive singular **bázjas*, however, gives *berġes*, with medial *-rġ-* preserved in the same way that Ringe and Taylor cite *herges* beside *here* (Ringe and Taylor 2014, 181). This points to paradigm choice rather than to an extra phonological rule.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bázją*	regular output: *bere*	*berige* / *berġe*	useful citation-form background, but not the Old English form here
genitive singular	**bázjas*	regular output: *berġes*	*berġes*	exact match for the conservative cell

bow — OE bēag

Derivation: citation reconstruction **béuganą*; form followed here **báug* > *bēag* (late analogy).

Derivation trace

Proto input: **báug*

Earlier Germanic changes	Old English changes	
West Germanic	OE Au Fronting	**báeug*
[no change]	OE Diphthong Leveling	**bēag*
Northwest Germanic		
[no change]		

Old English form: *bēag*

Reconstruction and comparative evidence

The inherited verb belongs to the class-II strong-verb family **béuganą* (Ringe and Taylor 2014, 55). Within that paradigm, however, the infinitive and the singular preterite continue different ablaut grades. The derivational input **báug* is the singular preterite cell, whereas the citation form **béuganą* is the infinitive.

Campbell's account of Old English class-II strong verbs treats the singular preterite *au* > *ēa* development as regular in this environment (Campbell 1959, 53). That is the phonological path relevant for *bēag*, whereas the analogical *ū* of the present stem belongs to the separate history behind *būgan* (Ringe and Taylor 2014, 55).

Old English evidence

Bosworth-Toller and Clark Hall both record *bēag* as a preterite form of *būgan* (Bosworth and Toller 1898, 122; Clark Hall 1960, 45). The form discussed here is therefore an attested Old English verbal form, not a reconstructed substitute for the infinitive.

The ordinary dictionary headword remains *būgan*, but the relevant comparison form for this entry is the singular preterite *bēag*. That is the paradigm cell in which the inherited **au* grade is preserved most directly.

Development to Old English

From **báug*, Anglo-Frisian fronting and the later leveling of the diphthong produce *bēag* (Campbell 1959, 53). No special analogical repair is needed for that cell. The form is the regular Old English outcome of the singular-preterite grade.

The analogical element in the wider lexeme belongs instead to the present stem seen in *būgan*. The derivational input differs from the citation form because the regular inherited pathway survives more transparently in the preterite than in the infinitive.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the regular singular preterite from the more familiar infinitival citation form.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**béuganą*	inherited present-stem history behind *būgan*	*būgan*	establishes the lexeme, but not the Old English form here
singular preterite	**báug*	regular output: *bēag*	*bēag*	exact match between input, output, and attested cell
past participial branch	participial **bugan-* type	later participial outcomes	bogen-type evidence	relevant to the paradigm, but not the clearest match for this entry

The singular preterite is the relevant comparison form. It gives a direct lautgesetzlich path to attested *bēag*, while the citation form *būgan* belongs to a paradigm whose present stem has already undergone later leveling.

cow — OE cȳ

Derivation: citation reconstruction **kōz*; form followed here **kūi* > *cȳ* (late analogy).

Derivation trace

Proto input: **kūi*

Earlier Germanic changes	Old English changes	
West Germanic	OE I Umlaut	**kȳi*
[no change]	OE High Vowel Apocope	**kȳ*
Northwest Germanic		
[no change]		

Old English form: *cȳ*

Reconstruction and comparative evidence

Kroonen reconstructs a root noun with the inherited alternation **kō-* ~ **ku-*, explicitly nom. **kōz*, obl. **kū-* (Kroonen 2013, 299). The citation form therefore belongs to the nominative singular, whereas the derivational input **kūi* belongs to the oblique stem.

Ringe and Taylor also posit a later PNWGmc nominative **kūaz* > **kūz* behind Old English *cū* (Ringe and Taylor 2014, sect. 3.1.3). That nominative history matters for the headword, but the form *cȳ* depends on the oblique **kū-* stem and a following **i*.

Old English evidence

Clark Hall lemmatizes the noun under *cū* and records gen.sg. *cū(e)*, *cȳ*, *cūs*, dat.sg. *cȳ*, nom.-acc.pl. *cȳ*, and dat.pl. *cūm* (Clark Hall 1960, 67). Ringe and Taylor likewise state that the root-noun *cū* exhibits dat.sg., nom.-acc.pl. *cȳ* < **cūi*, dat.pl. *cūm* < **cūm*, and apparently gen.sg. *cā* < **cūiz* (Ringe and Taylor 2014, sect. 6.6.1).

The dictionary headword is therefore *cū*, but *cȳ* is an established Old English paradigm form rather than a convenient reconstruction. It serves as dative singular and also as nominative-accusative plural. The genitive singular is less stable, with *cā*, *cȳ*, *cū(e)*, and *cūs* all appearing in the local source record.

Development to Old English

**kūi* is the dative singular of the oblique **kū-* stem. The following **i* triggers i-umlaut, so *ū* becomes *ȳ*, and loss of the final high vowel leaves *cȳ*. The development is **kūi* > **kȳi* > **kȳ* > *cȳ*.

This is the regular oblique-cell path recognized by Ringe and Taylor's paradigm statement (Ringe and Taylor 2014, sect. 6.6.1). It also explains why *cȳ* is the cleanest comparison form for the present entry, even though the ordinary headword is *cū*.

Paradigm comparison

A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*kōz	OE headword *cū* belongs to the nominative history of the lexeme	cū	useful background, but not the chosen comparison for *cȳ*
later generalized nominative	PNWGmc *kūaz* > kūz	inferred nominative *cū*	cū	explains the leveled headword, not the oblique target
dative singular oblique	*kūi	regular output: *cȳ*	cȳ	exact match between input, output, and paradigm cell
genitive singular oblique	*kūiz	Ringe-Taylor: apparently *cā*; Hall: *cū(e)*, *cȳ*, *cūs*	gen.sg. variable	too unstable to control the entry

The dative singular is the relevant comparison form. It gives a regular path to attested *cȳ*, while the broader Old English paradigm shows how far the oblique **kū-* grade spread beyond that one cell.

find — OE fundene

Derivation: citation reconstruction **fī́nθaną*; form followed here **fúnðanǭ* > *fundene* (late analogy).

Derivation trace

Proto input: **fúnðanǭ*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Fronting Early	**fúndænǭ*
PWGmc Dental Hardening	**fúndanǭ*	OE Unstressed Long Vowel Shortening	**fúndænæ*
Northwest Germanic		OE Unstressed AE Merger	**fúndene*
[no change]			

Old English form: *fundene*

Reconstruction and comparative evidence

The inherited verb is the strong verb **fínθaną*, continued by Old English *findan* (Ringe and Taylor 2014, 344). The form followed here, **fúnðanǭ*, belongs to the past-participial paradigm rather than to the infinitive. It represents an oblique singular form of the participle.

That distinction matters because the familiar dictionary form *funden* is not the form compared here. The derivational input instead models an attested participial form directly, rather than treating the infinitive or the ordinary dictionary headword as primary. It therefore reaches *fundene* in the form where the trace and the attested evidence match directly.

Old English evidence

Bosworth-Toller records *fundene* under *findan*, citing the form in *Beón þá* herigeata swa fundene (Bosworth and Toller 1898, 219). Clark Hall likewise preserves the participial stem in forms such as *funden* and *tō-fundennes* (Clark Hall 1960, 124).

The ordinary dictionary headword for the participle is *funden*, but the relevant comparison form for this entry is the attested oblique *fundene*. It is an Old English form in its own right, not a merely convenient probe.

Development to Old English

From **fúnðanǭ*, the participial oblique develops through regular loss and weakening of the final ending, yielding *fundene*. In that cell both the consonantism and the medial vowel history remain regular.

The broader participial paradigm then matters for interpretation. The more familiar nominative *funden* is the ordinary dictionary form, whereas the oblique form *fundene* is the attested form compared here.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the attested oblique participle from the more familiar nominative participial forms and keeps the cited Old English evidence alongside the regular derivational path.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**fínθaną*	inherited verb *findan*	*findan*	establishes the lexeme, but not the form compared here

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
nominative participial line	**fúnðanaz*	ordinary dictionary *funden* type	*funden*	important paradigm background, but not the form compared here
oblique participle compared here	**fúnðanǭ*	regular output: *fundene*	*fundene*	exact match between input, output, and attested cell

The oblique participle is the relevant comparison form. It matches the derivational input and Old English form directly, while the nominative participial headword remains a different presentation cell within the same paradigm.

fright — OE fyrhte

Derivation: citation reconstruction **furxtīn*; form followed here **fúrxtīnaz* > *fyrhte* (late analogy).

Derivation trace

Proto input: **fúrxtīnaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**fúrxtīn*
[no change]		OE I Umlaut	**fyrxtīn*
Northwest Germanic		NWGmc In Stem N Loss	**fyrxtī*
PGmc Final Z Deletion	**fúrxtīna*	OE Unstressed Long Vowel Shortening	**fyrxti*
		OE Med Unstressed I Lowering1	**fyrxte*

Old English form: *fyrhte*

Reconstruction and comparative evidence

The noun belongs to the inherited in-stem abstract **furxtīn*, the same family as Gothic *faurhtei* (Orel 2003, 120). The derivational input **fúrxtīnaz* is not a different lexeme but an oblique singular cell within that in-stem paradigm.

Ringe and Taylor treat the later nominative forms with *-u* or *-o* as analogically remodeled (Ringe and Taylor 2014, 395–96). The oblique in-stem forms therefore preserve the older history more directly. The comparison here uses that oblique line because it keeps the inherited pathway clearer than the better-known lemma forms do.

Old English evidence

Bosworth-Toller records *fyrhte* with textual attestation, and it also records nominative forms such as *fyrhtu* and *fyrhto* (Bosworth and Toller 1898, 160). Clark Hall separately preserves adjective and verb material under *fyrht* / *fyrhtan*, which helps keep the noun distinct from the adjectival family (Clark Hall 1960, 141).

The relevant comparison form is therefore the attested oblique *fyrhte*. The nominative lemma forms remain part of the Old English evidence, but the Old English form here of this entry is the oblique cell.

Development to Old English

From **fúrxtīnaz*, the oblique in-stem develops through the loss and weakening of the final ending, yielding *fyrhte*. The form compared here therefore follows the ordinary Old English reduction of the abstract ending in this paradigm.

The later nominative forms with *-u* or *-o* belong to a subsequent analogical reshaping of the paradigm. The Old English form here is earlier in that sense: it is the attested OE cell in which the inherited in-stem development remains most transparent.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the attested oblique form from the later remodeled nominative line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation in-stem headword	**furxtīn*	broader noun-class label	wider family context	useful lexeme label, but not the cell compared here
remodeled nominative line	nominative in-stem forms	*fyrhtu* / *fyrhto* type lemma forms	*fyrhtu* / *fyrhto*	genuine OE evidence, but later remodeled
selected oblique singular	**fúrxtīnaz*	regular output: *fyrhte*	*fyrhte*	exact match between input, output, and attested cell

The oblique in-stem form is the relevant comparison form. It yields attested *fyrhte* directly, while the more familiar nominative forms belong to a later analogical layer.

hammer — OE hameres

Derivation: citation reconstruction **xámaraz*; form followed here **xámaras* > *hameres* (late analogy).

Derivation trace

Proto input: **xámaras*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**xámæræs*
[no change]	OE Unstressed AE Merger	**xámeres*
Northwest Germanic		
[no change]		

Old English form: *hameres*

Reconstruction and comparative evidence

The inherited noun is the masculine a-stem **xámaraz*, reflected in Old English citation forms such as *hamor* and *hamer* (Kroonen 2013, 206; Orel 2003, 197; Clark Hall 1960, 160). The derivational input **xámaras* is the genitive singular of that same noun rather than a different lexeme.

The genitive matters because the citation tradition is already mixed in its unstressed vowel, while the oblique singular gives a cleaner comparison form. This is a cell choice within one paradigm, not a change of stem class.

Old English evidence

Bosworth-Toller directly records *hameres* in an Old English genitival phrase (Bosworth and Toller 1898, 78). Clark Hall preserves the simplex headword as *hamer* / *hamor* (Clark Hall 1960, 160).

Sievers-Brunner gives a paradigm line *hamor — hamores*, which shows that the oblique tradition itself was not entirely uniform (Sievers 1965, sect. 245). The relevant comparison form here is the attested genitive singular *hameres*.

Development to Old English

From **xámaras*, Anglo-Frisian brightening and the subsequent merger of unstressed *æ* with *e* yield *hameres*. The derivation of that oblique form is straightforward once the genitive singular cell is selected.

The noun as a whole retains a mixed citation tradition in *hamor* and *hamer*, and the selected oblique cell avoids making that variation carry the argument of the entry.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the attested genitive singular from the less stable citation tradition.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**xámaraz*	regular citation form *hamer* / *hamor*	*hamor* / *hamer*	good lexical background, but not the Old English form here
genitive singular	**xámaras*	regular output: *hameres*	*hameres*	exact match between input, output, and attested cell
later oblique tradition	oblique a-stem forms	*hamores* type evidence	*hamores*	attested background variant, but not the chosen comparison form

have — OE hæfeþ

Derivation: citation reconstruction **xabēną*; form followed here **xábēθi* > *hæfeþ* (late analogy).

Derivation trace

Proto input: **xábēθi*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**xæbǣθ*
PWGmc Early I Apocope	**xábēθ*	OE Velar Fricative Palatalization	**çæbǣθ*
Northwest Germanic		PGmc B Allophony	**çæβǣθ*
NWGmc Long E Lowering	**xábǣθ*	OE Unstressed Long Vowel Shortening	**çæβæθ*
		OE Unstressed AE Merger	**çæβeθ*

Old English form: *hæfeþ*

Reconstruction and comparative evidence

The verb belongs to the inherited class-III weak paradigm usually cited under **xabēną* and Old English *habban* (Kroonen 2013, 237; Ringe and Taylor 2014, 93). Within that paradigm, however, the infinitive and the singular present indicative do not continue the same stem. Ringe and Taylor distinguish a *-ja-* stem in the infinitive from a non-geminating -ai- / *-ē-* stem in the 2sg and 3sg present forms (Ringe and Taylor 2014, 93).

The derivational input **xábēθi* is therefore the 3sg present cell rather than a rephrasing of the infinitive. For the present analysis, that finite cell is the cleaner comparator for the inherited non-geminating stem.

Old English evidence

The ordinary Old English headword is *habban* (Clark Hall 1960, 157). Campbell's Anglian paradigm includes unsyncopated 3sg forms of the *hæfed* type, and the present paradigm therefore shows forms of the *hæf-* type that support the normalized target *hæfeþ* (Campbell 1959, sect. 762).

The target form is therefore a normalized finite cell rather than the ordinary dictionary lemma. It represents the inherited non-geminating present stem more directly than *habban* does.

Development to Old English

From **xábēθi*, the finite form yields *hæfeþ* regularly. Ringe and Taylor discuss this non-geminating present stem under *habban* (Ringe and Taylor 2014, 364). Campbell's Anglian paradigms include unsyncopated 3sg forms of the *hæfeþ* / *hæfed* type (Campbell 1959, sect. 762).

The wider lexeme is less straightforward only because the infinitive *habban* shows later leveling. That difference in paradigm history is what makes the 3sg present cell the more useful comparison form.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the analogically leveled citation form from the regular 3sg present line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*-ja-* stem of **xabēną*	citation form *habban*	*habban*	important headword, but shaped by later leveling

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
3sg present	**xábēθi*	regular output: *hæfeþ*	*hæfeþ*	exact match between input, output, and finite form compared here
syncopated finite tradition	same present stem	*hæfþ* type evidence	*hæfþ*	genuine later OE finite form, but not the normalized target used here

heaven — OE heofon

Derivation: citation reconstruction **xémenaz*; form followed here **xémonų* > *heofon* (late analogy).

Derivation trace

Proto input: **xémonų*

Earlier Germanic changes		Old English changes	
West Germanic		OE Med Unstressed U Lowering	**xéβonų*
[no change]		OE Velar Fricative Palatalization	**çéβonų*
Northwest Germanic		OE Back Mutation	**çéoβonų*
NWGmc Unstressed O Raising	**xémunų*	OE High Vowel Apocope	**çéoβon*
NWGmc Mn Dissimilation	**xéβunų*		

Old English form: *heofon*

Reconstruction and comparative evidence

The inherited noun belongs to the mn-stem family cited by Kroonen as **hemina-* ~ **hemna-* (Kroonen 2013, 220). The derivational input **xémonų* is an oblique singular form within that paradigm rather than the lexeme-level citation form **xémenaz*.

That difference matters for the West Saxon target. Ringe and Taylor give northern WGmc **hebun* > West Saxon and Northumbrian *heofon*, Mercian *heofen* (Ringe and Taylor 2014, 324). Campbell likewise gives *heofon* beside *hefen* in the same West-Saxon *u*-umlaut environment (Campbell 1959, sect. 210.1).

Old English evidence

Old English dictionaries record the standard West Saxon noun as *heofon*, alongside Anglian or Mercian *hefen* material (Clark Hall 1960, 188; Bosworth and Toller 1898, 43). Campbell also cites an earlier stage *hefzen* in the history of the word (Campbell 1959, sect. 381).

The target of this entry is the West Saxon citation form *heofon*. Its vowel history points toward the oblique stem rather than the front-vocalic nominative line.

Development to Old English

From _*xémonų_, the West Saxon line passes through the oblique-stem type reflected in northern WGmc _*hebun_ (Ringe and Taylor 2014, 324). Campbell's _heofon_ beside _hefen_ and earlier _hefzen_ show the later West-Saxon back-mutation and suffix reshaping behind _heofon_ (Campbell 1959, sect. 210.1; 1959, sect. 381).

The front-vocalic nominative line remains important as background because it explains the dialectal _hefen_ type. West Saxon _heofon_ reflects the oblique stem that was generalized into the nominative position.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the front-vocalic nominative line from the oblique stem selected for West Saxon _heofon_.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	_*xémenaz_	front-vocalic _hefen_ type outcome	_hefen_ / _heofen_	useful control, but not the West Saxon form used here
selected oblique singular	_*xémonų_	regular output: _heofon_	_heofon_	exact match between input, output, and target
older pre-OE stage	inherited oblique line	earlier _hefzen_ stage	_hefzen_	historical background for the same West Saxon development

live — OE lifeþ

Derivation: citation reconstruction _*libēną_; form followed here _*líbēθi_ > _lifeþ_ (late analogy).

Derivation trace

Proto input: _*líbēθi_

Earlier Germanic changes	
West Germanic	
PWGmc Early I Apocope	_*líbēθ_
Northwest Germanic	
NWGmc Long E Lowering	_*líbǣθ_

Old English changes	
PGmc B Allophony	_*líβǣθ_
OE Unstressed Long Vowel Shortening	_*líβæθ_
OE Unstressed AE Merger	_*líβeθ_

Old English form: _lifeþ_

Reconstruction and comparative evidence

The inherited verb belongs to the class-III weak family cited by Kroonen under _*libēn-_, reflected in Old English _libban_ (Kroonen 2013, 336). Ringe and Taylor show that the paradigm also contained a separate 3sg present stem, continued in late Northumbrian _lifed_, which they treat as an archaism (Ringe and Taylor 2014, 364).

The derivational input **líbēθi* therefore represents a finite present cell rather than the citation infinitive. That distinction matters because the ordinary later lemma tradition also includes remodeled forms such as *lifian*.

Old English evidence

The ordinary lemma tradition centers on *libban* and, in later remodeling, *lifian*. For this entry, however, the relevant comparison form is the archaic 3sg present attested as *lifed*, here normalized as *lifeþ* (Ringe and Taylor 2014, 364; Campbell 1959, sect. 762).

The target is thus a normalized finite form, not the ordinary dictionary lemma. Its value lies in preserving the older present-stem history more clearly than the remodeled lemma tradition does.

Development to Old English

From **líbēθi*, regular reduction of the final syllable and later weakening of the unstressed vowel yield *lifeþ*. The attested spelling *lifed* belongs to the same finite form in late Northumbrian orthography (Campbell 1959, sect. 762; Ringe and Taylor 2014, 364).

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the archaic finite cell from the ordinary infinitival and later remodeled lemma lines.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive line	**libēną*	OE *libban* headword tradition	*libban*	establishes the lexeme, but not the Old English form here
3sg present	**líbēθi*	regular output: *lifeþ*; attested *lifed*	*lifed*, normalized here as *lifeþ*	selected archaic finite cell
later remodeled present tradition	later class-II-type forms	*lifian* and related finite remodeling	*lifian*	genuine OE development, but secondary to the cell compared here

man — OE mannes

Derivation: citation reconstruction **mánnaz*; form followed here **mánnas* > *mannes* (late analogy).

Derivation trace

Proto input: **mánnas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**mánnæs*
[no change]	OE Unstressed AE Merger	**mánnes*
Northwest Germanic		
[no change]		

Old English form: *mannes*

Reconstruction and comparative evidence

The lexeme-level reconstruction is not uniform. Kroonen cites **mannan-*, and Orel has **mannz* (Kroonen 2013, 354; Orel 2003, 299). The derivational input **mánnas* belongs to a different level: it is the genitive-singular cell chosen for the Old English comparison.

That distinction matters because the target of this entry is not the ordinary citation form. The cell compared here is the one that keeps the geminate medial before the ending.

Old English evidence

Campbell gives the paradigm *mann*, *man* / *mannes* / *menn* (Campbell 1959, sect. 621). Sievers-Brunner likewise cites *man mannes* (Sievers 1965, sect. 226). He also explains that word-final simplification underlies forms such as *man* beside inflected *monnes* (Sievers 1965, sect. 231). Clark Hall keeps the dictionary headword under *mann* (Clark Hall 1960, 197).

The relevant comparison form is therefore the attested genitive singular *mannes*, not the citation lemma *mann*.

Development to Old English

Campbell's paradigm mann, *man* / *mannes* / *menn* confirms the selected genitive singular *mannes* (Campbell 1959, sect. 621). In the present analysis, **mánnas* develops through Anglo-Frisian brightening and later unstressed merger to *mannes*. In this cell the geminate remains medial before *-es*. The citation form behaves differently because word-final gemination was simplified in Old English (Sievers 1965, sect. 231).

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the citation-form line from the genitive singular.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mannăz*	expected citation-form outcome *man*	*mann* / *monn*	establishes the lexeme, but not the Old English form here
accusative singular	**mannął*	expected *man*	*man*	same word-final simplification as the nominative
dative singular	**mannăi*	expected *manne*	*manne*	preserves medial *nn*, but not the chosen cell
genitive singular	**mánnas*	regular output: *mannes*	*mannes*	exact match between input, output, and attested comparator

meed — OE meorde

Derivation: citation reconstruction **mizdō*; form followed here **mízdai* > *meorde* (late analogy).

Derivation trace

Proto input: **mízdai*

Earlier Germanic changes		Old English changes	
West Germanic		OE Breaking	**méordē*
PWGmc Ai Monophthongization	**mírdē*	OE Unstressed Long Vowel Shortening	**méorde*
Northwest Germanic			
NWGmc I Lowering	**mérdē*		

Old English form: *meorde*

Reconstruction and comparative evidence

The lexeme-level reconstruction is **mizdō*, but the derivational input **mízdai* is a dative-singular cell rather than the citation form. That distinction is important because the Old English evidence for the *meord* side is oblique.

The wider history of competing *mēd* remains disputed. Kroonen and Fulk explain it through some form of *z*-loss and compensatory lengthening (Kroonen 2013, 410; Fulk 2018, 69), while Orel keeps a doublet analysis (Orel 2003, 311). The comparison here concerns the attested oblique line *meorde*.

Old English evidence

The directly attested forms are obliques: *meorde* as a dative singular and *meorda* as a genitive plural (Bright 1971, 328; Bosworth and Toller 1898, 647). Lexicographers reconstruct a bare nominative *meord* from those obliques, while West Saxon prose more commonly shows the competing doublet *mēd* (Clark Hall 1960, 214; Bosworth and Toller 1898, 647).

The target of this entry is therefore the attested oblique *meorde*, not the reconstructed lemma *meord* and not the better-known West Saxon citation form *mēd*.

Development to Old English

Ringe and Taylor give the broader noun history as PGmc **mizdo* > PWGmce **mizdu* > OE *meord* ~ *méd* (Ringe and Taylor 2014, 99). The fuller oblique-cell path modeled here spells out the intermediate rhotacism, monophthongization, lowering, breaking, and unstressed-shortening steps needed for the selected dative-singular comparison.

This entry therefore follows the attested oblique line within the broader development. It does not depend on a full decision about the history of the competing *mēd* tradition.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the attested oblique target from the broader lemma history.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mizdō*	inferred lemma outcome *meord*	*meord*	useful background, but the bare lemma is reconstructed rather than directly attested

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
selected dative singular	**mízdai*	regular output: *meorde*	*meorde*	exact match between derivational input and attested target
genitive singular	**mizdōz*	regular output: *meorde*	*meorde*	converges on the same attested string, but the dat.sg. has the clearest direct support
genitive plural control	plural oblique line	attested *meorda*	*meorda*	confirms the broader oblique tradition, but not the chosen singular target

night — OE niht

Derivation: citation reconstruction **náxtz*; form followed here **náxti* > *niht* (late analogy).

Derivation trace

Proto input: **náxti*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**næxti*
[no change]	OE Breaking	**neaxti*
Northwest Germanic	OE I Umlaut	**niexti*
[no change]	OE Ws Palatal Umlaut	**nixti*
	OE High Vowel Apocope	**nixt*

Old English form: *niht*

Reconstruction and comparative evidence

Ringe and Taylor cite gen.sg. **nahtiz*, dat.sg. **nahti*, and nom.pl. **nahtiz* for the high-vowel side of the paradigm, and derive West Saxon *niht* from that side (Ringe and Taylor 2014, 240). The citation reconstruction **náxtz* therefore belongs to the nominative-like headword, while the derivational input **náxti* represents the dative-singular cell.

That distinction matters because the word later became the model for endingless datives. Ringe and Taylor explicitly explain forms such as *dœg* by analogy with dat. sg. *niht* < **nahti* (Ringe and Taylor 2014, 380).

Old English evidence

Clark Hall lemmatizes *niht* and cross-references forms such as *neaht*, *neht*, and *nieht* (Clark Hall 1960, 215). Campbell likewise preserves the fluctuation between *neaht* and *niht*, giving genitive *nihte, nihtes*, dative *niht, nihte*, nominative plural *niht*, and the contrasting plural-side forms represented by *neahtas* (Campbell 1959, sect. 628.3).

The comparison form used here is therefore an attested Old English *niht*, not a reconstructed substitute. The broader lexical record still preserves the non-umlauted side of the paradigm in *neaht*-type forms.

Development to Old English

Ringe and Taylor derive West Saxon *niht* from **nahti* via **nehti* and **neahti* (Ringe and Taylor 2014, 240). Campbell and Brunner preserve the contrasting non-umlauted *neaht*-type forms elsewhere in the paradigm (Campbell 1959, sect. 628.3; Sievers 1965, sect. 284).

The modeled path is therefore **náxti* > **næxti* > **neaxti* > **niexti* > **nixti* > *niht*.

Paradigm comparison

The comparison below sets the relevant forms side by side. It identifies the inherited cell that matches the attested Old English form.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*náxtz	expected non-umlauted outcome *neaht*	neaht	useful background, but not the comparison used for *niht*
selected dative singular	*náxti	regular output: *niht*	niht	exact match between input, output, and paradigm cell

rest — OE ræste

Derivation: citation reconstruction **rastō*; form followed here **rástōz* > *ræste* (late analogy).

Derivation trace

Proto input: **rástōz*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

PGmc Final Z Deletion	**rástō*

Old English changes

PWGmc Surviving Bimoric O Unrounding	**rástā*
Anglo Frisian Brightening	**ræstǣ*
OE Unstressed Long Vowel Shortening	**ræstæ*
OE Unstressed AE Merger	**ræste*

Old English form: *ræste*

Reconstruction and comparative evidence

Kroonen treats the noun as a feminine ō-stem **rastō-*, continued by Old English *ræst* (Kroonen 2013, 445). The form followed here, **rástōz*, therefore does not replace the lexeme-level headword. It identifies one oblique singular cell on the side of the paradigm that yields *ræste*.

The source tradition used here labels that cell specifically as genitive singular, but the broader local synthesis of the ō-stem paradigm shows that the oblique singulars converge on the same front-vocalic *ræste* side, in contrast to a nominative singular that would remain *rast*.

Old English evidence

The ordinary Old English citation form is *ræst* (Kroonen 2013, 445). Clark Hall likewise gives *ræst* (Clark Hall 1960, 239). Bosworth-Toller also preserves oblique uses of *ræste*, including prepositional examples such as on *ræste* and *tó ræste* (Bosworth and Toller 1898, 121).

The comparison form used here is therefore an attested oblique *ræste*, not a reconstructed surrogate. The dictionary headword *ræst* remains an equally real part of the Old English record.

Development to Old English

Once final *z is lost, the derivational input moves through the front-vocalic oblique side of the paradigm rather than the back-vocalic nominative side. In the modeled derivation, the surviving final long vowel is first exposed, unrounded, fronted, shortened, and then reduced to the final *-e* of *ræste*.

The key point is the paradigm split. Nominative **rastō* yields a regular *rast*, whereas the selected oblique input yields *ræste*. The later citation form *ræst* is best explained as leveling from that oblique *ræst-* stem.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the nominative citation form from the oblique singular chosen here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*rastō	expected regular outcome *rast*	ræst	useful background, but not the cell that matches attested oblique *ræste*
selected oblique singular	*rástōz	regular output: *ræste*	ræste	exact match between derivational input and attested OE oblique form

shoulder — OE sċuldrum

Derivation: citation reconstruction **skuldrō*; form followed here **skúldramiz* > *sċuldrum* (late analogy).

Derivation trace

Proto input: **skúldramiz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Sk Palatalization	**ʃúldrum*
NWGmc A To U Before M	**skúldrumiz*		
PWGmc Early I Apocope	**skúldrumz*		
Northwest Germanic			
PGmc Final Z Deletion	**skúldrum*		

Old English form: *sċuldrum*

Reconstruction and comparative evidence

The handbooks do not agree on the reconstruction of the Germanic word. Orel gives **skuldr(j)ō*, a feminine ō-/jō-stem, and explicitly notes that Old English *sculdor* is masculine beside OFrisian *skulder*, Middle Low German *schulder*, and Old High German *scultra*, *scultirra* (Orel 2003, 345).

Kroonen reconstructs **skuldra-*, a masculine a-stem, and derives the Old High German feminine forms from **skuldrjōn-* (Kroonen 2013, 478). Ringe and Taylor cite PWGmc **skuldru* for the Old English branch (Ringe and Taylor 2014, 142).

These forms imply different stem classes and different expectations for the Old English inflection. The question is which inflectional cell best aligns with the Old English evidence.

A dative/instrumental plural form **skúldramiz* aligns with the inherited plural ending that later yields Old English *-um*, and it corresponds directly to the attested dative plural discussed below.

Old English evidence

The ordinary Old English headword is *sculdor*. Clark Hall lemmatizes *sculdor* as the normal dictionary form (Clark Hall 1960, 257). Bosworth-Toller also preserves the dative plural *sculdrum* (Bosworth and Toller 1898, 85).

Bosworth-Toller's Supplement records a weak-feminine *sculdra*, an (Bosworth and Toller 1898, 699), so *sculdra* belongs to the Old English record beside the stronger masculine paradigm headed by *sculdor*. Brunner and Luick also record later spellings such as *sceoldor* and the i-mutated dative plural *scyldrum*, which reflect secondary phonological and analogical reshaping within Old English (Sievers 1965, sect. 92.2.a; Luick 1914--1940, 230).

The singular and plural evidence point to different parts of the paradigm. The relevant comparison form here is the attested dative plural *sċuldrum*. The spelling with *sċ-* is a normalized representation of the same Old English initial cluster.

Development to Old English

Proto-Germanic **skúldramiz* can be interpreted as a dative/instrumental plural form. In this environment the post-tonic *a* before *m* is raised to *u*, giving a form of the **skúldrumiz* type. Unstressed *u* is regularly preserved before *m*, especially in the dative plural ending *-um*: Campbell states this explicitly, and Hogg formulates the same condition for the dative plural inflexion (Campbell 1959, sect. 373; Hogg 1992, sect. 3.3.1.3). Brunner points in the same direction by excluding *m* from the environments in which medial *o* became general in West Saxon (Sievers 1965, sect. 44 Anm. 7).

Subsequent reduction of the ending removes the final **i* and **z*, so that the inflectional ending appears in Old English as *-um*. The initial cluster is written here as *sċ-*, and the development is **skúldramiz* > **skúldrumiz* > **skúldrum* > *sċuldrum*.

Paradigm comparison

A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
singular-oriented citation input	**skúldrō*	probe output: *sċoldor*	*sculdor*	fails: the singular output has root *o*, not the attested *u*
serious plural-based singular alternative	**skúldru*	probe output: *sċuldor*	*sculdor*	close formally, but it compares a plural-stage input with a singular form

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
dat./inst.pl. input	**skúldramiz*	regular output: *sċuldrum*	*sculdrum*	matches both the output and the dative plural comparison form
later weak-feminine singular	—	OE *sculdra*	*sculdra*	secondary doublet, useful as a control rather than the inherited target

The dative plural line is decisive because it matches both the output and the paradigm cell of Old English *sculdrum*. Singular-oriented candidates either lower the root vowel or compare unlike cells.

shove — OE sċēaf

Derivation: citation reconstruction **skéubaną*; form followed here **skáub* > *sċēaf* (late analogy).

Derivation trace

Proto input: **skáub*

Earlier Germanic changes	Old English changes	
West Germanic	OE Au Fronting	**skáeub*
[no change]	OE Diphthong Leveling	**skēab*
Northwest Germanic	PGmc B Allophony	**skēaβ*
[no change]	OE Sk Palatalization	**ʃēaβ*

Old English form: *sċēaf*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **skeuban-* ~ **skūban-* and cites Old English present forms *scēofan*, *scūfan* (Kroonen 2013, 444). Those present-system forms belong to the same verb family, but the comparison here uses the singular preterite **skáub*, not the infinitive.

Old English evidence

The ordinary dictionary verb is *scūfan*/*scēofan*, but the preterite itself is well attested. Bright gives the principal parts *scufan, sceaf, scufon, scofen* (Bright 1971, 347). Sweet gives the same paradigm (Sweet 1953, 29). The normalized form here is *sċēaf*, regularizing the attested spellings *sceaf* and prefixed *āsceaf*.

Development to Old English

From **skáub*, the development is straightforward. **au* fronts and levels to *ēa*, final **b* becomes a fricative and is written *f*, and initial **sk-* undergoes the usual Old English palatalized spelling in this environment. The derivation therefore gives **skáub* > **skáeub* > **skēab* > **skēaβ* > *sċēaf*.

Paradigm comparison

A paradigm comparison is needed here because the ordinary citation verb and *sċēaf* belong to different cells of the same strong paradigm. The comparison below is manual.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation infinitive	*skéubaną	inherited infinitive line *sċēofan*; present system also leveled *scūfan*	scēofan / scūfan	necessary background, but not the comparison used for *sċēaf*
1/3 sg. preterite	*skáub	documented regular output: *sċēaf*	sċēaf	direct match for the singular preterite
preterite plural	*skúbun	later leveled plural *scufon* beside expected *sċufun* under the corrected cascade	scufon	poorer comparison for the singular-preterite target
past participle	*skúbanaz	attested participial line *scofen*	scofen	valid alternative cell, but not the form compared here

span — OE spanne

Derivation: citation reconstruction **spannō*; form followed here **spánnai* > *spanne* (late analogy).

Derivation trace

Proto input: **spánnai*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**spánne*
PWGmc Ai Monophthongization	**spánnē*		
Northwest Germanic			
[no change]			

Old English form: *spanne*

Reconstruction and comparative evidence

Seebold gives Old English *spann* under this noun family (Seebold 1970, 450). The form followed here, **spánnai*, is therefore not a rival headword, but the specific dative singular form compared on the model of the feminine ō-stem paradigm (Sievers 1965, sect. 252; 1965, sect. 255.2).

Old English evidence

The reviewed lexicographic evidence more directly supports the citation noun *spann* than the exact form *spanne*. Clark Hall gives *spann* (Clark Hall 1960, 286), and *spanne* is accordingly treated as the regular dative singular form compared here rather than as a dictionary headword.

Development to Old English

Citation **spannō* yields *span*. The oblique cell **spánnai* therefore supplies the conservative comparison form: it preserves the medial geminate and yields *spanne*, while citation **spannō* gives the nominative background form.

Paradigm comparison

The comparison below sets the citation form beside the dative singular form compared here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*spannō	regular output: *span*	spann	useful citation-form background, but not the form compared here
dative singular compared here	*spánnai	regular output: *spanne*	spanne	exact match for that conservative form

thistle — OE þistles

Derivation: citation reconstruction **θéstilaz*; form followed here **θístilas* > *þistles* (late analogy).

Derivation trace

Proto input: **θístilas*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**θístilæs*
[no change]	OE L Adjacent Syncope	**θístlæs*
Northwest Germanic	OE Unstressed AE Merger	**θístles*
[no change]		

Old English form: *þistles*

Reconstruction and comparative evidence

Orel prints **þe(x)stilaz* for the lexeme (Orel 2003, 458). The comparative label **θéstilaz* therefore remains in view as the lexeme-level headword, while the derivational input **θístilas* is a specific genitive singular cell.

Old English evidence

The ordinary simplex headword tradition is broken *þistel* / *ðistel*. Clark Hall gives *ðistel* as the noun headword (Clark Hall 1960, 326). The Old English form here here is the genitive singular *þistles*, which preserves the same stem in an oblique form where the cluster is medial.

Development to Old English

Campbell's discussion of cluster nouns shows the contrast clearly. Simplex forms often develop a parasite vowel in word-final obstruent + sonorant clusters, while comparable medial clusters remain unbroken; his examples include *hrefn*, *tacn*, *wépn*, and *botm* beside forms with parasitic vowels elsewhere in the same lexical class (Campbell 1959, 151). The genitive singular **θístilas* therefore supplies the conservative comparison form: the cluster is medial and the regular development yields *þistles*, while the simplex nominative belongs to the broken headword tradition *þistel*.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*θéstilaz	computed output: *þistl*	þistel	useful citation-form background, but not the Old English form here
genitive singular	*θístilas	computed output: *þistles*	þistles	exact match for the conservative cell

make (iptv.2sg) — OE maca

Derivation: citation reconstruction **makōną*; form followed here **mákô* > *maca* (late analogy).

Derivation trace

Proto input: **mákô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**mækô*
[no change]	OE A Restoration	**makô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**maka*
[no change]		

Old English form: *maca*

Reconstruction and comparative evidence

The make-family belongs to the Old English class-II weak verbs. Campbell cites *lapian, macian* among verbs with restored *a* (Campbell 1959, sect. 159). Ringe and Taylor place the Germanic verb in the same class, comparing West Germanic continuants such as Old Frisian *makia*, Old Saxon *makon*, and Old High German *mahhon* (Ringe and Taylor 2014, 191).

The derivational input **mákô* is not the citation form of the lexeme but a finite paradigm cell. Ringe and Taylor give the class-II weak imperative singular as -a < **-ō*, which makes this cell the relevant comparison point for the Old English form treated here (Ringe and Taylor 2014, 314).

Old English evidence

The dictionary headword is *macian* (Clark Hall 1960, 193). The form compared here in this entry is therefore not the lemma but the imperative singular *maca*, chosen as a paradigm form beside the headword *macian* and the related finite form *macaþ*.

That distinction matters for the comparison. The lexical history of the verb is still the history of *macian*, but the finite cell isolates the regular outcome of trimoric **ō* more cleanly than the citation form does.

Development to Old English

From **mákô*, Anglo-Frisian brightening first gives **mækô*. Campbell cites *macian* among the class-II verbs with restored *a* (Campbell 1959, sect. 159). Ringe and Taylor's class-II weak imperative singular -a < **-ō* then supports the later finite ending (Ringe and Taylor 2014, 314).

The same development explains why earlier fronted forms of the *mæċa* type do not control the entry. Once trimoric **ô* is treated as a back-vocalic trigger for restoration, the imperative singular falls into line with the broader *macian* family.

Paradigm comparison

A paradigm comparison identifies which finite cell of the make-family matches the Old English form chosen here. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**mákōjaną*	comparative continuation *macian*	*macian*	ordinary headword of the verb, but not the finite form compared here
imperative singular	**mákô*	regular output: *maca*	*maca*	exact match between input, output, and selected paradigm form
present third singular companion	**mákōθi*	comparative companion *macaþ*	*macaþ*	useful family control, but not the target of this entry

make (3sg) — OE macaþ

Derivation: citation reconstruction **makōną*; form followed here **mákōθi* > *macaþ* (late analogy).

Derivation trace

Proto input: **mákōθi*

Earlier Germanic changes		Old English changes	
West Germanic		Anglo Frisian Brightening	**mækōθ*
PWGmc Early I Apocope	**mákōθ*	OE A Restoration	**makōθ*
Northwest Germanic		OE Late O Shortening	**makaθ*
[no change]			

Old English form: *macaþ*

Reconstruction and comparative evidence

Kroonen derives the Old English verb from **makōjan-* on the make-family base **maka-* (Kroonen 2013, 350). Ringe and Taylor likewise derive Old English *macian* from PWGmc **makon* through **mekojan* (Ringe and Taylor 2014, 191).

The derivational input **mákōθi* is therefore a finite 3sg cell of the same family, not the citation form of the verb.

Old English evidence

Clark Hall lemmatizes the verb as *macian* (Clark Hall 1960, 193). The relevant comparison form here is the normalized present-third-singular *macaþ*, set beside the dictionary headword and the related imperative singular *maca*.

Campbell's class-II paradigm makes the ordinary 3sg ending *-aþ*, while his dialect survey allows secondary *-e-* spellings in some traditions (Campbell 1959, sect. 356.4; 1959, sect. 757). *Macaþ* is thus the regular comparison form for the non-*j* 3sg cell.

Development to Old English

After early loss of final *-i*, **mákōθi* yields **mákōθ*. Anglo-Frisian brightening gives **mækōθ*, but Campbell lists *macian* among the class-II verbs with restored *a*, so the stem returns to *mak-* before the ending is reduced (Campbell 1959, sect. 159).

The ending then follows the ordinary class-II 3sg development. Campbell's lufas, *-aþ* (< *-ōsi*, *-ōþi)* and Ringe and Taylor's discussion of stable *a* in the finite non-*j* cells point to **makōθ* > **makaθ* > *macaþ* (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the make-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**makōjaną*	dictionary headword *macian*	*macian*	family background, but not the cell compared here
3sg present	**mákōθi*	regular output *macaþ*	*macaþ*	exact match
imperative singular companion	**mákô*	related finite form *maca*	*maca*	useful control, but not the target

bore (iptv.2sg) — OE bora

Derivation: citation reconstruction **burōną*; form followed here **búrô* > *bora* (late analogy).

Derivation trace

Proto input: **búrô*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**bóra*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**bórô*		

Old English form: *bora*

Reconstruction and comparative evidence

Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* among its continuants (Kroonen 2013, 85). Ringe and Taylor give the class-II weak imperative singular as -a < **-ō* (Ringe and Taylor 2014, 314).

The form followed here, **búrô*, is therefore an imperative cell of the same family, not the citation form of the verb.

Old English evidence

Clark Hall lemmatizes the verb as *borian* (Clark Hall 1960, 48). The comparison form here is the normalized imperative singular *bora*, used beside the headword and the related 3sg form *boraþ*.

The imperative is thus a paradigm form rather than a replacement for the dictionary lemma. It is the most direct Old English comparator for the non-*j* finite cell represented by **búrô*.

Development to Old English

Northwest Germanic lowering first gives **bórô* from **búrô*, and late shortening of the unstressed long vowel then yields **bóra*, whence *bora*.

Ringe and Taylor's class-II imperative singular -a < **-ō* points to exactly this type of outcome (Ringe and Taylor 2014, 314). The form compared here therefore isolates the regular finite-cell development more cleanly than the remodelled infinitive does.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected imperative cell from the bore-family lemma and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
imperative singular	**búrô*	regular output *bora*	*bora*	exact match
3sg present companion	**búrōθi*	related finite form *boraþ*	*boraþ*	useful control, but not the target

bore (3sg) — OE boraþ

Derivation: citation reconstruction **burōną*; form followed here **búrōθi* > *boraþ* (late analogy).

Derivation trace

Proto input: **búrōθi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Late O Shortening	**bóraθ*
PWGmc Early I Apocope	**búrōθ*		
Northwest Germanic			
NWGmc U Lowering	**bórōθ*		

Old English form: *boraþ*

Reconstruction and comparative evidence

Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* among its reflexes (Kroonen 2013, 85). The form compared here isolates the finite 3sg cell **búrōθi* rather than the infinitive.

Campbell's class-II pattern lufas, *-aþ* (< *-ōsi*, *-ōþi)* and Ringe and Taylor's account of stable *a* in the class-II 2sg and 3sg make this finite cell the relevant comparison form for the ending (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Old English evidence

Clark Hall lemmatizes the verb as *borian* (Clark Hall 1960, 48). The relevant comparison form here is the normalized present-third-singular *boraþ*, used beside the headword and the imperative singular *bora*.

Campbell's dialect survey allows secondary *-e-* and *-o-* spellings in 2sg and 3sg class-II forms, but the basic ending remains *-aþ* (Campbell 1959, sect. 757). *Boraþ* is therefore the regular comparison form for this non-*j* 3sg cell.

Development to Old English

Early loss of final *-i* first gives **búrōθ* from **búrōθi*. Northwest Germanic lowering then produces **bórōθ*, and late shortening of unstressed *ō* yields **bóraθ*, whence *boraþ*.

Campbell's class-II ending evidence and Ringe and Taylor's discussion of stable *a* in the finite non-*j* cells support exactly this sequence (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the bore-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjanq̨*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
3sg present	**búrōθi*	regular output *boraþ*	*boraþ*	exact match
imperative singular companion	**búrô*	related finite form *bora*	*bora*	useful control, but not the target

learn (iptv.2sg) — OE liorna

Derivation: citation reconstruction **liznōjaną*; form followed here **líznô* > *liorna* (late analogy).

Derivation trace

Proto input: **líznô*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

[no change]

Old English changes

OE Breaking	**líornô*
OE Unstressed Long Vowel Shortening	**líorna*

Old English form: *liorna*

Reconstruction and comparative evidence

Ringe and Taylor give Old English *liornian* ~ *leornian* from a learn-family base of the **lizn-* type (Ringe and Taylor 2014, 38), and Kroonen likewise keeps the weak verb as **liznōn-* (Kroonen 2013, 380). Fulk cites the same Old English family from **liznō-* (Fulk 2018, 127).

The derivational input **líznô* is a finite imperative cell of that family, not the citation form of the verb.

Old English evidence

Clark Hall gives the ordinary headword as *leornian* (Clark Hall 1960, 186). Brunner, however, explicitly records *leornian, nordh. auch liorna*, and Campbell notes that beside *leornian* Northumbrian forms with *io* occur where original *eo* and *io* remain distinct (Sievers 1965, sect. 417 Anm. 10; Campbell 1959, sect. 123 n. 2).

Liorna can therefore be treated as an attested Northumbrian finite form, while *leornian* remains the better-known dictionary headword.

Development to Old English

The form compared here develops regularly as **líznô* > **lírnô* by rhotacism, then **líornô* by breaking before *rn*, and finally **líorna* by late shortening of the unstressed long vowel.

Campbell's Northumbrian *io* evidence and Ringe and Taylor's explicit statement that no form of *liornian* stood in an i-umlauting environment support this stem shape (Campbell 1959, sect. 123 n. 2; Ringe and Taylor 2014, 247). The West-Saxon-looking *eo* forms belong to a different dialectal presentation of the same family.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected imperative cell from the learn-family infinitive and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**liznōjaną*	Northumbrian *liornian*; dictionary headword often *leornian*	*liornian* / *leornian*	family background, but not the cell compared here
imperative singular	**líznô*	regular output and Brunner's Northumbrian *liorna*	*liorna*	exact match
3sg present companion	**líznōθi*	related finite form *liornaþ*	*liornaþ*	useful control, but not the target

learn (3sg) — OE liornaþ

Derivation: citation reconstruction **liznōjaną*; form followed here **líznōθi* > *liornaþ* (late analogy).

Derivation trace

Proto input: **líznōθi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Breaking	**líornōθ*
PWGmc Early I Apocope	**lírnōθ*	OE Late O Shortening	**líornaθ*
Northwest Germanic			
[no change]			

Old English form: *liornaþ*

Reconstruction and comparative evidence

Ringe and Taylor give Old English *liornian* ~ *leornian* from a learn-family base of the **lizn-* type (Ringe and Taylor 2014, 38), and Kroonen likewise keeps the weak verb as **liznōn-* (Kroonen 2013, 380). The derivational input **líznōθi* is the finite 3sg cell of that family, not the citation form of the verb.

For the ending, Campbell's lufas, *-aþ* (< *-ōsi*, *-ōþi)* and Ringe and Taylor's discussion of stable *a* in the class-II 2sg and 3sg make the non-*j* 3sg cell the relevant comparison point (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Old English evidence

Clark Hall gives the ordinary headword as *leornian* (Clark Hall 1960, 186). Brunner records Northumbrian finite forms in *liorn-*, including *liorna* and the 3sg *liornes*, beside the West-Saxon-looking *leornian* tradition (Sievers 1965, sect. 417 Anm. 10). Campbell likewise notes Northumbrian forms with *io* beside *leornian* (Campbell 1959, sect. 123 n. 2).

The relevant comparison form here is the normalized 3sg *liornaþ*. The directly cited Old English evidence supports the finite stem *liorn-*; the exact *-aþ* ending follows the regular class-II 3sg pattern.

Development to Old English

The form compared here develops as **líznōθi* > **lírnōθi* by rhotacism, then **lírnōθ* after early apocope of final *-i*, then **líornōθ* by breaking before *rn*, and finally **líornaθ* > *liornaþ* by late shortening of the unstressed long vowel.

Campbell's Northumbrian *io* evidence and Ringe and Taylor's statement that no form of *liornian* stood in an i-umlauting environment support the stem, while Campbell's class-II ending evidence supports the final *-aþ* (Campbell 1959, sect. 123 n. 2; Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 247).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the learn-family infinitive and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**liznōjaną*	Northumbrian *liornian*; dictionary headword often *leornian*	*liornian* / *leornian*	family background, but not the cell compared here

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
3sg present	**líznōθi*	regular output *liornaþ*	*liornaþ*	exact match
imperative singular companion	**líznô*	regular output and Brunner's Northumbrian *liorna*	*liorna*	useful control, but not the target

lick (iptv.2sg) — OE licca

Derivation: citation reconstruction **likkōną*; form followed here **líkkô* > *licca* (late analogy).

Derivation trace

Proto input: **líkkô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Unstressed Long Vowel Shortening	**líkka*
[no change]		
Northwest Germanic		
[no change]		

Old English form: *licca*

Reconstruction and comparative evidence

Ringe and Taylor give PWGmc **li/ekkōn* continuing as Old English *liccian*, Old Saxon *likkon*, and Old High German *lecchon* (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* with the same Old English continuation (Orel 2003, 285).

Campbell's weak class-II discussion gives present forms such as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The form treated here is therefore not that remodeled infinitive but a finite cell in bare trimoric **-ō*.

Old English evidence

Bosworth-Toller lemmatizes the verb as *liccian* (Bosworth and Toller 1898, 614). Campbell cites *liccian* among Old English forms with preserved geminate *cc* (Campbell 1959, sect. 398.1). Brunner likewise cites *liccian* (Sievers 1965, sect. 45 Anm. 3). The Old English evidence therefore establishes the verbal headword and its consonantal frame securely.

The Old English form here in this entry is the imperative singular *licca*. It is a paradigm form chosen beside the headword *liccian* and the related present *liccaþ*, not a separately lemmatized citation word.

Development to Old English

With the stem *licc-* established, the remaining development is brief. Campbell's class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support late *-a* in this finite cell (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). The same stem consonantism that appears in *liccian* is preserved here, giving *cc* throughout the finite form.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian*	*liccian*	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**líkkô*	regular output *licca*	*licca*	exact match between the derivational input and the Old English form here
present third singular companion	**líkkōθi*	regular output *liccaþ*	*liccaþ*	useful family control, but not the target of this entry

lick (3sg) — OE liccaþ

Derivation: citation reconstruction **likkōną*; form followed here **líkkōθi* > *liccaþ* (late analogy).

Derivation trace

Proto input: **líkkōθi*

Earlier Germanic changes		**Old English changes**	
West Germanic		OE Late O Shortening	**líkkaθ*
PWGmc Early I Apocope	**líkkōθ*		
Northwest Germanic			
[no change]			

Old English form: *liccaþ*

Reconstruction and comparative evidence

Ringe and Taylor give PWGmc **li/ekkōn* continuing as Old English *liccian*, Old Saxon *likkon*, and Old High German *lecchon* (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* with the same Old English continuation (Orel 2003, 285).

The form compared here in this entry is the non-*j* present third singular **líkkōθi*, not the remodeled infinitive. Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80).

Old English evidence

Bosworth-Toller lemmatizes the verb as *liccian* (Bosworth and Toller 1898, 614). The same consonantal frame appears in Campbell's and Brunner's grammatical citations of *liccian* (Campbell 1959, sect. 398.1; Sievers 1965, sect. 45 Anm. 3). The Old English headword is therefore clear even though the entry here is not about the citation form.

The form treated here is the present third singular *liccaþ*. It is a selected paradigm form beside the lemma *liccian* and the related imperative *licca*, not a separately lemmatized headword.

Development to Old English

**líkkōθi* first loses final *-i*, giving **líkkōθ*. Campbell's class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support the regular 3sg outcome *-aþ* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). Because this ending never contains *-j-*, the form does not pass through an i-umlauted *-eþ* stage.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjanų*	regular output *liccian*	*liccian*	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**líkkô*	regular output *licca*	*licca*	useful family control, but not the target of this entry
present third singular	**líkkōθi*	regular output *liccaþ*	*liccaþ*	exact match between the derivational input and the Old English form here

show (iptv.2sg) — OE sċēawa

Derivation: citation reconstruction **skawōnų*; form followed here **skáwô* > *sċēawa* (late analogy).

Derivation trace

Proto input: **skáwô*

Earlier Germanic changes	Old English changes	
West Germanic	OE Aw Long Diphthong	**skḗawô*
[no change]	OE Sk Palatalization	**ʃḗawô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**ʃḗawa*
[no change]		

Old English form: *sċēawa*

Reconstruction and comparative evidence

Orel reconstructs the verb as **skawōjanan* and cites Old English *sceáwian* beside Old Frisian *skawia*, Old Saxon *skawōn*, and Old High German *scouwōn* (Orel 2003, 337). The derivational input in this entry is not that infinitive but the imperative singular **skáwô*, a finite class-II cell with imperative -a < **-ō* (Ringe and Taylor 2014, 314).

That distinction matters because the imperative singular provides the direct comparison for the Old English form treated here. The lexical history still belongs to the *sceáwian* verb, but the cell compared here isolates the finite *-a* outcome more clearly than the citation form does.

Old English evidence

Bright lists *scēawian* and explicitly gives the imperative singular *scēawa* under that headword (Bright 1971, 346). The form treated here is therefore an attested finite paradigm form, not a reconstructed convenience form.

The spelling used in this entry is normalized *sċēawa*, while Bright's glossary gives source spelling *scēawa*. The ordinary Old English headword remains *scēawian*; *sċēawa* is the imperative singular chosen beside it.

Development to Old English

Campbell lists *sceáwian* under the West Germanic **auw* developments (Campbell 1959, sect. 120). Ringe and Taylor's class-II weak imperative singular -a < **-ō* supports the late finite ending that yields *sċēawa* (Ringe and Taylor 2014, 314). The result is therefore the expected finite singular form of the *scēawian* family rather than an analogical replacement of the headword.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	regular output *sċēawian*	*scēawian*	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**skáwô*	regular output *sċēawa*	*scēawa* / normalized *sċēawa*	exact match between the derivational input and the Old English form here
present third singular companion	**skáwōθi*	regular output *sċēawaþ*	*sċēawaþ*	useful family control, but not the target of this entry

show (3sg) — OE sċēawaþ

Derivation: citation reconstruction **skawōną*; form followed here **skáwōθi* > *sċēawaþ* (late analogy).

Derivation trace

Proto input: **skáwōθi*

Earlier Germanic changes		Old English changes	
West Germanic		OE Aw Long Diphthong	**skḗawōθ*
PWGmc Early I Apocope	**skáwōθ*	OE Sk Palatalization	**ʃḗawōθ*
Northwest Germanic		OE Late O Shortening	**ʃḗawaθ*
[no change]			

Old English form: *sċēawaþ*

Reconstruction and comparative evidence

Orel reconstructs the verb as **skawōjanan* and cites Old English *sceáwian* beside Old Frisian *skawia*, Old Saxon *skawōn*, and Old High German *scouwōn* (Orel 2003, 337). The derivational input in this entry is the present third singular **skáwōθi*, a finite class-II cell with stable *a* in the 3sg ending (Ringe and Taylor 2014, 80).

Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The relevant comparison is therefore the 3sg cell itself, not an i-umlauted alternative.

Old English evidence

Bright lists the simplex headword *scēawian* and the imperative *scēawa*, and under *geond-scēawian* also records a third singular *sceawað* (Bright 1971, 383). The evidence thus establishes the *scēaw-* / *sceawað* finite-cell pattern directly.

The form written here as *sċēawaþ* is the normalized simplex comparison form for that weak class-II pattern. It is therefore not a dictionary headword but a finite comparison form aligned with the attested *scēaw-* evidence and the directly cited *sceawað* ending pattern.

Development to Old English

Campbell lists *sceáwian* under the same West Germanic **auw* development (Campbell 1959, sect. 120). **skáwōθi* therefore belongs to the *scēaw-* family before the class-II 3sg ending is applied. Campbell's chronology and Ringe and Taylor's stable-*a* discussion show that the class-II 3sg ending gives *-aþ*, not *-eþ* (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80). Because the ending never contains *-j-*, no i-umlaut applies.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	regular output *sċēawian*	*scēawian*	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**skáwô*	regular output *sċēawa*	*scēawa*	useful family control, but not the target of this entry

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
present third singular	**skáwōθi*	regular output *sċēawaþ*	normalized *sċēawaþ*; source-side pattern *sceawað*	exact match for the finite form compared here

3.9 Reconstructed Old English comparators

These entries use an explicitly reconstructed Old English-stage comparator for the branch being modelled. The relevant comparison is therefore later than the Proto-Germanic citation form but still belongs to the lexical derivation layer.

knob — OE *cnobba

Derivation: citation reconstruction **knúppaz*; form followed here **knúbbô* > **cnobba* (reconstructed Old English comparator).

Derivation trace

Proto input: **knúbbô*

Earlier Germanic changes		Old English changes	
West Germanic		OE Unstressed Long Vowel Shortening	**knóbba*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**knóbbô*		

Old English form: **cnobba*

Reconstruction and comparative evidence

The wider knob-family is not uniform. Kroonen's discussion of the Germanic n-stems points to related voiced and voiceless branches within this group (Kroonen 2011, 297). The citation reconstruction **knúppaz* represents the broader cognate-set headword, while the form followed here, **knúbbô*, represents the voiced weak-noun branch treated here.

That distinction matters because the Old English record is uneven. The better attested OE material belongs to the voiceless branch, but the present entry represents the reconstructed OE form that would continue the voiced branch behind later English knob.

Old English evidence

Clark Hall preserves Old English evidence of the *cnoppa* type (Clark Hall 1960, 79). Those forms are genuine Old English evidence, but they belong to the voiceless branch of the family.

The target **cnobba* is different in status. It is a **reconstructed Old English form**, not a directly attested one. The point of using it here is to give the voiced branch an explicit OE-stage representation instead of allowing the attested *cnoppa* branch to stand in for a different prehistory. The choice of **cnobba* is therefore a modeling and comparative decision rather than a settled point of Old English philology.

Development to Old English

From the weak-noun form followed here, **knúbbô*, the regular Old English outcome is **cnobba*, with Proto-Germanic *kn-* represented in Old English as *cn-* and with the expected weak-noun ending.

The entry therefore does not claim that **cnobba* is attested. Its claim is different: if the voiced weak-noun branch is the one to be represented, then **cnobba* is the regular Old English form corresponding to that branch.

Reconstruction status

The comparison below keeps apart the reconstructed target and the better-attested neighboring forms.

Form	Status	Relevance to this entry
**knúbbô* > **cnobba*	reconstructed OE form; regular derivation	reconstructed Old English form compared here
cnopp / *cnoppa*	attested OE branch	important control form, but belongs to the voiceless branch
cnæp	attested OE form from another family	not part of the present lexeme line

This remains the most review-sensitive item here, because the choice between reconstructed **cnobba* and attested *cnoppa* is still a comparator-policy question rather than a settled point of OE attestation.

reek — OE *rēac

Derivation: citation reconstruction **ráukiz*; form followed here **ráukaz* > **rēac* (reconstructed Old English comparator).

Derivation trace

Proto input: **ráukaz*

Earlier Germanic changes		Old English changes	
West Germanic		OE Au Fronting	**ráeuka*
[no change]		OE Diphthong Leveling	**rēaka*
Northwest Germanic		PWGmc Final Bare A Loss	**rēak*
PGmc Final Z Deletion	**ráuka*		

Old English form: **rēac*

Reconstruction and comparative evidence

The wider noun family is represented by **ráukiz* / **rauki-*, with Old English *rēc* as the attested noun reflex in the comparative dictionaries (Kroonen 2013, 446; Orel 2003, 338). The derivational input **ráukaz* is therefore not the lexeme-level headword, but the form used here for the Old English derivation.

Old English evidence

The attested noun is *rēc*, not **rēac*. Clark Hall records *rēc* as the noun and also preserves related forms such as *rēcels*; Kroonen likewise gives OE *rēc* under the noun family (Clark Hall 1960, 255; Kroonen 2013, 446). Clark Hall and Seebold also record verbal *rēac* as the preterite of *rēocan*, but that verbal form is separate from the noun treated here (Clark Hall 1960, 254; Seebold 1970, 380).

The Old English form here **rēac* is therefore a reconstructed West Saxon noun form, not a directly attested manuscript headword.

Development to Old English

From **ráukaz*, the regular West Saxon development gives **rēac*. The attested noun *rēc* belongs to the same lexical family, but reflects a later smoothed surface form rather than the regular noun target represented here.

Form note

The distinction here is between an attested noun headword *rēc* and a reconstructed regular West Saxon target **rēac*. The latter is treated as the modelling target, while the former remains philological background.

strew — OE *strīeġan

Derivation: **stráwjaną* > **strīeġan* (reconstructed Old English comparator).

Derivation trace

Proto input: **stráwjaną*

Earlier Germanic changes	Old English changes	
West Germanic	OE Awj Glide Formation	**stráujaną*
[no change]	OE Au Fronting	**stráeujaną*
Northwest Germanic	OE Diphthong Leveling	**strēajaną*
[no change]	OE Heavy Syllable Nasal Apocope	**strēajan*
	OE Secondary Nasalization	**strēająn*
	OE I Umlaut	**strīejąn*
	OE Weak Tail Reduction	**strīejan*
	OE J Strengthening After Front Diphthong	**strīeʒan*

Old English form: **strīeġan*

Reconstruction and comparative evidence

Kroonen cites the inherited weak verb as **straujan-* and gives Old English *streowian* as its dictionary continuation (Kroonen 2013, 483). Ringe and Taylor make the split within Old English explicit: the inherited class-I verb is continued by Anglian *strēgan*, while West Saxon *streowian* is a remodelled class-II verb (Ringe and Taylor 2014, sect. 6.1 n. 27).

The aw-series comparison is important here. Luick groups **strauwjan* with the same set as **hauwja-* and **kauwjan*, yielding Anglian *strēzan* beside West Saxon forms of the *hīez*, ciezan type (Luick 1914--1940, sect. 98). Fulk likewise allows an early West Saxon **striegan* directly from Proto-Germanic **straujana* (Fulk 2018, sect. 4.10 n. 1).

Old English evidence

The attested inherited Old English form is *strēgan* in Anglian. The attested West Saxon citation forms are *strewian*, *streowian*, and *strēawian*, which belong to the remodelled class-II branch (Ringe and Taylor 2014, sect. 6.1 n. 27; Campbell 1959, sect. 753.7).

The target **strīeġan* is therefore a **reconstructed Old English form**, not an attested manuscript lemma. It is the reconstructed West Saxon reflex of the inherited class-I verb, chosen to keep the inherited branch distinct from the better-attested remodelled West Saxon lemma.

Development to Old English

From **stráwjaną*, the inherited West Saxon development passes through **straujaną*, fronting and leveling to a **strēajan-* stage, i-umlaut to **strīejan*, and retention or strengthening of the glide after the front diphthong, written here as *ġ*. The resulting form is **strīeġan*.

This differs from Anglian *strēgan*, where smoothing removes the diphthongal sequence, and from West Saxon *strewian* / *streowian* / *strēawian*, where the verb has already been remodelled into class II (Fulk 2018, sect. 4.10 n. 1; Campbell 1959, sect. 753.7).

Reconstruction status

The comparison below sets the relevant forms side by side. It keeps apart the attested inherited branch, the attested remodelled branch, and the reconstructed West Saxon comparator.

Form or branch	Status	Relevance to this entry
strēgan	attested Anglian inherited class-I form	proves that the inherited verb survived into Old English
**strīeġan*	reconstructed West Saxon inherited class-I form; trace-supported	Old English form here
strewian / *streowian* / *strēawian*	attested remodelled West Saxon class-II forms	genuine OE evidence, but not the inherited branch modeled here

3.10 Known but unmodelled remodellings

These entries preserve cases where the historical remodelling is broadly understood, but the current deterministic transducer does not model that later reshaping directly.

fire — OE fȳre

Derivation: **fūri* yields regular *fȳr*; the Old English form here is *fȳre* (known but unmodelled remodelling).

Derivation trace

Proto input: **fūri*

Earlier Germanic changes		Old English changes	
West Germanic		OE I Umlaut	**fȳri*
[no change]		OE High Vowel Apocope	**fȳr*
Northwest Germanic			
[no change]			

Regular outcome: *fȳr*

Old English form: *fȳre*

Reconstruction and comparative evidence

Kroonen places the lexeme in a heteroclitic family **fōr* ~ **fun-* and explains the front-mutated West Germanic forms from an oblique form of the **fu(w)eri* type (Kroonen 2013, 151). The form followed here, **fūri*, therefore does not function as an arbitrary substitute for the headword: it represents the specific inherited cell that supplies the *i* needed for i-umlaut.

That distinction matters because the Old English target combines a regular inherited form *fȳr* with an attested analogical surface form *fȳre*.

Old English evidence

Bosworth-Toller records *fyr* as the noun 'fire' and also preserves oblique *fyre* in the Old English record (Bosworth and Toller 1898, 288). The first is the regular inherited outcome of the phonological development from the selected input; the second shows the later restoration of a final *-e* within the paradigm.

The entry therefore concerns the relation between a regular inherited oblique input and an attested Old English surface form that has undergone later morphological remodeling.

Development to Old English

From **fūri*, i-umlaut changes *ū* to *ȳ* (Hogg 1992, sect. 3.3.3.1). Subsequent loss of the final high vowel after a heavy syllable yields *fȳr* (Campbell 1959, sect. 345). The inherited phonology is complete at that point.

fȳre is later than that inherited output. Its final *-e* belongs to analogical restoration in the Old English paradigm rather than to the original Proto-Germanic ending. The form therefore remains a known but unmodelled remodelling: the deterministic phonology is regular, but the attested surface form includes later morphological rebuilding.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the lexeme-level etymological background from the inherited cell that actually produces the front-mutated form and from the later analogical surface result.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
lexeme-level heteroclitic headword	**fōr ~ *fun-*	comparative background only	fire family	explains the wider lexeme, but not the selected oblique input
inherited oblique cell	*fūri	regular output: *fȳr*	fȳr	regular inherited output from the derivational input
later analogical surface form	—	attested *fȳre* with restored *-e*	fȳre	genuine OE target, but not the direct phonological output

tap — OE tæppa

Derivation: **táppô* yields regular *tappa*; the Old English form here is *tæppa* (known but unmodelled remodelling).

Derivation trace

Proto input: **táppô*

Earlier Germanic changes	Old English changes	
West Germanic	Anglo Frisian Brightening	**tæppô*
[no change]	OE A Restoration	**tappô*
Northwest Germanic	OE Unstressed Long Vowel Shortening	**tappa*
[no change]		

Regular outcome: *tappa*

Old English form: *tæppa*

Reconstruction and comparative evidence

Orel gives the noun under **tappòn* and already connects it with Old English *tæppa* (Orel 2003, 402). The derivational input is therefore the inherited noun itself; the entry does not depend on a different lexeme-level proto or a different inherited noun cell.

Old English evidence

The Old English noun family is well attested. Orel gives *tæppa*, and Clark Hall records *tæppa* together with derivatives *tæppere* and *tæppestre* (Orel 2003, 402; Clark Hall 1960, 305). The target is therefore a real Old English noun form, not a reconstructed convenience spelling.

Development to Old English

From **táppô*, the regular inherited noun path gives *tappa*. The attested target *tæppa* therefore stands outside that regular phonological development.

The mismatch is historically intelligible, but it is not solved here by a new inherited input. A related j-verb pathway would give *teppan*, not the noun target *tæppa*. The entry accordingly remains a known but unmodelled case.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	*táppô	regular output: *tappa*	regular output, but not the target
attested OE target	—	*tæppa*	genuine target form, but analogically remodelled in the present classification
related j-verb background	*táppjaną	*teppan*	related formation, but not the noun target

3.11 Unexplained or deliberately unmodelled exceptions

These entries preserve a mismatch between the regular transducer output and the attested Old English form. They are retained as documented lexical exceptions rather than treated as evidence for further sound-change repair.

buck — OE bucc

Derivation: **búkkaz* yields regular *bocc*; the Old English form here is *bucc* (unexplained exception).

Derivation trace

Proto input: **búkkaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**bókk*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**bókkaz*		
PGmc Final Z Deletion	**bókka*		

Regular outcome: *bocc*

Old English form: *bucc*

Reconstruction and comparative evidence

Kroonen and Orel both reconstruct the word with a geminate stop, **bukkaz* (Kroonen 2013, 121; Orel 2003, 61). Orel also preserves parallel n-stem material behind Old English *bucca* (Orel 2003, 62). The derivational input therefore remains identical with the lexeme label: no alternative inherited cell accounts for the form.

Old English evidence

Old English preserves a mixed lexical picture. Campbell cites *bucca* in the exception set for this phonological environment (Campbell 1959, sect. 115). Clark Hall and Bosworth-Toller show that Old English has both *bucca* and *bucc* (Clark Hall 1960, 53; Bosworth and Toller 1898, 122). The a-stem citation form *bucc* is the target treated here, with *bucca* kept as genuine philological background from the same lexical family.

Development to Old English

From **búkkaz*, the regular inherited path gives *bocc*. That is the form expected under the ordinary lowering pattern in this environment. *bucc* therefore remains outside the deterministic phonology.

No accepted inherited cell repairs the mismatch. A high-vowel alternative would introduce i-umlaut and produce a *byċċ*-type form rather than the target. *bucc* is therefore best treated as a documented exception, not as a regular paradigm-cell survival.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**búkkaz*	regular output: *bocc*	regular output, but not the target
attested OE target	—	*bucc*	genuine target form, but unexplained in the present classification
parallel OE lexical background	—	*bucca*	related n-stem form, not the present target

fowl — OE fugol

Derivation: **fúglaz* yields regular *fogol*; the Old English form here is *fugol* (unexplained exception).

Derivation trace

Proto input: **fúglaz*

Earlier Germanic changes

West Germanic

[no change]

Northwest Germanic

NWGmc U Lowering	**fóglaz*
PGmc Final Z Deletion	**fógla*

Old English changes

PWGmc Final Bare A Loss	**fógl*
OE Epenthetic Vowel	**fógol*

Regular outcome: *fogol*

Old English form: *fugol*

Reconstruction and comparative evidence

The noun is the ordinary Germanic a-stem **fúglaz*, continued by forms such as Old Norse *fugl* and Old High German *fogal* (Kroonen 2013, 197; Orel 2003, 155). There is no stem-class or paradigm-cell dispute behind this entry. The comparative headword and the derivational input are the same.

The difficulty lies instead in the stressed root vowel. Under the regular West Germanic and Old English development, that *u* should lower before the following non-high vowel, yielding an *o*-vocalism (Ringe and Taylor 2014, 42–43; Campbell 1959, 43).

Old English evidence

Old English dictionaries record the noun as *fugol*, with variant spelling *fugel* (Bosworth and Toller 1898, 282; Clark Hall 1960, 138). The target is therefore an attested ordinary Old English noun, not a reconstructed or selectively chosen paradigm form.

The attested word already contains the crucial problem. Its medial *-o-* is the ordinary parasite vowel of Old English cluster phonology, but the root *fu-* retains *u* where the regular history predicts *fo-* (Campbell 1959, 150).

Development to Old English

From **fúglaz*, the regular cascade yields *fogol*: the root vowel lowers before the following non-high vowel (Ringe and Taylor 2014, 42–43; Campbell 1959, 43), final *z* is lost, and the cluster is resolved by the usual medial vowel (Ringe and Taylor 2014, 345; Campbell 1959, 150). That is the expected inherited outcome.

The attested Old English noun is *fugol*, not *fogol*. Luick and later handbooks treat this preservation of *u* as a small inherited residue, not as a categorical sound law (Luick 1914--1940, 148; Ringe and Taylor 2014, 47). The item therefore remains a genuine lexical exception rather than the output of a recoverable regular mechanism.

Expected and attested forms

The comparison below sets the relevant forms side by side. It distinguishes the regular prediction from the attested Old English noun.

Form	Status	Relevance to this entry
fogol	computed regular output from **fúglaz*	establishes the expected inherited outcome
fugol	attested Old English form	Old English form here; preserves unexplained root *u*
fugel	attested variant spelling	secondary spelling variant of the attested noun

The unresolved point lies only in the root vowel. The medial *-o-* is regular, but no accepted lautgesetzlich pathway has been found from **fúglaz* to attested *fugol*.

rust — OE rust

Derivation: **rústō* yields regular *rost*; the Old English form here is *rust* (unexplained exception).

Derivation trace

Proto input: **rústō*

Earlier Germanic changes		Old English changes	
West Germanic		OE High Vowel Apocope	**róst*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**róstō*		
NWGmc Final Long O Raising	**róstu*		

Regular outcome: *rost*

Old English form: *rust*

Reconstruction and comparative evidence

The comparative dictionaries do not support a single citation reconstruction uniformly. Orel cites **rustaz* sb.m./f. with Old English *rust* and Old Saxon and Old High German *rost* (Orel 2003, 308). The form **rústō* therefore stands here as a competing citation reconstruction rather than as the best-supported inherited headword.

That disagreement does not remove the central problem. Whether one starts from **rústō* or from source-supported **rustaz*, the regular citation-form history points toward *rost*, not toward the attested Old English noun.

Old English evidence

The Old English noun is attested, not reconstructed. Clark Hall gives *rūst* m. (Clark Hall 1960, 245), and Bosworth-Toller records *rúst* (? and rust) (Bosworth and Toller 1898, 677). The form is normalized here as *rust* from that attested record.

Those dictionary entries also matter morphologically. They support a masculine noun, which aligns better with Orel's **rustaz* than with the competing **rústō* preserved in the header.

Development to Old English

Under Campbell's regular lowering of stressed *u* before a following mid or low vowel, the citation-form input gives *rost*, not *rust* (Campbell 1959, sect. 115). The same lowering would also affect comparative citation-form reconstructions such as **rustaz*.

A high-vowel comparator such as instrumental-type **rústu* would yield *rust* regularly, but that does not explain the attested citation form of the noun. No accepted regular pathway from the citation form to attested *rust* has been established.

Expected and attested forms

The comparison below sets the regular inherited outcomes beside the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**rústō* > *rost*	computed regular output from one competing citation reconstruction	shows that this citation reconstruction does not reach attested *rust*
**rustaz* > *rost*	expected regular output from the source-supported citation reconstruction	shows that correcting the stem class does not solve the vowel problem
**rústu* > *rust*	regular high-vowel comparator	useful negative control, but not a defensible citation-form solution
rust	attested Old English noun, normalized from *rūst* / *rúst* / *rust*	attested Old English form; the citation-form development remains unexplained

wolf — OE wulf

Derivation: **wúlfaz* yields regular *wolf*; the Old English form here is *wulf* (unexplained exception).

Derivation trace

Proto input: **wúlfaz*

Earlier Germanic changes		Old English changes	
West Germanic		PWGmc Final Bare A Loss	**wólf*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**wólfaz*		
PGmc Final Z Deletion	**wólfa*		

Regular outcome: *wolf*

Old English form: *wulf*

Reconstruction and comparative evidence

The inherited noun is an a-stem: Kroonen gives **wulfa-*, and Ringe and Taylor list PGmc **wulfaz* among the inherited words that preserve *u* in Old English beside Old High German *wolf* (Kroonen 2013, 638; Ringe and Taylor 2014, 47). Campbell accordingly names Old English *wulf* as an exception to the regular lowering of stressed *u* before a following non-high vowel (Campbell 1959, sect. 115).

The older literature often notices that the exceptional words cluster near labials. Bülbring lists *full*, *wulle*, and *wulf* together, but he also says that the ordinary rule still gives *o* in comparable forms such as *folc* and *bolt* (Bülbring 1902, sect. 116). Luick rejects a categorical labial blocker on the same grounds and prefers a lexical or analogical account instead (Luick 1914--1940, 148).

Old English evidence

Campbell treats *wulf* as part of the exceptional *u* set (Campbell 1959, sect. 115). Sievers-Brunner notes that oblique *wulfe* continues *wulfi* or older *wulfai* (Sievers 1965, sect. 160).

That warning matters because the surviving oblique forms do not supply a clean regular route back to bare *wulf*. They belong to the same lexeme, but they do not remove the explanatory problem presented by the citation form.

Development to Old English

Under the ordinary Northwest Germanic lowering of stressed *u* before a following non-high vowel, the citation-form input would point toward *o*-vocalism (Ringe and Taylor 2014, 42–44). The compact trace shows exactly that path: **wúlfaz* > **wólfaz* > **wólfa* > *wolf*.

A high-vowel oblique input would behave differently. There the following high vowel would block the lowering of *u*, but the same environment would also trigger i-umlaut, so the regular control result would be *wylf* or *wylfe*, not bare *wulf*. The attested noun therefore remains unexplained at the citation-form level.

Expected and attested forms

The comparison below sets the regular inherited outcomes beside the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**wúlfaz > wolf*	computed regular output from the citation form	shows the regular development expected from the inherited a-stem
OHG wolf	comparative regular cognate	confirms that the *o*-vocalism is the ordinary outcome
**wúlfi / *wúlfis > wylf / wylfe*	expected high-vowel control forms	shows why oblique high-vowel cells do not solve the noun's vowel history
wulf	attested Old English noun	attested Old English form; the preservation of *u* remains unexplained

wool — OE wull

Derivation: **wúllō* yields regular *woll*; the Old English form here is *wull* (unexplained exception).

Derivation trace

Proto input: **wúllō*

Earlier Germanic changes		**Old English changes**	
West Germanic		OE High Vowel Apocope	**wóll*
[no change]			
Northwest Germanic			
NWGmc U Lowering	**wóllō*		
NWGmc Final Long O Raising	**wóllu*		

Regular outcome: *woll*

Old English form: *wull*

Reconstruction and comparative evidence

The inherited form is a feminine ō-stem **wúllō*. In the ordinary phonological history of West Germanic, stressed *u* lowers before a following non-high vowel, so the regular Old English outcome is an *o*-form. Campbell's discussion of the parallel adjective *full*, with OHG *foll* as the regular comparator, shows that the handbooks treat this as a genuine exception cluster rather than as a place where the rule itself is doubtful (Campbell 1959, sect. 115).

Bülbring likewise lists *wulle* among the traditional *u*-preserving exceptions (Bülbring 1902, sect. 116). The comparative evidence therefore establishes two things at once: the regular result should be *woll*, and Old English still has a lexical exception of the *wull* / *wulle* type.

Old English evidence

The Old English target is given here as *wull*, a normalized lexeme form. Handbook discussion often cites *wulle*, the feminine weak form of the noun (Bülbring 1902, sect. 116). Both point to the same lexical item and to the same exceptional preservation of root *u*.

The OE evidence therefore does not remove the problem. It confirms that the language has a *u*-form where the regular phonology would have produced *o*.

Development to Old English

From **wúllō*, the regular sequence is lowering of stressed *u* before a non-high vowel, followed by the ordinary later reductions of the ending. The regular outcome is therefore *woll*.

That regular derivation is not the attested Old English form. Luick rejects a simple phonological rule that would protect this word alone, and Ringe and Taylor state the larger problem plainly: we do not really know why **u* failed to lower in forms of this sort (Luick 1914--1940, 148; Ringe and Taylor 2014, sect. 2.3.1).

What remains unexplained

The comparison below sets the regular result beside the attested lexical exception.

Form	Status	Relevance to this entry
**wúllō* > *woll*	regular output	shows what the deterministic sound laws produce
wull / *wulle*	attested OE exception	attested Old English form to be recorded
high-vowel escape from another paradigm cell	unsupported for this noun	rejected because the feminine ō-stem paradigm supplies no suitable escape cell

3.12 References

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Index verborum

Old English forms

Proto-Germanic forms

Proto-West Germanic forms

Pre-Old-English and model-internal forms

Old Norse forms

Old High German forms

Old Frisian forms

Gothic forms

Old Saxon forms

Dutch forms

German forms

Modern English linguistic forms